

THE ELEMENTS OF MANAGEMENT



*Being a Plain Account of Planning, Organising, Directing and
Controlling*

T. TAN, S. NILL

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*To the one I once wanted gone at any cost,
whom I thought was the worst in the room,
only to find myself becoming the one who
wanted to protect her. RU–MGT 30 • Nazifa
Binte Kamal.*

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

This book teaches The Elements of Management one idea at a time, from the very beginning.

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

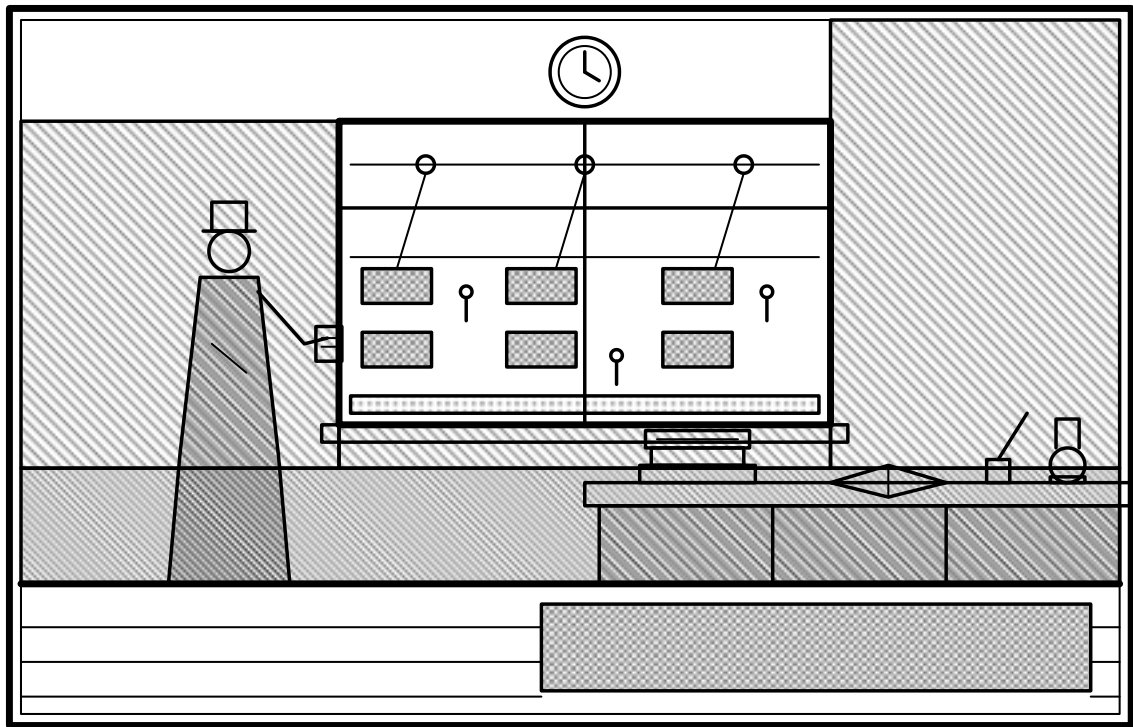
A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.



A mill manager's office, with a window over the factory floor

CHAPTER I

What Management Is

Who decides what everyone does, and who answers when the work goes badly?



This chapter introduces the central word of the book: management. It explains what management is, what its main features are, why every organisation needs it, and what a manager answers for. It also shows you how to read the rest of the book.

This chapter needs no earlier knowledge.

A family wedding shows that willing helpers are not enough.

A bookshop meets the same puzzle every day, and the answer has a name.

The book has a fixed way of teaching, and you can learn it now.

You will need: nothing. This is the first chapter, and it assumes no earlier knowledge of the subject.

Think of a family wedding. Someone must choose the date and the guest list. Someone must hire the hall, buy the food and collect the money. Many relatives are willing to help, and each one works hard. Yet on the morning of the wedding, nobody can say who decides what happens next, or who answers if the food is not enough. Willing people alone do not make the day succeed. Something more is needed.

Now picture a small bookshop with seven employees. Every day the same questions return. Which books should we order? Who serves the customers, and who counts the cash? What happens when a school asks for a large order by Friday? If nobody has the task of answering such questions, the employees work hard but not together. The shop loses time, money and customers. If someone has that task, the same seven people achieve far more. The work that answers these questions has a name: management. This chapter builds the idea slowly, one step at a time.

You will find the same signs in every chapter. A new word appears in bold, and its meaning follows at once. A short note in the side column repeats that meaning. At the end of each section, a paragraph beginning *Before moving on.* repeats the main points. Short questions marked *Pause* let you test yourself before you continue. Cases about a bookshop called Meghna and Sons are labelled *Case file, illustrative.*, because the shop is an illustration and not a real organisation. Where you see the note *Go slowly here*, read each sentence twice.

* * *

1.1 THE IDEA OF MANAGEMENT

GO SLOWLY HERE.

A household already contains the idea we need.

Management. The work of planning, organising, leading and controlling people and resources to reach an organisation's goals.

Management is the name for this careful deciding, arranging, guiding and checking.

Management. Word in use. In daily life, to manage can

We begin with something every reader knows: a household. A family has a limited income, a few members who can work, a home and some wishes. They want food on the table, school fees paid and the rent settled. Someone must decide how to spend the money. Someone must decide who does which task. Someone must also check, at the end of the month, whether the plan worked. When nobody does these things, bills are missed and tasks are repeated. When someone does them with care, the same income lasts longer.

The same work takes place in a business, a school or a hospital, and there it has a special name. **Management** is the work of planning, organising, leading and controlling people and resources to reach an organisation's goals. In a bookshop, management includes choosing which books to order and checking that the day's takings are correct.¹

You may know the verb *to manage* from daily speech, as in "she manages on a small income". That meaning is about coping. The
¹ People also use the word management for the managers as a group. This book uses *management* only for the work, and *manager* for the person.

The definition uses four verbs that you may not yet know as subject words. Here they are in plain words. *Planning* means deciding in advance what to do. *Organising* means arranging people and tasks. *Leading* means guiding and encouraging people. *Controlling* means checking results and correcting any difference from the plan. Chapter 3, 'The Functions of Management and How Management Is Judged', teaches all four. The definition also uses the words organisation and resources. The next two paragraphs explain them.

PAUSE AND CHECK 1.P1 Name the four kinds of work in the definition of management. For each one, give an example from a family household.

First, the word organisation. A roadside seller of mangoes works alone and needs no group. A bookshop with seven employees is different, because the work is too large for one person. We call such a group an **organisation**. An organisation is a group of people working together in a planned way to reach goals they could not reach alone. A school, a hospital, a bank and a village cricket club are all organisations. People also say institution or enterprise. This book uses one word, organisation, for all of them.

To reach its goals, an organisation needs things to work with. A **resource** is anything an organisation uses to reach its goals, such as people, money, materials and information. A bookshop's resources include its books, its cash, its shelves, its staff and its list of school customers. People are a resource too, though a special one, because only people can decide and act.

Some resources can be touched, such as books and shelves. Others cannot. The list of school customers is information, and the skill of the staff belongs to the people resource. A manager must think about all of them, because each one helps the organisation to reach its goals.

Now we reach the last part of the definition: managers reach goals through people. Think of a shop owner who sells books alone. The owner does the work with her own hands. Once a second person joins the shop, a new job appears. Someone must see that both people do their work and that the shop's results are good. A *result* is what the work produces, such as books sold or customers served. The person who has this job is a **manager**. A manager is a person who is responsible for the work of other people and for the results of their work. The word *responsible* has a precise meaning, which Section 1.4 explains. For now, read it as answerable. The person who directs the sales assistants and answers for the shop's sales is a manager.

This idea stands at the centre of the whole book. A manager's own hands rarely make the product or serve the customer. The manager's work is to see that other people do it well, together, and toward the goals. A good sales assistant who becomes a manager must therefore learn a new skill: succeeding through the effort of

Four verbs in the definition need a plain-word preview.

Organisation. A group of people working together in a planned way to reach goals they could not reach alone.

A group of people with a plan is an organisation.

Resource. Anything an organisation uses to reach its goals, such as people, money, materials and information.

Resources are the things an organisation works with.

Resource. Word in use. In daily life, a resource often means a natural source of wealth, such as water or coal. In management, a resource is anything the organisation uses, including people and money.

Not every resource can be touched.

Manager. A person who is responsible for the work of other people and for the results of their work.

A manager gets results through the work of other people.

Results through people is the idea that separates management from doing the work oneself.

results through other people. *You should now be able to say that management is the work of directing people and resources toward an organisation's goals.*

* * *

1.2 NATURE OF MANAGEMENT

Do a bookshop and a hospital share one kind of management?

Management is found everywhere, and it never finishes.

Management exists because people work together.

Management always aims at goals.

The table sets the four features side by side.

Is the management of a bookshop a different kind of work from the management of a hospital, a bank or a school? The question is practical. If the work were different each time, a book like this one would have little to teach. The answer is that the *nature* of management, meaning its main features, is shared. Four features appear in every organisation. We take them one at a time.

The first feature is that management is universal and continuing. *Universal* means found everywhere. A family shop, a government office, a factory in Chattogram and a hospital in Edinburgh all need someone to decide, arrange, guide and check. The details differ, and we return to that point below, but the work is of the same kind. *Continuing* means never finished. A manager does not plan once and stop. Customers change, prices change and employees change, so the manager returns again and again to the same four kinds of work, for as long as the organisation lives. Think of a household cook: the meal is eaten, and tomorrow the cook plans again. A bookshop is the same. The books ordered on Monday are sold by Friday, and on Saturday the order must be placed again.

The second feature is that management is a group activity. A person who works entirely alone has nobody to direct. A group is different. Tasks must be shared, effort must be combined, and disagreements must be settled. Management is the work that does all three. In the bookshop, the sales assistants, the cashier and the stock clerk each perform a part. Management is what joins the parts into one shop. Without it, there is a collection of individuals and not an organisation.

The third feature is that management is goal-directed. It does not exist for its own sake. A manager always works toward the goals of the organisation, such as a sales target for the year. A manager who is busy but moves toward no goal is not performing management.

The table below shows the four features together. Read each row from left to right: the feature, its meaning, and an example from a bookshop.

FOUR FEATURES OF MANAGEMENT

FEATURE	MEANING	BOOKSHOP EXAMPLE
Universal	Found in every organisation	The owner decides which books to order
Continuing	Never finished	Each new day brings new orders to place
Group activity	Needs people working together	Assistants, cashier and clerk each perform a part
Goal-directed	Aims at goals	The shop aims to sell more

after day, the continuing work of management is the reason. The owner sees all four features at once, because the owner answers for the whole shop and for every goal it has set.

Each feature has a limit. Universal does not mean identical: a hospital manager and a bookshop manager use the same four kinds of work, but the details differ, because patients are not customers. Continuing does not mean that the work is the same every day. Group activity raises a harder question. Many students find it confusing, because a one-person business seems to have no group. Such an owner leads nobody, but still decides what to buy, arranges the stall and checks the day's takings. These are management on a very small scale.

Before moving on. Management is universal, continuing, a group activity and goal-directed. It is found in every organisation, it never finishes, it joins the efforts of many people, and it aims at goals. *You should now be able to say* that the nature of management means these four features.

* * *

1.3 PURPOSE AND IMPORTANCE

Imagine two bookshops on one street. They buy from the same suppliers and charge the same prices. After a year, one shop is busy and the other must close. The books did not make the difference. The difference lay in how each shop used its resources: its people, its money, its stock and its time. This section asks two questions. What is management for, and why does it matter?

First, a bridge from daily life. Suppose a family has Tk 20,000 for one month. Rent takes Tk 8,000 and food takes Tk 9,000. Together these use Tk 17,000, and Tk 3,000 remains. The remaining money cannot pay for school books, transport and savings all at once. Someone must choose, and the choice has a result for everyone in the family. An organisation faces the same situation on a larger scale, with more people and more choices.

The purpose of management is to help the organisation reach its goals by using its resources well. Using resources well means that the organisation does not waste the time of its people, its money or its materials. A shop that orders books nobody buys wastes money. A shop where two employees perform the same task while a customer waits wastes time. Management tries to prevent both.

The next figure gives a picture of this purpose. Read it from left to right. The box on the left holds the resources. The dashed arrow below it carries the goals. Both arrows enter the middle box, where managers decide, arrange, guide and check. The thick arrow on the right leaves the middle box and reaches the results. If the results match the goals, the purpose has been met.

Every feature has a limit, and the limits prevent mistakes.

The recap repeats the four features.

Two shops, the same books and the same street can have very different futures.

A small household budget shows what limited resources force a manager to decide.

The purpose of management is to reach goals by using resources well.

The figure shows how managers turn two inputs into results.

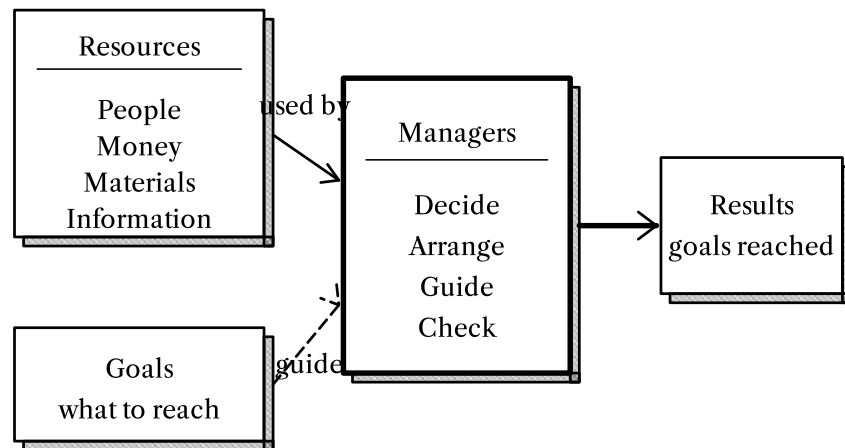


FIG. 1.1. *Managers turn resources and goals into results*

Four conditions explain why every organisation needs management.

Good management benefits owners, employees, customers and society.

The same good management looks different to the manager, the owner and the employee.

The recap repeats purpose, need and importance.

Why does every organisation need management? Four conditions explain it. First, resources are limited, so someone must decide how to use them. Second, people have different wishes, so someone must bring their efforts toward common goals. Third, conditions change, because customers, prices and rules are not fixed, so someone must watch and adjust. Fourth, many tasks depend on one another, so someone must see that they fit together. Without management, each of these four problems remains unsolved.

Importance means how much something matters. Management matters to four groups. For the owner, good management protects the money invested and helps the organisation succeed. For employees, it gives clear tasks, fair treatment and steadier work. For customers, it means goods and service of the promised quality, delivered when promised. For society, successful organisations provide jobs and useful goods. Poor management reverses each of these benefits.

Each person in the bookshop sees the benefit in a different place. The owner sees it in the year's sales and in the care of the stock. The manager sees it in a working day that runs without delay. The employee sees it in knowing, each morning, which task is his or hers. The customer sees it in the book found quickly. All four are looking at the same management from four positions. When one of these views turns bad, the others usually follow, because the work is joined together.

Before moving on. The purpose of management is to reach goals by using resources well. Organisations need it because resources are limited, people differ, conditions change and tasks depend on one another. It matters to owners, employees, customers and society. *You should now be able to say why an organisation without management would waste its resources.*

* * *

is responsible for the till must count the cash correctly and explain any difference.

A manager answers for results. It is not enough to say that the employees worked hard. If the sales target is missed, the manager must explain why, and say what will change. The manager's responsibility covers the work of other people, as the definition of a manager says.

A manager answers to owners, employees and customers. The table below shows what each group expects. Read each row as: the group, then what the manager answers for.

WHAT A MANAGER ANSWERS FOR

TO WHOM	WHAT THE MANAGER ANSWERS FOR
Owners	Results, and the care of the owners' money and goods
Employees	Clear tasks, fair treatment and pay on time
Customers	Goods and service as promised

Responsibility looks different from the manager's own chair. The manager cannot say that a mistake belongs to the assistant who made it, because the manager answers for the assistant's work. Many people find this unfair at first. It becomes reasonable when you remember the definition: a manager is a person who gets results through other people, and the results belong to the person who directs the work.

Responsibility is not the same as authority. Authority means the right to give orders and make decisions. Chapter 14, 'Authority, Delegation and Coordination', teaches it. A manager may be given a duty without receiving the right to act. That gap causes trouble, and the later chapter explains how to avoid it.

Before moving on. Responsibility is a duty to perform a task and to answer for it. A manager answers for results to owners, employees and customers.

* * *

WORKED CASE: MEGHNA AND SONS

This worked case applies the ideas of the chapter to one documented organisation. It does not model a written answer. It shows one way of reasoning, step by step. *Situation.* Read the case file below first.

Case file, illustrative. Meghna and Sons is a family-run bookshop in Dhanmondi, Dhaka, founded on 4 April 1996 by Nurul Meghna, the owner-manager. Position on 1 January 2026: seven staff are Rafiq Meghna, store manager; Tariq Meghna, accounts and online orders; three sales assistants; one cashier; and one stock clerk. Resources: stock Tk 3,500,000; fittings Tk 600,000; seven staff. The sales target for 2026 is Tk 15,840,000, which is 10% above the 2025 sales of Tk 14,400,000.

Question. Which parts of the case file are the organisation, the resources, the manager and the goal? How was the target reached?

A manager answers for results, and not only for effort.

A manager answers to three groups, and each expects something different.

From the manager's own side, responsibility can feel heavy, and it needs support.

Responsibility is different from authority, which later chapters explain.

The recap repeats responsibility and its three audiences.

The worked case applies the definitions to a documented shop.

The case file records the shop's people, resources and sales target.

The question asks us to identify the organisation, the resources, the manager and

Step two: the resources are the three items the file lists.

Step three: the definition of a manager decides who counts, and a title alone does not.

Step four: the goal is the sales target, and its arithmetic can be checked.

Step five: the table of Section 1.4 shows who the manager answers to in this shop.

The conclusion joins the four parts into one picture.

Three common slips are worth naming before the check.

The check confirms that the target is a sensible increase.

The summary gathers the chapter into five sentences.

Four short cases take you one step at a time from the case file to new settings.

Step 2. The resources are the stock, the fittings and the seven staff. The stock and fittings are materials, and the staff are people.

Step 3. A manager is a person who is responsible for the work of other people and for the results of their work. Nurul fits this definition, because she is the owner-manager of a shop with seven staff. Rafiq holds the title of store manager. The case file does not say which employees report to him, so we cannot apply the definition to him without more facts. A title is a clue, and the definition is the test.

Step 4. The goal is the 2026 sales target. We check how it was found. Ten per cent of Tk 14,400,000 is $\text{Tk } 14,400,000 \times 10 \div 100 = \text{Tk } 1,440,000$. Adding this to the 2025 sales gives $\text{Tk } 14,400,000 + \text{Tk } 1,440,000 = \text{Tk } 15,840,000$. The figure matches the case file.

Step 5. Use the table of Section 1.4. Nurul is both the owner and the manager, so she answers to herself as owner for the results and to the employees for fair treatment. She also answers to customers for goods and service as promised. A family shop therefore joins two roles in one person, which is common in small organisations.

Conclusion. At Meghna and Sons, management is the work of directing the seven staff, the stock and the fittings toward a sales target of Tk 15,840,000. The owner-manager is the clear manager under the definition.

Common slips. Readers often name the title as the test of a manager, forget that people are a resource, or confuse the sales target with the 2025 sales. The title is only a clue, the staff are a resource, and the target is the 2025 sales plus ten per cent.

Check. A tenth of Tk 14,400,000 is Tk 1,440,000, because dividing by ten moves the digits one place. The target is the 2025 sales plus that amount, so it is ten per cent higher. The answer is sensible.

* * *

SUMMARY

Management is the work of planning, organising, leading and controlling people and resources to reach an organisation's goals. An organisation is a group working together in a planned way. Resources are everything it uses. A manager gets results through other people and is responsible for their work. Management is universal, continuing, a group activity and goal-directed. Its purpose is to reach goals by using resources well. A manager answers to owners, employees and customers, and this duty is different from authority, which is the right to give orders.

THINK IT THROUGH

Each case below moves a little further from the shop in the case file. Use only the definitions of this chapter.

1.T1 Use the case file. Name one manager at Meghna and Sons, and list the resources the case file records.

1.T3 Why does a one-person stall still need management, although it has no employees?

1.T4 A village cricket club has members, a small fund, a ground and a wish to win its league. Name the organisation, two resources, and the person who would be the manager.

WHERE PEOPLE GO WRONG

Wrong: Management means giving orders. *Correct:* Giving orders is one small part of leading. Management also includes deciding what to do, arranging the work and checking the results.

Wrong: Only large businesses need management. *Correct:* Management is universal. A family shop, a stall and a village club all have resources and goals, so they all need it.

Wrong: Anyone whose job title includes the word manager is a manager. *Correct:* The test is the definition. A manager is responsible for the work of other people and for the results of their work. A title is only a clue.

Wrong: Responsibility and authority are the same thing. *Correct:* Responsibility is a duty to perform a task and to answer for it. Authority is the right to give orders and make decisions.

Wrong: A manager who works hard has done the job. *Correct:* A manager answers for results. Hard work that does not move the organisation toward its goals does not meet the manager's responsibility.

Five common misunderstandings are stated and corrected.

A second misunderstanding limits management to large businesses.

A third misunderstanding trusts the job title.

A fourth misunderstanding confuses a duty with a right.

A fifth misunderstanding confuses effort with results.

IN HISTORY

Robert Owen (1771–1858) was a Welsh-born manufacturer. On 1 January 1800 he became manager and part-owner of the New Lanark cotton mills in Scotland, beside the River Clyde near the town of Lanark. David Dale had founded the mills in the 1780s.

Owen did not manage only machines and output. He managed the people who worked there. He limited working hours and raised the standards of housing and cleanliness in the village. He opened a village store and a fund for the sick. He also established schools, including an infant school in 1816. When the mills stopped for four months, he continued to pay wages. His mills became famous, and visitors from many countries came to see how they were run.

Owen's reforms cost money, and some of his partners disagreed with the spending. Yet the mills continued to succeed as a business. His story shows a manager who answered for output and for people at once, and who also had to answer to the owners. It also shows that responsibility has more than one direction. A manager's duty reaches upward to owners, sideways to the work, and downward to the employees.

Robert Owen managed a mill and its people together, and the case still teaches.

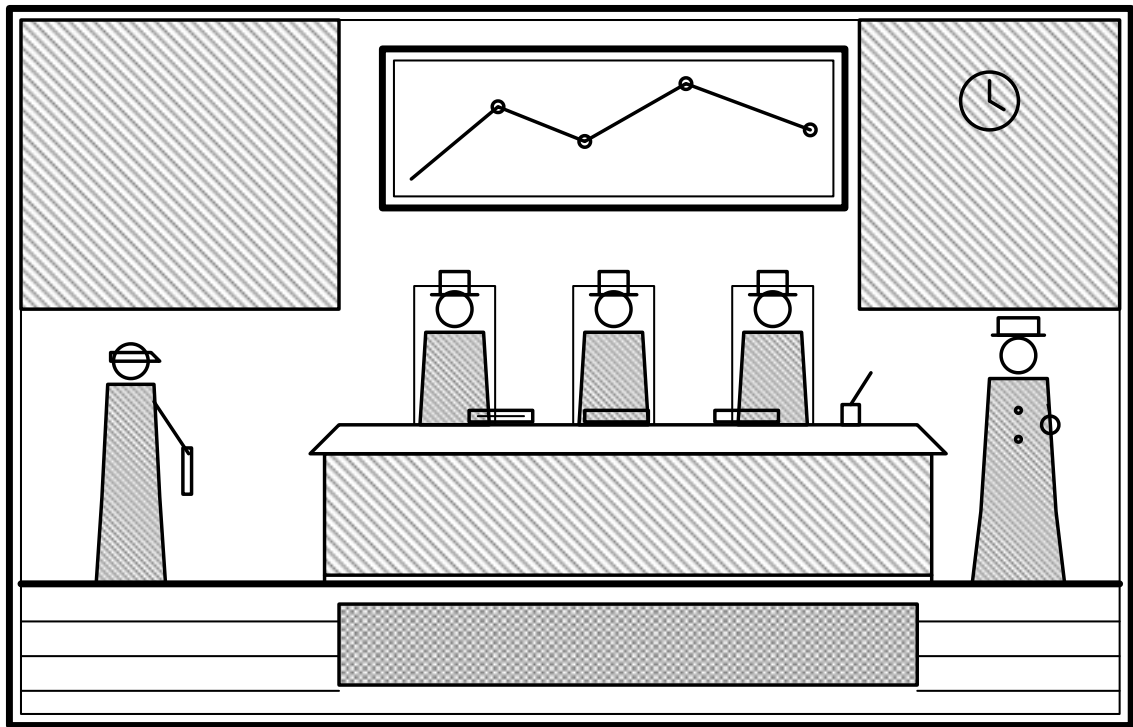
Owen managed people as well as output.

His partners disliked the cost, yet the business continued to succeed.

REVIEW QUESTIONS

1.1 Name four resources that a hospital uses, and say which kind of

Three questions return to the main ideas of the chapter.



A railway company boardroom, with a station master and a foreman standing apart

CHAPTER II

Managers and Their Skills

Does one kind of ability fit every kind of manager?



This chapter shows that managers work at different levels, and that each level calls for a different mix of abilities. It names three levels of manager, three kinds of skill, and the resources and roles that managers handle. It also asks what makes a manager effective.

This chapter builds on the first chapter.

Three jobs in one shop need three different abilities.

The puzzle asks whether managers at every level need these abilities equally.

You will need: Chapter 1 'What Management Is', section 1.1 'The Idea of Management'; Chapter 1 'What Management Is', section 1.4 'A Manager's Responsibility'. *Recall:* management, organisation and manager, Chapter 1, section 1.1; responsibility, Chapter 1, section 1.4.

Imagine the owner of a small bookshop on an ordinary Monday. A shelf has fallen and must be repaired before customers arrive. A school has asked for next year's textbooks, and the owner must decide how many copies to order. Two employees are quarrelling about who closes the till, and neither will speak to the other. Each problem belongs to management, yet each calls for a different ability. Mending a shelf needs skill with tools. Planning next year's orders needs a view of the whole shop. Settling a quarrel needs an understanding of people.

Now the puzzle. Should the owner of a large chain of bookshops spend the day mending shelves? Should the supervisor of one counter spend the day planning the chain's future? Most readers feel that the answer is no, but cannot yet say why. The reason is that managers stand at different levels, and the abilities they need change from one level to the next. This chapter explains the levels first. It then explains the abilities, and last it shows what managers handle and what they do.

* * *

2.1 LEVELS AND KINDS OF MANAGER

GO SLOWLY HERE.

A school shows three levels in a familiar setting.

Level. Word in use. In daily life, a level can mean a floor of a building or a height. In management, a level is a step in the chain of managers, from the top of the organisation to the bottom.

In this book, level means a step in the chain of managers.

Top manager. A manager at the highest level, who sets the direction of the whole organisation.

The top manager sets the direction of the whole organisation.

Middle manager. A manager between top and

We begin with a school that you know. At the top, a principal decides what kind of school it will be. Below the principal, the head of each subject turns that direction into plans for the teachers of the subject. At the bottom, teachers guide pupils in the classroom day after day. Nobody finds this strange. The school has levels, and each level has a different job. A business has the same pattern. Management is not one flat group of equals; it is arranged in steps.

In this book the word *level* always means a step in that chain. It does not describe how important or how clever a person is. A manager at a lower level may be cleverer than one above, and the level still describes only the position in the chain.

At the highest level stands the **top manager**. A top manager is a manager at the highest level, who sets the direction of the whole organisation. In a bank, the managing director is a top manager. In a small family shop, the owner-manager is one. Top managers decide what the organisation will do and what it will not do. Two levels lie below them, and the next two paragraphs name them.

Below the top stands the **middle manager**. A middle manager is a manager between top and first-line managers, who turns the direction into plans for the first-line managers. A department is a part of an organisation that does one kind of work, such as accounts or sales.

At the lowest level stands the **first-line manager**. A first-line manager is a manager who directly supervises employees doing the day-to-day work. The supervisor of the cashiers at a bank counter is a first-line manager. The people under this manager are not themselves managers, so this level is the closest to the customer and to the daily work.

One can picture the same ladder in a railway company of the 1800s. The directors, meeting in a boardroom, set the direction of the whole company. A station master directed the work of one station. A foreman supervised a gang of workers on the track. Each of these three saw a different part of the business, and each answered for a different part of its results.

Case file, illustrative. Roles at Meghna and Sons in 2026. Top: Nurul sets prices, hires and approves loans. Middle: Rafiq makes the weekly stock plan and supervises 5 staff; Tariq handles accounts and online orders. First-line work: Rafiq directly supervises 3 sales assistants, 1 cashier and 1 stock clerk.

Read the case file with care, because Rafiq appears twice. His weekly stock plan is middle-level work, because it turns the owner's direction into a plan. His direct supervision of five employees is first-line work. Many students find this confusing, because they expect each person to have one level. In a small organisation the levels are thin, and one person often covers two of them. The level belongs to the work, and not only to the person.

PAUSE AND CHECK 2.P1 In the case file, at which level does Nurul work? At which two levels does Rafiq's work stand?

Levels describe height in the chain. Managers also differ in width. A *functional manager* is in charge of one kind of work, such as accounts, sales or stock. A *general manager* is in charge of several kinds of work together, and answers for a whole unit. A department head in a large company is a functional manager, and the head of a whole branch is a general manager. At Meghna and Sons, Tariq's work lies in two specialised areas, accounts and online orders, while Nurul answers for the whole shop. Height and width are separate ideas, so a middle manager can be either kind.

The levels also look different from different viewpoints. An employee takes daily directions from a first-line manager and may rarely meet the top manager. A customer meets the first-line level most often, at the counter or on the shop floor. The owner expects the top manager to answer for the whole organisation, as Chapter 1 explained when it described responsibility.

Before moving on. A top manager sets the direction of the whole organisation. A middle manager turns the direction into plans for departments. A first-line manager directly supervises the employees who do the daily work. A functional manager looks after one kind of work, and a general manager looks after several. *You should now be able to say* what each of the three levels does.

First-line manager. A manager who directly supervises employees doing the day-to-day work.

The first-line manager directly supervises the employees who do the daily work.

A railway company of the 1800s gives a historical picture of the three levels.

The case file shows all three levels in one small shop.

One person can stand at two levels, as Rafiq does in the case file.

Managers also differ in the width of their area, apart from their level.

Each level looks different from the employee's side, the customer's side and the owner's side.

The recap repeats the three levels and the two kinds of width.

Technical skill. The ability to use the tools, methods and knowledge of a specific job.

Technical skill is the ability to do a specific job with its own tools and methods.

Human skill. The ability to work with, understand and motivate other people.

Human skill is the ability to work well with people.

Conceptual skill. The ability to see the organisation as a whole and to understand how its parts connect.

Conceptual skill is the ability to see the whole organisation and how its parts connect.

The mix of skills changes with level, but one skill stays large at every level.

Katz's pattern is easiest to see by following one person up the levels.

Step one: first-line managers need the most technical skill.

Step two: top managers need the most conceptual skill and less technical skill.

Management needs the same three kinds of ability, and Robert Katz named them in 1955.

The first kind is **technical skill**. Technical skill is the ability to use the tools, methods and knowledge of a specific job. A stock clerk who counts and shelves books quickly and correctly has technical skill. So does an accountant who keeps a correct record of money. Technical skill is mainly about working with things and methods. It can be learned by training and practice, and a person can see at once whether the work is done well.

The second kind is **human skill**. Human skill is the ability to work with, understand and motivate other people. To *motivate* means to give a person a reason to begin acting and to keep acting; Chapter 16, 'Motivation', teaches it fully. A first-line manager who notices that a tired cashier needs a short break, and who explains a new rule in a way that the cashier accepts, shows human skill. Human skill is about working with people, and not with things. It also includes listening, because a manager who does not listen cannot understand an employee's difficulty.

The third kind is **conceptual skill**. Conceptual skill is the ability to see the organisation as a whole and to understand how its parts connect. Suppose a publisher delivers books late. A manager with conceptual skill sees more than a late delivery. Late books delay school orders, late school orders damage the shop's name, and a damaged name reduces future orders. Seeing this chain of effects is the skill. It is about working with ideas and with the links among them.

PAUSE AND CHECK 2.P2 A manager notices that a new employee is afraid to ask questions, and arranges a weekly short talk with the employee. Which of the three skills does this show? Give a reason.

Now the main question of the section: does every manager need all three skills equally? Katz found that every manager needs all three, but not in the same amounts. The amounts change with the level. This is the idea of the *skill mix by level*, and we build it in four steps.

Follow one person upward to see the pattern. A cashier who counts money quickly is promoted to supervise the cashiers. Her technical skill is large and helps her at once, because she can teach and check. Later she becomes manager of the whole counter area, and then of the branch. At each step, she uses less of her own counting skill and more of her skill with people and with ideas. The cashier did not lose the skill; the work asked for a different mix.

First, consider the first-line manager. The team under this manager does the daily work, such as selling, counting or repairing. To teach a new employee, to judge whether work is good, and to help when something goes wrong, the supervisor must understand the job. A first-line manager therefore needs a large amount of technical skill, and only a small amount of conceptual skill, because the daily work is close and the supervisor can see it directly.

Second, consider the top manager. A top manager's decisions affect the whole organisation, and the top manager seldom does

each level, the results depend on people. The table below gathers the three steps. Read each row from left to right: the level, then the size of need for each skill.

SKILL NEED BY LEVEL

LEVEL	TECHNICAL SKILL	HUMAN SKILL	CONCEPTUAL SKILL
Top manager	Small	Large	Large
Middle manager	Medium	Large	Medium
First-line manager	Large	Large	Small

The figure below shows the same pattern. Read it from top to bottom. Each row is one level, drawn wider as it goes down. The three bars on the right show how much of each skill that level needs. A longer bar means a larger need. Notice that the bar for human skill has the same length in every row.

The figure shows the same pattern as bars, and it is read from top to bottom.

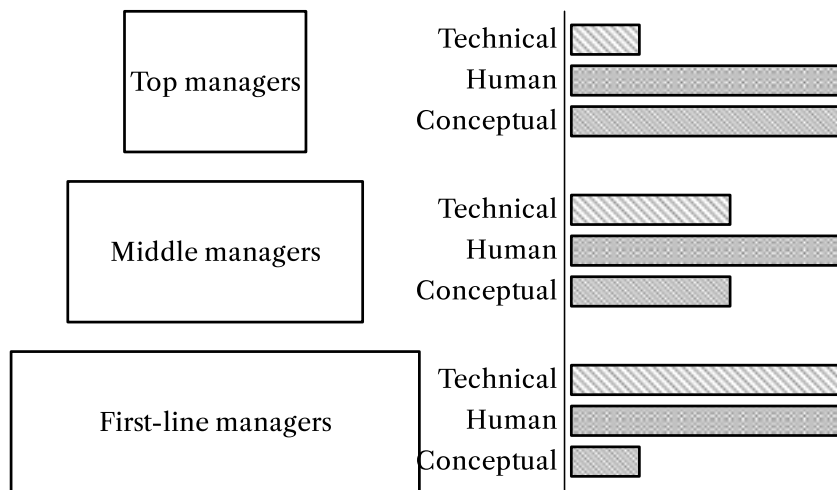


FIG. 2.1. Three levels of management and the skills each needs

IN SHORT

The need for technical skill falls as we go up the levels, the need for conceptual skill rises, and the need for human skill stays large at every level.

Fourth, the exceptions. The table shows a general pattern, and not a law. In a very small organisation, the top manager may still perform technical work every week, because there are too few people to share it. A skilled specialist who is promoted often arrives with plenty of technical skill and little human skill, and must learn the second one. Finally, skills can be learned. A person who is weak in one skill is not fixed in that weakness, and an organisation can train its managers.

Step four: the pattern has exceptions, and a beginner should know them.

Before moving on. Technical skill uses the tools and methods of a job. Human skill works with people. Conceptual skill sees the whole organisation. The mix changes with level, but human skill stays large at every level. *You should now be able to say which skill each level needs most.*

The recap repeats the three skills and the pattern by level.

2.3 MANAGERS AND RESOURCES

A manager handles resources, and the four kinds named in Chapter 1 each need different care.

Role. Word in use. In daily life, a role is a part played by an actor. In management, a role is a set of tasks that a manager is expected to perform.

A role is a set of tasks that a manager is expected to perform.

Resources and skills fit together, because each resource calls for a different skill.

Four plain roles describe most of what managers do.

A first-line manager and a top manager use the same four roles in different proportions.

Qualities describe what a person is like, and they support skills.

Chapter 1 defined a resource as anything an organisation uses to reach its goals, such as people, money, materials and information. Managers at every level handle all four kinds, though in different proportions. People need direction and fair treatment. Money needs a record and a careful use. Materials, such as stock and fittings, need storage and protection. Information, such as sales records and customer lists, needs to be accurate and to reach the right person in time. A first-line manager mostly handles people and materials. A top manager spends more time on money and information that concern the whole organisation.

Managers also perform *roles*. In this book a role is a set of tasks that a manager is expected to perform. It is a useful word, because it describes what a manager does and not the manager's title. One person can perform several roles in the same hour.

Resources and skills fit together. Handling people calls mainly for human skill. Handling stock, fittings or a record of money calls mainly for technical skill. Handling information so that it reveals how the parts connect calls mainly for conceptual skill. A manager who sees these links can decide which skill to practise when a new kind of problem appears.

Four plain roles describe most of the daily work. In the first, the manager directs people: assigns tasks, answers questions and settles disagreements. In the second, the manager deals with people outside the organisation, such as suppliers, customers and lenders. In the third, the manager handles information: receives it, checks it and passes it to those who need it. In the fourth, the manager decides how to use the organisation's resources. The mix of roles changes with level, in the same way as the mix of skills. A first-line manager spends most of the day in the first role, and a top manager spends more time in the second and fourth.

Consider two managers in one bank on the same morning. The cashier supervisor spends most of the hour in the first role, because a new employee needs help at the counter and a customer is waiting. The managing director spends the same hour in the second and fourth roles: meeting a lender, and deciding whether the bank can afford a new branch. Both managers perform all four roles during a week. The difference lies in the amount of time given to each, and that difference follows from the level.

Qualities are different from skills. A skill is what a person can do, and a quality is what a person is like. Fairness, honesty, reliability and a willingness to learn are qualities that help an effective manager. They matter because employees follow directions more readily from a manager they trust. Qualities do not replace skills, and skills do not replace qualities. A kind but unskilled manager cannot repair a failing process, and a skilled but unfair manager loses the trust of the team.

WORKED CASE: SKILLS IN ONE WEEK AT MEGHNA AND SONS

This worked case applies the skills of this chapter to one documented week. It does not model a written answer. It shows one way of reasoning, step by step. *Situation.* Read the case file below first.

Case file, illustrative. Time use of Nurul, the top manager of Meghna and Sons, in a typical week of 48 hours: 20 hours planning and dealing with publishers; 16 hours on the shop floor; 12 hours on staff and money matters.

Question. Which skill does each block of Nurul's week mainly use? Does the pattern fit the usual mix for a top manager?

Reasoning. We take the three blocks one at a time.

Step 1. Planning, and dealing with publishers, 20 hours. Planning the shop's year means seeing the whole shop and how its parts connect, so this block uses mainly conceptual skill. Dealing with publishers also needs human skill, because it means working with people outside the shop.

Step 2. The shop floor, 16 hours. Selling, shelving and serving customers are the methods of specific jobs, so this block uses mainly technical skill.

Step 3. Staff and money matters, 12 hours. Working with the staff uses human skill. Keeping a record of money uses some technical skill. We call this block mainly human skill.

Common slips. Readers often give each block of the week to one skill only. Most blocks use more than one skill, so we name the main skill and note the second. Readers also forget that the pattern for a top manager is a general pattern, and not a rule that every top manager must follow.

Step 4. The usual pattern says that a top manager needs a small amount of technical skill. Nurul spends 16 of 48 hours on technical work, which is one third, because $16 \div 48 = 1 \div 3$. This is a larger share than the pattern suggests. The reason is the size of the shop. With few employees, the top level and the shop floor overlap, in the same way that Rafiq's work reaches two levels.

Step 5. The same week can be seen from other viewpoints. The employees see the 16 hours on the shop floor as help at busy times, and as a manager who knows the work. The publishers see the 20 hours as dealing with the person who decides the shop's orders. Nurul, as owner, knows that all three blocks answer to the same duty: results for the shop. Each viewpoint shows a different side of the same week.

Conclusion. Nurul's week uses all three skills. Conceptual skill takes the largest block, as the pattern predicts for a top manager. The large technical block is an exception explained by the small size of the organisation.

Check. The three blocks add to the week: $20 + 16 + 12 = 48$ hours. Each hour has been given to one skill, so nothing is missing or counted twice.

The worked case applies the three skills and the pattern by level to one week.

The case file records how Nurul used a typical week of 48 hours.

The question asks which skill each block uses and whether the pattern fits a top manager.

The reasoning takes each block in turn.

Step one: planning and publishers call mainly for conceptual skill.

Step two: the shop floor calls for technical skill.

Step three: Staff and money matters call for human skill and some technical skill.

A common slip is to give every hour to one skill only.

Step four: we compare the week with the usual pattern for a top manager.

Step five: the week looks different from the owner's side and the employees' side.

The conclusion states what the week shows and where the pattern bends.

The check confirms that the three blocks add to the whole week.

changes with level, and human skill stays large at every level. Managers handle people, money, materials and information, and they perform several roles. Qualities such as fairness support these skills but do not replace them. In a small organisation, such as Meghna and Sons, one person may work at two levels, and the top manager may still perform technical work.

THINK IT THROUGH

Four short cases take you from the case file to new settings.

Each case below moves a little further from the case file. Use only the ideas of this chapter.

2.T1 Use the case file. At which level does Nurul work, and at which levels does Rafiq's work stand?

2.T2 Suppose Meghna and Sons opens a second shop with its own manager, who answers to Nurul and directs a small team. At which level does the new manager stand, and why?

2.T3 A bus company has a chairman, a depot manager and a driver supervisor. Place each at a level, and say which skill each one needs most.

2.T4 A sales assistant who sells books very well is promoted to supervise the other sales assistants. Which of the three skills is the new supervisor most likely to lack at first? Use Chapter 1 to explain why that skill matters.

WHERE PEOPLE GO WRONG

Five common misunderstandings are stated and corrected.

A further misunderstanding mixes up the three skills.

A second misunderstanding gives each person one fixed level.

A third misunderstanding treats level as a measure of worth.

A fourth misunderstanding treats skill as a gift.

Wrong: Top managers need no technical skill, and first-line managers need no conceptual skill. *Correct:* Every manager uses all three skills. Only the amounts differ from one level to the next.

Wrong: A person who is good with people has conceptual skill. *Correct:* The skills are different. A person can be excellent with people and still fail to see how the parts of an organisation connect.

Wrong: Each person stands at exactly one level. *Correct:* The level belongs to the work. In a small organisation, one person often performs work at two levels, as Rafiq does.

Wrong: A higher level means a more valuable person. *Correct:* Level describes a position in the chain of managers. It says nothing about how valuable or how clever a person is.

Wrong: Managers are born with their skills. *Correct:* Skills can be learned. A manager who is weak in one skill can improve it through training and practice.

IN HISTORY

Robert Katz named the three skills in 1955.

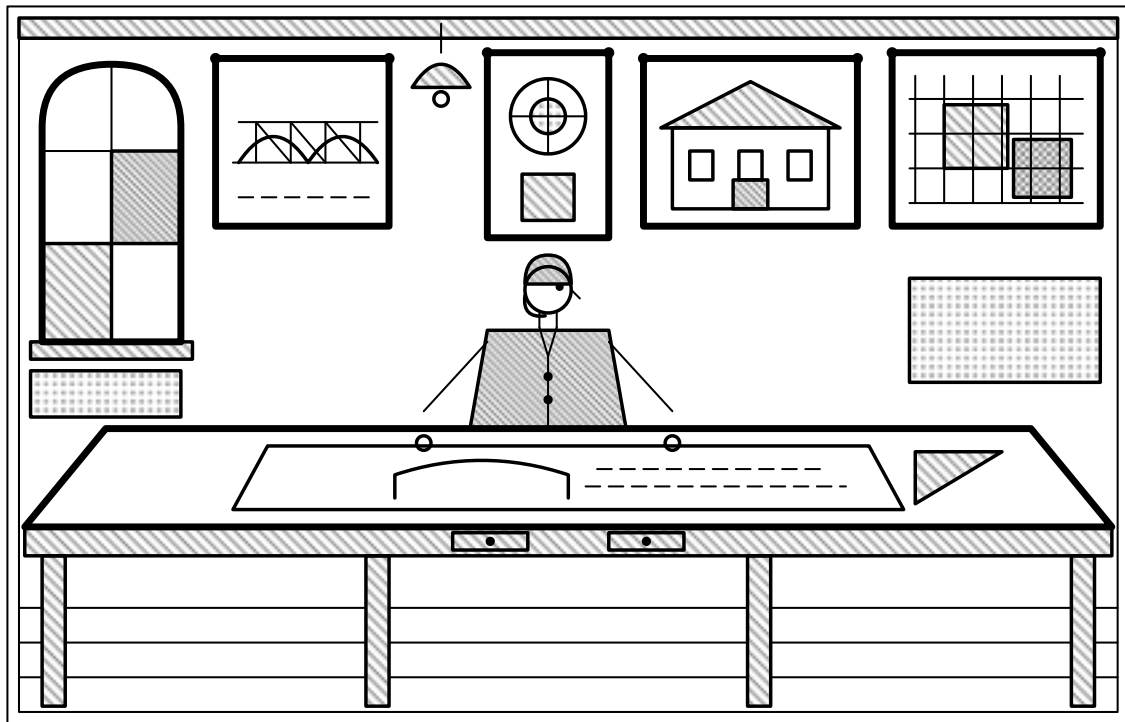
In 1955, Robert Katz published an article in the *Harvard Business Review* called "Skills of an Effective Administrator". The journal is an American magazine for business managers. Katz used the word administrator for the person; this book uses manager. He began from a view close to Chapter 1: a person who directs the work of others and answers for the results.

organisation. Human skill is needed at every level. Later teachers have kept this pattern in many books. It is the pattern that this chapter has used. The article appeared in volume 33, issue 1, of the journal, on pages 33 to 42. Its importance lies in its plain claim: a manager's success depends on what the manager can do, and what a manager can do can be taught.

REVIEW QUESTIONS

- 2.1 Place each job at a level: the head of a bank's whole network of branches; the manager of the loans department of one branch; the supervisor of the cashiers at the counter.
- 2.2 Explain why a top manager needs more conceptual skill than a first-line manager.
- 2.3 Name the four kinds of resource that a manager handles, and give a short example of each from a bookshop.
- 2.4 Explain the difference between a functional manager and a general manager, with one example of each.

Four questions return to the main ideas of the chapter.



A drawing office where every line on the wall plans was drawn before the first stone was cut

CHAPTER III

The Functions of Management and How Management Is Judged

Four kinds of work, three ways to ask: was it done well?



This chapter names the four kinds of work that make up management and shows how they repeat in a cycle. It then teaches three measures of results: productivity, effectiveness and efficiency. It ends by asking whether management is a science, an art or a profession, and by introducing the principle of management.

YOU WILL NEED

Chapter 1 "What Management Is", section 1.2 "Nature of Management"; Chapter 2 "Managers and Their Skills", section 2.1 "Levels and Kinds of Manager".

Two cooks, two different failures: which one managed better?

Each cook failed in a different way, and each way needs its own name.

Behind the feast stand four kinds of work, and a question: do they happen once, or repeat?

The chapter answers three questions: what managers do, how results are judged, and what kind of knowledge management is.

Two cooks are asked to prepare a wedding feast for 200 guests. The first cook finishes on time, and all 200 guests are served. Afterwards the kitchen shows a large loss: rice thrown away, trays burnt, and many hours of paid overtime. The second cook wastes nothing. The costs are low and the pots are clean. But the food runs out after 150 guests, and 50 guests go home hungry. Which cook managed better?

Most readers answer that neither did well. The first cook reached the goal at a very high cost. The second cook used resources with great care but did not reach the goal. These are two different failures, and a manager who cannot tell them apart will repair the wrong one. To judge management, we need exact words for the different ways in which it can succeed or fail.

A second puzzle stands behind the first. Before the feast, somebody decided the menu and the day. Somebody divided the tasks among the helpers. Somebody kept the helpers working willingly through a hot afternoon, and somebody checked the pots while the cooking was in progress. These are four different kinds of work. Do they happen once, in order, like the steps of a recipe? Or do they come round again and again?

There is also a third question. Is cooking a feast a science, with fixed rules that give the same result each time? Is it an art, learnt by practice and feel? Or is it a profession, like medicine, with a licence and a code of conduct? This chapter answers all three questions. It names the four kinds of work, gives three measures for judging results, and asks what kind of knowledge management really is.

* * *

3.1 THE MANAGEMENT PROCESS

GO SLOWLY HERE.

A family journey contains all four kinds of work, though nobody in it has studied management.

Recall: nature of management, Chapter 1, section 1.2 "Nature of Management".

Chapter 1 said what management is; this chapter says what managers do.

Before we use the technical words, consider a family travelling to a village for a wedding. Someone chooses the day, the route and the bus. Someone assigns the tasks: who buys the tickets, who packs the sweets, who looks after the grandparents. Someone keeps everyone cheerful when the bus is late. And someone counts the family at every stop, to be sure that nobody is left behind. No member of this family has studied management. Yet the family has used all four kinds of work that this section names.

In Chapter 1 you learned that management means reaching an organisation's goals through people and resources. That statement is correct, but it is broad. It does not tell us what a manager does between nine in the morning and five in the afternoon. This section divides the broad statement into four kinds of work and shows how they join together.

Together, the four kinds of work form the **management process**. The management process is the continuing cycle of planning, organising, leading and controlling that managers repeat. The word *cycle* matters: when the last kind of work ends, the first begins again. In a printing works, a process is a fixed method for one product. Here, it means this one repeating cycle of managerial work.

The four kinds of work form one repeating cycle called the management process.

Process. Word in use. In daily life, a method of making something, such as a printing process. In management, a repeating cycle of kinds of work.

Management process. The continuing cycle of planning, organising, leading and controlling that managers repeat.

Each of the four kinds of work is called a function of management.

Function of management. One of the four main kinds of work managers do: planning, organising, leading or controlling.

Each of the four kinds of work has its own name, and we call each one a **function of management**. A function of management is one of the four main kinds of work managers do: planning, organising, leading or controlling. Here, *function* means a main kind of work. It does not mean a social event, and it does not mean the rule linking two numbers in algebra.

PLANNING

Planning means choosing goals and deciding in advance how to reach them. It comes first because it gives direction to every other kind of work. The family chooses the day and the route before anyone packs a bag. A plan can be wrong, but without any plan nobody can say what the other work is for.¹

Planning means choosing goals and deciding in advance how to reach them.

Case file, illustrative. Meghna and Sons sold Tk 14,400,000 in 2025. The sales target for 2026 is Tk 15,840,000, which is 10% above that figure. Fixing this number is an act of planning, because it states the result the shop intends to reach.

The shop's 2026 sales target is an act of planning.

ORGANISING

Organising means arranging tasks, people and resources so that the plan can be followed. It answers one question: who does what, using which resources? A plan without this arrangement stays a wish. The family decides who buys the tickets and who packs the sweets, and that decision is organising.²

Organising means arranging tasks, people and resources so that the plan can be followed.

Case file, illustrative. In 2025 Meghna and Sons employed 7 people: Rafiq Meghna as store manager, Tariq Meghna for accounts and online orders, 3 sales assistants, 1 cashier and 1 stock clerk. Dividing the shop's work among these seven people is organising.

The shop's seven jobs are an act of organising.

LEADING

Leading means guiding, encouraging and influencing employees so that they work willingly toward the goals. Plans and arrangements do not carry themselves; people carry them, and people differ in energy, skill and mood. When the bus is late, the family needs

Leading means guiding and encouraging employees so that they work willingly toward the goals.

¹ Planning is taught in full in Chapter 7, "Planning: Meaning, Nature and Steps", section 7.1 "What Planning Is".

² Organising is taught in Chapter 12, "Organizing and Types of Organization", section 12.1 "What Organizing Is".

³ Leading is taught in Chapter 17, "Leadership", section 17.1 "Leader and Manager". A closely linked idea, motivation, is taught in Chapter 16, "Motivation", section 16.1 "What Motivation Is".

person. At Meghna and Sons, the sales target of Tk 15,840,000 gives a measure. When 2026 ends, actual sales can be compared with it, and any gap shows what needs correction.⁴

A CYCLE, NOT A LINE

The four functions do not run once, in a straight line, and then stop. Controlling produces information, and that information changes the next plan. In daily work the functions also overlap: a manager who holds a short meeting may be organising, leading and checking in the same ten minutes. Their weight differs by level. A top manager usually spends more time on planning, and a first-line manager more time on leading and checking daily work.

Other books use different lists, for example adding a fifth function called staffing. The four used here cover the same ground, because the work of finding and keeping employees falls within organising and leading. The list is a tool for thinking, not a law of nature.

Each person sees the cycle from a different position. The owner asks whether the cycle ends in the goals. The manager lives inside the cycle. The employee feels mostly its leading and controlling. The customer sees none of it directly and sees only the results: books on the shelf, a fair price and a correct bill.

Before moving on. The management process is the repeating cycle of four functions: planning, organising, leading and controlling. Each function is a main kind of work. They overlap and repeat, and their weight changes with the level of the manager. *You should now be able to say* what the four functions are, and why they form a cycle and not a line.

* * *

3.2 PRODUCTIVITY

GO SLOWLY HERE.

Two market stalls show why a bigger total is not always a better result.

Output is what an organisation produces; input is what it uses to produce it.

Productivity is output divided by input.

Productivity. The amount of output an organisation obtains for each unit of input it uses.

Imagine two stalls in a market. The first stall has 2 sellers and sells 40 melons in a morning. The second stall has 3 sellers and sells 54 melons. The second stall sold more, so is it better run? Many students answer yes at once and then feel unsure, because the sellers are not equal in number. The unease is correct. We must compare the results with what was used to obtain them.

We need two plain words. *Output* is what an organisation produces or sells, such as melons, loaves or books. *Input* is what it uses to obtain that output, such as people's time, money, materials and machines. Now the puzzle resolves. The first stall sold 40 melons using 2 sellers: $40 \div 2 = 20$ melons for each seller. The second sold 54 melons using 3 sellers: $54 \div 3 = 18$ melons for each seller. The smaller stall obtained more output for each unit of input.

This ratio has a name. **Productivity** is the amount of output an organisation obtains for each unit of input it uses. At the stalls, productivity is 20 melons per seller for the first and 18 for the second. In words: productivity equals output divided by input. In symbols, $P = O \div I$, where P is productivity, O is output and I is input.

⁴ Controlling is taught in Chapter 18, "Controlling", section 18.1 "What Controlling Is".

error. Use the same period for both figures. Use the same units when comparing two years.

Case file, illustrative. In 2025 Meghna and Sons sold 41,000 copies with 7 staff, which is 5,857 copies per employee. The forecast for 2026 is 45,100 copies with 7 staff, which is 6,443 copies per employee (rounded). The full arithmetic appears in the worked example later in this chapter.

The shop's productivity is forecast to rise from 5,857 to 6,443 copies per employee.

RAISING PRODUCTIVITY

Productivity can be raised in three ways. Output can grow while input stays the same, as in the shop's forecast, where copies rise and the staff remain 7. Input can fall while output stays the same. Or output can grow faster than input. A manager chooses among these ways, and each has costs. Reducing input may harm service, and growing output may need longer hours, which are themselves more input.

Productivity rises when output grows, input falls, or output grows faster than input.

Productivity has limits. It counts quantity and says nothing about quality: a shop could sell more copies by selling worn copies. It does not say whether the output is the right output. And a higher total is not the same as higher productivity. If the stall with 54 melons had hired many more sellers, its total would grow while its output for each seller fell. These limits explain why productivity is only one of three measures in this chapter.

Productivity shows quantity only, and a bigger total can hide a lower ratio.

PAUSE AND CHECK 3.P1 A shop sells 600 books in one week with 3 employees, and 800 books in the next week with 5 employees. Did productivity rise or fall?

Before moving on. Productivity is output divided by input, measured in one period with fixed units. It rises when output grows faster than input. It measures quantity and not quality. *You should now be able to say why a larger total does not always mean higher productivity.*

Productivity is output divided by input, and a larger total does not always mean a higher ratio.

* * *

3.3 EFFECTIVENESS AND EFFICIENCY

Return to the two cooks. The first reached the goal at a high cost. The second used resources carefully but missed the goal. Productivity alone cannot describe this difference. Two further words can, and they are easily confused, so we take them one at a time.

Productivity cannot separate the two cooks; two further words can.

Effectiveness is how far an organisation reaches its goals, whatever the cost. The first cook was effective: all 200 guests were served. The question of effectiveness ignores cost completely. It asks only how close the result came to the goal.

Effectiveness asks whether the goal was reached.

Effectiveness. How far an organisation reaches its goals, whatever the cost.

Efficiency is how little input an organisation uses to reach a given result. For the same result, less input means higher efficiency. The first cook, who wasted rice and paid for overtime, was not efficient. Efficiency is about the cost of reaching a result, and it says nothing about whether the right result was chosen.

Efficiency asks how little input was used to reach a given result.

Efficiency. How little input an organisation uses to reach a given result.

Case file, illustrative. Effectiveness check: the 2026 sales target is

The shop's target tests

FOUR SITUATIONS COMPARED

SITUATION	GOAL REACHED?	INPUT USED
Effective and efficient	Yes	Little
Effective, not efficient	Yes	Much, with waste
Efficient, not effective	No	Little
Neither	No	Much, with waste

Two short questions separate the words: was the goal reached, and what did it cost?

A short test separates the words. For effectiveness, ask: was the goal reached? For efficiency, ask: how much input did reaching it cost? A manager can be effective without being efficient, as the first cook was. A manager can also be efficient without being effective, as an organisation is when it does the wrong job very cheaply. Good management aims at both.

IN SHORT

Effectiveness asks whether the goal was reached. Efficiency asks how much input was used to reach it. Productivity measures output for each unit of input.

PAUSE AND CHECK 3.P2 A charity delivers all 500 food parcels it planned to deliver, but spends twice its budget. Is it effective, efficient or both?

Effectiveness is goals reached; efficiency is least input for a result.

Before moving on. Effectiveness is how far goals are reached, whatever the cost. Efficiency is how little input a result needs. The two can differ, and a manager needs to judge each.

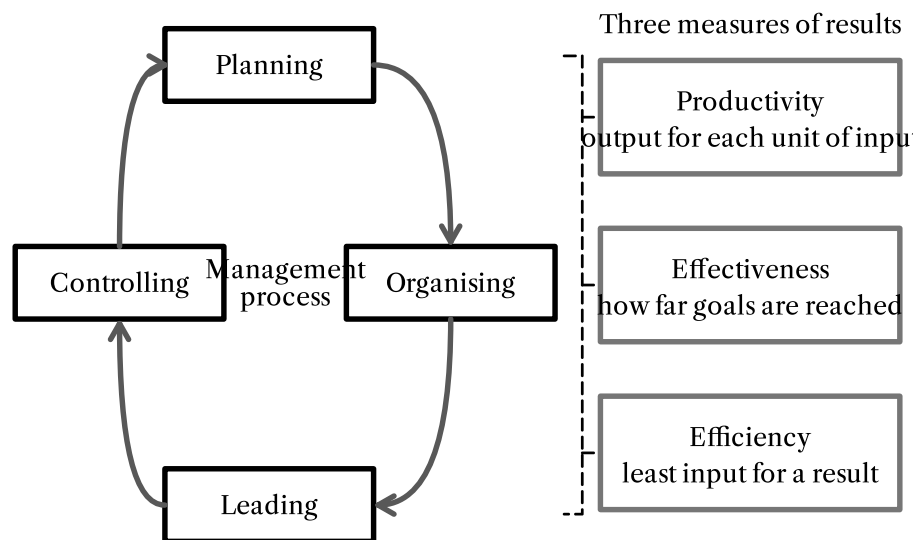


FIG. 3.1. *Functions as a cycle, with the three measures beside it*

The figure shows the cycle on the left and the three measures on the right.

To read the figure, begin with the left side. The four boxes are the functions, and each arrow leads to the next, so that the last arrow returns to the first. The right side lists the three measures with which the results of that cycle are judged.

* * *

3.4 SCIENCE, ART OR PROFESSION

Is management a science, an art or a profession? The question matters because each answer asks something different of managers. We take the three claims in turn. The second figure compares them.

A science is knowledge built by observation and repeated testing, whose rules give the same result each time. Management has scientific parts. A measured rule can be tested and repeated. But management cannot be a full science, because people differ, conditions change, and controlled experiments are seldom possible.

Case file, illustrative. Stock reordering at Meghna and Sons follows a measured rule: reorder when 30 days of cover remain. Any employee can apply this rule and reach the same result.

An art is personal skill in using knowledge, gained by practice and shown in judgement. No rule gives the answer in every case, and the manager must judge. Management needs this skill in every function.

Case file, illustrative. The shop-window design for the textbook season depends on Rafiq's judgement.

A **profession** is an occupation that requires long training, a body of knowledge and a code of conduct enforced by an association. Medicine is a clear example: in most countries a doctor must register with a medical council before treating patients. Management has a body of knowledge, but it fails the other tests in most countries. In the United Kingdom, the Chartered Management Institute awards a chartered title to managers who choose to seek it, but employers do not require it.

Case file, illustrative. No licence or governing body is required to manage Meghna and Sons.

Management can be called a science, an art or a profession, and each claim demands something different.

A science rests on tested knowledge, and management has only some of it.

Science. Word in use. In daily life, a school subject such as physics. Here, knowledge built by observation and repeated testing, whose rules give the same result each time.

The shop's reorder rule is the scientific part of its management.

An art is personal skill in using knowledge, and management needs it everywhere.

Art. Word in use. In daily life, painting, music or other creative work. Here, personal skill in using knowledge, gained by practice and shown in judgement.

The shop-window design is the artistic part of its management.

A profession has three tests, and management meets them only in part.

Profession. An occupation that requires long training, a body of knowledge and a code of conduct enforced by an association.

No licence or governing body is required to manage the shop.

What each claim demands		
Science	Art	Profession
Tested knowledge Rules that give the same result each time	Personal skill Gained by practice Judgement in each new situation	Long training Body of knowledge Code of conduct enforced by an association
What management shows		
Some measured rules	Judgement in every function	No licence required

Management is partly a science, largely an art, and only partly a profession.

Before moving on. Management uses tested knowledge as a science does, and personal judgement as an art does. It is only partly a profession, because it usually needs no licence and no governing association.

* * *

Management matters because it directs effort, saves resources and prepares an organisation for change.

A principle of management is a broad guide drawn from experience, to be adapted and not obeyed like a law.

Principle of management.
A broad guide to action drawn from experience, to be adapted and not applied like a law.

Management is important, and its principles are guides to adapt.

3.5 IMPORTANCE AND PRINCIPLES IN BRIEF

Management is important for three plain reasons. It directs the effort of many people toward common goals. It helps an organisation use limited resources with less waste. And it prepares an organisation to meet change, such as new laws, rivals and technologies. Without it, the same people and materials produce less, and the cost falls on owners, employees and customers.

Over many years, managers have collected broad guides drawn from experience. Each is a **principle of management**. A principle of management is a broad guide to action drawn from experience, to be adapted and not applied like a law. Water boils at 100 °C at sea level every time. A principle such as "divide work among specialists" may suit a factory and fit badly in a shop with seven people.⁵

Before moving on. Management directs effort, saves resources and prepares for change. A principle of management guides action and must be adapted to the situation.

* * *

The first example is small and clean, so that only the method is new.

WORKED EXAMPLE AND WORKED CASE

WORKED EXAMPLE: PRODUCTIVITY

Many students find rounding confusing here, because a result such as 5,857.14 cannot be a number of copies in daily life. We round at the end and never in the middle. The table below sets out the working in six parts. Read it from the top: the first column names the part, and the second column gives its content.

WORKED EXAMPLE 1: COPIES PER EMPLOYEE

PART	WORKING
Given	2025: 41,000 copies sold by 7 employees. 2026 forecast: 45,100 copies by 7 employees.
Find	Copies per employee in each year, and the change.
Formula	In words: productivity = output ÷ input. In symbols: $P = O \div I$, where P is productivity, O is output in copies and I is input in employees.
Steps	2025: $P = 41,000 \div 7 = 5,857.14$, so 5,857 copies per employee. 2026: $P = 45,100 \div 7 = 6,442.86$, so 6,443 copies per employee. Change: $6,443 - 5,857 = 586$ copies per employee. Percentage change: $586 \div 5,857 = 0.1000$. which is 10.0%.

⁵ A well-known list of such principles is taught in Chapter 4, "The Evolution of Management Thought", section 4.2 "Administrative Principles".

In the second example, output is measured in taka, and the figures do not divide evenly.

The second example is messier, because output is measured in taka.

WORKED EXAMPLE 2: SALES PER EMPLOYEE

PART	WORKING
Given	Sales 2025: Tk 14,400,000. Sales target 2026: Tk 15,840,000. Employees: 7 in each year.
Find	Sales per employee in each year.
Formula	Productivity = sales ÷ employees.
Steps	2025: $14,400,000 \div 7 = 2,057,142.86$, so Tk 2,057,143. 2026: $15,840,000 \div 7 = 2,262,857.14$, so Tk 2,262,857.
Answer	Sales per employee are planned to rise from Tk 2,057,143 to Tk 2,262,857.
Check	$2,057,143 \times 1.10 = 2,262,857$, which is the 10% rise in the target.

Four slips are common. The first is dividing input by output, which turns the ratio upside down. The second is mixing periods, such as monthly output with yearly input. The third is forgetting units: write copies per employee, not copies. The fourth is rounding in the middle of the working. Money also hides a trap. If prices rise 5%, sales per employee rise even when no more copies are sold.

Four common slips: an upside-down ratio, mixed periods, missing units and early rounding.

WORKED CASE: EFFECTIVE, EFFICIENT OR BOTH?

This case applies an idea to a situation. The table gives the five parts in order.

The case table applies the two words to the shop's own facts.

WORKED CASE: TWO FACTS, TWO MEASURES

PART	WORKING
Situation	Meghna and Sons set a 2026 sales target of Tk 15,840,000. Before 2 March 2026, staff spent 3 hours a day searching for stock. After the shelves were relabelled on 2 March 2026, they spent 1 hour.
Question	Which fact tests effectiveness, and which tests efficiency?
Reasoning	The target is a goal, so comparing actual sales with Tk 15,840,000 tests effectiveness. Search time is input. The result, finding the stock, is unchanged, so less input for the same result shows efficiency. The saving is $3 - 1 = 2$ hours a day, which is about 67% of the former search time.
Conclusion	The relabelling raised efficiency. Effectiveness cannot be judged until the sales for 2026 are known.
Check	If sales miss the target while search time stays at 1 hour, the shop is efficient but not effective. The two judgements are separate.

CHANGING THE SITUATION

Rung 1, a familiar case. The shop as described: a clear target, a shorter search for stock, and 7 employees.

Rung 1 is the familiar case: the shop as it stands.

Rung 2, the same case reshaped. Suppose the shop sells 45,100 copies, but only after hiring 2 more employees. The goal is reached, so the shop is effective. Productivity is $45,100 \div 9 = 5,011$ copies per employee, which is below the 5,857 of 2025.

Rung 2 reshapes the case: the target is met, but with more employees.

Rung 4 joins the cycle and the three measures in one chain of events.

Rung 4, a case that combines earlier ideas. Nurul sets the target (planning). Shelves are relabelled (organising and leading). Search times are compared month by month (controlling). The target tests effectiveness, the relabelling raises efficiency, and the monthly comparison closes the cycle, so that the next plan can use the result.

* * *

The chapter in brief.

SUMMARY

Management work falls into four functions: planning, organising, leading and controlling. They form a repeating cycle called the management process. Results are judged by three measures: productivity (output for each unit of input), effectiveness (how far goals are reached) and efficiency (how little input a result needs). Management uses a science and an art, but it is only partly a profession. A principle of management is a guide to adapt, not a law.

THINK IT THROUGH

3.T1 A delivery firm promises to deliver 300 parcels in a day. It delivers all 300, but 4 vans each drive twice the distance needed, because the routes were not planned. Explain why the firm was effective but not efficient, and name the function whose improvement would help most.

3.T2 Meghna and Sons is forecast to reach 6,443 copies per employee in 2026. Suppose the shop hires 1 more employee and sells 45,100 copies. Find the new productivity and say whether it is higher or lower than in 2025.

WHERE PEOPLE GO WRONG

Mistake: the functions happen once, in a fixed order.

Mistake: "The four functions are steps in a recipe, done once in order." Correction: they repeat and overlap, and controlling gives the information for the next round of planning.

Mistake: more output always means higher productivity.

Mistake: "More output always means higher productivity." Correction: productivity is output for each unit of input. If input grows faster than output, productivity falls.

Mistake: effectiveness and efficiency are two names for one idea.

Mistake: "Effective and efficient mean the same." Correction: effectiveness asks whether the goal was reached; efficiency asks how little input was used.

Mistake: management is a profession like medicine.

Mistake: "Management is a profession, like medicine." Correction: in most countries a manager needs no licence and answers to no single governing association.

Mistake: a principle of management is a law.

Mistake: "A principle must be followed in every case." Correction: a principle is drawn from experience and must be adapted to the situation.

IN HISTORY

The Toyota Production System

Taiichi Ohno (1912 to 1990) joined the Toyota company in 1943 and worked

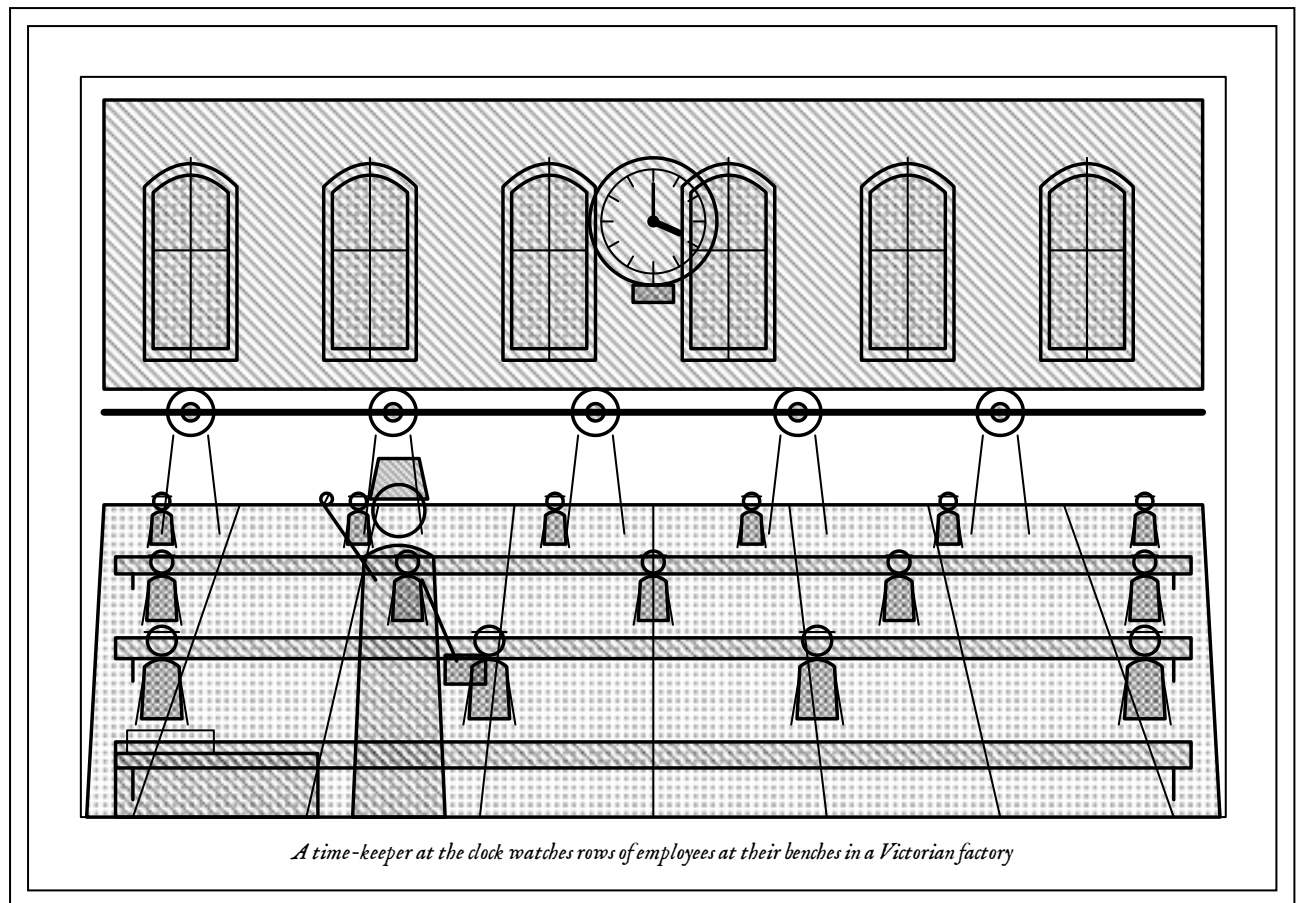
complete system of production over the following decades, aimed at removing waste.

The system is called the Toyota Production System. It was developed between the 1940s and the 1970s, and its aim was the same result with less input, which is the meaning of efficiency in this chapter. Ohno became an executive vice president of Toyota in 1975 and retired in 1978. His book on the system first appeared in Japan in 1978, and an English edition followed in 1988. Observers noted that Toyota did well in the oil crisis of the 1970s, when many rivals suffered.

Ohno's system was a landmark in efficiency: the same result from less input.

REVIEW QUESTIONS

- 3.1 Name the four functions of management in order, and give one example of each from a bookshop.
- 3.2 A bakery made 1,800 loaves in a week with 4 bakers. After buying a new oven it made 2,100 loaves with the same 4 bakers. Find the productivity before and after, in loaves per baker, and the percentage change.
- 3.3 Explain why a rise in total output does not always mean a rise in productivity. Give a small numerical example.
- 3.4 Define effectiveness and efficiency, and describe one situation that is efficient but not effective.
- 3.5 State the three parts of the definition of a profession, and say which part is missing when a person manages a small shop.
- 3.6 In two sentences, explain why a principle of management is not like a law of physics.



CHAPTER IV

The Evolution of Management Thought

Can the work of a shop be studied like a science?



This chapter follows the ideas of management from the early factories to the modern organisation. You will meet Taylor, Fayol, Weber and Mayo, learn what each believed, and see how their ideas still shape a shop floor, an office and a busy kitchen.

YOU WILL NEED

This chapter uses two earlier ideas. The first is whether management can claim to be a science, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.4 'Science, Art or Profession'. The second is the list of activities that every manager performs, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

Two shops with the same books and staff can earn very different results.

For most of history nobody studied management; people learned by watching.

Thinkers who share one main idea are placed together in one school.

School. Word in use. In daily life, a place where children learn. In management, a group of thinkers who share one main idea about how to manage.

Picture two bookshops in the same street. They stock the same titles, charge the same prices and employ the same number of people. Yet at the end of the year one shop has sold far more books than the other. The books have not changed, so something else has. What is it? Many students first answer "luck" or "a better owner". Both answers may hold some truth, but neither can be taught, and nobody can learn luck.

For thousands of years, people managed armies, farms, trading houses and building sites without a book of management to read. A young person learned by watching an older one, and good methods stayed secret inside families. This worked while work was done in small groups. It failed when factories gathered hundreds of people under one roof, because no one could learn fast enough by watching. Owners needed ideas that could be written down, tested and passed on. This chapter tells how such ideas were found, and in what order.

Historians place thinkers who share one main idea in the same school of thought. We shall meet several schools, one after another. The timeline below is a map of the road. Read it from top to bottom: each row is one school, each bar shows the years when its ideas were most active, and the scale along the bottom gives the dates. Do not worry about the names yet; each is explained in its own place.

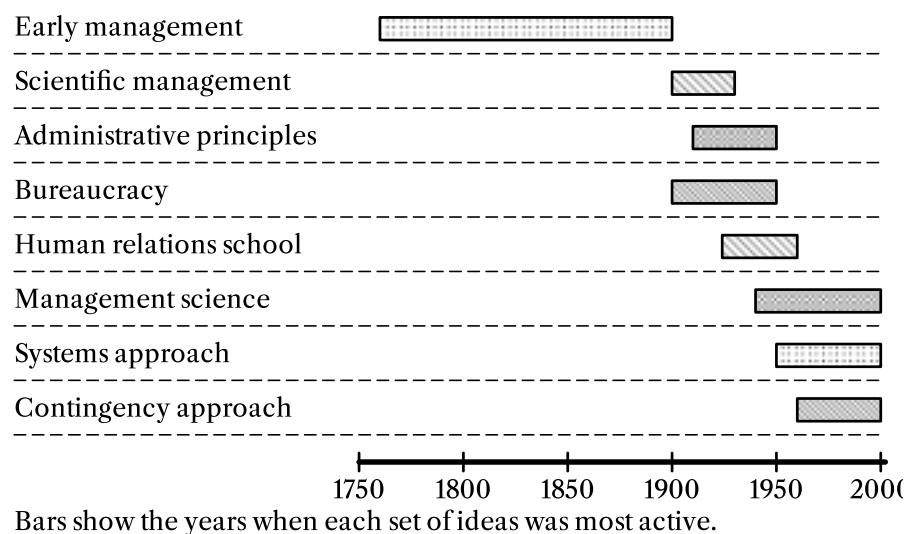


FIG. 4.1. *Timeline of the schools of management thought*

4.1 EARLY MANAGEMENT AND SCIENTIFIC MANAGEMENT

Why did management ideas appear when they did? Because factories created new problems. From about 1760 in Britain, machines and factory buildings brought many employees together (older books call them workers), and an owner could no longer watch every task. Three early thinkers deserve one line each. Adam Smith, a Scottish thinker, wrote in 1776 that dividing a job into small steps raised output, and he used a pin factory as his example. Robert Owen, a Welsh-born manager of cotton mills at New Lanark in Scotland from 1800, argued that good conditions for employees helped the business. Charles Babbage, an English mathematician, wrote in 1832 that the steps of a task should be studied and measured.

Each early thinker offered one useful idea, but none joined the ideas into a complete method. That step came from Frederick Winslow Taylor (1856 to 1915), an American engineer who worked in steel works in Pennsylvania and published *The Principles of Scientific Management* in 1911. Taylor noticed that employees often worked more slowly than they could. Each person followed a personal rule of thumb, which means a method learned by habit and not tested by measurement. Some employees also feared that faster work would reduce the number of jobs. Taylor asked a plain question: what is the best way to do this task, and how long should it take?

His answer was **scientific management**. Scientific management means studying work by measurement, to find the one best method and to train employees in it. For example, a foreman who once chose how to load a wagon by habit would, under this idea, test several ways and measure which was fastest. The word "scientific" matters: Taylor wanted facts from measurement to replace opinion and habit.¹

The main tool of scientific management is the **time and motion study**. A time and motion study means timing each movement of a task, to remove wasted effort and to set a fair standard. Taylor timed tasks with a stopwatch. Frank Gilbreth and his wife Lillian Gilbreth, American engineers working in the early 1900s, studied the movements of the hands and body and removed those that gave no benefit. Together the two ideas answer two questions: which movements are needed, and how long they should take.

Case file, illustrative. Method change 1 March 2026: unloading a carton of books was timed at 14 minutes before and 9 minutes after a fixed shelf-position list. The shop unloads 10 cartons a day, so the saving is $5 \times 10 = 50$ minutes a day.

Taylor summarised his ideas in four principles. First, replace rule-of-thumb methods with a science for each element of a job. Second, choose each employee with care, then train and develop that person, instead of leaving the employee to learn alone. Third,

Factories created problems that older ways of managing could not solve.

Taylor noticed that employees worked slower than they could, and asked why.

Scientific management means finding the one best method by measuring work.

Scientific management.

The study of work by measurement, aimed at finding the one best method and training workers in it.

Timing each movement removes wasted effort and sets a fair standard.

Time and motion study.

Timing each movement of a task to remove wasted effort and set a fair standard.

The shop timed the unloading of a carton before and after a fixed shelf list.

Taylor set out four principles that turn the idea into rules for managers.

¹ Whether management can claim to be a science is discussed in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.4 'Science, Art or Profession'.

a task is repeated many times, as in unloading cartons, and works poorly when every task is different. The owner sees higher productivity; the employee may feel constantly watched.

Section 4.1 in four lines.

Before moving on. Scientific management studies work by measurement to find one best method. A time and motion study times each movement. Taylor's four principles replace habit with science, train the employee, share the work with the employee, and divide responsibility. Its limit is that it treats employees as parts of a machine.

* * *

4.2 ADMINISTRATIVE PRINCIPLES

Taylor studied the single task; Fayol studied the whole organisation.

Taylor looked at one employee at one bench. But who looks at the whole organisation: the owner's decisions, the way departments connect, the lines of authority? A French mining engineer asked exactly this question, and his answer has lasted more than a hundred years.

WHO WAS FAYOL?

Fayol was a mining engineer who ran a large company and wrote down what he learned.

Henri Fayol (1841 to 1925) trained as a mining engineer in France. In 1888 he became managing director of a French mining and iron company that was in difficulty. In 1916 he published *Administration industrielle et générale* in French; the English translation, *General and Industrial Management*, appeared in 1949. Fayol wrote from his own experience as a senior manager, so his view starts at the top, where Taylor's view started at the bench.

Fayol sorted a company's activities into six groups, and managing is one of them.

Fayol saw six kinds of activity in any industrial organisation: technical (making things), commercial (buying and selling), financial (finding and using money), security (protecting people and property), accounting (recording and counting), and managerial. He said that managing is a separate activity that needs its own study. He described it through five activities: planning, organising, commanding, coordinating and controlling.² These five are an early form of the management process.

Fayol believed management can be taught, so he wrote principles, not laws.

Law. Word in use. In daily life, a rule made by the state and enforced by courts. In science, a statement that always holds. Fayol's principles are guides that bend with the situation.

Fayol held one belief that mattered greatly: management can be taught. If it can be taught, it needs principles, which are general guides that managers can learn and apply. He listed fourteen and said that the list was not complete. He called them principles and not laws, because a law in science holds every time, while a principle must bend with the situation.

THE FOURTEEN PRINCIPLES

Fayol's fourteen principles are short rules about work, authority, fairness and order.

The table lists Fayol's fourteen principles with a plain meaning for each. Read across a row: the name is on the left, and what it asks of a manager is on the right.

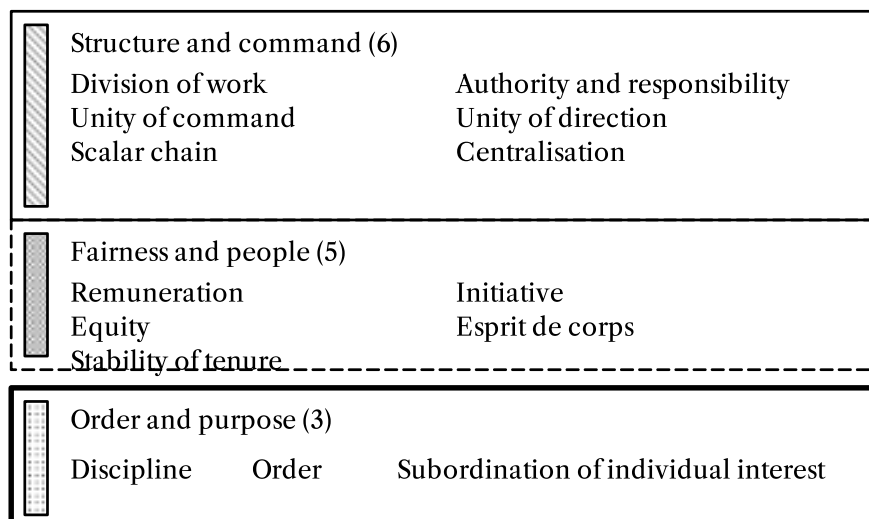
² The management process is set out in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

FAYOL'S FOURTEEN PRINCIPLES IN PLAIN WORDS

PRINCIPLE	WHAT IT ASKS
Division of work	Give each employee a smaller part of the whole job, so that skill grows with practice
Authority and responsibility	Match the right to give orders with the duty to answer for results
Discipline	Make clear agreements, and apply penalties fairly when they are broken
Unity of command	Each employee receives orders from one manager only
Unity of direction	One plan and one head for each group of activities with the same goal
Subordination of individual interest	The goal of the organisation comes before the interest of one person
Remuneration	Pay fairly, so that employee and organisation are both satisfied
Centralisation	Decide how much decision making stays at the top and how much is shared
Scalar chain	A line of authority runs from the top to the lowest level, with a clear path for messages
Order	A place for everything and everyone, and everything and everyone in its place
Equity	Treat employees with kindness and justice
Stability of tenure	Keep employees long enough to learn, and avoid needless turnover
Initiative	Let employees think out a plan and carry it through
Esprit de corps	Build team spirit and harmony among employees

Fourteen items are hard to remember in a list, so the diagram sorts them into three groups. Read it from top to bottom: each band is one group, the strip at its left edge has its own hatching, and the lines inside the band are the principles it holds.

Sorting the fourteen into three groups makes them easier to remember.



Six principles concern how work and authority are arranged.

Five principles concern fairness and the treatment of people.

Three principles protect order and the common goal.

The shop posted a written list of rules.

Written rules apply the principles of discipline and order.

Fayol's principles come from one man's experience, and some pull against each other.

Taylor began at the bench, Fayol at the top, and both believed management can be taught.

Section 4.2 in three lines.

The first group, structure and command, has six principles: division of work, authority and responsibility, unity of command, unity of direction, scalar chain and centralisation. These describe how an organisation is arranged and how orders travel. Take unity of command. If an employee receives opposite orders from two managers, the employee must disobey one of them, and confusion follows. The principle prevents this by giving each employee one manager.

The second group, fairness and people, has five principles: remuneration, equity, stability of tenure, initiative and *esprit de corps*, a French phrase that means team spirit. These ask managers to treat employees as people with needs. A fair wage, a kind manner and a steady job all encourage loyalty, and loyalty helps the organisation as much as it helps the employee.

The third group, order and purpose, holds discipline, order and the subordination of individual interest. These keep the organisation orderly and pointed toward one goal. For example, if one sales assistant prefers a private method of shelving but the shop needs one shared method, the goal of the shop comes first.

Case file, illustrative. Rule list posted: 14 rules for the shop.

A written rule list is one way for a shop to apply discipline and order, two of Fayol's principles. The shop's 14 rules and Fayol's fourteen principles share a number by chance only.

THE LIMITS OF FAYOL'S PRINCIPLES

Fayol's work is a landmark because it gave managers a shared language. It also has limits. He drew on his own experience in one industry, and he did not test the principles by measurement. Some principles pull against each other. Unity of command gives clear orders, but in an emergency an employee may need to act on the word of whichever manager is present. Division of work raises skill, but it can make work dull. Fayol knew this and advised judgement: the principles are guides, and a manager must decide how far to apply each one.

Compare the two thinkers. Taylor began at the bench and worked upward; Fayol began at the top and worked downward. Both believed that management could be studied and taught, and both believed in principles. Together they give the first complete picture of management as a subject of study: Taylor supplies the method for the task, and Fayol supplies the principles for the whole organisation.

PAUSE AND CHECK 4.P1 An employee in a shop is told by two managers to do two different jobs at the same hour. Which of Fayol's principles is being broken, and what should the shop do?

Before moving on. Fayol, a French mining engineer, wrote in 1916 that managing is a separate activity with principles that can be taught. He gave fourteen principles in three groups: structure and command, fairness and people, and order and purpose. They are guides and not laws, and some of them pull against each other.

4.3 BUREAUCRATIC AND BEHAVIOURAL THOUGHT

Max Weber (1864 to 1920), a German scholar of society, studied how large organisations could treat people fairly. He described **bureaucracy**. Bureaucracy means a system of management built on written rules, clear levels of authority, and appointment by skill and not by family favour. In daily speech the word often means slow paperwork, but Weber meant a design for fairness and steady work. A shop with a written rule list and clear roles takes a small step in this direction.

Rules and measurement were not the whole story. Studies at the Hawthorne Works of the Western Electric Company, near Chicago, led thinkers such as Elton Mayo to the **human relations school**. The human relations school is a school of thought holding that employees' feelings and social bonds strongly affect output. For example, an employee who feels respected by the group may work harder than one who feels ignored, even when pay is the same. The studies are told in the section 'In History' at the end of this chapter.

Behavioural science grew from this school. Behavioural scientists are psychologists, sociologists and others who study how people behave, and they test their ideas by careful research and not by opinion. Their findings help managers understand why employees act as they do. Taylor trusted method and pay; Mayo and the behavioural scientists looked at people.

Before moving on. Bureaucracy rests on written rules, clear levels of authority and appointment by skill. The human relations school holds that feelings and group bonds affect output. Behavioural science studies human behaviour by research.

* * *

4.4 MODERN APPROACHES

During the Second World War (1939 to 1945), teams of scientists in Britain and the United States used mathematics to plan the use of ships, aircraft and supplies. After the war, businesses adopted the same methods, and the **management science approach** was born. The management science approach means using mathematical models and data to solve management problems. For example, a shop can use past sales data to decide how many copies of a title to order. It works well when a problem can be measured, but it cannot measure feelings or judgement.

The systems approach, popular from the 1950s, views an organisation as a set of connected parts that take in resources, change them, and deliver results. In a bookshop, buying, shelving, selling and accounts are parts of one chain: a delay in one part appears later in another. The approach reminds managers to look at the whole and at the links between the parts.

The earlier thinkers each looked for one best way. From the 1960s, researchers found that no single method suited every situ-

Weber described a way of organising by written rules, levels and skill.

Bureaucracy. A system of management built on written rules, clear levels of authority and appointment by skill.

Studies of employees' feelings and group bonds led to a new school.

Human relations school. A school of thought holding that workers' feelings and social bonds strongly affect output.

Behavioural science tests ideas about human behaviour by careful research.

Section 4.3 in three lines.

Management science uses mathematics and data to solve problems.

Management science approach. The use of mathematical models and data to solve management problems.

The systems approach sees an organisation as connected parts.

The contingency approach says the best way depends on

The comparison shows what each approach does best and where it fails.

The table compares the approaches in this chapter. Read a row from left to right: it gives the school, its main idea, what it does well and where it fails. The timeline near the start of the chapter shows the order of these ideas in time.

COMPARING THE MAIN APPROACHES

APPROACH	MAIN IDEA	DOES WELL	LIMIT
Scientific management	One best method found by measuring	Raises output on repeated tasks	Treats employees as machine parts
Administrative principles	General principles for the whole organisation	Gives a shared language	Principles can conflict
Bureaucracy	Written rules and levels of authority	Fairness and steady work	Can be slow and rigid
Human relations school	Feelings and group bonds affect output	Values people	Can understate pay and method
Management science	Mathematics and data	Solves measurable problems	Cannot measure feelings
Contingency approach	Best way depends on the situation	Fits real conditions	Gives no single answer

Modern managers borrow from several approaches and choose by situation.

No approach has replaced the others. In practice, managers borrow from several, and the contingency approach tells them to choose according to the situation. A busy shop may time its unloading, write a few clear rules, listen to its employees and count its sales, all in one week.

Section 4.4 in three lines.

Before moving on. The management science approach uses mathematics and data. The systems approach looks at connected parts. The contingency approach holds that the best way depends on the situation, so managers borrow from many approaches.

* * *

WORKED CASE: TIMING THE UNLOADING OF CARTONS

This worked case applies scientific management to the shop's unloading.

This worked case applies the chapter to the shop's own record. It shows how to reason from facts, step by step. Many students find the school-naming step confusing, because the same record can fit more than one school. The steps below separate the arithmetic from the naming.

The situation is the shop's record of the method change.

Situation. Meghna and Sons timed the unloading of one carton of books before and after a fixed shelf-position list. The time was 14 minutes before and 9 minutes after. The shop unloads 10 cartons a day. A list of 14 rules for the shop was also posted.

The question asks for the school of thought and the time saved.

Question. Which school of thought does the method change belong to, and how many minutes does the shop save in a day?

The reasoning first finds the saving, then names the

Reasoning. First, find the saving for one carton: 14 minutes minus 9 minutes is 5 minutes. Second, find the saving for the day: 5

Conclusion. The shop saves 50 minutes a day, which is 5/6 of an hour. The change belongs to scientific management, supported by a rule list that applies Fayol's discipline and order.

Check. Before the change, 10 cartons took $10 \times 14 = 140$ minutes. After the change, they took $10 \times 9 = 90$ minutes. The difference is 140 minutes minus 90 minutes, which is 50 minutes. The answer agrees with the first calculation.

* * *

The conclusion gives the saving and the school.

The check repeats the arithmetic from the other side.

SUMMARY

Management thought grew as factories grew. Taylor's scientific management measured work to find one best method, using time and motion study. Fayol looked at the whole organisation and gave fourteen principles in three groups. Weber described bureaucracy, and Mayo's followers formed the human relations school. Later thinkers added the management science, systems and contingency approaches. No approach replaced the others; managers choose by situation.

The chapter in a few lines.

THINK IT THROUGH

These four questions start with the shop's case and move to new settings. Each builds on the one before.

4.T1 In the shop's method change, which of Taylor's four principles does the fixed shelf-position list best show, and why?

4.T2 Suppose the unloading time after the change had been 11 minutes and not 9 minutes. How many minutes a day would the shop save?

4.T3 A busy kitchen must prepare many orders in a rush, the cooks differ in skill, and orders change every minute. Which school of thought suits it best, and what would you borrow from another school?

4.T4 A shop times its employees for every task and also wants them to feel part of one team. Where might scientific management and the human relations school pull against each other, and how could a manager reduce the conflict?

Four questions that change the situation step by step.

WHERE PEOPLE GO WRONG

Some students think Taylor wanted only faster work at any cost. This is half true. He also proposed careful selection, training and higher pay for employees who reached the standard. The fair criticism is different: his method treated employees as parts of a machine and ignored their feelings.

Some students think bureaucracy always means delay and waste. In daily speech it often does. In Weber's sense it is a design with written rules, clear levels and appointment by skill, meant to give fairness and steady work. Delay appears when such a design is applied without judgement.

Taylor is wrongly remembered as caring only for speed.

Bureaucracy in Weber's sense is not the same as slow paperwork.

The Hawthorne Studies began with lighting and ended with group bonds.

Harvard researchers joined and studied small groups and interviews.

The Studies moved attention to attention, groups and feelings, though later writers dispute some claims.

Five questions that review the chapter.

IN HISTORY

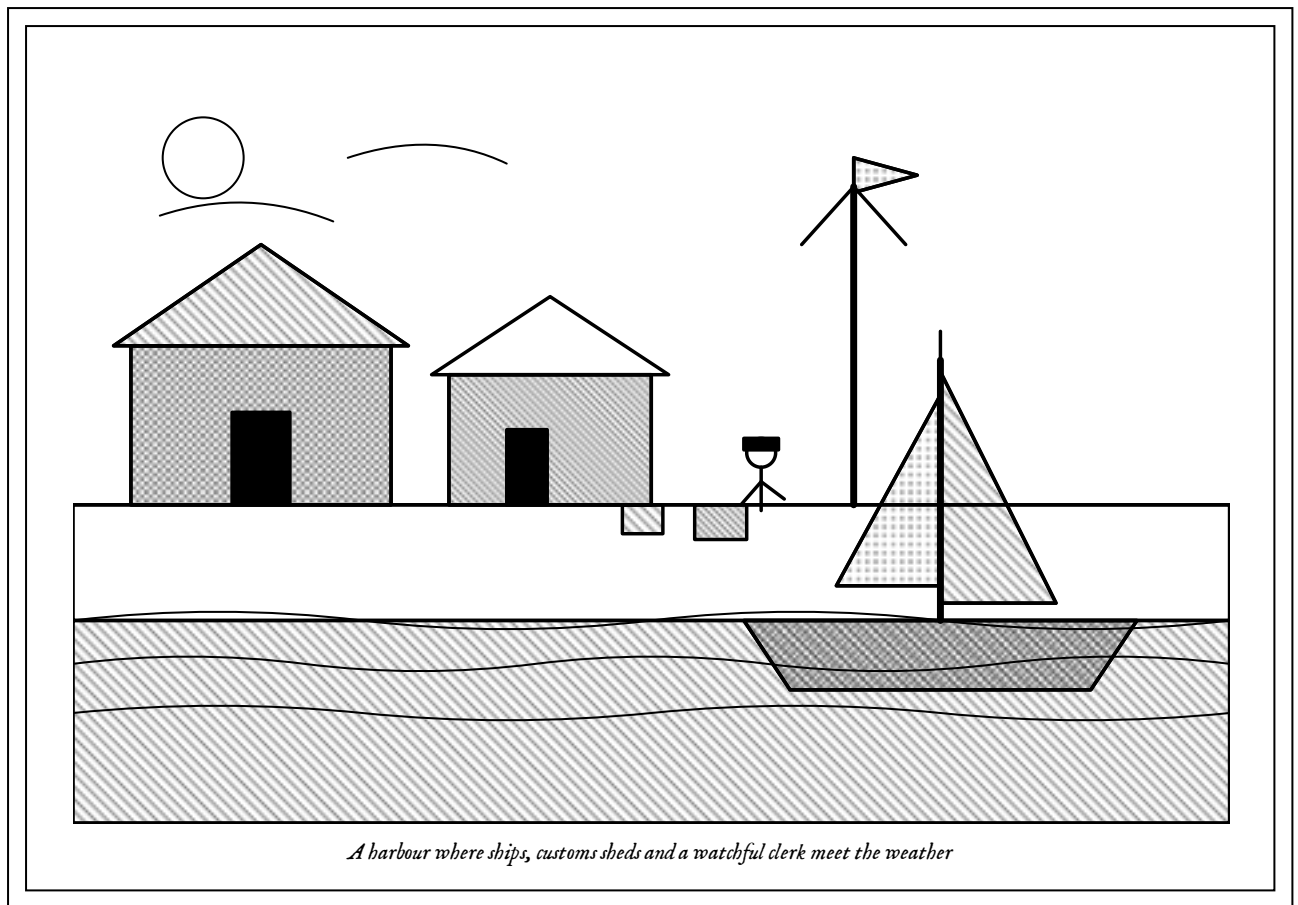
In 1924 the Western Electric Company, which made telephone equipment, began studies at its Hawthorne Works near Chicago. With the National Research Council, an American scientific body, it tested whether the quality and amount of light changed how much employees produced. These illumination studies ran until 1927, and their results were puzzling.

Researchers from Harvard University, among them Elton Mayo and Fritz Roethlisberger, then joined the work. They studied a small group of women assembling telephone relays in a test room while conditions changed. An interview programme later asked employees about their feelings about work, and another study observed the habits of a group of men wiring equipment. The studies lasted until about 1932.

The researchers concluded that attention from managers and the bonds within a group affected output. This conclusion helped to create the human relations school. Later scholars have questioned how far the original data supports the claim now called the Hawthorne effect, and many experts doubt it. The studies remain important because they turned attention toward the social side of work.

REVIEW QUESTIONS

- 4.1 Match each name to its idea: Taylor, Fayol, Weber, Mayo. The ideas are: written rules and levels of authority; one best method found by timing; fourteen principles for the whole organisation; group bonds affect output.
- 4.2 State Taylor's four principles in your own words.
- 4.3 Define the contingency approach, and give one example from a shop.
- 4.4 Name the three groups of Fayol's principles, and give one principle from each group.
- 4.5 Give one strength and one limit of the management science approach.



CHAPTER V

The Environment of Organisations

Can any organisation live apart from the world around it?



Every organisation stands inside a larger world of customers, rivals, laws and machines. This chapter divides that world into its parts, shows how managers study it, and explains the ways in which they respond to it.

Two earlier ideas support this chapter.

A clerk checks the wind before he opens his books.

The office controls its books, but not the forces that decide its fortune.

Environment. Word in use. In daily life, the natural world of air, land and water. In management, everything inside and around an organisation that affects how it works.

The question of the chapter: what else must managers examine?

The shop door divides what an owner can change from what an owner cannot.

Internal environment. Conditions inside an organisation, such as its people, structure and culture, which managers can change directly.

The internal environment is everything inside the door

You will need two ideas from earlier chapters. The first is the management process, taught in Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.¹ The second is the organisation, taught in Chapter 1, 'What Management Is', section 1.1 'The Idea of Management'.² Read those sections again if either idea feels uncertain.

Picture a busy harbour in the middle of the nineteenth century. At the head of the quay stands a shipping office, and on its roof a weather vane turns. Each morning a clerk climbs to the window and studies the vane before he opens the ledgers. The ledgers tell him what the office owns and owes. The vane tells him something the ledgers cannot: whether the ships will arrive on time.

The clerk's habit hides a puzzle. The office has careful books, a sound building and skilled workers. Yet its fortune depends on matters it does not control. The wind decides when the cargo arrives. Officers at the customs shed decide how long goods wait and what tax they pay. Merchants decide whether to send their goods through this harbour or another. Over some decades, iron steamships began to replace sailing ships. A firm that had mastered sail could lose its trade without making one error in its own books.

Here is the question for this chapter. If an organisation can be managed perfectly inside its own walls and still fail, what else must its managers examine? The answer is the world around the organisation, which managers call its environment. We will name its parts, learn to sort its forces, and see how managers watch it and respond to it. Later we will meet a real company, Nokia, whose managers faced this problem on a very large scale. We will also read the records of a small Dhaka bookshop, Meghna and Sons, which met the same problem on a small scale.

* * *

5.1 INSIDE AND OUTSIDE THE ORGANISATION

Think of the door of a shop. Inside, the owner can move the shelves, change who works at the counter and decide the opening hours. Outside, the owner cannot move the road, cannot change the weather and cannot order a customer to buy. This ordinary picture gives us the first distinction of the chapter. Some conditions a manager can change directly. Other conditions a manager cannot change directly, and must learn to meet in other ways.

The conditions inside the door form the **internal environment**. The internal environment means the people, the structure and the culture inside an organisation, which managers can change directly. Structure means the way work and authority are divided among people. Culture means the shared habits and beliefs of a group. In a bookshop, the internal environment includes the

¹ Recall: management process, Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

² Recall: organisation, Chapter 1, 'What Management Is', section 1.1 'The Idea of Management'.

The departure of 2 employees belongs to the internal environment. The owner and the store manager can respond by hiring, by training, or by dividing the work again. Notice the word *directly*. Managers did not cause the departures, but they can act on the result with their own authority. That power to act directly is what marks a condition as internal.

The forces outside the door form the **external environment**. The external environment means all outside forces, such as customers, rivals, laws and technology, that affect an organisation. A manager cannot give these forces an order. A manager can, however, study them and prepare. A customer who changes her tastes, a rival who lowers a price and a government that issues a new rule all belong here. Each of them reaches the organisation from beyond the door.

An employee leaving is an internal change, because managers can act on it directly.

External environment. All outside forces, such as customers, rivals, laws and technology, that affect an organisation.

The external environment is every outside force that affects the organisation.

IN SHORT

Internal environment: a manager can change it directly. External environment: a manager can study it, prepare for it, and sometimes influence it, but cannot command it.

PAUSE AND CHECK 5.P1 A new bookshop opens two streets away from Meghna and Sons. Is this change internal or external? Give one reason.

The line between the two environments is called the boundary. It is not a wall, because goods, money, information and people cross it every day. It also follows control, not distance. A supplier in the next street is outside the boundary, because the owner cannot direct that supplier. A branch in another town is inside the boundary if the same organisation owns it and directs it.

Before moving on. The internal environment is what managers can change directly. The external environment is what they can only study, prepare for or influence. The boundary between them follows control, not distance.

The line between inside and outside is a boundary, and it follows control, not distance.

The recap fixes the two environments and the boundary between them.

* * *

5.2 COMPONENTS OF THE EXTERNAL ENVIRONMENT

Suppose a manager writes down every outside force that touches a bookshop in one year. The list would include tastes, prices, laws, machines, rivals, weather and world events. A list that long is hard to use. Managers therefore sort outside forces into groups, and watch each group in turn. Writers divide the groups in different ways. This book uses five, and the groups are a convenience for thinking, not a law of nature.

Before we meet the five groups, one more distinction helps. Some outside forces are people and bodies that deal directly with the organisation. They form the **task environment**. The task environment means the outside groups that deal directly with an organisation, such as customers and suppliers. For a bookshop, the

The outside forces are too many to study as one heap, so managers sort them.

Task environment. The outside groups that deal directly with an organisation, such as customers and suppliers.

The task environment holds

nearer a force stands to the centre, the more directly it deals with the organisation.



FIG. 5.1. *External environment layers around the organisation*

The economic environment is the state of money, prices, incomes and trade.

The social and cultural environment is the tastes, beliefs and habits of the people served.

The political and legal environment is the rules and actions of public bodies.

The technological environment is the methods and machines available to the organisation and its rivals.

The global environment is the part of the outside world beyond the country's borders.

The first group is the economic environment: the state of money, prices, incomes and trade around the organisation. When families have less to spend, they buy fewer books. When rivals lower their prices, customers compare. Bank interest rates, the cost of paper and the general level of employment all belong here. A manager watches the economic environment to learn what customers can afford and what the organisation must pay.

The second group is the social and cultural environment: the customs, beliefs, tastes and education of the people the organisation serves. A society that values schooling buys many school books. A society in which more people read on screens buys fewer printed ones. These forces change slowly, and they differ from one community to another, so a manager must study the people actually served and not assume that all customers think alike.

The third group is the political and legal environment: the rules and actions of governments, courts and public bodies. Taxes, shop licences, rules on imported goods and decisions about what schools teach all belong here. These rules differ from country to country, so this book names the country whenever it names a rule.³ A decision by an education authority, for example, can change which books every school in a country must buy.

The fourth group is the technological environment: the methods and machines available to the organisation and its rivals. The steamship changed the harbour. The internet changed the bookshop. The same technology can bring a threat and an opportunity together. A website lets a distant rival reach the shop's customers, and it also lets the shop itself reach customers who never enter the door.

The fifth group is the global environment: forces that come from beyond the borders of the country. A shop that buys foreign books

³ A tax on imported books may exist in one country and not in another. A manager must always check the rules of the country where the organisation operates.

The table below places the five groups side by side. Each row names a group, says in plain words what it covers, and gives an example for a bookshop. Read across a row to see one group, and down a column to compare the groups.

The table sorts the five groups by what each covers.

THE FIVE GROUPS OF WIDER FORCES

GROUP	WHAT IT COVERS	EXAMPLE FOR A BOOKSHOP
Economic	Money, prices, incomes, trade	A fall in family spending on books
Social and cultural	Tastes, beliefs, habits, education	A shift from printed books to screens
Political and legal	Rules and actions of public bodies	A new tax or a change in school books
Technological	Methods and machines	Online selling and phone payment
Global	Forces from beyond the country	The price of foreign money

One caution applies to the table. The groups overlap. A rival's online price cut is an economic force, because it changes prices. It is also a technological force, because online selling made it possible. It also belongs to the task environment, because the rival competes for the same buyers. A manager does not need to decide which single box is correct. The sorting is useful when it makes the manager notice every way in which the event reaches the organisation.

The groups overlap, so one event can touch several of them.

Case file, illustrative. Outside changes recorded at Meghna and Sons: online price cuts of about 15% on textbooks since March 2025; textbook sales of Tk 4,300,000 in 2025 against Tk 5,100,000 in 2024; and a curriculum change from January 2026.

The case file records the outside changes at Meghna and Sons.

Each recorded change reaches the shop through a different route. The online price cuts come from rivals and from technology. The fall in textbook sales shows how those forces reached the shop's own results. The curriculum change is a decision of public authority, and it falls within the political and legal group. In the Worked Case at the end of the chapter we shall sort these changes step by step.

Each outside change reaches the shop through a different group.

Before moving on. The task environment is the set of outside groups that deal directly with the organisation. The wider forces fall into five groups: economic, social and cultural, political and legal, technological, and global. One event may belong to several groups at once.

The recap fixes the five groups and the task environment.

* * *

5.3 MANAGING THE ENVIRONMENT

A manager who knows that a rival has cut prices has learned something useful. The next question is what to do about it. Managers have three broad ways of meeting the external environment: they can watch it, they can fit the organisation to it, and they can try to change it. We take them one at a time, and then we join them into a single plan of response.

Knowing the outside world is useful only if managers act on what they learn.

Environmental scanning. Regularly collecting and studying information about outside changes that could affect the organisation.

Scanning is the regular watching of outside change, like the clerk and his weather vane.

Scanning cannot predict the future, but it reduces surprise.

Adapting means changing the organisation to fit an outside force.

Influencing means acting on the outside force itself, within the limits of law and honesty.

A strategic response is a chosen combination of watching, adapting and influencing.

The case file records how Meghna and Sons responded on 3 March 2026.

The price-match list is a case of adapting, because the shop changed its own prices.

The harbour clerk's morning habit has a name in management. **Environmental scanning** means regularly collecting and studying information about outside changes that could affect the organisation. A shop owner who reads rivals' online prices every week is performing it. So is a store manager who asks the sales assistants what customers requested and could not find. The word *regularly* matters. A single look is a glance; a habit of looking is scanning.

Scanning has limits. It cannot predict the future, because many outside changes have no warning. It also produces more information than anyone can use, so a manager must decide which signals matter. Its value lies in reducing surprise. An organisation that watches the outside world meets a change early, when many responses remain possible. An organisation that does not watch meets it late, when few remain. Scanning is also part of planning, one of the functions in the management process.

The second way is adapting: changing something inside the organisation so that it fits an outside force. A shop that sees buyers moving online may add an online order service. A shop that sees a new school curriculum may change the books it keeps on its shelves. Adapting works because the internal environment is the part that managers can change directly. Its weakness is that the organisation always follows the outside force and never leads it.

The third way is influencing: acting on the outside force itself, so that it changes in the organisation's favour. A group of shops may join a trade association and put its case to a public body. A shop may advertise to change what customers prefer. Influence has strict limits. A small organisation seldom has the power to change a law or a market. All influence must also stay within the law and within honest dealing, because an organisation that cheats to change its environment damages itself.

Managers rarely use one way alone. They choose a combination, and we call that choice a strategic response. A strategic response means the broad choice a manager makes about how the organisation will meet an outside force.⁴ The choice depends on two questions. How strong is the force? How much power does the organisation have to change it, or to change itself? A strong force and a weak organisation usually call for adapting. A weak force may call only for watching.

Case file, illustrative. Response of 3 March 2026: Meghna and Sons issued a price-match list covering 200 titles. Price-matching means selling a listed title at the price a rival charges for it. The record shows the earlier sequence: online price cuts of about 15% on textbooks since March 2025; textbook sales of Tk 4,300,000 in 2025 against Tk 5,100,000 in 2024.

The price-match list is a case of adapting. The shop did not try to change the rivals' prices, and it did not try to change the law. It changed something it controls directly, which is its own price list. It also limited the change to 200 titles, which is a choice about where the pressure was strongest. From the viewpoint of the owner, the

⁴ A strategy is a broad plan for reaching a goal. Plans are taught later in Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'. Here, the phrase *strategic response* needs only the plain meaning given in the text.

management taught earlier. A response that wins customers but loses money has not solved the problem.

Before moving on. Scanning means watching the outside world on a regular basis. Adapting means changing the organisation to fit an outside force. Influencing means acting on the outside force itself. A strategic response is the chosen combination of these, matched to the strength of the force and the power of the organisation.

The recap fixes the three ways of meeting the environment.

* * *

WORKED CASE: SORTING THE OUTSIDE CHANGES OF 2025 AND 2026

This section applies the ideas of the chapter to the records of Meghna and Sons. It does not give a model answer. It shows one way of reasoning, step by step, so that you can follow each decision and then repeat it on a new situation.

The worked case applies the chapter's ideas to the records of Meghna and Sons.

Situation. The shop's records for 2025 and 2026 list four changes. First, online price cuts of about 15% on textbooks since March 2025. Second, textbook sales of Tk 4,300,000 in 2025 against Tk 5,100,000 in 2024. Third, a curriculum change from January 2026. Fourth, the departure of 2 of the 7 employees in 2025. A fifth record is a response: the price-match list for 200 titles, issued on 3 March 2026.

The situation lists every change in the record.

Question. For each change, decide whether it is internal or external. For each external change, name the group or groups to which it belongs. Then decide whether the response of 3 March 2026 is adapting or influencing.

The question asks for three sortings of each change.

Reasoning. Begin with the boundary test from section 5.1: can the managers change this condition directly? The departure of 2 employees fails the test as a cause, but passes it as a result, because the owner can hire and train. It is internal. The price cuts do not pass the test. The owner cannot command a rival's price list, so they are external. The curriculum change is external for the same reason. The fall in textbook sales is a result recorded in the shop's own books, but its cause lies partly outside the door, so we treat it as the effect of external forces.

Reasoning begins with the boundary test: can the manager change it directly?

Now we sort each external change into groups, using the five groups and the task environment. The table below shows the result. Read each row from left to right: the change, the inside or outside decision, and the groups that fit.

The table records the sorting of each change.

SORTING THE RECORDED CHANGES

CHANGE	INSIDE OR OUTSIDE	GROUPS THAT FIT
Online price cuts of about 15%	Outside	Economic, technological, task environment
Curriculum change from January 2026	Outside	Political and legal; task environment (schools)
Fall in textbook sales	Result in the shop's books	Effect of the forces above

The conclusion states the sorting in plain sentences.

The check tests the arithmetic and guards against a false conclusion.

Conclusion. Three of the four recorded changes are external or result from external forces, and one is internal. The shop met the price cuts by adapting. The record does not show scanning as a regular habit, so the next question for the managers is how they will learn of the next change early.

Check. We test the numbers. The fall in textbook sales is Tk 5,100,000 minus Tk 4,300,000, which equals Tk 800,000. As a share of the 2024 figure, Tk 800,000 divided by Tk 5,100,000 equals about 0.157, or 15.7%. That is close to the 15% price cut, but it does not prove that the price cuts caused the fall, because other forces may also have acted. The curriculum change began in January 2026, after the 2025 fall, so it cannot explain that fall. A careful manager keeps these facts apart.

* * *

The summary restates the chapter in five sentences.

SUMMARY

The internal environment is what managers can change directly. The external environment is what they can only study, prepare for or influence. The boundary follows control, not distance. Outside forces fall into five groups: economic, social and cultural, political and legal, technological, and global. The groups of forces that deal directly with the organisation form the task environment. Managers meet the environment by scanning, by adapting and by influencing, and they combine these ways in a strategic response.

Four situations test the chapter, from familiar to combined.

THINK IT THROUGH

Each question below changes the situation a little more than the one before. Answer them in order.

5.T1 A school near a bookshop moves its book list to a new publisher. Is this change internal or external for the shop? Name the group that fits best.

5.T2 Suppose the shop's own stock clerk, and not the school, proposed the new list. Does your sorting change? Explain using the boundary.

5.T3 A government raises the tax on imported books. Name the group, and say whether the change is inside or outside a bookshop's boundary. Give one response that is adapting and one that is influencing.

5.T4 A small clinic finds that 2 of its 5 nurses have left, a new clinic has opened across the road, and a new rule requires extra records. Sort the three changes. Then explain why watching them is part of planning in the management process.

WHERE PEOPLE GO WRONG

Mistake one: believing that the environment is only what lies outside

Some students believe that the environment means only what lies outside the organisation. This is incomplete. The internal environ-

environment at once. The groups are tools for noticing, not boxes with rigid walls.

Some students believe that scanning predicts the future. It does not. It collects information about changes that have begun, so that managers meet them early. The value of scanning is a smaller surprise, not certainty.

Mistake four: believing that scanning predicts the future.

IN HISTORY

Nokia is a company from Finland. For many years it was the largest maker of mobile phones in the world by number sold. In 2007 the American company Apple released the iPhone, a phone that worked as a small computer with a touch screen. Phones that used Google's Android software soon followed. Nokia chose Microsoft's Windows Phone software for its phones, and its first Windows Phone devices reached customers in late 2011.

Nokia, a leader in mobile phones, sold its phone business in 2013 after a shift in its market.

On 3 September 2013, Nokia announced that it would sell its devices and services business to Microsoft, the American software company. The total price was €5.44 billion. Of that sum, €3.79 billion paid for the business and €1.65 billion paid for a licence to use Nokia's patents. About 32,000 Nokia employees were to move to Microsoft. On 19 November 2013, Nokia's shareholders approved the sale by a very large majority of the votes cast.

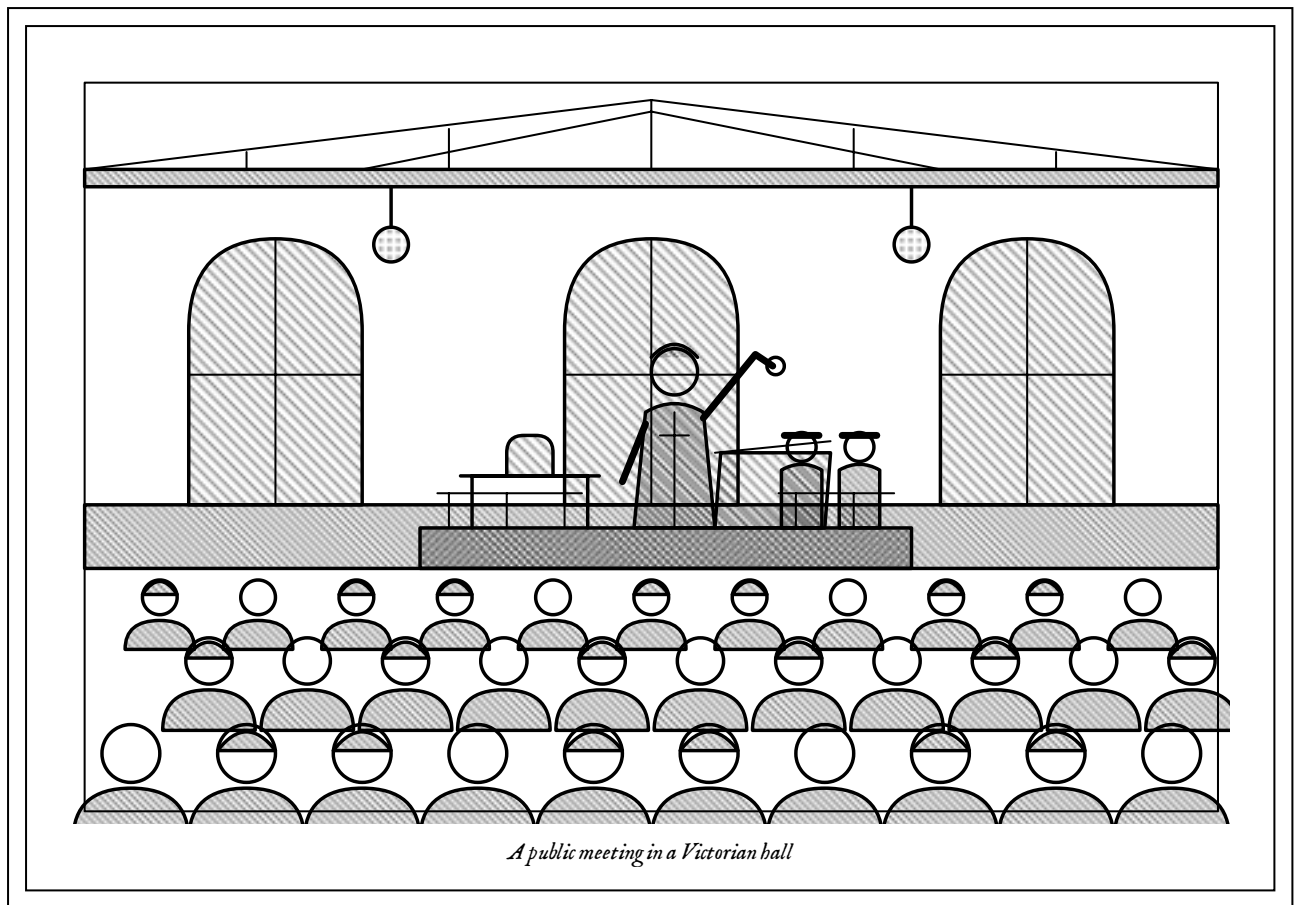
The sale was announced on 3 September 2013 and approved by shareholders on 19 November 2013.

Writers still debate why the change happened. Some stress the company's software and the speed of its decisions. Others stress the force of the new competitors. We do not need to settle that debate here. The case shows three ideas from this chapter. A technological change altered what customers wanted. Rivals in the task environment moved quickly. A strategic response that comes late may leave few choices. Nokia remained in business after the sale, with its network equipment business and its patents.

Writers still debate the causes, but the case shows the environment at work.

REVIEW QUESTIONS

- 5.1 Sort these forces as internal or external: a rival's price cut; a new store manager; a change in school book rules; the shop's own opening hours.
- 5.2 In your own words, state the difference between the internal environment and the external environment. Explain how the boundary decides which is which.
- 5.3 Name the five groups of wider forces. Give one example of each for a bookshop.
- 5.4 Explain the difference between scanning, adapting and influencing. Say which one the price-match list of 3 March 2026 is.



CHAPTER VI

Social Responsibility, Ethics and Sustainability

What does an organisation owe to people it never invited to its table?



This chapter asks what an organisation owes to society beyond its profit. It explains the duty of organisations to society, the moral rules that guide business choices, and the goal of leaving the world fit for later generations. A bookshop in Dhaka and a disaster at Savar serve as cases.

You will need: Chapter 5, 'The Environment of Organisations', section 5.1 'Inside and Outside the Organisation'; Chapter 1, 'What Management Is', section 1.4 'A Manager's Responsibility'.

Imagine a cheap offer that no customer could detect, and ask whether you would accept it.

The easy answer becomes hard once we ask who else is affected.

Every organisation lives among people whom it affects without asking them.

The chapter asks three questions and ends at a building in Savar.

A family acts freely inside its flat but shares the stairs with its neighbours.

Recall: Chapter 5, 'The Environment of Organisations', section 5.1 'Inside and Outside the Organisation' — external forces from society.

An organisation lives in a larger block of flats, which is society.

The duty to act for society's good as well as for profit has a name.

Corporate social responsibility. An organisation's duty to act for the good of society as well as

Imagine that you own a small shop. On Monday a supplier offers you goods at a price far below the usual one. The goods look the same as those you buy now. Your customers would not notice any difference, no inspector will visit, and your profit would rise. Do you accept the offer?

Many people answer at once, but the question is harder than it first appears. Who made these goods, and were those people paid fairly? Who would lose if you bought them? Is the low price lawful? If nobody ever discovers your choice, does it still matter? Each of these questions reaches people who never signed a contract with you.

Every organisation meets this kind of question. It buys from some people, sells to others and employs others. It also uses air, water, roads and land that belong to everyone. Its choices travel far beyond its own door. A manager who reads only the price list sees a small part of the result.

This chapter asks three questions. First, what does an organisation owe to society? Second, how do we decide what is right and wrong when a choice is hard? Third, how can an organisation continue without using up what later generations will need? We shall meet a small bookshop in Dhaka that faces such a choice on 18 June 2026. We shall also visit Savar, near Dhaka, where in 2013 a disaster showed the cost of failing to ask.

* * *

6.1 SOCIAL RESPONSIBILITY

Begin with an everyday picture. A family lives in a block of flats. Inside its own door, the family decides what to cook, when to sleep and how loud to play music. The stairs, the roof and the water tank, however, are shared. A family that leaves rubbish on the stairs harms neighbours who never agreed to it. A good neighbour keeps the stairs clean without being forced to.

An organisation lives in a much larger block of flats, which we call society. Chapter 5 showed that society is one of the outside forces that press on every organisation. This chapter turns the picture round. It asks what the organisation owes to society in return.

The picture has a name in management. **Corporate social responsibility** means an organisation's duty to act for the good of society as well as for profit. A bookshop that refuses to sell stolen goods, pays its employees what it promised and sends its waste paper to recycling is acting on this duty. The word *corporate* once pointed to large companies. The duty, however, applies to every

small example of care for the natural environment, and section 6.3 returns to the idea.

The same duty looks different from each position. The owner asks what it costs now and what trust it earns later. The manager asks how to reach the goals for profit while treating employees, customers and suppliers fairly. An employee sees the duty in wages, safety and respect. A customer sees it in honest prices and honest goods. A neighbour sees it in noise, waste and the state of the street. Each of these people can judge whether the duty is being met.

Not everyone agrees that organisations should carry the duty. The disagreement is old, and each side has a serious case. Milton Friedman, an American economist, argued in 1970 that the only social responsibility of a business is to use its resources to increase its profit while keeping to the law.¹ Others answer that society gives organisations their permission to operate, and that society can withdraw the permission.

The table below sets out three questions in the debate. Read each row from left to right. The first cell names the question, the second gives the argument for the duty, and the third gives the argument against it.

ARGUMENTS FOR AND AGAINST SOCIAL RESPONSIBILITY

QUESTION	ARGUMENT FOR	ARGUMENT AGAINST
Whose money is spent?	Spending on society protects the good name of the organisation and so its profit over time.	The money belongs to the owners, who did not choose to spend it on society.
Who decides?	Managers know their own effects and are best placed to limit harm.	Managers are not elected, so they should not decide what society needs.
What if the organisation does nothing?	Society may reply with stricter laws and lost customers.	Governments and charities exist for social needs, and the organisation should pay its taxes.

Neither column wins in every case. The argument for stresses time: harm to society often returns to the organisation as lost trust or stricter rules. The argument against stresses ownership and skill: managers are trained to run an organisation, not to solve every social problem. Most modern writers hold that both points are true. For them the question is how much responsibility an organisation should accept, not whether it should accept any.

A model is a simplified picture of how something works. Keith Davis, an American management scholar, offered one that links power to duty.² We call it the Davis model. It has five steps, and each step rests on the one before it.

Step 1: an organisation holds social power. It employs people,

¹ Friedman published this argument in the *New York Times Magazine* in 1970. Many writers have replied to it since.

² Davis wrote on this subject from 1960 onwards. Later textbooks present his ideas as a short list of steps.

Each person affected sees the duty from a different place.

Thinkers disagree about whether an organisation should carry this duty at all.

The main arguments for and against appear side by side in a table.

Each side rests on a reasonable claim, and most modern writers accept both.

Davis built a model in five steps that links an organisation's power to its duty.

Steps 1 and 2: social power

Steps 3 to 5: open dealing, costs counted and priced, and help beyond the organisation's own business.

Step 3: the organisation deals openly with society. It listens to people outside and tells them what it does. Step 4: it counts the costs and benefits to society before it acts, and it adds the social costs to its price, so that those who gain from a product also pay for its effects. Step 5: it helps to solve social problems beyond its own activity, where it has the ability to do so.

From power (left) to responsibility and action (right)

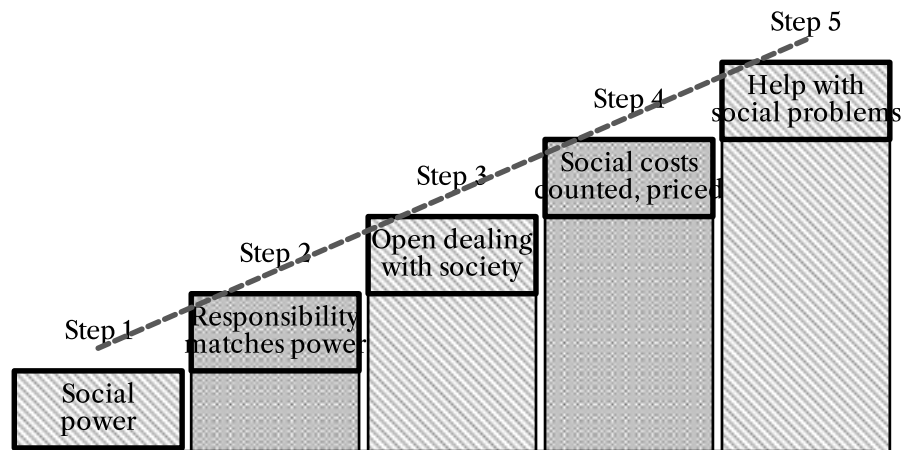


FIG. 6.1. Davis model as five steps from power to responsibility

The figure is read from the lowest step on the left to the highest on the right.

Read the figure from the lowest step on the left to the highest on the right. Each step is one part of the model, and its height shows that it rests on the steps below. The dashed line joins the steps in order. The numbers and words carry the meaning, so the figure reads the same in grey.

The model guides thought but does not measure results.

The Davis model has limits. It does not say how much cost is too much, or which social problem an organisation should help first. It assumes that society can see what an organisation does, and in many places society cannot. It gives a smaller organisation a lighter duty, which is fair, but not a duty of zero. A bookshop with seven employees has little power, yet it still owes something to the people it employs and to the neighbours on its street.

The recap of section 6.1.

Before moving on. Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. Thinkers disagree about it. One side stresses the owners' money, and the other stresses society's permission and the long-term good name of the organisation. The Davis model climbs five steps, from power to responsibility and on to help beyond the organisation's own business.

* * *

6.2 BUSINESS ETHICS

A first question: can an act be lawful and still be wrong?

Ask two questions about any choice. Is it lawful? Is it right? Most people see that the answers can differ. Telling a friend a hurtful lie breaks no law, yet it may be wrong. In the other direction, an old law may require something that many people now think unfair.

Business ethics means the moral rules that decide what is right and wrong in business conduct. A moral rule is a belief about how people ought to treat one another, such as "do not deceive" or "keep your promises". A shop that gives a customer the correct change, even when nobody would notice a mistake, is following such a rule. Chapter 1 explained that a manager answers for results. Ethics adds that a manager also answers for the way the results were reached.

People form their beliefs about right and wrong in many places: in the family, in religion, in the culture around them, in a profession and in the workplace itself. Because these sources differ, ethical beliefs also differ among people and among countries. The same is true of law. A manager who works across borders must therefore name the place whose rule applies before judging an act.

The table below sorts three acts by what the law and ethics say. Read across each row: the act comes first, then the law, then the ethical view. The text after the table explains a fourth case that the table leaves out.

THREE ACTS JUDGED BY LAW AND BY ETHICS

ACT	LAW	ETHICS
Paying employees the wages that were promised	Required	Right
Praising a book's good points while saying nothing of its badly faded print	Often allowed	Doubtful, because it misleads the buyer
Selling a stolen book	Forbidden	Wrong

The first and third rows show law and ethics agreeing. The second row shows them parting: the act may be lawful, but many people would still judge it wrong. A fourth case runs the other way. An organisation may give its employees paid leave to attend a funeral when no law requires it. The act is ethical without being required. For this reason a careful manager asks "Is it right?" after asking "Is it lawful?", and never in place of it.

Some choices have no painless answer. An **ethical dilemma** means a choice in which every option has a cost to some person's rights or interests. Suppose a shop owner can survive a bad year only by dismissing one of two employees or by cutting the pay of both. Each option harms someone. The owner cannot avoid harm, and can only choose which harm to accept and how to reduce it.

A dilemma is not the same as a temptation. In a temptation, one option is plainly wrong, and the difficulty is the wish to take it. A shop owner who is tempted to sell stolen books for a high profit does not face a dilemma. The right answer is known, and only the

Recall: Chapter 1, 'What Management Is', section 1.4 'A Manager's Responsibility' — responsibility.

Business ethics is the set of moral rules that judge conduct at work.

Business ethics. The moral rules that decide what is right and wrong in business conduct.

Ethics. Word in use. In daily life, a person's private sense of right and wrong. In management, shared rules of conduct that guide choices at work and can be written down.

Ethical beliefs come from many sources and differ among places.

A table sorts three acts by what the law and ethics say about them.

Law and ethics overlap but are not the same thing.

Some choices have no painless answer.

Ethical dilemma. A choice in which every option has a cost to some person's rights or interests.

A dilemma differs from a temptation, where one option is plainly wrong.

Good reasoning about a dilemma follows steps that any manager can use.

Organisations can promote ethical conduct in several practical ways.

Reporting channels, rewards and checks complete the set of ways.

Ethics seen by owner, manager, employee and customer.

The recap of section 6.2.

A farmer who cuts every tree gains one season and loses all later harvests.

Sustainability is the wiser farmer's rule applied to an organisation.

Sustainability. Running an organisation so that it meets today's needs without harming later generations'

A manager can reason through a dilemma in steps. First, state the facts and the choices. Second, list every person affected and the cost each choice puts on that person. Third, ask which moral rules apply, such as honesty or fairness. Fourth, ask whether the law allows each choice. Fifth, choose, and be ready to explain the choice to the people who bear its cost. The steps do not remove the difficulty, but they stop a manager from choosing by habit.

Managers cannot order people to be good, but they can build conditions in which good conduct is likely. Several ways are in common use. The first is a written code of ethics, which states the rules the organisation expects its people to follow. The second is the example of leaders, because employees watch what managers do far more closely than what managers say. The third is training, in which employees practise real dilemmas before they meet them.

Three more ways support the first three. A reporting channel lets an employee raise a concern about wrong conduct without fear of punishment. The organisation should reward ethical conduct and discipline wrong conduct in the same way for every person, whatever their rank. Regular checks of accounts and records then show whether the rules are followed in practice. Each way is weak alone. A code that leaders ignore teaches employees that the code is for show.

Each person sees ethics from a different place. The owner wants a name that customers trust. The manager wants rules that are clear enough to apply on a busy day. An employee wants to know that a refusal to do wrong will not cost a job. A customer wants to know that the price, the product and the promise are honest.

Before moving on. Business ethics is the set of moral rules that decide what is right and wrong in business conduct. Law and ethics overlap but are not the same, and a lawful act can still be wrong. An ethical dilemma is a choice in which every option has a cost to someone, and it differs from a temptation. Organisations promote ethics through codes, example, training, reporting channels, fair rewards and checks.

* * *

6.3 SUSTAINABILITY

Picture a farmer with an orchard of mango trees. In one hungry year the farmer cuts every tree and sells the wood. The money feeds the family this year, but there will be no fruit next year or in any year after it. A wiser farmer picks the fruit, plants new seedlings and leaves the orchard as healthy as it was found.

The wiser farmer's rule has a name. **Sustainability** means running an organisation so that it meets today's needs without harming later generations' ability to meet theirs. The United Nations asked a group called the World Commission on Environment and Development to study the question. Its report *Our Common Future*, published in 1987, made this meaning widely known.³ A bookshop

³ The commission is often called the Brundtland Commission, after its chair, Gro Harlem Brundtland of Norway.

A writer on business, John Elkington of Britain, gave this wider judgement a name in 1994. The **triple bottom line** means a way of judging an organisation by its profit, its effect on people and its effect on the natural environment. The phrase "bottom line" comes from accounts, where the last line shows the profit. Elkington added two more last lines, so that an organisation reports on people and on the natural environment as it reports on profit.

Case file, illustrative. Record for 2025, Meghna and Sons: profit Tk 750,000; 7 employees (store manager, accounts and online orders, 3 sales assistants, 1 cashier, 1 stock clerk); 380 kg of unsold paper sent to recycling. Read as a triple bottom line, profit fills the first line, the 7 employees give a first fact for the second line, and the recycled paper gives a first fact for the third. The record holds no measure of pay, safety or respect, so it cannot complete the judgement.

How can a customer or a student tell whether an organisation is responsible? The best answer is to look for evidence instead of claims. A responsible organisation reports what it does, including its failures. It pays its employees what it promised and keeps their workplace safe. It lets independent people check its suppliers. It keeps to the law and to its own code of ethics, and it admits its mistakes and repairs them.

This test has limits. A handsome report can hide poor conduct, and an organisation can say that it is green while doing little. The useful question is always "Who checked this claim?" Standards also differ by place. Rules on waste, wages and building safety vary from one country to another, so we judge an organisation against the rules of the place where it works and against the higher rules that it sets for itself.

Before moving on. Sustainability is running an organisation so that it meets today's needs without harming later generations' ability to meet theirs. The triple bottom line judges an organisation by profit, by its effect on people and by its effect on the natural environment. We recognise a responsible organisation by evidence that an independent person has checked.

* * *

WORKED CASE: THE PHOTOCOPIED TEXTBOOKS

We now apply the ideas of this chapter to one decision in the case file. The worked case has five parts: Situation, Question, Reasoning, Conclusion and Check. It shows how the ideas combine. It is an application and not a model answer.

Case file, illustrative. Ethical choice, 18 June 2026, Meghna and Sons. A supplier offered 500 photocopied copies of a textbook at Tk 180 each against Tk 260 for genuine copies, a saving of $500 \times 80 = \text{Tk } 40,000$. The shop declined.

Question. Should the shop have accepted the offer? To answer,

⁴ Copyright law differs from country to country. A student should read the copyright law of the country concerned before judging a real case.

The triple bottom line judges an organisation by three results instead of one.

Triple bottom line. A way of judging an organisation by its profit, its effect on people and its effect on the natural environment.

A single year's record fills the three lines only in part.

A responsible organisation shows evidence and not only claims.

Fine reports can hide poor conduct, so the question to ask is who checked.

The recap of section 6.3.

The worked case applies the chapter's ideas in five parts to one decision.

The situation: a supplier offers cheaper photocopied copies, and the shop declines.

The question is whether the

Tests 3 and 4: the model and the triple bottom line weigh the gain against the risk.

The conclusion and the check agree that the shop should decline.

Rung 2 reshapes the case: the books are genuine old stock.

Rung 3 moves the case to a garment buyer in another setting.

Rung 4 combines the ideas into a choice with costs to several people.

A short summary of the chapter's five defined terms and their links.

sale, because they receive nothing from it. The customers may also receive a poorer product than they believe they are buying. The shop would bear the cost of a damaged name.

Reasoning, tests 3 and 4: the Davis model and the triple bottom line. Davis step 3 asks whether the shop could tell its customers openly how it obtained the books. Step 4 asks that the social costs be counted, and here they fall on the authors and the publisher. For profit, the saving of Tk 40,000 compares with the 2025 profit of Tk 750,000 as $40,000 \div 750,000 = 0.053$, or about 5.3 per cent. Against this, schools bought Tk 3,600,000 of the 2025 sales of Tk 14,400,000, which is 25 per cent of sales. A school that learned of unauthorised copies might stop buying.

Conclusion. The shop should decline. The offer probably fails the law, fails the moral rules, hides a cost from the authors and the publisher, and risks a much larger share of sales than it gains. *Check.* All four tests point the same way, and the case file records that the shop declined. The shop's own record therefore agrees with the reasoning.

CHANGING THE SITUATION

The same ideas help in new situations. In the second rung, suppose the supplier offers genuine copies of an earlier edition at Tk 180 because the edition is being cleared. The law test is passed. The moral rule of honesty now asks whether customers know that the edition is old.

In the third rung, a clothing buyer in another country is offered garments at a low price from a factory that no one has inspected. The Davis model says that the buyer's power creates a duty, and step 4 says that the safety cost must be counted and not passed to the factory's employees.

In the fourth rung, a small organisation learns that its supplier keeps prices low by dismissing employees without notice. Dropping the supplier may raise its own costs. The choice is a dilemma, so we list those affected, apply the Davis model and judge the result by the triple bottom line.

* * *

SUMMARY

Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. The Davis model links that duty to power in five steps. Business ethics gives the moral rules for judging conduct, and an ethical dilemma is a choice in which every option has a cost to someone. Sustainability asks an organisation to meet today's needs without harming later generations. The triple bottom line judges an organisation by profit, people and the natural environment.

THINK IT THROUGH

6.T1 A small printing shop is offered a rush order. It can finish on

6.T2 Using only the facts given in this chapter, judge Meghna and Sons by the triple bottom line. State what each of the three lines shows, and name two facts that you would still need before you could finish the judgement.

WHERE PEOPLE GO WRONG

Mistake: "If an act is legal, it is ethical." Law sets the minimum that society enforces. Many lawful acts mislead or harm people, and a careful manager asks whether an act is right as well as whether it is lawful.

Mistake: "Social responsibility means giving money to charity." Gifts can be part of it, but the duty begins with the organisation's own conduct. An organisation that gives money while it pays unfairly or ignores safety has not met the duty.

Mistake: "Sustainability is only about protecting nature." Sustainability also covers people and money. An organisation that harms its employees or loses all its money cannot continue, and the triple bottom line counts all three.

Mistake: "Every difficult choice is an ethical dilemma." A dilemma is a choice in which every option has a cost to someone. When one option is plainly wrong and the difficulty is the wish to take it, the situation is a temptation.

Mistake: "Only large organisations need to think about this." Power grows with size, so a large organisation owes more. A small shop owes less, but its choices still reach employees, customers and neighbours, and the duty is never zero.

Mistake one: believing that whatever is lawful must also be right.

Mistake two: believing that social responsibility means giving money away.

Mistake three: believing that sustainability concerns only nature.

Mistake four: believing that every hard choice is a dilemma.

Mistake five: believing that only large organisations carry the duty.

IN HISTORY

On 24 April 2013, the Rana Plaza building collapsed at Savar, near Dhaka, in Bangladesh. The building housed five garment factories that made clothing for brands sold in many countries. More than 1,100 people died, most of them garment employees, and more than 2,000 others were injured. The disaster was not the first in the industry. Earlier fires in garment factories had already killed many employees.

In May 2013, brands and retailers signed the Accord on Fire and Building Safety in Bangladesh together with international trade unions. The Accord was a legally binding agreement. It required independent inspections of supplier factories by qualified engineers, and it required the results to be made public. Reports on the events state that pressure from the media and from customers led more companies to sign after the collapse.

The events show this chapter's ideas at work. Buyers hold power over their suppliers, which Davis step 2 says creates a duty. Low prices can push costs of safety onto employees, which Davis step 4 says must be counted. The agreement made the dealing between buyers and suppliers more open, as Davis step 3 asks. Finally, the loss of life shows what a triple bottom line would record on the line for people.

Rana Plaza in 2013 showed the human cost of ignoring safety and responsibility.

Pressure from the public led brands and unions to sign a binding agreement.

The events show the chapter's ideas at work in a real case.

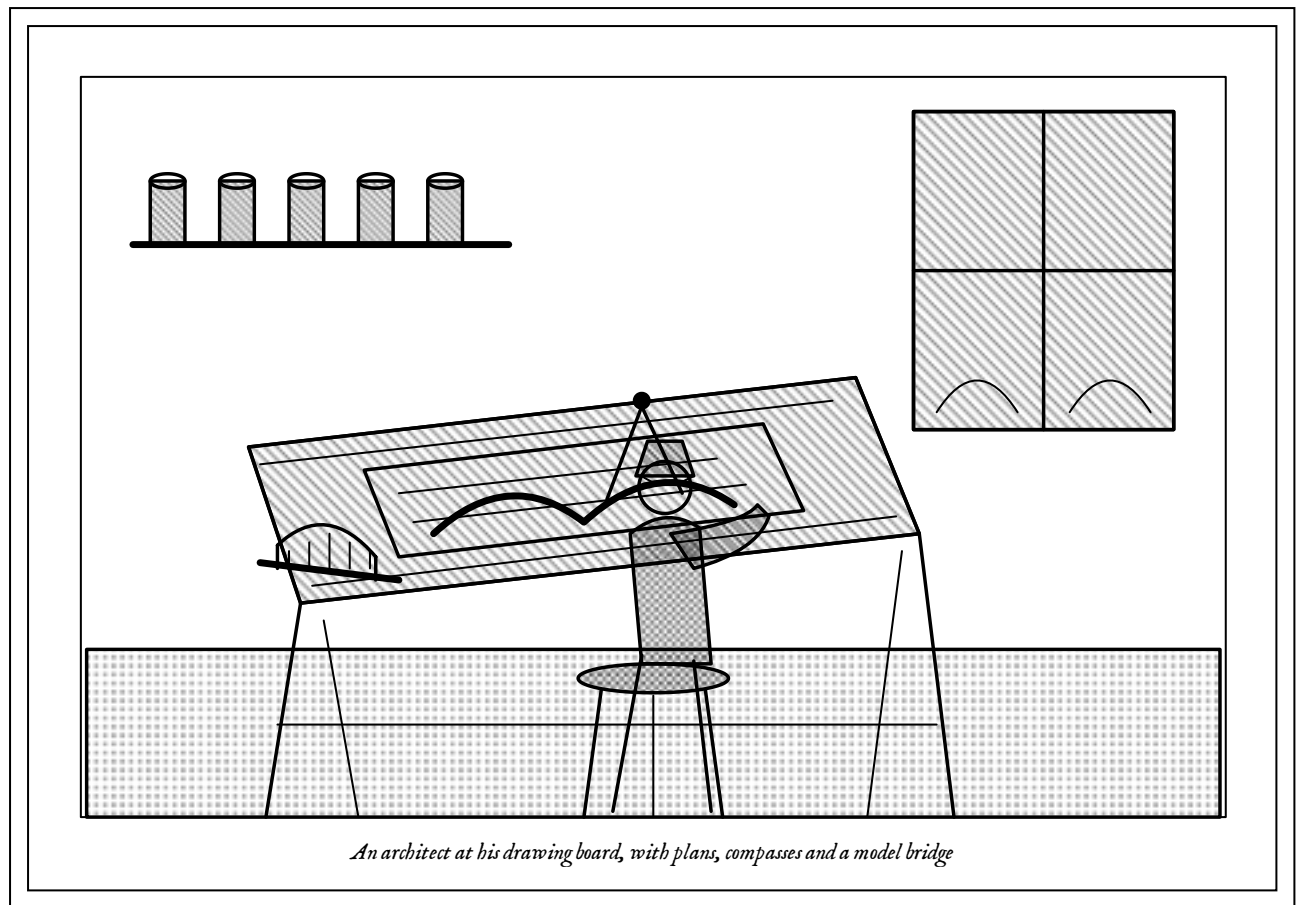
6.2 Give one argument for and one argument against the idea that organisations should carry a duty to society. State which owner, manager or employee would most likely make each argument.

6.3 Name the five steps of the Davis model in order. For step 2, say what the iron law of responsibility warns.

6.4 Explain the difference between an ethical dilemma and a temptation, with one example of each.

6.5 Sort each act as legal, ethical or both, and give a reason: (a) paying employees on the day promised; (b) selling a stolen book; (c) giving employees paid leave for a funeral when no law requires it.

6.6 A rival bookshop cuts the safety of its storeroom to lower its price. Using ideas from this chapter, explain why this is a problem for society and for the rival, and say what a manager at your own shop should do.



CHAPTER VII

Planning: Meaning, Nature and Steps

How does a wish turn into something you can actually do?



This chapter explains what planning is and why every organisation needs it. You will learn the five Steps that turn a wish into a workable plan, and you will study the reasons why plans fail. The ideas are built slowly, from a household journey to the order book of a shop.

YOU WILL NEED

This chapter builds on two earlier ideas. The first is the list of activities that every manager performs, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'. The second is how output is compared with the resources used, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.2 'Productivity'.

A family must reach a wedding in another city before the ceremony starts.

GO SLOWLY HERE.

Wanting to arrive is not the same as knowing how to arrive.

Plan. Word in use. In daily life, an intention, as in "I plan to rest." In management, a statement of what will be done, by whom and by when.

A shop faces the same puzzle on a larger scale, with money and deadlines.

Answering that question is the work of planning, which this chapter teaches.

Imagine that a family has been invited to a wedding in a city 300 kilometres away. The ceremony starts at noon on Saturday. Nobody has decided how to travel, what to carry, or when to leave. On Friday evening, one person says, "We will manage somehow." Is that sentence enough? Think about the questions that remain. Which road? Which vehicle? Who pays for fuel? What happens if the road is closed by rain? Each question has an answer, but the family has not found the answers yet.

The family wants one thing, which is to arrive on time. Wanting is the easy part. The difficult part is deciding, before the journey begins, what will be done and in what order. A family that sits down on Wednesday, chooses the road, counts the hours, and agrees who will drive has made a plan. A family that waits until Saturday morning has only a wish.

Now move from a family to a shop. A bookshop must order thousands of textbooks weeks before the school season opens. If the owner waits until the shelves are empty, the books will arrive too late, and customers will buy elsewhere. If the owner orders too many, money is locked in books that do not sell. The puzzle is the same as the family's, but the cost of a wrong answer is greater. This chapter answers a simple question that has no simple answer: how does a manager decide, in advance, what to do?

We shall build the answer in layers. First we define planning and see what it does. Then we learn five steps that take a manager from a wish to a decision. Last we study why plans fail, because a plan that fails teaches as much as one that works. Many students find this chapter slow, because the ideas feel obvious after they are explained. They are not obvious before. Take your time.

* * *

7.1 WHAT PLANNING IS

PLANNING DEFINED

Begin with the plan itself, because planning is the act of making one. A **plan** is a written or spoken statement of what will be done, by whom and by when, to reach an objective. Here an objective

¹ A full account of objectives is given in Chapter 9 'Objectives and Management by Objectives', section 9.1 'Objectives'. Until then, read objective as the result that you have chosen to reach.

A plan states what will be done, by whom and by when.

Plan. A written or spoken statement of what will be

process.² Planning comes first because the other activities depend on it. An organisation cannot arrange its people, direct their work or check results until it knows what it wants to achieve.

Recap. A plan states what will be done, by whom and by when. Planning is the activity of deciding this in advance. You should now be able to say: "A plan names the action, the person and the time."

You should now be able to say what a plan contains.

THE CASE OF THE TEXTBOOK SEASON

Let us test the idea on a real record. The shop's textbook season is the busiest time of the year. Nurul Meghna, the owner-manager, cannot wait until customers arrive, because the supplier needs time to deliver. The case file below records the first facts of the season.

Meghna and Sons must choose an order size long before the season opens.

GO SLOWLY HERE.

Case file, illustrative. Planning for the textbook season: the season starts 2 January 2026. The order deadline is 12 December 2025. The expected demand is 1,800 textbook sets at an average of Tk 2,400 each, which is Tk 4,320,000.

The case file gives the season date, the order deadline and the expected order.

Read the record slowly. The deadline of 12 December comes before the season start of 2 January, so the shop must decide first and sell later. The value of the order comes from one multiplication: 1,800 sets × Tk 2,400 = Tk 4,320,000. This sum is large for a family shop. A wrong plan would cost real money, and this is why the shop plans before it acts.

Multiplying the expected sets by the average price gives the value of the order.

Recap. The shop orders on 12 December and sells from 2 January. Between those dates it must trust its plan. You should now be able to say: "Planning is needed because action and result are separated by time."

You should now be able to say why the shop must plan before the season.

THE NATURE OF PLANNING

What features does planning have in every organisation, from a shop to a government? Five are worth learning, and the table sets them side by side. Read it row by row: the left column names a feature, and the right column says what it means.

Planning has five features that describe it wherever it is found.

GO SLOWLY HERE.

THE NATURE OF PLANNING

FEATURE	WHAT IT MEANS
Planning looks forward	It concerns the future, and the future is not known
Planning is for a purpose	It aims at an objective, and without an objective there is nothing to plan
Planning chooses	It selects one course of action and leaves the others
Planning is needed at every level	The owner, the manager and the employee all make plans of different size
Planning is continuous	Plans are reviewed and renewed as conditions change

Take the second and third features. Planning is for a purpose, because a plan is a tool for reaching an objective. Planning chooses, because the family cannot take two roads at once. These features also explain a common confusion. Planning is not a guess about the

The five features show that planning is forward-looking, purposeful, choosing, shared and continuous.

² The management process is set out in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

You should now be able to name the five features.

Planning serves several purposes, from reducing guesswork to making control possible.

GO SLOWLY HERE.

Planning improves productivity and effectiveness by directing effort to the aim.

Planning has limits, and the last section of this chapter returns to them.

You should now be able to list the purposes of planning.

Section 7.1 in three lines.

Planning follows five steps in order, and each step needs the one before it.

in size, but each one answers the same three questions of action, person and time.

Recap. Planning looks forward, serves a purpose, chooses, happens at every level and never ends. You should now be able to say: "Planning is a reasoned decision about the future, taken in advance."

THE PURPOSES OF PLANNING

Why plan at all? The family could drive and see what happens. Four purposes answer the question. First, planning reduces uncertainty, because it makes the manager look at the future before it arrives. Second, planning gives direction: every employee knows the aim and the part each person plays. Third, planning saves effort, because it removes wasted work and repeated decisions. Fourth, planning makes control possible, because a manager can check results only against something that was planned.

These purposes connect to two ideas from Chapter 3. Planning raises productivity, which compares output with the resources used.³ It also raises effectiveness, which asks whether the right aim was reached. A shop that orders the correct number of books wastes less money, so it is more productive. It also meets its customers' needs, so it is more effective.

Planning has its limits too, and an honest account must say so. It costs time and money. It can make an organisation rigid if the plan is followed after conditions have changed. These limits do not remove the value of planning. They warn the manager to plan with care, and section 7.3 examines them.

Recap. Planning reduces uncertainty, gives direction, saves effort and makes control possible. You should now be able to say: "Without a plan there is nothing to check results against."

PAUSE AND CHECK 7.P1 Think of a task you did recently without a plan. What would the plan have contained: the action, the person and the time?

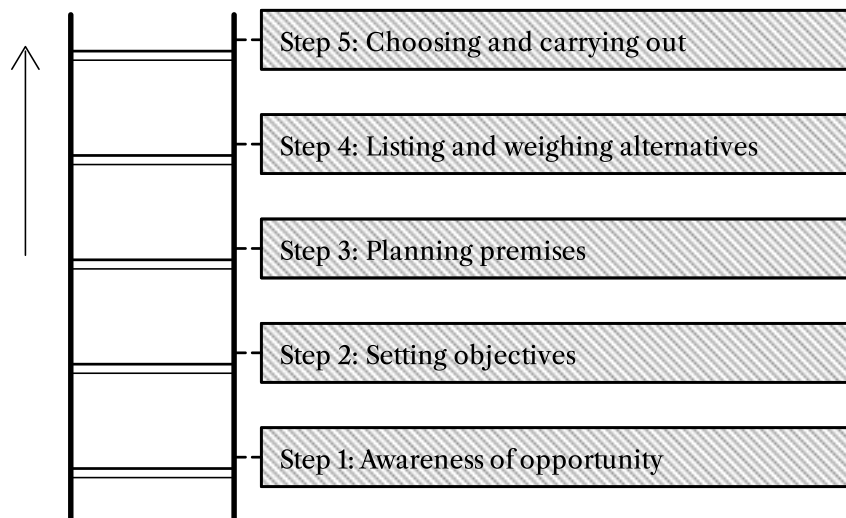
Before moving on. A plan states what will be done, by whom and by when. Planning is the advance decision that produces it, and it is forward-looking, purposeful, choosing, shared and continuous. Its purposes are less uncertainty, direction, saved effort and the possibility of control.

* * *

7.2 STEPS IN PLANNING

Is there a method for planning, or does each manager invent one? Managers have found that good plans usually pass through five steps in the same order. The ladder below shows them. Read it from the bottom rung to the top: each rung is one step, and a manager climbs one rung before reaching the next. Missing a rung is like missing a stair in the dark; the next step becomes unsafe.

³ Measures of performance are explained in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.2 'Productivity'.



Start at the bottom rung and climb in order.

FIG. 7.1. *The five Steps of planning as a ladder*

STEP ONE: AWARENESS OF OPPORTUNITY

Planning begins before any plan exists. It begins when a manager notices something: a chance to gain, or a problem to remove. We call this awareness of opportunity. The family in our opening puzzle noticed that a wedding invitation had arrived. The shop notices that the textbook season is approaching. Awareness asks the manager to look at the organisation, at its customers and at the world outside, and to ask what could be done better.

An everyday bridge helps. A person who sees that a bus fare has risen thinks of cycling to work. The rise in fare is the signal, and the thought of cycling is the opportunity. Notice that no decision is made at this stage. Awareness only identifies a possibility. It fails when a manager is too busy to look up from daily tasks, and then an opportunity passes unseen.

Recap. The first step is noticing a chance or problem. You should now be able to say: "Planning starts with seeing something worth acting on."

The first step is to notice a chance or a problem before others do.

GO SLOWLY HERE.

Awareness is a habit of looking, and it prepares the ground for the next step.

You should now be able to say what awareness of opportunity is.

STEP TWO: SETTING OBJECTIVES

Once an opportunity is seen, the manager decides what result to aim for. This is setting objectives. The aim must be stated so exactly that everyone can tell whether it was reached. "Do well in the season" is a wish. "Order 1,800 textbook sets by 12 December" is an objective. Chapter 9 explains objectives in full; for now, remember that a plan cannot be built without one.

Think of the viewpoints. The owner wants a profitable season. The store manager wants every shelf ready on the first day. The customer wants the right book, in stock, at a fair price. A good objective brings these viewpoints together in one sentence that all can read. When the objective is vague, each person fills the gap with a personal guess, and the guesses disagree.

Recap. The second step states the result wanted, exactly and with a date. You should now be able to say: "An objective is stated so exactly that everyone can tell whether it was reached."

The second step is to decide exactly what result is wanted, and by when.

A clear objective tells every person where the effort is going.

You should now be able to say why the second step comes second.

A planning premise is an assumption about future conditions on which a plan is built.

Premise. Word in use. In daily life, a building and its land, as in "the shop premises". In planning, an assumption stated at the start of an argument or a plan.

Planning premise. An assumption about future conditions on which a plan is built.

Premises are a choice of which assumptions matter, so a manager picks the few that count most.

A forecast is a reasoned estimate of what will happen, used as a base for a plan.

Forecast. A reasoned estimate of what will happen in the future, used as a base for a plan.

The shop's case file records its main premise and its supplier lead time.

The lead time is the gap between ordering and receiving, and it fixes the deadline.

You should now be able to say what a planning premise is and why it is written down.

The fourth step is to list the possible ways to reach the objective and compare them.

GO SLOWLY HERE.

premise. A family might assume that the road will be open and that fuel prices will stay the same. These assumptions are the ground on which the plan is built.

A **planning premise** is an assumption about future conditions on which a plan is built. Suppose a family assumes that it will rain on Saturday. That assumption changes the plan: umbrellas are packed and extra time is allowed. If the assumption is wrong and the day is dry, little harm is done. If the assumption is the opposite, and the family expects sun in a monsoon, the plan fails on the road.

The manager cannot list every possible condition, so the manager picks the few assumptions that would change the plan most. Of all the things that might change, a shop owner asks which ones would make the order too large or too small. Prices, customer numbers and delivery time usually matter most. Choose the few important assumptions and write them down, so that everyone who uses the plan shares them.

One of the best sources of premises is a **forecast**. A forecast is a reasoned estimate of what will happen in the future, used as a base for a plan. For example, a shop may estimate next season's sales from the sales of the last three seasons. The word "reasoned" matters: a forecast rests on evidence such as past records, and it is more than a hope. Even so, a forecast can be wrong, and a careful manager reviews the forecast when facts change.

Case file, illustrative. Premise for the textbook season: publishers raise prices by 5%. Supplier lead time: 21 days.

Read the record carefully. The lead time of 21 days is the gap between placing an order and receiving the books. Count 21 days from the order deadline of 12 December: December has 31 days, so $12 + 21 = 33$, which is 2 January. The books ordered on the last day arrive on the first day of the season. This shows why a deadline must be set backward from the date when the result is needed.

Recap. A planning premise is an assumption about future conditions on which a plan is built, and a forecast is a good source for one. You should now be able to say: "A premise is what we assume about the future, written down so that everyone shares it."

PAUSE AND CHECK 7.P2 The shop's plan assumes that publishers raise prices by 5%. Say what would happen to the order value if publishers raised prices by 10% and the shop did not change its plan.

STEP FOUR: LISTING AND WEIGHING ALTERNATIVES

With the objective and premises written down, the manager asks how the objective can be reached. There is almost never one way. A family can drive, take a bus or hire a car. We call each possible course of action an alternative. Listing alternatives forces the manager to think beyond the first idea, which is often not the best

Weighing is not done in the dark. The manager tests each alternative against the premises. If the premise says that publishers raise prices by 5%, an alternative that delays the order may cost more, because later orders might be priced higher. An alternative that looks cheap today may look poor under the premises. This is why the premises are written before the alternatives are weighed.

Recap. An alternative is one possible course of action. The manager lists several and weighs each against the premises. You should now be able to say: "A good plan chooses from several alternatives, not from one."

Weighing alternatives uses the premises as the standard of comparison.

You should now be able to say what an alternative is.

STEP FIVE: CHOOSING AND CARRYING OUT

The last step has two parts. First the manager chooses the alternative that best reaches the objective under the premises. A choice means giving up the others, and this is difficult, because each alternative has some attraction. Second, the manager carries the plan out. This means turning the choice into tasks: who does what, with what resources, by what date. A choice that is never carried out is no better than no choice.

Many students find this step easy to skip, because it feels like action and not planning. In fact it belongs to planning, because the plan states who will do what and by when. If employees do not know the plan, they cannot follow it. The manager must communicate the plan and provide the money, time and tools to carry it out. Then the manager watches the results, and the control of those results is taught in Chapter 18.

After the plan is carried out, results appear. A good result confirms the premises; a poor result shows that a premise was wrong. Either way the manager sees something new, and this is the first step again: awareness. Planning therefore does not end. It circles back, which is why the fifth feature of planning, continuity, is true.

Before moving on. The five steps are awareness of opportunity, setting objectives, stating planning premises, listing and weighing alternatives, and choosing and carrying out. Each step needs the one before. Results lead back to awareness, so planning is a loop.

The fifth step is to choose one alternative and put it into action.

Carrying out needs people to understand the plan and to be given the means to act.

Planning is a loop, because results lead to new awareness.

Section 7.2 in four lines.

* * *

7.3 WHY PLANS FAIL

Suppose a family makes a careful plan and still arrives late. Does this mean that planning is useless? No. A plan that fails is a lesson. The causes of failure repeat across organisations, and a manager who knows them can avoid most. Three causes appear again and again, and a fourth cause is the limit of planning itself.

The first cause is a wrong premise. If the plan assumes that the road is open and the road is closed, the plan fails however well it was made. A shop that assumes steady prices and meets a large price rise will find its order cost higher than planned. This is why premises are written down and checked: a wrong premise is hard to find when it is hidden.

The second cause is a vague objective, or a plan that is not com-

Many plans fail, and the causes are known and can be avoided.

Wrong premises are the most common cause of failure.

A plan fails when the

Plans can also fail from rigidity, cost and the difficulty of knowing the future.

Section 7.3 in three lines.

Planning also has limits. The future cannot be known, so every forecast can be wrong. Planning takes time and money, and a very detailed plan may cost more than it saves. A plan that is followed too strictly stops a manager from using a new chance. The answer is not to stop planning but to plan in proportion: more detail where mistakes are costly, less detail where conditions change fast, and a regular review in every case.

Before moving on. Plans fail because of wrong premises, vague or unshared objectives, and lack of commitment. Planning has limits: the future is uncertain, it costs time and money, and it can become rigid. A manager reviews plans often.

* * *

WORKED CASE: CHECKING THE TEXTBOOK PLAN

This worked case tests the shop's plan against its premise.

This worked case follows the standard order for a descriptive chapter: Situation, Question, Reasoning, Conclusion and Check. It applies the five steps to the shop's record. It is a study of the idea and not a model answer.

The situation collects the facts from the shop's case file.

Situation. The season starts on 2 January 2026. The order deadline is 12 December 2025. The shop expects to sell 1,800 textbook sets at an average of Tk 2,400, which is Tk 4,320,000. The premise is that publishers raise prices by 5%. The supplier lead time is 21 days.

The question asks whether the budget still holds under the premise.

Question. If the premise is right, how much will the order cost, and does the deadline still allow the books to arrive on time?

The reasoning finds the new price, the new order value and the arrival date.

Reasoning. First, find the new average price. Five per cent of Tk 2,400 is $2,400 \times 5 \div 100 = \text{Tk } 120$, so the new price is $2,400 + 120 = \text{Tk } 2,520$. Second, find the new order value: $1,800 \times \text{Tk } 2,520 = \text{Tk } 4,536,000$. Third, find the increase: $\text{Tk } 4,536,000 - \text{Tk } 4,320,000 = \text{Tk } 216,000$. Fourth, check the date: 12 December plus 21 days is 2 January, so the books arrive on the day the season starts.

The conclusion tells the manager what to prepare.

Conclusion. If the premise is right, the order costs Tk 4,536,000, which is Tk 216,000 more than the first estimate. The deadline allows the books to arrive on 2 January, with no spare days. A careful manager would raise the budget and consider ordering earlier.

The check repeats the increase by a second route.

Check. Five per cent of Tk 4,320,000 is $4,320,000 \times 5 \div 100 = \text{Tk } 216,000$. This equals the increase found in the third step, so the two routes agree.

* * *

SUMMARY

The chapter in a few lines.

Planning is deciding in advance what to do, by whom and by when. A plan states these things to reach an objective. Planning looks forward, serves a purpose, chooses, happens at every level and never ends. It follows five steps: awareness of opportunity, setting objectives, stating planning premises, listing and weighing alternatives,

THINK IT THROUGH

The questions begin with the shop and move to new settings.

Four questions that change the situation step by step.

7.T1 The shop's season starts on 2 January 2026 and the supplier lead time is 21 days. By what date must the order be placed, and why?

7.T2 Suppose the shop built its plan on the premise that prices stay the same, but publishers raise prices by 5%. Show how much more the order costs, and say what the manager should have done.

7.T3 A family restaurant plans to open a second branch. Name the five steps of planning and give one question the owner should ask at each step.

7.T4 A school plans a sports day outdoors in the rainy season, and the plan assumes dry weather. List two alternatives the school could weigh and say how each handles the premise.

WHERE PEOPLE GO WRONG

Some students think that a plan promises what will happen. It does not. A plan is a decision about what to do, based on premises. When the premises change, the plan must change.

A plan is not a guarantee about the future.

Some students think that only owners plan. In fact an employee who decides the order of tasks for the morning is planning too. The size of the plan changes with the level; the three questions of action, person and time stay the same.

Planning is not only for senior managers.

Some students jump from the first idea to action. The five steps exist because each one removes a kind of error. Without objectives the aim is unclear; without premises the future is ignored; without alternatives the choice is the first idea and not the best one.

Skipping steps produces weak plans.

Some students believe that writing the plan completes the work. A plan that is not carried out, and not reviewed, is only a document. Planning ends when results have been compared with the plan.

A written plan is not useful if nobody follows it.

IN HISTORY

On 25 May 1961, President John F. Kennedy told the United States Congress that the nation should commit itself to landing a person on the Moon and returning that person safely to Earth before the decade ended. At that time the country had sent a person into space for only a few minutes. The goal was bold, and it was stated with a date.

A president set a goal and a national team built the plan to reach it.

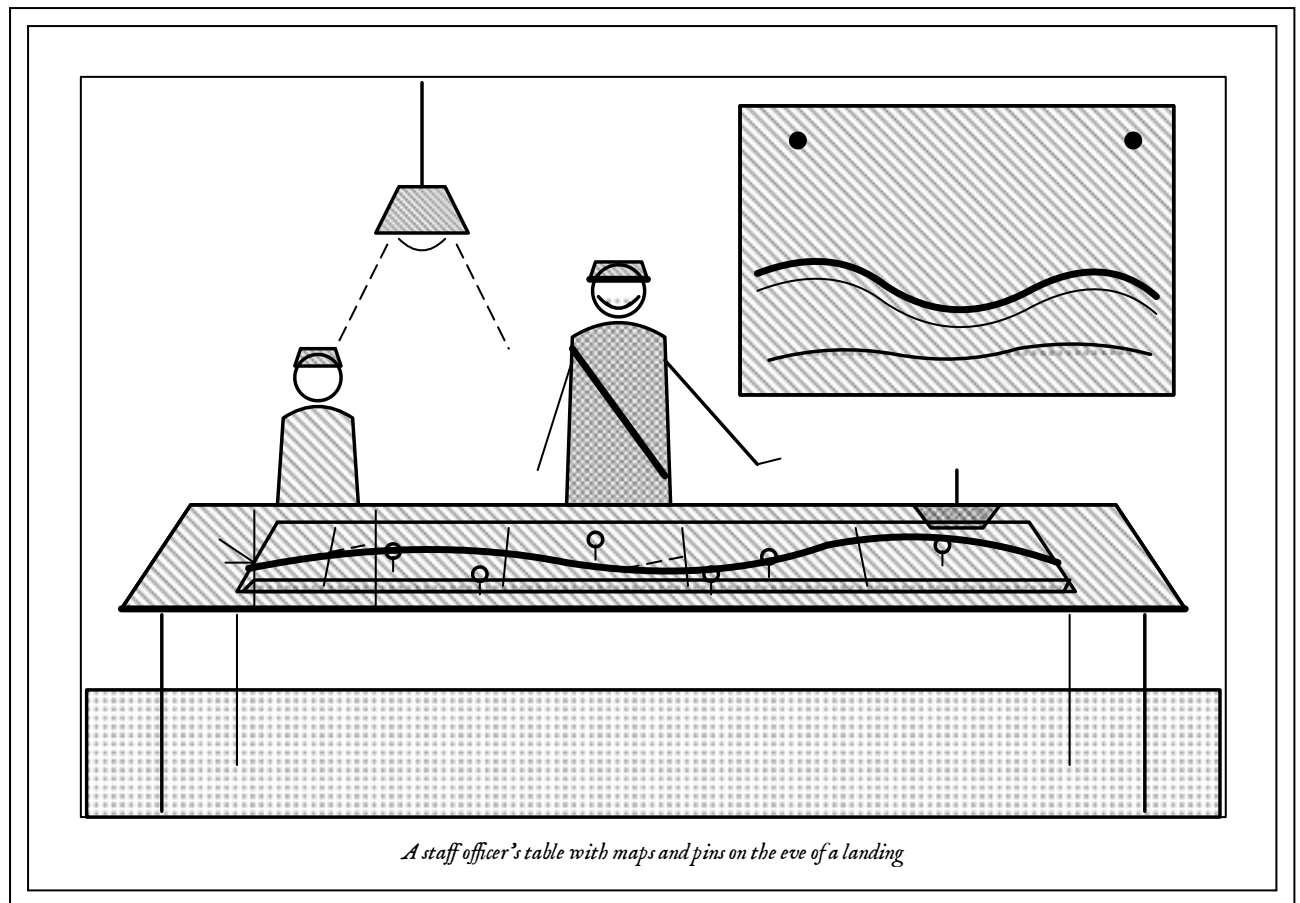
A goal of this size cannot be reached by wishing. NASA, the space agency of the United States, had to study possible methods, estimate costs and time, build rockets and train astronauts. It listed and weighed different ways of reaching the Moon and chose the method of lunar-orbit rendezvous, in which a small landing craft leaves a larger spacecraft that stays in orbit. The work followed the same pattern that this chapter has described: an objective, premises, alternatives, a choice and then action.

Planning turned the goal into many steps carried out by many people.

REVIEW QUESTIONS

Five questions that review the chapter.

- 7.1 Put the five steps of planning in order, and say in one sentence what happens at each.
- 7.2 Define a plan and a planning premise, and give one example of each from a shop.
- 7.3 What is the difference between a forecast and a planning premise?
- 7.4 Explain why the shop cannot wait until the shelves are empty before it decides on its order.
- 7.5 Name three reasons why plans fail, and say how a manager can reduce each one.



CHAPTER VIII

Kinds of Plans and Making Planning Work

Must a manager think everything out again each morning?



This chapter sorts plans into kinds: those used again and again, and those used once. It shows how plans differ by length of time and by level, and it explains what makes planning work in practice. The examples come from a bookshop's returns, a book fair and a yearly budget.

YOU WILL NEED

This chapter builds on two earlier ideas. The first is what planning is and why it is done, in Chapter 7 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'. The second is the list of activities that every manager performs, in Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

A shop must answer the same customer question many times and a rare question once.

A rare event, such as a book fair, needs its own careful thought.

The answer is to use different kinds of plan for repeated and one-time situations.

Picture the counter of a bookshop on a busy Saturday. A customer asks to return a book. Ten minutes later, another customer asks the same thing. By the end of the day, six customers have asked. Must the manager think the problem through six times, and perhaps give six different answers? That would waste time, and it would be unfair, because one customer might be helped and another refused for the same reason.

Now picture a different day. The shop is invited to run a stall at a book fair that lasts four weeks and will not be repeated. No earlier answer exists for this event, because it has never happened before. The manager must think it through once, with great care, and then the thinking will be finished. Is it sensible to treat the two days in the same way? Surely not.

The puzzle shows that plans are not all alike. Some are made once and used many times, so that the manager can save thought for harder problems. Others are made once for one occasion and then set aside. In Chapter 7 we learned how to make a plan by following five steps. This chapter asks a new question: what kinds of plan exist, and how can a manager make planning work in daily life? We begin with the main division.

* * *

8.1 KINDS OF PLAN

Plans divide into those used repeatedly and those used once.

How can we sort the many plans that a shop makes? A useful first step is to ask whether the plan will be used again. If it will, it belongs to one group. If it will not, it belongs to the other. Almost every plan fits one group or the other, and the sorting question is easy to ask.¹

STANDING PLANS

A standing plan is made once and used again for situations that repeat.

Standing plan. A plan made once and used again and again for situations that repeat.

Standing plans save thought, give fair treatment and let employees decide alone.

The first group is the **standing plan**. A standing plan is a plan made once and used again and again for situations that repeat. For example, a shop that decides how to answer returned books once, and then applies the same answer to every customer, uses a standing plan. The word "standing" tells us that the plan remains in place, like a post that stays upright from day to day.

Standing plans give three benefits. They save time, because the manager does not decide the same question again. They give fair treatment, because every customer meets the same answer. They ¹ The meaning of a plan and the steps of planning are set out in Chapter 7 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'.

SINGLE-USE PLANS

The second group is the **single-use plan**. A single-use plan is a plan made for one occasion or project and not used again. A book fair that happens once, a move to a new building and the opening of a branch are all examples. Once the occasion is over, the plan has no further use, although the manager may learn from it when making the next one.

The two groups answer different needs. Think of the owner's viewpoint. The owner wants the repeated questions answered cheaply and the rare event answered carefully. The employee's viewpoint is different: the employee wants to know quickly what to do about a returned book, and has little need to know about a book fair that will not return. A well-run shop therefore keeps a small collection of standing plans for daily work, and writes a new single-use plan for each large event.

PAUSE AND CHECK 8.P1 A shop plans a one-day sale to clear old stock. Is its plan a standing plan or a single-use plan? What would change your answer?

A single-use plan is made for one occasion and then set aside.

Single-use plan. A plan made for one occasion or project and not used again.

Single-use plans fit rare events, while standing plans fit repeated ones.

POLICY, PROCEDURE AND RULE

Standing plans come in three common forms. They differ in one respect: how much freedom each leaves to the person who follows it. A policy leaves the most freedom, a procedure leaves less, and a rule leaves none. The table below sets them side by side. Read it row by row: the left column names the form, the middle column says what it does, and the right column gives a shop example.

Three kinds of standing plan differ in how much freedom they leave.

THREE FORMS OF STANDING PLAN

FORM	WHAT IT DOES	SHOP EXAMPLE
Policy	Gives a general guide and sets limits for choices	Returns are accepted only within a set time
Procedure	Lists the steps in order for a repeated task	Check the receipt, check the book, record the exchange
Rule	Requires one action in one situation, with no choice	No exchange without a receipt

A **policy** is a general guide for decisions, setting limits within which choices are made. For example, a shop that decides "we exchange books within a fixed number of days" has set a policy. Within that limit the employee can decide: a customer who returns a damaged book on the last day may be helped, while one who returns an old book may be refused. A policy gives direction and leaves judgement.

A policy is a general guide that sets limits within which choices are made.

Policy. A general guide for decisions, setting limits within which choices are made.

A **procedure** is a fixed series of steps to be followed for a repeated task. Where a policy says what is allowed, a procedure says how to carry it out. For example, the exchange procedure may say: ask for the receipt, check the date, check that the book is undamaged, give the new book, and write the exchange in the record. The steps are always the same, and the order always matters.

A procedure is a fixed series of steps for a repeated task.

Procedure. A fixed series of steps to be followed for a repeated task.

A rule is the narrowest standing plan. It says that in one situation

A rule requires one action and

The shop's returns policy is a standing plan for a repeated situation.

The returns policy shows how one standing plan holds both a policy and a rule.

A programme is a set of linked single-use plans, and a project is one part of it.

Programme. Word in use. In daily life, a television or radio show, or a printed list of events. In planning, a set of linked plans, usually single-use, that together reach a larger objective.

The book fair stall is a single-use plan with a start date and an end date.

The fair is a project because it begins, ends and is not repeated.

A budget states expected income and spending in money terms for a period.

Budget. A plan that states expected income and spending in money terms for a period.

Budgets can be standing or single-use, depending on the period they cover.

The case file gives the book fair's budget and the yearly sales budget.

Comparing expected sales with budgeted spending shows whether the event is worth planning.

Section 8.1 in four lines.

Case file, illustrative. Plans 2026, standing: returns policy, an exchange within 7 days with a receipt.

Read the record carefully. "Exchange within 7 days" sets a limit, and so it is a policy. "With a receipt" is a condition that allows no choice, and so it behaves as a rule. The record does not list the steps, and a procedure for them could be added. The case shows that real plans often mix the three forms in one sentence, and the reader's task is to see each part.

PROGRAMMES AND PROJECTS

Single-use plans also come in two sizes. A programme is a larger set of linked plans that together reach a large objective, and a project is one defined piece of work within it. For example, a shop that plans a year of school visits may treat the whole year as a programme and each visit as a project. Each project has a start, an end and a person in charge.

Case file, illustrative. Plans 2026, single-use: a 28-day book fair stall in February 2026.

The book fair stall is a project, because it has a start, an end and a fixed length of 28 days, and it will not be repeated in the same form. It is a single-use plan. Notice how this differs from the returns policy, which has no end date and applies every day.

BUDGETS

One more kind of plan is used by almost every organisation. A **budget** is a plan that states expected income and spending in money terms for a period. For example, a family might write that it expects to receive a fixed income this month and plans to spend part of it on food. A budget turns the plans of the shop into figures, so that the manager can see whether the shop can afford them.

A budget may cover a short event or a whole year. The yearly budget is renewed every year, so it is repeated in form, although the figures change. A budget for one event is used once. This shows that the two main groups are guides and not walls: a plan may have features of both.

Case file, illustrative. Plans 2026, book fair budget: spending Tk 240,000 and expected sales Tk 900,000. Yearly budget: sales Tk 15,840,000.

Read the record. The book fair has two figures: Tk 240,000 of planned spending and Tk 900,000 of expected sales. The expected sales are larger than the spending, so the event seems worth planning, although the cost of the books themselves must still be counted. Notice that the budget does not guarantee the sales. It states what the manager expects and plans for.

Before moving on. A standing plan is used again; a single-use plan is used once. Policy, procedure and rule are standing plans, from the most free to the least free. A programme holds linked projects, and a budget states income and spending in money.

8.2 TIME SPAN AND LEVEL

A second way to sort plans is by time. How far into the future does the plan reach? A long-range plan looks several years ahead, often three to five years, or more. A short-range plan looks ahead for a year or less. The shop's plan to open a second branch in five years is long-range. Its plan for next week's staff timetable is short-range. The two are linked, because the short-range plans are the steps that lead toward the long-range plan.

A third way to sort plans is by level. A strategic plan sets the main direction of the whole organisation, such as which customers to serve and which new business to enter. An operational plan states how one part of the organisation will carry out its daily work, such as how many people stand at the counter. Strategic plans are usually made by the owner or top managers and look far ahead. Operational plans are made by lower managers and employees and look a short way ahead. The planning steps of Chapter 7 apply at every level.

The figure arranges the plans of this chapter in a grid. Read it as follows: the two columns show whether a plan is used again or used once, and the two rows show whether it looks far ahead or only a short way. Find a plan by choosing its column and its row. The placing is a guide and not a law, because a policy may last for a few months and a project may last for years.

Plans can be sorted by how far ahead they look.

Plans can also be sorted by level: strategic plans set direction, operational plans do the daily work.

The grid places the main plans by repeat use and by time span.

	Used again	Used once
Long-range	Policy Yearly budget	Programme Opening of a branch
Short-range	Procedure Rule	Project Book fair stall

Dotted and hatched squares only separate the four cells; the words in each cell carry the meaning.

FIG. 8.1. Plans arranged by time span and by repeat use

Why does the shop need both? A long-range plan alone is too distant to guide the week, and a short-range plan alone has no direction. Suppose the owner wants to open a second branch in five years. That aim is long-range. To reach it, the shop needs a yearly budget, a plan for saving money, and a plan for training a manager for the new branch. These are short-range plans, each a step on the road. A manager checks that every short-range plan points the same way as the long-range one.

The same fitting is needed between levels. The owner's strategic plan says which customers the shop wishes to serve, for example

Short-range plans are the steps that lead toward long-range plans.

Strategic and operational plans fit together like a frame and its pictures.

Section 8.2 in three lines.

Before moving on. Long-range plans look several years ahead and short-range plans a year or less. Strategic plans set direction and operational plans organise daily work. The grid helps to sort plans, but its boundaries are not fixed.

* * *

8.3 MAKING PLANNING EFFECTIVE

A plan that is well made can still fail if the organisation does not support it.

Knowing the kinds of plan is the first half of the lesson. The second half is making them work. Chapter 7 showed that plans fail for many reasons. Four habits help a manager to prevent failure, and each one is explained here with a shop example.

SUPPORT FROM THE TOP

A plan needs the visible support of the owner or senior managers.

The first habit is support from the top. If the owner makes a plan but is seen to ignore it, employees will ignore it too. Support means that senior managers approve the plan, give it money and time, and follow it themselves. In a family shop, this means the owner follows the returns policy in the same way as an assistant must. If the owner makes an exception for a friend, the policy loses its force.

Without support, a plan competes with daily work and loses.

Consider the viewpoints. The owner supports a plan by following it. The store manager supports it by giving employees time to learn it. The employee supports it by using it and by reporting what does not work. When one of these three withdraws support, the plan weakens, because people watch what their leaders do and not only what they say.

CLEAR OBJECTIVES

A plan is useful only if its objective is exact and shared.

The second habit is clear objectives. An objective is the result that has been chosen as the target. If the objective is vague, such as "have a good book fair", then each person decides privately what good means, and the efforts pull in different directions. An exact objective states the result and the date: for example, to reach the expected sales of the fair by the end of its 28 days.

Exact objectives let a manager check results against the plan.

An exact objective also makes later checking possible. If the stall's expected sales are known in advance, the manager can compare the actual sales with them after the first week, and again after the fourth. Without an exact objective there is nothing to compare. The control of results is taught in Chapter 18.

COMMUNICATION AND FLEXIBILITY

Employees must know the plan, and the plan must be able to bend.

The third habit joins two ideas that seem opposite. Communication means that everyone affected by the plan knows it: what it says, why it exists and who does what. A standing plan that is kept in the owner's head cannot guide the assistants. The manager should put it in writing, explain it, and answer questions.

Flexibility keeps a plan useful when conditions change.

Flexibility means that the plan can be changed when conditions change. A plan that cannot bend becomes a trap. Suppose the returns policy allows exchange within seven days, and a large school order arrives with a longer delay between delivery and use.

OVERCOMING THE LIMITS OF PLANNING

The fourth habit is to accept that planning has limits and to reduce them. The future cannot be known, so every plan rests on assumptions that may prove wrong. Planning takes time and money, and a plan that is too detailed may cost more than it saves. Plans can also lead managers to follow them after they stop fitting the facts.

Three actions help. First, review plans on a fixed date, for example at the start of each season, and ask whether the planning premises still hold. Second, plan in proportion: give more detail where an error costs much, such as a large order, and less detail where the cost is small. Third, keep a margin, which means a small reserve of money or time to meet surprises. A shop that plans for a large order with no reserve is one price change away from trouble.

A short test helps a manager to decide whether a plan is ready. Ask four questions. Do the senior managers support it? Is the objective exact, with a date? Do all the affected employees know it? Is there a fixed day to review it and change it? If any answer is no, the plan has a weak point, and it is cheaper to repair the weak point before the plan starts than after it fails.

PAUSE AND CHECK 8.P2 The shop's owner makes a new plan for the book fair but tells nobody except the store manager. Which of the four habits is weak, and what should the owner do?

Before moving on. Planning works when the top supports it, objectives are exact and shared, employees know the plan, and the plan can change. The limits of planning are reduced by regular review, proportion and a margin.

* * *

WORKED CASE: SORTING THE SHOP'S PLANS FOR 2026

This worked case follows the standard order for a descriptive chapter: Situation, Question, Reasoning, Conclusion and Check. It applies the ideas of the chapter to the shop's own record, and it is a study of the idea and not a model answer.

Situation. The shop has three plans for 2026. The first is a returns policy: exchange within 7 days with a receipt. The second is a 28-day book fair stall in February 2026, with a budget of Tk 240,000 and expected sales of Tk 900,000. The third is a yearly budget with sales of Tk 15,840,000.

Question. Which plans are standing and which are single-use, and what do the figures show about the book fair?

Reasoning. First, sort the plans by repeat use. The returns policy applies every day and has no end date, so it is a standing plan. The book fair stall has a start and an end and will not be repeated, so it is a single-use plan. The yearly budget is renewed each year in form, so it behaves as a standing plan, although its figures change. Second, compare the figures of the fair. The expected sales exceed the planned spending by Tk 900,000 – Tk 240,000 = Tk 660,000.

Planning has limits, and a manager can reduce them but not remove them.

Three actions reduce the limits: review often, plan in proportion and keep a margin.

A manager should test a plan against the four habits before using it.

Section 8.3 in four lines.

This worked case sorts the shop's plans and tests their figures.

The situation lists the three plans in the 2026 case file.

The question asks for the kind of each plan and what the figures show.

The reasoning sorts the plans first, then compares the figures.

The check multiplies back to the starting figure.

Check. If 5.7% of Tk 15,840,000 is calculated, $15,840,000 \times 5.7 \div 100$ is about Tk 902,880, which is close to Tk 900,000. The small difference arises because 5.7% is a rounded figure, so the answer is sensible.

* * *

The chapter in a few lines.

Plans are sorted by repeat use into standing plans and single-use plans. Standing plans include policy, procedure and rule, and single-use plans include programmes and projects. A budget states income and spending in money terms. Plans are also sorted by time into long-range and short-range, and by level into strategic and operational. Planning works when the top supports it, objectives are exact, people know the plan, and the plan can change.

Four questions that change the situation step by step.

The questions begin with the shop and move to new settings.

8.T1 The shop's returns policy allows exchange within 7 days with a receipt. A customer returns a book on day 8 with a receipt. Which part of the standing plan decides the case, and what could the manager do within the policy?

8.T2 Suppose the book fair had expected sales of Tk 600,000 and the same budget of Tk 240,000. By how much would the expected sales exceed the spending, and what would you advise?

8.T3 A school canteen serves lunch every day and also plans a once-a-year sports day. Give one standing plan and one single-use plan for the canteen.

8.T4 Explain when a standing plan should be changed. Give two signs that a policy has grown old.

A policy, a procedure and a rule are not the same.

Some students use the three words as if they meant one thing. They differ in freedom: a policy guides and leaves judgement, a procedure fixes the steps, and a rule leaves no choice. Reading the sentence for how much choice it leaves usually shows which form it is.

A single-use plan is not a weak or unimportant plan.

Some students think that a plan used once matters less. A single-use plan, such as a landing on a coast, may be the most important plan an organisation ever makes. It is called single-use because it is used once, not because it is small.

A standing plan does not last for ever.

Some students believe that once a standing plan is made, it stays. Conditions change, and the plan must be reviewed. A plan that is never reviewed becomes a habit that no longer fits the facts.

A budget does not guarantee the sales it expects.

Some students read the expected sales in a budget as a promise. They are an estimate, based on premises. The actual figures may differ. and the manager must compare them with the budget as the

SUMMARY

THINK IT THROUGH

WHERE PEOPLE GO WRONG

IN HISTORY

On 6 June 1944, during the Second World War, Allied forces landed on the coast of Normandy in northern France. The operation was called Operation Overlord, and it began with the largest landing from the sea in history. The landing was not repeated. It was a single-use plan on a very large scale.

Planning began in 1943, when a joint British and American staff under Lieutenant-General Frederick Morgan was told to prepare a plan for an invasion in 1944. In late 1943, General Dwight D. Eisenhower was named Supreme Commander. The first target date was in the spring of 1944, and it was later moved to June, and the final day was set by the commanders after studying conditions.

The operation also needed a second plan, a campaign of deception called Operation Bodyguard, meant to hide where and when the landing would be. Both plans were prepared together. The history shows what this chapter has taught: a very large single-use plan needs exact objectives, support from the top, communication among many groups, and the ability to change.

The Allied landings in Normandy were planned as a single operation on a very large scale.

Planning began more than a year before the landing.

The plan shows how a single-use plan joins many standing and single-use parts.

REVIEW QUESTIONS

8.1 Sort these plans into standing and single-use: a returns policy, a book fair stall, a yearly budget, the opening of a new branch.

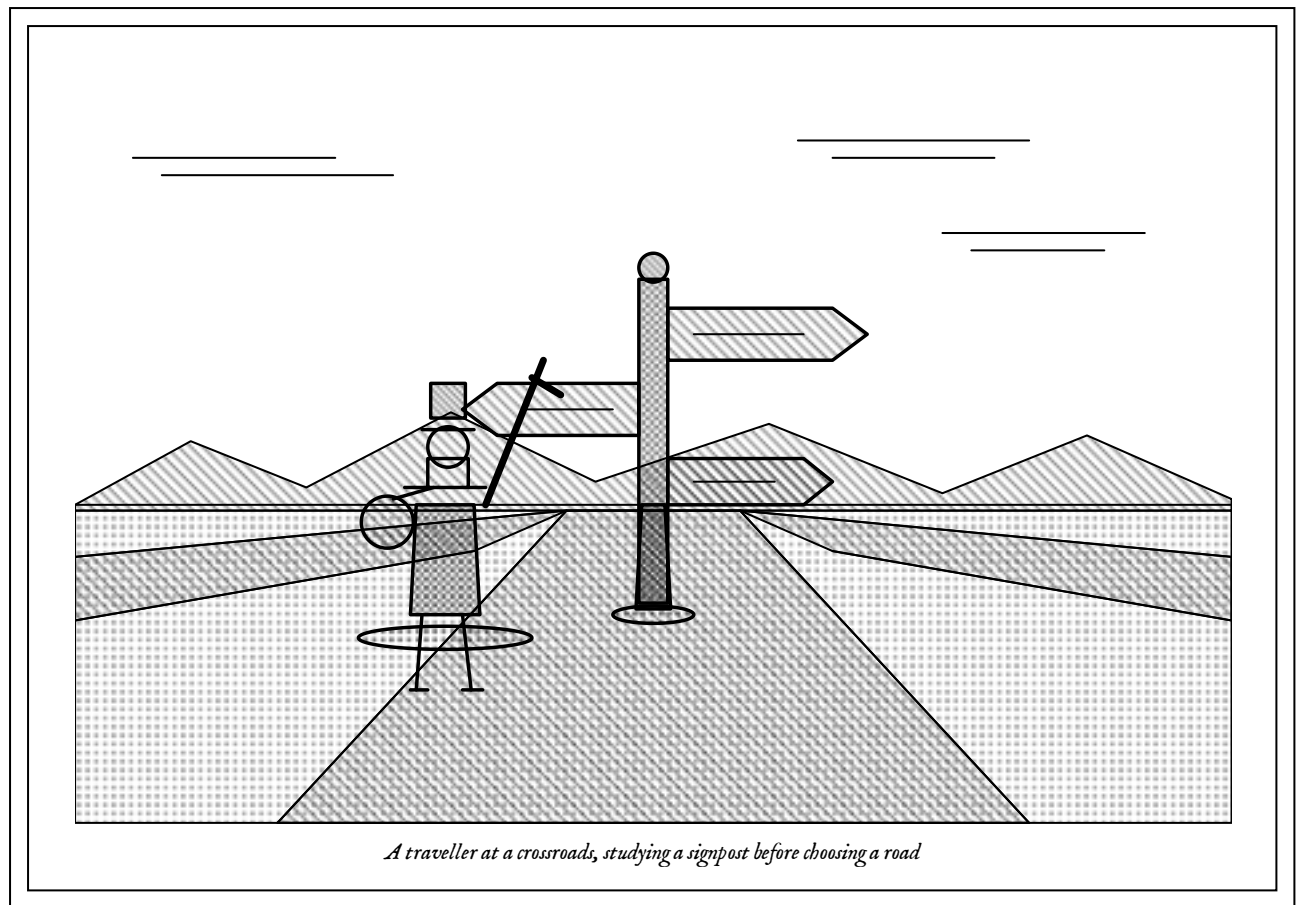
8.2 Define policy, procedure and budget, and give one shop example of each.

8.3 What is the difference between a long-range plan and a short-range plan, and between a strategic plan and an operational plan?

8.4 Explain why support from the top matters to a plan.

8.5 State two ways in which a manager can reduce the limits of planning.

Five questions that review the chapter.



CHAPTER IX

Objectives and Management by Objectives

Where are we going, and how will we know when we have arrived?



This chapter explains what an objective is, how to write one that can be measured, and how managers and employees can agree objectives together. It follows one bookshop through the objectives set for 2026 and ends with the system that Peter Drucker described in 1954.

YOU WILL NEED

Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'; and Chapter 8, 'Kinds of Plans and Making Planning Work', section 8.1 'Kinds of Plan'. Chapter 7 explains what a plan is and why managers make plans. Chapter 8 describes the kinds of plan, such as standing plans and single-use plans. This chapter starts where those two end: every plan needs a place to reach.

Two students make two promises, and only one of the promises can be checked.

A shop has the same problem as the first student: a wish with no edge cannot be steered towards or checked.

A plan is a road, and a road is useful only after someone names the place it leads to.

This chapter answers one question in three steps and then shows the system that joins them.

A traveller needs a named destination before any road makes sense.

Imagine two students at the start of a school year. The first says, "I want to do well this year." The second says, "I will score at least 70 out of 100 in every subject in the final tests, and I will study for two hours on each weekday evening." One year passes. Can the first student say whether she kept her promise? She cannot. The word "well" has no edge, so any result can be called well or badly, depending on her mood. The second student can answer in one minute, by reading her results and her study record.

Now put a shop in place of the student. The owner says, "We must improve the business." The stock clerk hears this and decides to tidy the storeroom. The cashier decides to serve customers faster. The store manager decides to buy more stock. Each person works hard, and each person chooses a different road. A year later, nobody can say whether the business improved, because nobody said what improvement would look like.

In Chapter 7 you studied the plan, which is a chosen way to reach a result. A plan is a road. A road helps only when someone has named the town at its end. Picture a traveller at a crossroads with four roads and a signpost whose arms carry no names. The traveller has energy and a good map, and still cannot choose. An organisation can be in the same position: busy every day, yet unable to say what the work is for.

The question for this chapter is therefore: how does a wish become a result that a whole organisation can steer by and check? The destination has a formal name, *objective*, which section 9.1 defines. Section 9.1 also shows which parts of an organisation's work deserve an objective. Section 9.2 shows how to write one so that anyone can tell afterwards whether it was reached. Section 9.3 describes a system, called management by objectives, in which managers and employees agree objectives together. Along the way we follow a bookshop in Dhaka through the objectives it agreed for 2026. Many students find the word *objective* confusing, because it also has an everyday meaning. We deal with that at once, so that it does not trouble you later.

* * *

9.1 OBJECTIVES

Return to the traveller at the crossroads. Suppose each arm of the signpost now carries a name and a distance: "Market town, 12 kilometres." The traveller can choose a road, can measure progress

An **objective** means a specific result that an organisation or a person intends to reach within a stated time. For example, "sell 500 copies of one school textbook before 30 June" is an objective, because it names a result, a number and a date. In everyday talk people say goal or aim. In this book, objective is the formal word, and goal appears only in plain talk. An organisation usually holds several objectives at once, and together they decide which plan it chooses from the alternatives open to it.

Why does an organisation need objectives? There are four reasons. Objectives give direction, because every person knows the destination. They join effort, because the cashier and the stock clerk steer towards the same place. They motivate, because people work harder towards a result they can see and reach. They supply a measure: when the year ends, the result can be set beside the objective. Chapter 18, 'Controlling', teaches how that comparison is used.

Objectives share several features, and each feature helps you test whether a statement deserves the name. First, an objective names a result, not an activity. "Hold a sale" is an activity. "Sell 2,000 copies in the sale" is a result. Second, an objective states a time. A result with no date can always be postponed. Third, an objective carries a measure where one is possible; section 9.2 returns to this. Fourth, objectives form a network. The organisation's objectives lead to objectives for each department, and these lead to objectives for individual employees. Fifth, objectives can conflict. A shop that cuts its stock to save money may find that it cannot also promise every customer that every title is on the shelf. The manager balances the objectives and does not pretend that the conflict is absent.

Each person sees an objective from a different position. The owner sees a result that must be paid for and protected. The manager sees a task to be shared among people. The employee sees a measure of what is expected, and so of what counts as a good year. The customer rarely sees the objective itself, yet feels its effect, for example in better service at the counter.

An organisation does many things, and it cannot give an objective to every action. Where should it place its effort? The answer is the **key result area**, which means a part of the organisation's work where results are vital to success. For a bookshop, sales are one key result area, because a shop that does not sell cannot survive. Stock is another, because too little stock loses sales and too much stock ties up money. Customer service, returns and profit are further

Recall: planning, Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'.

An objective is a specific, dated result that an organisation intends to reach.

Objective. Word in use. In daily life, "objective" can describe a fair opinion that does not depend on feelings, and it is also the lens of a microscope. In management, it is a noun: a result to be reached.

Objective. A specific result an organisation or person intends to reach within a stated time.

Objectives give direction, join effort, motivate and supply a measure for later checks.

Objectives have five features: a result, a time, a measure, a network and a possible conflict.

Objective. Word in use. See the note above: in management, a result to be reached.

The same objective looks different to the owner, the manager, the employee and the customer.

Key result areas are the few parts of the work where results decide success.

Key result area. A part of the organisation's work where results are vital to success.

¹ Tk is the symbol for the taka, the currency of Bangladesh. Large amounts are written with commas in groups of three digits.

Before moving on: objectives are results with a time, and key result areas say where to place them.

Meghna, the store manager, agreed them on 5 January 2026. Each objective lies in one key result area: sales, profit, returns, customer service and stock.

Before moving on. An objective is a specific result to be reached within a stated time. It differs from an activity, which is only something done. Objectives give direction, join effort, motivate and supply a measure. A key result area is a part of the work where results decide success, and the five objectives in the case file lie in five such areas.

PAUSE AND CHECK 9.P1 Name two key result areas for a small school, and say why a result in each area decides whether the school succeeds.

* * *

9.2 SETTING OBJECTIVES

Take a vague wish and sharpen it, one step at a time, until anyone can check it.

Start with a wish that many organisations write: "Improve customer service." Can anyone check this statement in December? No one can. Now sharpen it in four moves. First, choose the key result area: customer service. Second, choose a measure for that area: the number of hours of training each employee receives. Third, add the number: 14 hours. Fourth, add the period: the year 2026. The sharpened statement reads: "14 hours of customer-service training per employee in 2026." It is one of the five objectives in the case file, and a person can check it by counting hours.

Good objective-setting follows six steps, from the key result area to agreement with the person responsible.

The example contains the whole technique, which has six steps. Step one: choose the key result area. Step two: state the result you want, in words that name a result and not an activity. Step three: attach a measure, which is the unit used to count progress, such as taka, hours or percentage. Step four: fix the number and the date. Step five: test the objective. Can it be reached with the people, money and time available, and does it ask for more than last year, so that effort is needed? Step six: agree the objective with the person who must deliver it. An objective that the responsible person has never discussed is only an instruction.

A target is the number in an objective, and the measure tells you what the number counts.

The words measure, number and target need care. The measure is the unit, such as taka of sales. The target is the number set for that unit, such as Tk 15,840,000. Together they make an aim that can be measured. The table below shows four vague wishes beside the measurable objectives from the case file. Read each row from left to right: the first column gives a wish, the second gives the objective that sharpens it, and the third names the measure.

Target. Word in use. In daily life, a mark that you aim at, such as the centre of a board. In management, the number or level written into an objective, such as 3.2 times for stock turnover.

FOUR VAGUE WISHES SHARPENED INTO THE OBJECTIVES IN THE CASE FILE

VAGUE WISH	MEASURABLE OBJECTIVE	MEASURE
Sell more	Sales of Tk 15,840,000 in 2026	Taka of sales
Keep returns low	Returns below 2% of sales	Share of sales

the 2025 result, an increase of 10%.² The profit objective is Tk 150,000 above the 2025 result, an increase of 20%.

Two exceptions deserve attention. First, not every result can be counted directly. The quality of advice from a sales assistant is hard to put in a number, so organisations often count something close to it, such as hours of training. The count is a substitute and should be treated with care. Second, a number can mislead. A manager who chases a sales target may reduce prices until profit disappears. This is why an organisation sets several objectives together, so that each one guards against the harm that another might cause.

Objectives also link levels. The owner's sales objective becomes an objective for the store manager, who divides it among customer groups and among employees. For illustration only, and not as part of the case file, suppose sales to walk-in customers, schools and online buyers each rise by 10% above their 2025 values of Tk 9,000,000, Tk 3,600,000 and Tk 1,800,000. The three new values are Tk 9,900,000, Tk 3,960,000 and Tk 1,980,000, and they add to Tk 15,840,000. Each person who owns one part can see how it supports the whole.

Before moving on. To set an objective, choose the key result area, state a result, attach a measure, fix a number and a date, test whether it can be reached, and agree it with the person responsible. The measure is the unit; the target is the number. Be careful with results that resist counting, and set several objectives together so that one does not damage another.

* * *

9.3 MANAGEMENT BY OBJECTIVES

Imagine two shops with identical objectives. In the first, the owner writes a number on a sheet for each employee and hands the sheet over. In the second, the owner meets each employee, discusses the number, changes it where the employee shows a good reason, and both people agree it. Which shop will reach its objectives with more commitment? Most people answer the second, and that answer is the seed of this section.

In 1954 Peter Drucker, a writer on management born in Vienna and working in the United States, published *The Practice of Management*. In it he described a way of managing in which objectives are agreed between the manager and each employee, and in which each employee then judges his or her own progress. Drucker used the phrase "management by objectives and self-control".³ Later writers shortened it to MBO. Section 'In History' near the end of this chapter tells more of his story.

Management by objectives means a system in which managers and employees agree objectives together, then judge performance

² A percentage means a part in every 100. An increase of 10% on Tk 14,400,000 is $14,400,000 \times 10 \div 100 = \text{Tk } 1,440,000$, and $\text{Tk } 14,400,000 + \text{Tk } 1,440,000 = \text{Tk } 15,840,000$.

³ Self-control here means that employees check their own results against the agreed objectives, and do not wait for a manager to tell them how they are doing. It does not mean control by the self in the sense of controlling one's temper.

A measurable aim has limits: some results resist counting, and a number can mislead.

Objectives link level to level, from the owner to the manager to each employee.

Before moving on: a good objective names a result, a measure, a number and a date, and the person responsible agrees it.

Two shops can hold the same objectives on paper and still behave very differently.

Peter Drucker gave the idea its name in 1954, in a book about the practice of management.

Management by objectives is a system: agree objectives

IN SHORT

Management by objectives is a cycle. Agree objectives together, work towards them with freedom to choose the means, review progress during the year, appraise the results at the end, and use what is learned to agree the objectives for the next year.

The case file shows the first stage of the cycle: two people agreeing objectives.

The process has six stages, and the last stage leads back to the first.

Recall: kinds of plan, Chapter 8, 'Kinds of Plans and Making Planning Work', section 8.1 'Kinds of Plan'.

In stage three, each person chooses plans and means that fit the agreed objective.

Performance appraisal is the stage where results are compared with the agreed objectives.

Performance appraisal. A review of how well an employee has done against agreed standards.

To read the ring, start at the top and follow the arrows clockwise; the last arrow returns to the start.

Case file, illustrative. On 5 January 2026, Nurul Meghna and Rafiq Meghna agreed the objectives for the year: sales of Tk 15,840,000; profit of Tk 900,000; returns below 2% of sales; 14 hours of customer-service training per employee; and stock turnover raised from 2.8 to 3.2 times. The file records the agreement between these two people. It does not record later stages of the cycle.

THE PROCESS OF MBO

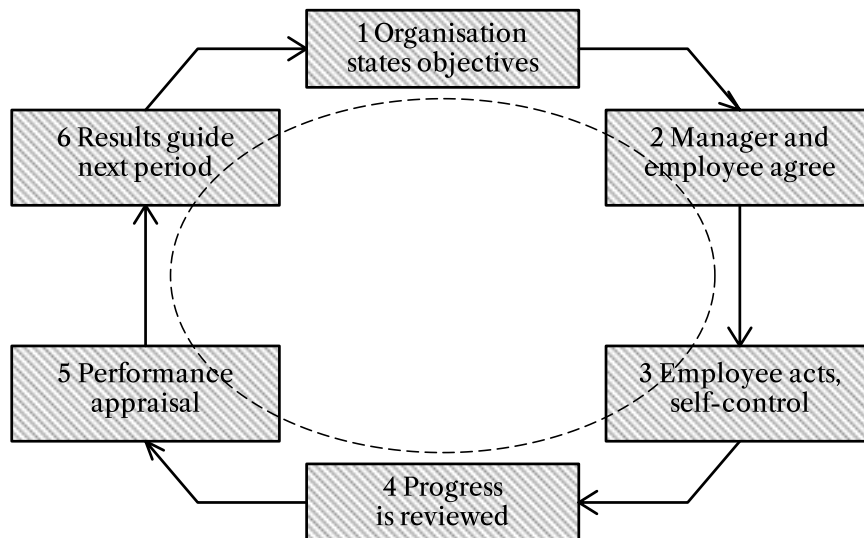
The process of management by objectives has six stages. In stage one, the organisation states its own objectives for the period. In stage two, each manager and employee agree the objectives for his or her own work, so that these support the organisation's objectives. In stage three, each person plans the means and acts. During this stage the employee has freedom to choose how to work, and checks his or her own progress; this is the self-control that Drucker named. In stage four, manager and employee review progress during the period, and they change an objective if conditions have changed. In stage five, performance appraisal takes place at the end of the period. In stage six, the results guide the objectives for the next period, and the cycle begins again.

In stage three a person chooses the means. The kinds of plan from Chapter 8 are tools here. A standing plan, such as a policy on how to handle a customer's complaint, repeats from year to year. A single-use plan, such as a project to fit security tags in one month, serves one objective once. Every plan rests on planning premises, which are the assumptions about the future that the plan accepts. An objective of Tk 15,840,000 in sales assumes that demand from schools and walk-in customers will stay about as strong as before. If that assumption fails, the review in stage four is the moment to say so.

A **performance appraisal** means a review of how well an employee has done against agreed standards. In management by objectives, the standard is the objective agreed at the start, so the appraisal asks one question: what was the result, compared with the number we agreed?⁴ For example, an employee who agreed to receive 14 hours of training and received 10 can be told the gap and asked why it arose. The appraisal is also a time to plan help, such as more training.

Figure 9.1 shows the six stages as a ring. To read it, begin at the box at the top and follow the arrows in the direction of the clock. Each box is one stage. Each arrow shows which stage comes next. After the last box, an arrow returns to the first box, which shows

⁴ A standard is a level of performance fixed in advance as the measure of what is acceptable. Chapter 18, 'Controlling', teaches it in full. Here it means only the agreed objective.

FIG. 9.1. *The MBO cycle as a ring*

PAUSE AND CHECK 9.P2 The last arrow of the ring returns to the first stage. Explain why a cycle suits management by objectives better than a straight line.

THE BENEFITS OF MBO

Management by objectives offers five benefits. It brings clarity, because each person knows what the organisation expects. It brings commitment, because people who help to set an objective feel responsible for it. It improves coordination, because each person's objectives support the objectives above. It gives a fair basis for appraisal, because the judgement uses a number agreed in advance and not a general impression. It helps to develop people, because the review shows where an employee needs training. Each person gains in a different way. The owner gains control without checking every action. The manager gains a sound way to share work. The employee gains knowledge of what counts as a good year. The customer gains from a shop whose people all work towards the same result.

MBO brings clarity, commitment, coordination, a fair basis for appraisal and a way to develop people.

THE WEAKNESSES OF MBO

No system is free of faults, and management by objectives has several. It takes time, because meetings and records are needed for every employee. It favours results that are easy to count, so quality may suffer, as you saw in section 9.2. It can push attention towards the short term, since a year is the usual period. It can make objectives rigid: an employee may keep following an objective after conditions have made it useless. It can be unfair when an employee is judged on a result that depends on events outside his or her control. It can fail when top managers do not support it, or when they use the cycle to place pressure on people. A system that works only on paper has the weaknesses of MBO and none of its benefits.

MBO has weaknesses: heavy paperwork, a bias towards what is easy to count, short-term focus and rigid objectives.

The table below sets four benefits beside the risk that goes with each. Read each row from left to right: the first column gives a benefit, and the second column gives the way that benefit is lost

The table sets each benefit against the weakness that most often accompanies it.

FOUR BENEFITS OF MBO AND THE MATCHING RISK

BENEFIT	MATCHING RISK
Clarity of expectations	Objectives chosen because they are easy to count
Commitment from agreement	Objectives imposed, so agreement is only on paper
Fair basis for appraisal	Results judged that the employee cannot control
Coordination across levels	Heavy paperwork that takes time from the work

Before moving on: MBO is a cycle of agreed objectives, self-control, review and appraisal, with real benefits and real risks.

Before moving on. Management by objectives is a system in which managers and employees agree objectives together and judge performance against them. Its six stages are: state the organisation's objectives, agree individual objectives, act with self-control, review progress, appraise performance, and begin the cycle again. Its benefits are clarity, commitment, coordination, fair appraisal and development. Its weaknesses are paperwork, a bias towards what is easy to count, a short-term focus, rigid objectives and pressure when it is used badly.

* * *

WORKED CASE: MAKING THE STOCK OBJECTIVE MEASURABLE

The worked case tests the stock objective in the case file from wish to number.

The situation states the objective and the figures that give it meaning.

The question asks whether the objective can be measured and what stock it implies.

The 2025 figures show where the shop starts, and the formula shows what the number means.

The conclusion and the check show that the objective is measurable and that its arithmetic holds.

Case file, illustrative. We now apply the ideas of this chapter to one objective. We take it in the order *Situation, Question, Reasoning, Conclusion, Check*, so that each step is announced and explained.

Situation. The case file records the wish that Meghna and Sons should hold stock more efficiently. The objective agreed on 5 January 2026 is stock turnover raised from 2.8 to 3.2 times. Here stock means the books that the shop holds for sale. Stock turnover means the number of times in a year that the shop sells its average stock, found by dividing cost of sales by average stock. Cost of sales is Tk 9,800,000, and average stock is Tk 3,500,000.

Question. Is the objective measurable? And if cost of sales stays at Tk 9,800,000, how much average stock does the shop need to reach 3.2 times?

Reasoning. First we confirm the starting point. The formula in words is: stock turnover equals cost of sales divided by average stock. With the numbers: $9,800,000 \div 3,500,000 = 2.8$. So 2.8 is the 2025 starting point, and the objective is measurable, because it names a measure (times per year), a target (3.2) and a period (2026). Next we find the stock that gives 3.2. Rearrange the formula: average stock equals cost of sales divided by turnover. Then $9,800,000 \div 3.2 = 3,062,500$. The shop therefore needs average stock of Tk 3,062,500. The reduction is $3,500,000 - 3,062,500 = \text{Tk } 437,500$.

Conclusion. The objective is measurable. If cost of sales stays at Tk 9,800,000, reaching 3.2 times means holding average stock of Tk 3,062,500. which is Tk 437,500 less than in 2025. *Check.*

SUMMARY

An objective is a specific result to be reached within a stated time. Key result areas show where objectives matter most. To set an objective, name a result, a measure, a target and a date, and agree it with the person responsible. Management by objectives is a system in which managers and employees agree objectives together and judge performance against them, and its cycle runs from agreement to action, review and appraisal, and back to agreement. The system has real benefits and real risks.

The summary states the chapter in a few lines.

THINK IT THROUGH

A school with 600 pupils and 30 teachers wants to adopt management by objectives. Its head teacher says only: "We want to be a better school." Use the ideas of this chapter to answer both questions below.

The problem asks you to apply the chapter to a school.

9.T1 Write two measurable objectives for the school. For each, name the key result area, the measure, the target and the period.

9.T2 Describe two ways in which management by objectives could fail in the school, and say what the head teacher could do to prevent each.

WHERE PEOPLE GO WRONG

The first misunderstanding is that an activity is an objective. "Hold a book fair" says what is done and not what is reached. Correction: name the result, such as the number of copies sold. The second is that a vague word is enough. "Better" and "improve" have no edge. Correction: add a measure, a target and a date.

Six common misunderstandings about objectives are stated and corrected here.

The third misunderstanding is that more objectives are better. Twenty objectives divide attention, and the key results are lost among them. Correction: choose the few that decide success. The fourth is that management by objectives means a manager hands out numbers. Correction: the system depends on agreement, and an imposed number is only an order. The fifth is that objectives never change. Correction: review them during the period, and change one when conditions have changed. The sixth is that a measured result is always the whole truth. Correction: a number can hide poor quality, so read objectives together and add judgement.

Further misunderstandings concern the number of objectives, the way they are set and their permanence.

IN HISTORY

Peter Drucker was born in Vienna on 19 November 1909 and died on 11 November 2005. He moved to the United States, worked as a writer and adviser on management, and published *The Practice of Management* in 1954. The book treated management as a practice that can be studied and taught. In it he described management by objectives and self-control, and he proposed that a business set objectives in a small number of key areas. His list had eight, among them market standing, innovation and new facilities.

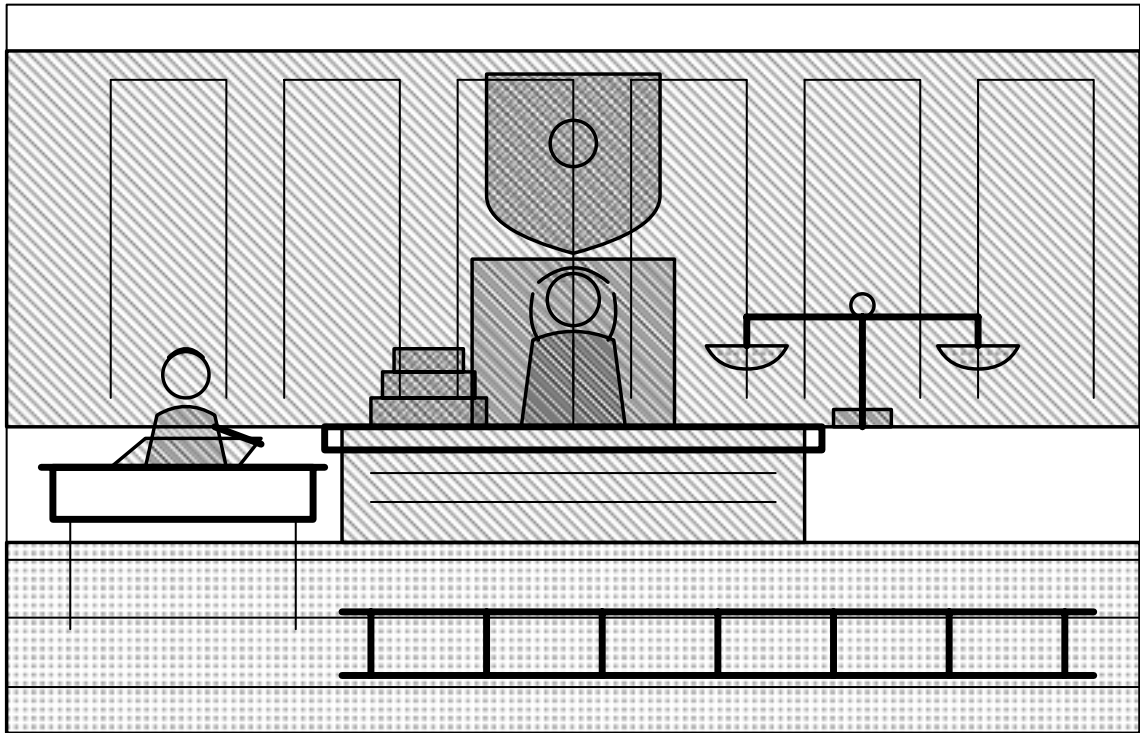
Drucker published management by objectives in 1954, and many organisations adopted it afterwards.

towards what is easy to count, and misuse as a tool of pressure. These faults are in section 9.3. Drucker's central point remains useful: people work better when they know the result expected and have helped to define it.

REVIEW QUESTIONS

The review questions test each main idea of the chapter.

- 9.1 Define objective in your own words, and say how an objective differs from an activity.
- 9.2 Rewrite each wish as a measurable objective, adding a measure, a target and a period: "Sell more books to schools", "Keep stock low", "Make staff more helpful". Use figures of your own.
- 9.3 In the case file, stock turnover is cost of sales divided by average stock. If cost of sales is Tk 9,800,000, find the average stock that gives a turnover of 3.2 times.
- 9.4 List the six stages of the MBO process in order, and name the stage in which performance appraisal takes place.
- 9.5 Give two benefits and two weaknesses of management by objectives.



A magistrate at the bench with his clerk

CHAPTER X

Decision Making: Process and Conditions

How do you choose when you cannot see what comes next?



This chapter explains how managers decide. It shows what a decision is, follows the steps from seeing a problem to checking the result, and then describes the three conditions in which every decision is made: certainty, risk and uncertainty. A second shop in Dhaka and a launch in Florida serve as cases.

You will need: Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'; Chapter 9, 'Objectives and Management by Objectives', section 9.2 'Setting Objectives'.

A simple choice about travel hides a method that managers use every day.

A short choice already holds several separate acts.

A manager makes such choices with more at stake and with others affected.

The chapter teaches the method first and the conditions second.

It is seven in the morning, and your class begins at nine. You can walk, take a bus or hire a rickshaw. You pause for a moment, think of the cost and the time, and choose. By the time you reach the door you have made a decision. Did you do anything that could be written down as a method?

Look again at what you did. You noticed that you had to be somewhere by nine. You asked what mattered to you: the cost, the time, or both. You listed the ways to go, compared them and picked one. Then you went, and you learned something from the journey. Most people perform all of these acts in a few seconds and never name them.

A manager performs the same acts under heavier conditions. The choice may cost thousands of taka, or change the work of seven employees. Other people will live with the result, and the manager cannot test it in advance. A fast habit that works for a bus ride can fail badly when the stakes are high.

This chapter teaches the method first, in clear steps that a manager can follow and check. It then describes the three conditions in which every decision is made. We follow one real choice, made by a bookshop in Dhaka on 20 August 2026. We also visit a launch pad in Florida, where in January 1986 a different choice ended in disaster.

* * *

10.1 WHAT A DECISION IS

GO SLOWLY HERE.

The idea of a decision begins with an everyday choice of one thing from several.

Recall: Chapter 7, 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is' — planning.

Rina's choice has the shape of every decision: several options and one choice.

Decision making. Choosing one course of action from several, in order to solve a problem or use an opportunity.

Decision. Word in use. In daily life, any choice that a person has made, or the act of settling a question. In management, a choice of one course of action from several, made to solve a problem or use an opportunity.

Every plan is made of

Begin with something small. Rina is a student, and on Friday she has Tk 500 to spend. She can buy a book, buy a cinema ticket or keep the money. She cannot do all three with Tk 500. She must choose.

Her choice has a name in management. **Decision making** means choosing one course of action from several, in order to solve a problem or use an opportunity. Rina has three courses of action, and she picks one. A shop owner who chooses between two sites for a new shop does the same thing with larger sums. Notice the word *several*: with only one course of action open, nothing is chosen, and no decision exists.

Chapter 7 taught that planning means deciding in advance what

management a decision is the choice of one course of action from several, made to solve a problem or use an opportunity. The second meaning always has a purpose and always follows a comparison.

You should now be able to say: a decision is a choice of one course of action from several, made for a purpose.

Why do managers decide at all? A decision starts when a manager sees that something needs to change. We use the word problem for a gap between what is and what should be. A stock clerk reports that sales of school books are falling, and the gap between actual and wanted sales is a problem. We use the word opportunity for a chance to do better than before. A school asks the shop to supply all its textbooks, and that offer is an opportunity.

The two start differently. A problem is usually loud: complaints, losses and shortages press the manager to act. An opportunity is quiet. Nobody complains about a chance that was missed, so a manager must look for it. For this reason, careful managers watch the world outside the organisation and not only the troubles inside it. Both a problem and an opportunity lead to the same act, which is a decision.

Return to Rina. Suppose a friend tells her that the cinema has a half-price show on Friday. Before the friend spoke, she had no problem, only a plan to buy a book. The friend's news is an opportunity, and it makes her choice worth making again. A decision can start with good news as well as with bad.

In one line: a decision begins when a manager sees a gap to close or a chance to use, and the manager then chooses one course of action from several.

Managerial decision making has a special nature. Rina spends her own Tk 500, and only she bears the result. A manager spends the organisation's resources, and many people bear the result. This difference gives managerial decisions four features, and each one is explained below.

First, the manager answers for the decision to owners, employees and customers. Second, managers decide with incomplete facts, because the future cannot be seen and time is short. Third, decisions are linked: choosing to open a second shop leads to later choices about staff, stock and prices. Fourth, a decision often changes the work of other people, so how the manager shares the choice matters as much as the choice itself.

Each person sees a decision from a different place. The owner asks whether the choice is worth the money. The manager asks how to carry it out. An employee asks how the work will change. A customer asks only whether the shop will continue to serve well. A wise manager considers all four views before choosing.

Before moving on. Decision making is choosing one course of action from several, in order to solve a problem or use an opportunity. A problem is a gap to close and an opportunity is a chance to use; both start a decision. Managers decide for an organisation, with incomplete facts, and their choices affect many people.

You should now be able to say: a manager decides when a prob-

The first idea is secure when the reader can state it in one sentence.

Every decision starts when someone sees a problem or an opportunity.

A problem is felt as pain, but an opportunity is easy to miss.

Rina's Friday shows an opportunity: a free offer that she would not have noticed.

The recap of the two ideas so far.

Managerial decisions differ from private ones in four ways.

The first feature is that the manager answers for the choice to others.

Each position sees the decision differently.

The recap of section 10.1.

A sentence to test the reader's

10.2 THE DECISION PROCESS

GO SLOWLY HERE.

A bus ride shows each step of the method before we use it on a business.

Step 1 is to state the problem or opportunity in plain words.

Step 2 is to decide how the best choice will be judged.

Decision criterion. A standard used to judge which alternative is best.

Criteria should be set before the options are examined.

Step 3 is to list the possible courses of action.

Step 4 is to test each alternative against each criterion.

A table shows which alternatives pass which tests.

Step 5 is to choose the alternative that best meets the criteria.

A method is easier to learn on a small case. Return to the morning at the start of the chapter, and give it numbers. Rina must reach class by 8:55, and she lives 40 minutes away by bus. She has little money. We will follow her through each step, and then apply the same steps to a shop.

Step 1: name the problem or opportunity. Rina writes it plainly: "I must be in class by 8:55 and I must not spend much." A vague statement leads to a vague decision. Many students find this step easy to skip, because the problem feels obvious. A decision made on the wrong problem can be perfect in every other way and still fail.

Step 2: set the criteria. A **decision criterion** means a standard used to judge which alternative is best. Rina has two criteria: arrive by 8:55, and spend no more than Tk 50. Each criterion is a test that an option must pass. Criteria come from the goals that the decision serves. Chapter 9 showed how a goal becomes a measurable objective, and a good criterion often comes straight from an objective.

The order matters. Set the criteria before you look at the options. If you see the options first, you may choose the criteria that favour the option you already like. A manager who sets the criteria first can defend the choice later, because the standard was fixed before the result was known.

Step 3: develop the alternatives. An alternative is one possible course of action. Rina lists three: walk, take the bus and hire a rickshaw. A manager should seek more than two or three, and should look beyond the first idea that comes to mind. A common error is a list of one real option and two weak ones, which only appears to give a choice. Include the option of doing nothing or waiting, because it too has a cost.

Step 4: evaluate the alternatives. To evaluate means to test each alternative against each criterion and record the result. Rina's walk takes 70 minutes and costs nothing, so she arrives at 9:10 and fails the first test. The bus takes 40 minutes and costs Tk 20. The rickshaw takes 25 minutes and costs Tk 100. The table below sets these results side by side.

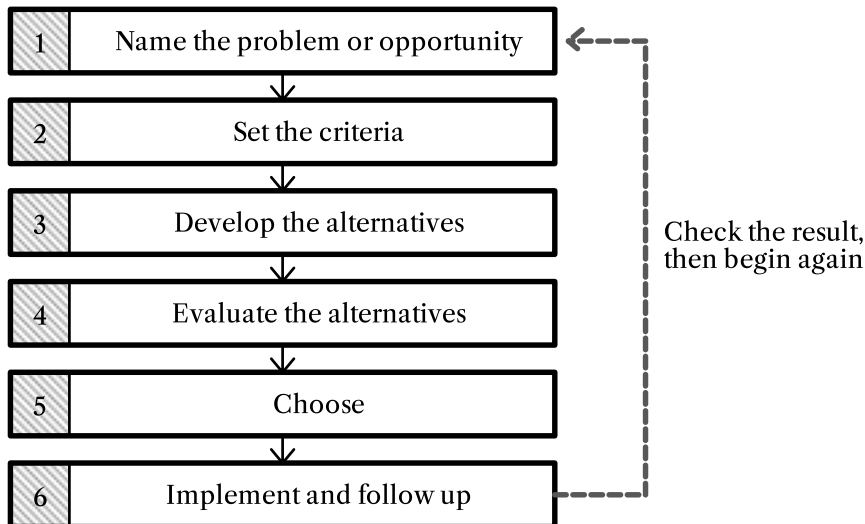
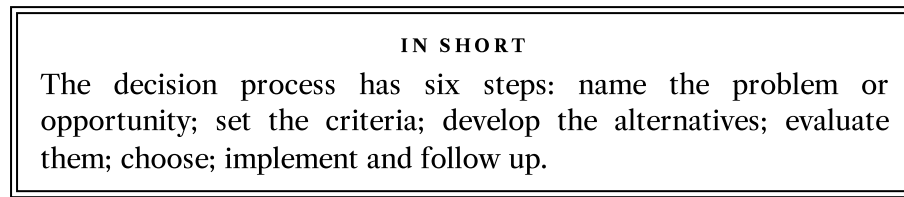
Read the table one row at a time. The first cell names an alternative, the next two cells give its results, and the last cell shows whether it passes both criteria. The criteria are on the top row.

RINA'S THREE ALTERNATIVES TESTED AGAINST TWO CRITERIA

ALTERNATIVE	ARRIVAL TIME	COST	PASSES BOTH CRITERIA?
Walk	9:10	Tk 0	No: arrives after 8:55
Bus	8:40	Tk 20	Yes
Rickshaw	8:25	Tk 100	No: costs more than Tk 50

Step 5: choose. Only the bus passes both tests, so Rina chooses it. Choosing is often the shortest step. The hard work has already been done in the earlier steps. When no alternative passes all the criteria, or when several pass, the manager must rank the criteria

choice is made. It is finished when the results are checked and the lesson is stored.



The figure shows the six steps in order, with a loop back from the last step to the first.

FIG. 10.1. *Decision Steps as a flow*

Read the figure from the top box downwards. Each box is one step, and each arrow shows that one step leads to the next. The dashed line on the right returns from step 6 to step 1. It means that checking the result may reveal a new problem, so the process begins again.

Steps 1 to 5 take place in thought and on paper. Only step 6 changes anything outside the manager's head. This is why managers should spend care on the first five steps and not rush toward the sixth. A mistake caught on paper costs nothing, and the same mistake caught after implementation may cost a great deal.

The process has limits. Real managers do not always follow the six steps in strict order. They may return to step 2 after seeing the alternatives, because a new option shows that a criterion was missing. Time may be short, and some information may cost more to gather than the decision is worth. The steps are a guide for thinking, and they work best when a manager treats them as a checklist and not as a ritual.

PAUSE AND CHECK 10.P1 Rina finds that the bus will be 20 minutes late on one day. Which step of the process does she return to, and what could she change?

Before moving on. A decision criterion is a standard used to judge which alternative is best. The process has six steps: name the problem or opportunity, set the criteria, develop alternatives, evaluate them, choose, then implement and follow up. Criteria are set before options are examined. Following up closes the loop and

The figure is read from top to bottom, then back to the top by the dashed line.

Steps 1 to 5 are done in the mind and on paper; only step 6 changes the world.

The process has limits that a careful manager must know.

The recap of section 10.2.

10.3 CONDITIONS OF DECISION MAKING

GO SLOWLY HERE.

Every decision faces a different amount of knowledge about the future.

Certainty means the manager knows the exact result of each alternative.

Certainty. A condition in which the result of each alternative is known in advance.

Risk means results are unknown but their chances can be estimated.

Risk. A condition in which the result is not known but its chances can be estimated.

Risk. Word in use. In daily life, a danger of harm. In management, a condition in which the result is not known but its chances can be estimated, and the result may be good or bad.

Risk depends on records or experience that allow a chance to be estimated.

Uncertainty means that neither the results nor their chances can be estimated.

Uncertainty. A condition in which neither the results nor their chances can be estimated.

A table sets the three conditions side by side with their tests.

Why do two decisions of the same size feel so different? One may be calm and the other frightening. The difference lies in how much the manager knows about what will happen after the choice. Management names three conditions, from the most knowledge to the least. We take them one at a time, with a small example for each.

The first condition is **certainty**, which means a condition in which the result of each alternative is known in advance. Rina knows that the bus fare is Tk 20, and she knows that the bus takes 40 minutes. A shop that signs a rent of Tk 38,000 a month knows what it will pay each month. Certainty is rare in business, but it does exist for costs that a contract fixes.

The second condition is **risk**. Risk means a condition in which the result is not known but its chances can be estimated. A tea stall owner has kept records for 10 market days. It rained on 4 of them. The owner cannot know whether tomorrow will be wet, but can estimate that the chance of rain is 4 in 10, or 0.4. The word *risk* has an everyday meaning of danger. In management it means that the result is unknown and its chances can be estimated, and the result may be good or bad.

How does a manager estimate a chance? The usual source is past records or experience of similar cases. The owner's 10 days of records give a rough chance of 0.4. More records would give a better estimate. When the records are few, the estimate is weak, and a manager should say so honestly.

The third condition is **uncertainty**, which means a condition in which neither the results nor their chances can be estimated. Suppose the tea stall owner wants to open in a town where she has no records and no comparison. She cannot say what the sales might be, and she cannot say how likely each result is. Many students find the difference between risk and uncertainty confusing, because both involve not knowing. The test is whether a chance can be estimated: if yes, it is risk; if no, it is uncertainty.

The table below compares the three conditions. Read each row from the left. The first cell names the condition, the second says what the manager knows, and the third gives the tea stall example.

THE THREE CONDITIONS OF DECISION MAKING

CONDITION	WHAT THE MANAGER KNOWS	TEA STALL EXAMPLE
Certainty	The result of each alternative	The rent for the month, fixed by a contract
Risk	The possible results and their chances	Rain on 4 of the last 10 market days
Uncertainty	Neither the results nor their chances	A town where the owner has no records

Each condition asks for a different attitude. Under certainty, a manager can compare the known results and choose the best. Under risk, a manager weighs the chances and accepts that the result may fall short. Under uncertainty, the manager must use judgement, gather more information if time allows, and prefer choices that can be changed later if they prove wrong.

Consider the second-shop decision of 20 August 2026, which the next section works in full. The rents and fitting costs are fixed amounts and so are certain. The expected sales are estimates, and the case file gives no chances for them. We therefore cannot call them a risk, and the sales belong closer to uncertainty. The condition shapes how firmly the owner should trust the numbers.

PAUSE AND CHECK 10.P2 A manager of a small shop must choose a new supplier. The supplier's price list is fixed for a year, but nobody can say how many customers will want the new books. Name the condition for the price and the condition for the number of customers, and give a reason for each.

Before moving on. Certainty is a condition in which the result of each alternative is known in advance. Risk is a condition in which the result is not known but its chances can be estimated. Uncertainty is a condition in which neither the results nor their chances can be estimated. A single decision may hold all three.

You should now be able to say: what the manager knows under each condition, and how to tell risk from uncertainty.

* * *

WORKED CASE: THE SECOND SHOP

We now apply the whole chapter to one decision in the case file. The worked case has five parts: Situation, Question, Reasoning, Conclusion and Check. It applies the ideas to a situation. It is an application, not a model answer.

Case file, illustrative. Decision, 20 August 2026, Meghna and Sons: open a second shop? Option 1, Mirpur: rent Tk 38,000 a month, fitting Tk 650,000, expected sales Tk 5,400,000 a year. Option 2, Uttara: rent Tk 52,000 a month, fitting Tk 720,000, expected sales Tk 5,100,000 a year. Option 3: wait a year.

Question. Which option should the shop prefer on the figures in the case file? The case file does not state the owner's goal. For this exercise only, we assume that the owner wants the most sales for the money spent in the first year.

Reasoning, steps 1 and 2. The opportunity is a second shop, and the choice to be made is which site, or whether to wait. From the assumed goal we set two criteria. Criterion A: expected sales for each taka spent in the first year on rent and fitting; a higher figure is better. Criterion B: expected sales less one year's rent; a higher figure is better. Both are fixed before we look at the totals.

Reasoning, step 3. The three alternatives are Mirpur, Uttara and waiting. Waiting has no cost for fitting or rent, and it earns nothing from a second shop. The case file gives no other figure for it. We

Managers meet each condition differently.

The conditions apply to the second-shop decision in the case file.

The recap of section 10.3.

A sentence to test the reader's grasp of the section.

The worked case applies the six steps and the three conditions to one decision.

The situation lists three options with their costs and expected sales.

The question asks which option best meets the owner's criteria.

Steps 1 and 2: the problem is a choice of site, and two criteria follow from the assumed goal.

Step 3: the three alternatives are already listed in the case file.

A table collects the results so that the options can be compared.

expected sales for each taka spent. Criterion B: 5,100,000 – 624,000 = Tk 4,476,000.

The table collects the results. Read each row from the left: the option, the first-year cost, the two criteria. The last row shows that waiting cannot be scored.

MIRPUR AND UTTARA ON THE TWO CRITERIA

OPTION	FIRST-YEAR COST	A: SALES PER TAKA SPENT	B: SALES LESS RENT
Mirpur	Tk 1,106,000	4.88	Tk 4,944,000
Uttara	Tk 1,344,000	3.79	Tk 4,476,000
Wait a year	Tk 0	Not scored	Not scored

Steps 5 and 6: Mirpur wins on both criteria, and the follow-up compares actual with expected sales.

Reasoning, steps 5 and 6. Mirpur is higher on both criteria, so on these figures it is the better site. If the owner chose Mirpur, the follow-up would record the actual sales and costs in the first year and compare them with Tk 5,400,000 and Tk 1,106,000. The size of any gap would show how far the estimates could be trusted. The condition matters here: rent and fitting are certain, but expected sales hold no estimated chance, and so belong nearer uncertainty.

The conclusion states what the figures support and what they leave open.

Conclusion. On the assumed goal, Mirpur is the better site, because it costs less and expects more sales. The figures do not say whether waiting is better, since they do not measure it. The case file records the options and does not record the final choice. *Check.* Mirpur has the lower rent, the lower fitting cost and the higher expected sales, so every figure points the same way. It follows that both criteria must favour Mirpur, and they do.

Rung 2 reshapes Rina's case so that no option passes both tests.

CHANGING THE SITUATION

The same method works in new situations. Rung 1 is Rina's bus ride, solved above. In rung 2, suppose the bus fare rises to Tk 60. Now no alternative passes both criteria. Rina must rank her criteria and decide which counts more, and the process returns to step 2.

Rung 3 moves the method to a school with a very different setting.

In rung 3, a school head must choose a place for a class visit. The head sets criteria for cost, distance and safety, lists four places, tests each one and chooses. The steps are the same, though the people and the sums are different. Chapter 7 showed that planning places the choice inside a larger plan, and here the visit is that plan.

Rung 4 combines the process with the three conditions and with planning.

In rung 4, a small clinic must decide whether to buy a new machine. The price is certain. The number of patients who will need it can be estimated from last year's records, so that part is a risk. The reaction of a rival clinic cannot be estimated, so that part is uncertainty. The owner uses all six steps and treats each part according to its condition.

* * *

SUMMARY

A short summary of the chapter's five defined terms and the process.

Decision making is choosing one course of action from several, in order to solve a problem or use an opportunity. The process has

THINK IT THROUGH

10.T1 A manager must choose between two suppliers of paper. Supplier A offers a fixed price for the year. Supplier B offers a lower price, but its delivery times are not known because it is new. Name the condition that applies to each part of the choice, and say which step of the process would tell the manager what to find out first.

10.T2 In the second-shop case, the expected sales are estimates only. Suppose the owner learns that the Mirpur figure is far too high. Which criteria change, which do not, and what should the owner do next?

WHERE PEOPLE GO WRONG

Mistake: "The decision is the moment when I choose." The choice is one step of six. The steps before it decide what is chosen from, and the step after it decides whether the choice worked.

Mistake: "A good result means the decision was good." A good decision can end badly through bad luck, and a poor decision can end well. Judge the method used and the facts known at the time, and then check the result.

Mistake: "I must wait until I know everything." Full knowledge is rare, and waiting has a cost of its own. A manager should gather the information that is worth its cost and then decide.

Mistake: "Risk and uncertainty mean the same." Under risk the chances can be estimated, and under uncertainty they cannot. The tea stall with 10 days of records faces risk, and the stall in a new town without records faces uncertainty.

Mistake: "I will decide on the criteria when I see the choices." The criteria then tend to favour the option already liked. Fix them first.

Mistake one: believing that a decision is the moment of choosing.

Mistake two: judging a decision by its result alone.

Mistake three: believing that more information always gives a better decision.

Mistake four: treating risk and uncertainty as the same thing.

Mistake five: setting criteria after seeing the options.

IN HISTORY

On the evening of 27 January 1986, engineers at Morton Thiokol, the company that made the rocket boosters for the United States space shuttle, spoke by telephone with managers of NASA, the national space agency. The shuttle Challenger was due to launch the next morning from Florida, and the forecast was for unusually cold weather. The engineers feared that the rubber seals in the boosters would not work in the cold. They had followed damage to such seals on earlier flights.

Thiokol's managers at first supported a delay, but after discussion with NASA they reversed that advice, and the launch went ahead on 28 January 1986. Challenger broke apart 73 seconds after liftoff, and all seven crew members died. A presidential commission led by William Rogers, a former United States Secretary of State, reported in June 1986. It judged the decision process flawed, and it found that the concerns of the engineers had not been fully passed to the senior managers who approved the launch.

The case shows the chapter's ideas at work. The criterion of

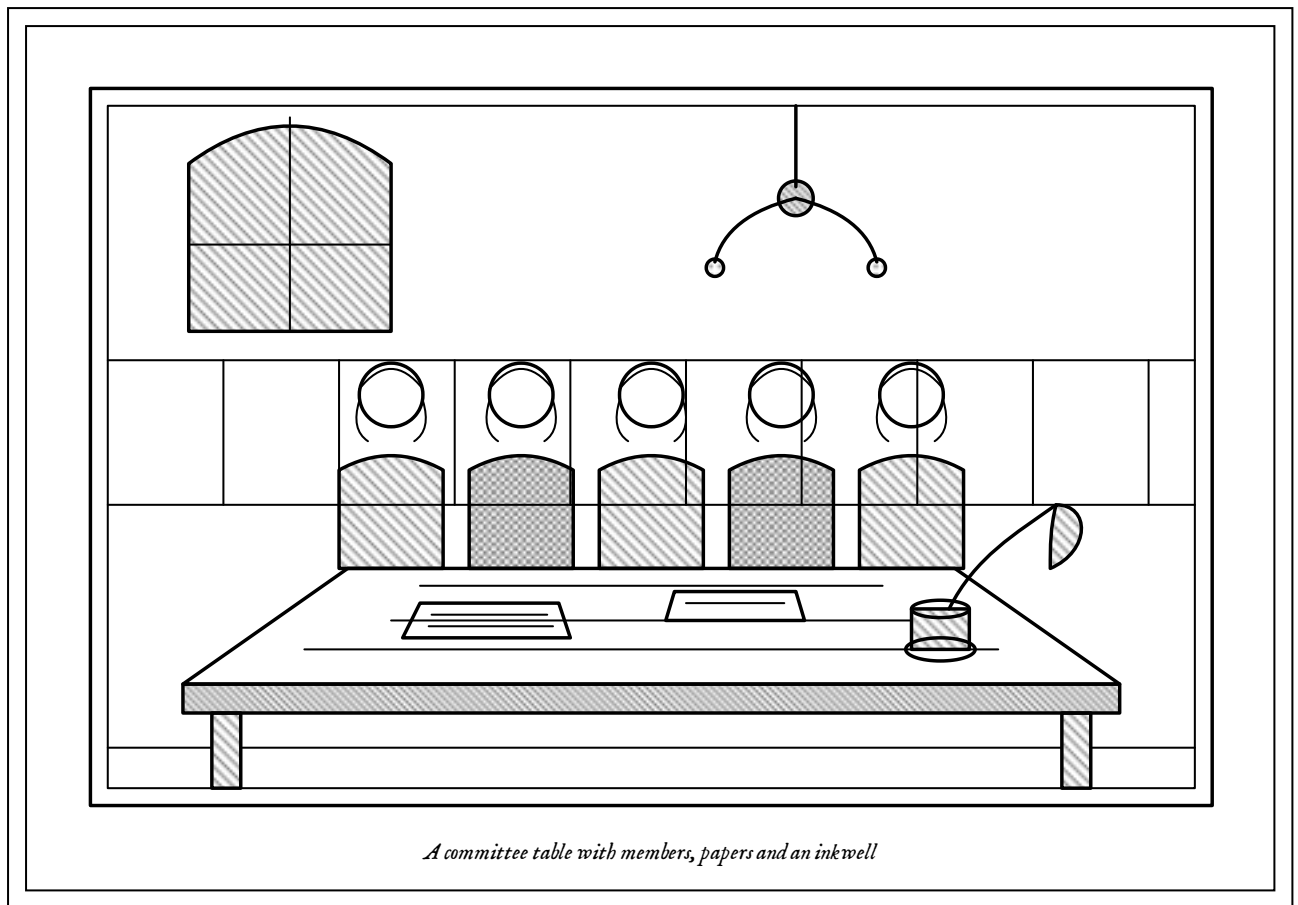
The launch of Challenger in January 1986 shows a decision under pressure and uncertainty.

The managers reversed the first advice and the launch went ahead.

The case shows each part of the chapter at work

REVIEW QUESTIONS

- 10.1 Define decision making in your own words and give one example from a bookshop.
- 10.2 Explain the difference between a problem and an opportunity, and say which is easier to miss and why.
- 10.3 List the six steps of the decision process in order. For step 2, say why the criteria are set before the alternatives are examined.
- 10.4 Define decision criterion. Give two criteria that a shop owner might use to choose between two suppliers.
- 10.5 Classify each part of the following as certainty, risk or uncertainty, with a reason: (a) the monthly rent fixed by a lease; (b) the chance of rain based on 10 days of records; (c) the sales of a new kind of product that no shop has sold before.
- 10.6 A manager has records showing that 3 of the last 12 school orders arrived late. What condition is this, and what is the estimated chance of a late order as a fraction?
- 10.7 The bus fare in Rina's case rises to Tk 60. Explain which step of the process she must repeat, and what she must decide.



CHAPTER XI

Decision Tools and Group Decisions

How can a manager make a doubtful choice less doubtful?



Managers seldom know what will happen after they decide; this chapter measures that doubt with numbers, compares choices on one diagram, uses computers for facts, and shows when groups should decide.

This chapter builds on two sections of Chapter 10.

A manager must choose before the future shows itself.

Waiting is also a choice, and it has its own cost.

Numbers can describe a doubt without removing it.

You will need: Chapter 10 'Decision Making: Process and Conditions', section 10.3 'Conditions of Decision Making' (certainty and risk); and Chapter 10 'Decision Making: Process and Conditions', section 10.2 'The Decision Process'.

Imagine that you own a small shop, and a larger shop becomes free in the next street. The rent is high. If customers come, you will earn a good profit. If they do not, you will lose money for a year. You cannot ask the future what will happen. Yet you must answer today, because another trader may take the shop tomorrow.

Many people in this position do one of two things. They decide by feeling, and say that the shop looks right. Or they refuse to decide, and say they need more time. Both reactions are understandable. Neither is satisfying, because neither can be explained to a bank, a partner or a family member who asks the plain question: why this choice?

This chapter offers a better way. It does not remove the doubt, because nothing can. It gives the doubt a size. You will learn to say how likely each result is, to weigh each result by its likelihood, and to draw all the choices on one page. You will then see how a computer can supply facts, and when it is wiser to decide together than alone.

* * *

11.1 PROBABILITY AND DECISION TREES

PROBABILITY

If you toss a fair coin, how likely is it to land heads up? Most people answer: one chance in two. That answer is already a probability. A probability is a number from 0 to 1 that shows how likely an event is. A probability of 0 means the event cannot happen. A probability of 1 means it is certain. A probability of 0.5 means the event is as likely to happen as not. In plain talk we say *chance*; when we mean the number, we say *probability*.

Where does the number come from? There are two sources. The first is counting. A tea-stall owner notes that rain fell on 3 of the last 10 working days, and rain keeps customers away. The probability of rain is 3 divided by 10, which is 0.3.¹ The second source is judgement. When no records exist, an experienced person gives an honest estimate, such as 0.6.

One rule always holds. The probabilities of all possible results of one event add up to 1, because something must happen. If rain has probability 0.3, then no rain has probability $1 - 0.3 = 0.7$. Check by adding: $0.3 + 0.7 = 1$. When your own numbers do not add up to 1, one of them is wrong.

Now the limit. You met risk in Chapter 10, and probabilities are the numbers that describe it. A manager who plans a new venture, such as a second shop, has no records of that venture. The probab-

¹ A probability may also be written as a fraction (3/10) or a percentage (30 per cent). The three forms mean the same. This chapter uses decimals, because they multiply easily.

A probability is a number from 0 to 1 that shows how likely an event is.

A probability comes from counting past records or from honest judgement.

The probabilities of all possible results of one event add up to 1.

Recall: risk and certainty, Chapter 10, section 10.3 'Conditions of Decision Making'.

EXPECTED VALUE

Suppose a friend offers you a game. A fair coin is tossed once. If it lands heads, your friend pays you Tk 100. If it lands tails, you pay your friend Tk 20. Would you play? Most people say yes, but cannot say why. The reason is that the game is worth something to you, and we can find out how much. Think about playing ten times. You would expect about five heads and five tails.

Five heads bring $5 \times 100 = \text{Tk } 500$. Five tails cost $5 \times 20 = \text{Tk } 100$. The net gain over ten plays is $500 - 100 = \text{Tk } 400$. Divided by ten plays, this is Tk 40 for each play. No single play pays Tk 40, because each play pays Tk 100 or costs Tk 20. But Tk 40 is the average, and it tells you what the game is worth.

That average has a name. **Expected value** means the average result of a choice, found by weighting each possible result by its chance. In the game above, the expected value is Tk 40. The word *expected* does not mean that the result will happen. It means the result you would see on average, if the same choice were made many times.²

The method has one rule. In words: expected value equals the sum, over every possible result, of the chance of that result multiplied by the result. In symbols: $EV = (p_1 \times x_1) + (p_2 \times x_2) + \dots$. Here EV stands for expected value, p stands for the probability of a result, and x stands for the result in taka. The small numbers 1 and 2 label the first and second results. A gain is written as a positive number and a loss as a negative number. The probabilities must add up to 1.

Given. Heads (probability 0.5) pays Tk 100. Tails (probability 0.5) costs Tk 20, written as -20. *Find.* The expected value of one play. *Formula.* $EV = (p_1 \times x_1) + (p_2 \times x_2)$. *Steps.* Heads: $0.5 \times 100 = 50$. Tails: $0.5 \times (-20) = -10$. Add: $50 + (-10) = 40$. *Answer.* The expected value is Tk 40 for each play. *Check.* The probabilities add up ($0.5 + 0.5 = 1$), and Tk 40 lies between the worst result (-20) and the best (100). It also equals the Tk 400 we found over ten plays, divided by ten.

Now a messier case. A trader plans a stall at a three-day fair. A busy fair (probability 0.5) gives a profit of Tk 4,000. A normal fair (probability 0.3) gives Tk 1,500. A wet fair (probability 0.2) gives a loss of Tk 1,000. *Find* the expected value. *Formula.* $EV = (p_1 \times x_1) + (p_2 \times x_2) + (p_3 \times x_3)$. *Steps.* Busy: $0.5 \times 4,000 = 2,000$. Normal: $0.3 \times 1,500 = 450$. Wet: $0.2 \times (-1,000) = -200$. Add: $2,000 + 450 - 200 = 2,250$. *Answer.* The expected value is Tk 2,250. *Check.* $0.5 + 0.3 + 0.2 = 1$, and 2,250 lies between -1,000 and 4,000.

Three slips are common. The first is to forget the minus sign on a loss, which makes a bad choice look good. The second is to use probabilities that do not add up to 1. The third is to mix units, for example profit in thousands of taka in one line and in taka in the next. Write every result in the same unit before you multiply.

² Some books use the longer name *expected monetary value* when all results are sums of money. This book uses *expected value*.

A fair way to describe an uncertain choice is the average result over many tries.

GO SLOWLY HERE.

Over ten plays the game pays Tk 400, which is Tk 40 for each play.

Expected value. The average result of a choice, found by weighting each possible result by its chance.

Value. Word in use. In daily life, how much something is worth or how important it is. Here, a number in taka: the average result of a choice.

Expected value is the average result of a choice, weighted by chance.

The formula multiplies each result by its chance and adds the products.

The first example uses small numbers so that the method is the only new thing.

A second example has three results and is more realistic.

The common slips are wrong signs, probabilities that miss 1, and mixed units.

Before moving on: expected value is an average, not a promise of any single result.

When a choice leads to chances, which lead to more results, a picture helps.

Decision tree. A diagram of choices and chance events, drawn as branches, showing the value of each path.

Tree. Word in use. In daily life, a plant with a trunk and branches. Here, a diagram whose lines divide like branches.

A decision tree draws choices and chance events as branches.

A small table shows the parts of a tree and what to do at each.

We read a tree from left to right, but we calculate from right to left.

Recall: risk, Chapter 10, section 10.3 'Conditions of Decision Making'.

The case file gives a tree for the second shop.

PAUSE AND CHECK 11.P1 A seller may earn Tk 500 with probability 0.8, or lose Tk 1,000 with probability 0.2. Find the expected value.

Before moving on. Expected value means the average result of a choice. Multiply each result by its probability, then add. Gains are positive and losses are negative. The probabilities must add up to 1, and no single play need produce the expected value itself.

DECISION TREES

What if a manager has several choices, and each choice leads to several chance results? The sums are easy one at a time, but the whole problem is hard to hold in the head. Managers therefore draw it. Think of a road map: one road leaves the town, then divides, and divides again. A picture of such roads shows every route at once.

A **decision tree** is a diagram of choices and chance events, drawn as branches, showing the value of each path. It is called a tree because the lines divide as branches do, though we draw it on its side, from left to right. Its parts are few. A square marks a *decision point*, where the manager chooses. A circle marks a *chance point*, where events decide the result. Each line is a branch. Each branch that ends the story carries a result in taka.

The table below lists each part, what it means, and what you do at that part. Read each row from left to right: the first cell names the part, the second explains it, and the third tells you the step to carry out.

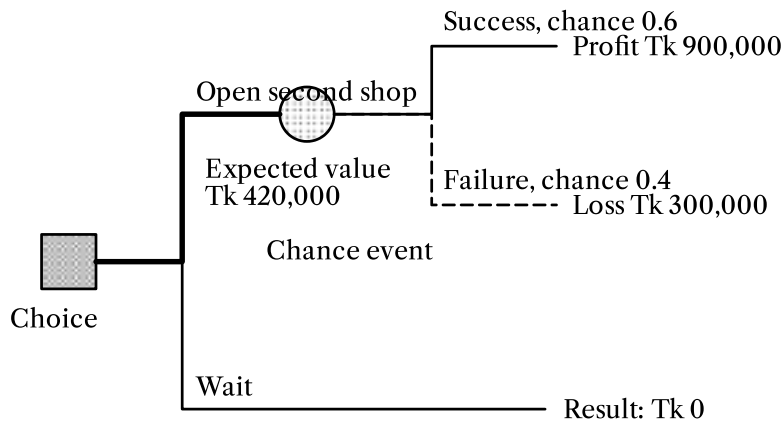
THE PARTS OF A DECISION TREE

PART	MEANING	WHAT YOU DO THERE
Square	A choice the manager makes	Pick the branch with the highest value
Circle	A chance event	Calculate the expected value of its branches
Branch	One choice or one possible result	Label it with its probability, if it is a chance branch
End of branch	The final result in taka	Write profit as positive and loss as negative

The order matters. The story runs from left to right, because the choice comes first and the results come later. The calculation runs in the opposite way. Start at the right-hand ends, where the results are known. Work back to each circle and find its expected value. Then come to the square and compare the values of the branches. This backward step is called folding back the tree.

Case file, illustrative. Meghna and Sons considered a second shop. The tree has two choices. Opening the second shop leads to a chance event: success with probability 0.6 and a profit of Tk 900,000, or failure with probability 0.4 and a loss of Tk 300,000. Waiting leads to a result of Tk 0. Expected value of opening = $0.6 \times 900,000 + 0.4 \times (-300,000) = 540,000 - 120,000 = \text{Tk } 420,000$

waiting, and it ends at Tk 0. The thick line marks the choice with the higher value.



Square: a choice. Circle: a chance event.
Thick line: the choice with the higher value.
Dashed line: the failure result.

FIG. 11.1. *Decision tree for the second shop*

Now the exceptions. A tree is only as good as its numbers, and the numbers 0.6 and 0.4 are estimates. A tree also needs every result to be written in taka, so it ignores matters that have no price, such as family strain or the shop's name among its customers. Finally, a high expected value does not remove the chance of the loss. The result of Tk 420,000 will never happen: the shop will earn Tk 900,000 or lose Tk 300,000.

Before moving on. A decision tree shows choices as squares, chance events as circles and results at the branch ends. We fold it back from right to left, finding an expected value at each circle, and then choose at the square. Its answer is only as good as its estimates.

* * *

11.2 DECISION SUPPORT SYSTEMS

THE SYSTEM ITSELF

A shop owner wants to know which books sell best each week. Counting through paper receipts takes days, and by then the answer is old. A computer can give the same answer in seconds, and can answer follow-up questions as well. Managers have used this idea in every kind of organisation, from banks to hospitals.

A **decision support system** is a computer-based tool that helps managers study data and compare alternatives. A program that stores a firm's sales records and answers questions about them is one example. Such a system usually has three parts: stored data, tools that analyse the data, and a way to ask *what if* questions, such as how profit would change if a price rose by ten per cent.

Case file, illustrative. The sales software of Meghna and Sons lists the top 20 titles each week.

The tree's answer depends on its numbers and ignores what has no price.

Before moving on: a decision tree is read left to right and calculated right to left.

Hand counting takes days, and by then the answer is old.

Decision support system. A computer-based tool that helps managers study data and compare alternatives.

A decision support system is a computer tool for studying data and comparing alternatives.

The case file shows a small system at work.

stored data are wrong, the answers will be wrong. A system costs money to buy and time to learn. And it cannot see matters that were never recorded, such as a new competitor opening next door.

INFORMATION FOR DECISIONS

Data are only useful when they become information that fits the decision. Good information for a decision has four qualities. It is *accurate*, so that the manager can trust it. It is *timely*, so that it arrives before the decision is due. It is *relevant*, so that it bears on this choice and not another. And it is *understandable*, so that the person using it knows what it means.

A fifth point is about cost. More information is not always better. Collecting it takes time and money, and the Tk spent on it should be less than the improvement it brings to the decision. For the owner, this means asking what a fact would change. For an employee who prepares a report, it means asking what the reader needs to decide.

Before moving on. A decision support system is a computer tool that helps managers study data and compare alternatives. It supports the manager and does not replace the manager. Its value depends on information that is accurate, timely, relevant and understandable, and that is worth what it costs.

* * *

11.3 DECIDING IN GROUPS

THE GROUP DECISION

Who should decide: one person, or several? A single manager can act quickly, but sees the problem from one side. A team sees more sides, but must talk. Both ways are used in every organisation, and the skill lies in choosing the right one for the problem.

A **group decision** is a choice reached by several people discussing and agreeing, instead of one person alone. A committee that votes on a new rule makes a group decision. A manager who announces a rule after hearing advice does not, because the manager alone still chooses.

Case file, illustrative. A group of three, Nurul, Rafiq and Tariq, took 3 days to reach a decision. Nurul alone took 1 day.

ADVANTAGES AND DISADVANTAGES

The advantages of group decisions are real. A group holds more information than one person, so more alternatives appear. Members who help to choose usually understand the decision and are more willing to carry it out. The disadvantages are also real. A group is slower, as the case file shows: it took two days more than Nurul alone. It uses the time of several employees. One loud

Information helps a decision only when it has certain qualities.

Information has a cost, so a manager should gather only what is worth its price.

Before moving on: a system supports decisions with information that is accurate, timely, relevant and understandable.

A manager must choose whether to decide alone or with others.

Group decision. A choice reached by several people discussing and agreeing, instead of one person alone.

A group decision is a choice that several people reach together.

Recall: the decision process, Chapter 10, section 10.2 'The Decision Process'.

The case file records how long each way took.

A group brings more knowledge and more support, and it costs more time.

A GROUP COMPARED WITH ONE MANAGER

POINT	GROUP DECISION	ONE MANAGER DECIDING
Speed	Slower	Faster
Range of ideas	Wider	Narrower
Support from those affected	Stronger	Weaker, unless explained
Responsibility	Shared and less clear	Clear
Cost in time	Higher	Lower

The table suggests a rule. Choose a group when the problem is important, unclear, and affects many people, and when time allows. Choose one manager when the problem is urgent or simple, or when one person holds the authority and the knowledge. The exception is that no rule fits every case, and a manager may still choose to consult before deciding alone.

A group suits important, unclear problems; one manager suits urgent, clear ones.

OTHER FACTORS IN DECISIONS

A decision tree uses taka and probabilities. Real decisions also depend on other factors. The first is time: some decisions cannot wait for a long study. The second is cost: a study can cost more than the decision is worth. The third is importance: a small choice deserves a small effort. The fourth is the manager's own values and attitude to risk, for two managers may read the same expected value and choose differently. The fifth is authority: who is allowed to decide, and whose agreement is needed.

Money values and probabilities are not the only things that shape a decision.

One last fact explains much of this. Managers rarely have time or information to compare every alternative. They usually stop when they find one that is good enough. The section *In History*, later in this chapter, tells who named this habit.

Managers rarely find the best answer; they stop at a good one.

Before moving on. A group decision is a choice reached by several people discussing and agreeing. It brings more ideas and support, and it costs time and can blur responsibility. Time, cost, importance, values and authority also shape a decision.

Before moving on: a group decision gains knowledge and support but costs time, and several factors besides money shape any choice.

* * *

WORKED CASE: CHOOSING BETWEEN TWO SITES

Situation. Selim runs a stationery stall and may rent one of two sites. At Site A, success (probability 0.5) gives a profit of Tk 80,000, and failure (probability 0.5) gives a loss of Tk 20,000. At Site B, success (probability 0.8) gives a profit of Tk 30,000, and failure (probability 0.2) gives a loss of Tk 10,000. Waiting gives Tk 0. *Question.* Which choice has the highest expected value, and is that choice automatically the right one?

Selim must compare two sites and waiting, using expected value.

Reasoning. We fold back the tree from the right. Site A: $0.5 \times 80,000 = 40,000$, and $0.5 \times (-20,000) = -10,000$, so the expected value is $40,000 - 10,000 = \text{Tk } 30,000$. Site B: $0.8 \times 30,000 =$

We fold back the tree: first each circle, then the square.

* * *

The summary gathers the main ideas.

SUMMARY

A probability is a number from 0 to 1, and the probabilities of one event add up to 1. Expected value is the average result of a choice: multiply each result by its probability and add. A decision tree draws choices and chance events as branches, and we fold it back from right to left. A decision support system supplies data and comparisons, and the manager decides. A group decision gains knowledge and support but costs time.

THINK IT THROUGH

11.T1 The case file shows that a group of three took 3 days, and Nurul alone took 1 day. Name two situations in which the extra 2 days are worth it, and one in which they are not.

11.T2 In the second-shop tree, suppose the probability of success falls to 0.4 and the probability of failure rises to 0.6, with the same profit and loss. Find the new expected value of opening. Does opening still beat waiting?

11.T3 Find the probability of success at which opening the second shop has the same expected value as waiting. Write p for this probability, so that failure has probability $1 - p$.

WHERE PEOPLE GO WRONG

A common error is to treat expected value as the result that will happen.

Another error is to think that the highest expected value is always best.

A third error is to trust a tree more than its numbers deserve.

A last error is to think that groups are always better, or that a system decides.

Some students believe that an expected value of Tk 420,000 means the shop will earn Tk 420,000. It will not. The shop earns Tk 900,000 or loses Tk 300,000. The expected value is an average over many tries, and it is a guide for a choice made once.

Some students believe that the choice with the highest expected value must always be taken. A choice with a large possible loss may be too dangerous for a small firm, as the worked case shows.

Some students believe that a tree is exact because it uses arithmetic. The arithmetic is exact, but the probabilities are estimates, so the answer is no better than they are.

Some students believe that a group always decides better than one manager, or that a decision support system makes the decision. A group costs time and a system only supports. Each has a place, and neither replaces judgement.

IN HISTORY

Herbert Simon showed in 1947 that managers settle for a good enough choice.

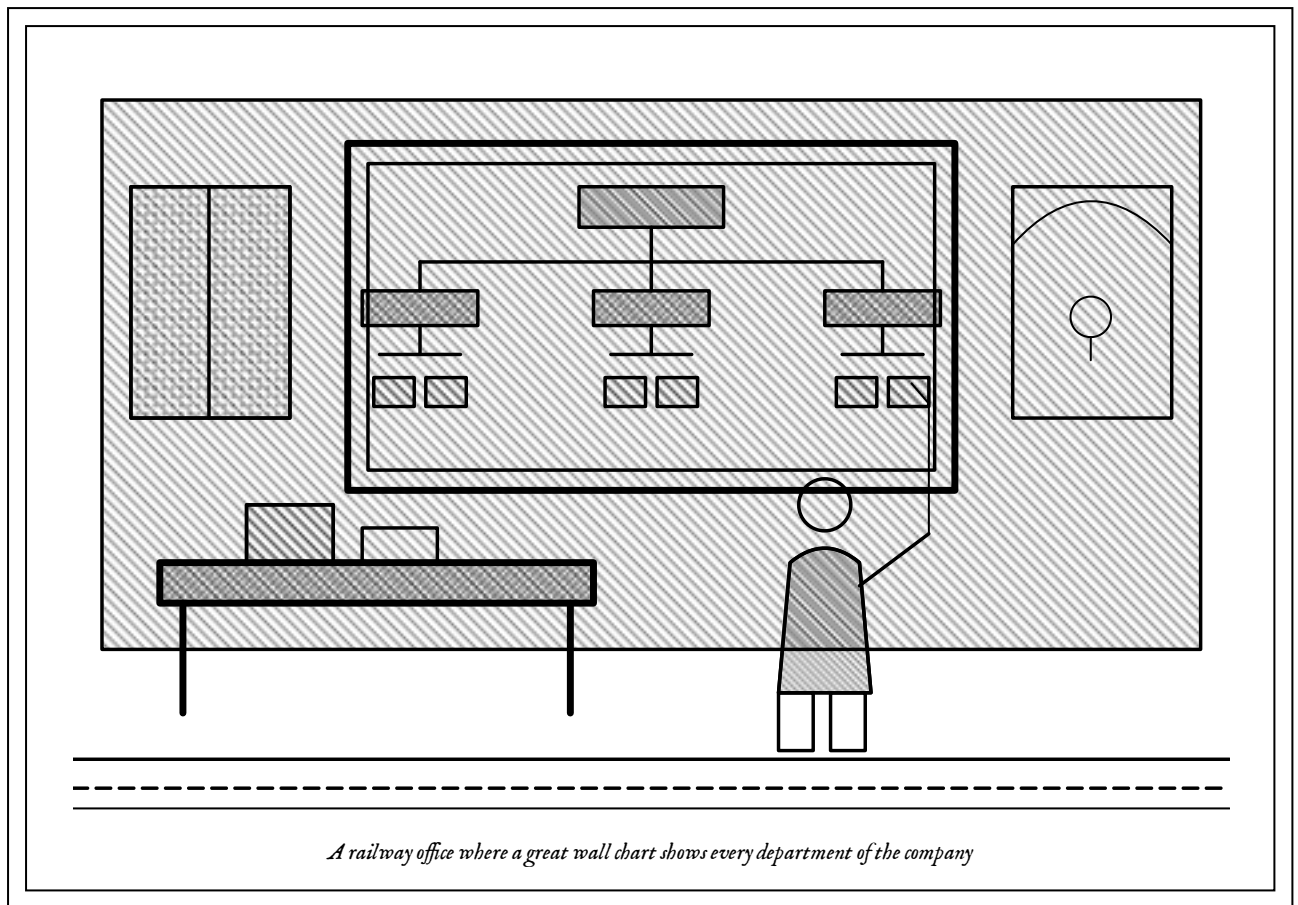
In 1947 the American social scientist Herbert Simon, born in Milwaukee in 1916, published *Administrative Behavior*. He had studied at the University of Chicago and earned a doctorate in political science there in 1943. Many economists of the time pictured a decision maker who weighs every alternative and picks the best. Simon replied that real managers cannot do this. They have limited

In 1978 Simon received the Nobel prize in economics, and later in his career he turned to artificial intelligence by computer. His idea fits this chapter. A tree and a decision support system widen what a manager can examine. A group widens what is known. But each costs time, so managers still stop when the choice is good enough.

His idea explains why tools and groups help, but cannot make the search endless.

REVIEW QUESTIONS

- 11.1 A probability is 0.25. What does this number mean, and what is the probability that the event does not happen?
- 11.2 Find the expected value: probability 0.7 of a profit of Tk 50,000, and probability 0.3 of a loss of Tk 80,000.
- 11.3 Find the expected value: probability 0.2 of Tk 100,000, probability 0.5 of Tk 40,000, and probability 0.3 of a loss of Tk 60,000.
- 11.4 Choice X gives Tk 200,000 with probability 0.6 and -Tk 100,000 with probability 0.4. Choice Y gives Tk 70,000 for certain. Which has the higher expected value, and what else might a manager weigh?
- 11.5 In a decision tree, what do a square and a circle show, and in which direction do we draw the tree and in which do we calculate it?
- 11.6 Describe two ways in which a decision support system helps a manager, and one limit.
- 11.7 Give two advantages and two disadvantages of a group decision.
- 11.8 Name three factors, other than money and probability, that shape a decision.
- 11.9 Using Chapter 10, say why a decision tree suits a decision under risk, and what changes when the manager has certainty.



CHAPTER XII

Organizing and Types of Organization

Who does what, and who answers to whom?



A goal cannot be reached by good intentions alone; someone must decide who does which task. This chapter explains organizing, the steps by which managers divide and link work, and the difference between the official structure and the friendships that grow beside it.

Two earlier ideas support this chapter.

A railway with no division of duties would stop at its first emergency.

The puzzle is that everyone sees the danger and no one is told to act.

The chapter will name the arrangement, show how it is made, and meet its unofficial twin.

A household shows that many tasks need a plan of who does what.

Organizing. Arranging tasks, people and resources into a working structure so that goals can be reached.

Organizing is the function that arranges work, people

You will need two ideas from earlier chapters. The first is the management process, taught in Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.¹ The second is the organisation, taught in Chapter 1, 'What Management Is', section 1.1 'The Idea of Management'.² Read those sections again if either idea feels uncertain.

Imagine a railway company in the middle of the nineteenth century. Its track runs for hundreds of miles, and every day trains leave and arrive at stations far from the head office. On a winter morning a telegraph message reaches the head office: a bridge near a distant town has been damaged, and a passenger train is due to cross it in two hours. Several people in the office read the message. Each of them knows that something must be done. Each of them believes that another person will do it.

This is the puzzle of the chapter. The company has money, engines, track and willing workers, yet it may fail on this morning, because no one has been told that the message is his or her responsibility. Good intentions do not move a train. What the company lacks is an arrangement: a known answer to the question *who does what, and who answers to whom?* An organisation that cannot answer that question for every task will meet the same danger again and again, however skilled its people may be.

In this chapter we study that arrangement. First we name it and say why it exists. Then we follow the steps by which a manager builds it. Finally we meet two forms that it takes: the one managers draw on paper, and the one that people form among themselves without any instruction. We shall use the records of a small Dhaka bookshop to practise, and a famous railway chart of 1855 to see where the idea first took a clear shape.

* * *

12.1 WHAT ORGANIZING IS

Think first of a household that is preparing a wedding meal. There are dishes to buy, to cook and to serve, guests to welcome, and a bill to pay. If every relative does whatever seems useful at the moment, some tasks are done twice and others not at all. A sensible family leader says: you buy, you cook, you serve, you welcome. The meal succeeds because the work was divided and each person knew the share that was his or hers. Management uses the same idea on a larger scale.

The function that does this has a name. **Organizing** means arranging tasks, people and resources into a working structure so that goals can be reached. A resource is anything an organisation uses to do its work, such as money, rooms, machines and time. When a bookshop owner decides who will buy books, who will

¹ Recall: management process, Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

² Recall: organisation, Chapter 1, 'What Management Is', section 1.1 'The Idea of Management'.

Organizing is an activity, and it leaves a result behind. We call the result the **organizational structure**, which means the pattern of jobs, groups and reporting lines that divides and links the work. A reporting line is the path that shows which person answers to which other person. In a bookshop the structure answers questions such as these: Who orders books? Who may refuse a refund? To whom does the cashier bring a problem? The structure exists whether or not anyone has drawn it.

Why do organisations organize at all? The first purpose is to turn a large goal into jobs of a size that one person can perform. No one can sell the books, count the stock and keep the accounts at the same moment. The second purpose is to avoid waste. When each task has an owner, it is neither forgotten nor done twice. The third purpose is to link the parts, so that the work of the buyer reaches the seller in good order. These three purposes serve the aim of the whole management process, which is to reach goals with productivity and effectiveness.

The same arrangement looks different from each viewpoint. The owner sees a way to reach the shop's goals without doing every task personally. The manager sees a tool for checking whether any task has been left without an owner. The employee sees a fixed answer to the question *what is my job, and to whom do I turn?* The customer, who never sees the structure, feels its results when the right person answers a question without delay.

Organizing also has a nature that is worth knowing, because it prevents three errors. First, organizing is continuous. A structure that suited a shop with three employees may fail when the shop has seven, so managers return to it whenever the work changes. Second, organizing serves goals. A structure is judged by whether it helps the organisation reach its goals, and not by whether it looks neat on paper. Third, organizing deals with people as well as tasks. A perfect plan of duties can fail if the people who must perform it do not understand or accept it.

There are limits to what organizing can do. A structure cannot repair a poor plan, because it only arranges the work that the plan has set. Too much structure can also slow an organisation: if every small question must travel through several reporting lines, customers wait and employees lose interest. A good structure is therefore the lightest one that still makes every task and every reporting line clear.

IN SHORT

Organizing is the function that arranges tasks, people and resources. Its result is the organizational structure. Its purpose is to make every task owned, every owner known and every part linked to the others.

Before moving on. Organizing is the function of arranging tasks, people and resources into a working structure. The structure is the pattern of jobs, groups and reporting lines. Its purpose is to give every task an owner and to link the parts. It is continuous, it serves

Organizational structure. The pattern of jobs, groups and reporting lines that divides and links the work.

The result of organizing is an organizational structure.

Structure. Word in use. In daily life, a building or other object that has been put together. In management, the pattern of jobs and reporting lines inside an organisation.

The purpose of organizing is to turn a goal into jobs that people can perform.

From each viewpoint, organizing gives something different.

Organizing has a nature: it is continuous, it serves goals, and it follows people and work.

Exceptions: a structure can help in one setting and harm in another.

The recap fixes the function, its result and its purpose.

Step one asks: what work must be done?

The case file records the ten duties of Meghna and Sons in 2026.

Ten duties and seven people show that some people must hold more than one duty.

Step two asks: which tasks belong together?

There is no single correct grouping, so the manager must judge the shop's own situation.

Step three asks: who will do each group of tasks?

Assigning a duty is incomplete unless the person also has the means and the authority to perform it.

example, a small bookshop, so that you can see how one step leads to the next. Many students find this process confusing at first because the steps seem to blend together. They become clear when each step is asked a single question.

Step one: identify the tasks. The question of this step is *what work must be done?* The manager lists every task that the shop must perform to reach its goals, from the largest to the smallest. The list should come from the plan and not from the names of the people who are available, because a list built around people will hide the tasks that nobody has yet considered. At this step the manager does not ask who will do each task. Only the work is listed.

Case file, illustrative. Duties recorded at Meghna and Sons for 2026, ten in number: buying, receiving, shelving, selling, cashier, accounts, online orders, returns, school accounts and marketing. The ten duties are assigned to 7 people.

The record gives a useful fact: there are ten duties and seven people. Ten divided among seven cannot give one duty to each person, so at least some people hold more than one duty. This is normal in a small organisation. It also shows why the next two steps matter, because the manager must decide which duties belong together in one person's hands.

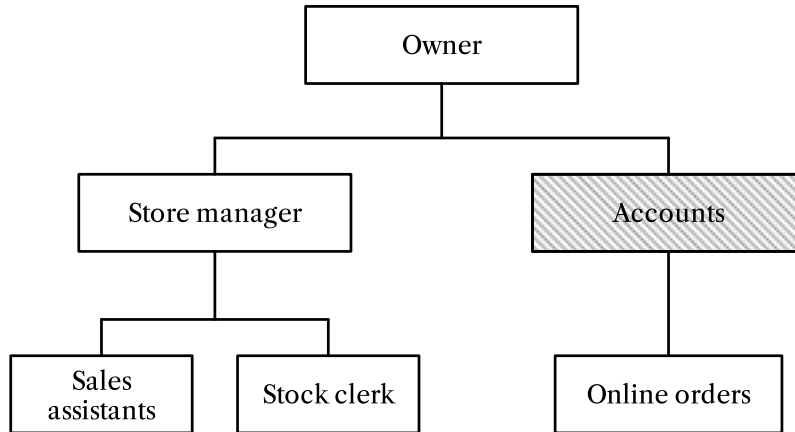
Step two: group the tasks. The question of this step is *which tasks belong together?* Tasks belong together when they need the same skill, use the same place, or depend closely on one another. Receiving books, shelving books and handling returns all involve the stockroom and the physical books, so a manager may place them in one group. Selling and the cashier's work both happen at the counter, so they may form another group. A group of tasks is the first step from a list to a pattern.

Another manager might group the same ten duties in a different way. Online orders and school accounts both deal with customers who never walk into the shop, and a manager could place them together. The best grouping depends on the people available and the work to be done. This is a place where judgement is needed, and the guide is the purpose of organizing: whichever grouping makes tasks owned, linked and free of waste serves the shop best.

Step three: assign the duties. The question of this step is *who will do each group of tasks?* The manager now turns to people. A duty is a task that a named person is expected to perform. The manager matches each group to the person with the skill and the time to perform it. In the base record of the shop, for example, Tariq Meghna already handles accounts and online orders. That assignment shows how a manager keeps related tasks together in one person's hands.

Assigning a duty is not complete until the person has what is needed to perform it. A cashier who must handle returns needs a rule that says how much money may be refunded without asking anyone. The right to take such decisions is called authority, which was taught earlier as the right to command or to decide. A duty without authority leaves the person unable to act, and authority without a duty leaves the organisation without an owner for the task

When the four steps are complete, the manager can draw the result. An **organization chart** is a diagram showing the jobs in an organisation and who reports to whom. Each job is shown in a box, and lines join the boxes to show the reporting lines. The figure below shows a small example. To read it, begin at the top box, which holds the highest authority. Follow a line downward from a box to meet the jobs that report to it. A line always joins two boxes and means *reports to*, read from the lower box to the upper box.



Each line means reports to, read upward

FIG. 12.1. *An organization chart with lines of authority*

An organization chart has two limits. It shows only the official lines and nothing of the friendships and habits that also guide behaviour; we shall meet those in the next section. It can also become out of date, because a chart drawn in January may not describe the shop after two people leave in March. A chart is a record of a decision, and the manager must update the record when the decision changes.

Students often make two slips in this process. The first is to skip step one and begin with the people, which makes the list of tasks fit the people instead of the goals. The second is to draw the chart first and decide the duties later, which produces boxes with no real work in them. Keeping the order of the steps prevents both slips.

PAUSE AND CHECK 12.P1 A shop has three sales assistants and one cashier. Name the step in which the manager decides that all four report to the store manager.

Before moving on. The four steps are to identify the tasks, group the tasks, assign the duties and set the relationships. Assigning a duty means giving the person the means and the authority to perform it. The organization chart is the drawing that records the result.

* * *

12.3 FORMAL AND INFORMAL ORGANIZATION

Walk into a bookshop at noon and watch for a moment. The off

Organization chart. A diagram showing the jobs in an organisation and who reports to whom.

The organization chart is a drawing of the result of the four steps.

Chart. Word in use. In daily life, a table or picture that shows numbers, such as a chart of rainfall. In management, a diagram of jobs and reporting lines.

A chart shows the official lines only, and it can become out of date.

Common slips in the process: skipping a step, or building the chart around people.

The recap fixes the four steps and the chart.

A shop has an official pattern

Formal organization. The official structure of jobs, rules and lines of authority that management sets up.

The formal organization is the official structure that management sets up.

The formal organization gives order, stability and clear authority, but it can be slow.

Informal organization. The network of friendships and social groups that forms among employees on its own.

The informal organization is the network of friendships that forms without management's instruction.

The case file records an informal group at Meghna and Sons.

The lunch group is informal, because no manager created it and it appears on no chart.

People join informal groups to meet needs that the official structure does not meet.

the store manager walks between them. Now watch where people stand during a quiet hour. Some of them gather in small groups, talk about books and families, and pass along advice that no one has ordered them to give. A second pattern has formed beside the first. Both patterns shape how the shop works, and a manager who sees only the first will often be surprised by the second.

The first pattern is the **formal organization**. It means the official structure of jobs, rules and lines of authority that management sets up. It is deliberate, which means that managers choose it on purpose. It is usually written down, in job descriptions, rules and the organization chart. Everything we studied in section 12.2 produces a formal organization, because the four steps are decisions that management takes and records.

The formal organization has real strengths. It gives each employee a known job and a known person to ask. It lasts when individuals come and go, because the jobs remain even when the people change. It supports the control of work, since someone is answerable for each task. Its weakness is that it cannot describe everything people do. Official rules cannot foresee each small situation, and an organisation that relies only on its rules can become slow and stiff.

PAUSE AND CHECK 12.P2 A shop's job description says that the stock clerk reports to the store manager. Is this part of the formal or the informal organization? Give one reason.

The second pattern is the **informal organization**. It means the network of friendships and social groups that forms among employees on its own. No manager designs it, no document records it, and it has no official leader. It appears wherever people work together over time. Employees who share a bus route, a hometown or an interest in poetry will talk to one another more than the chart would predict, and a group of friends is the usual result.

Case file, illustrative. In 2026, 3 sales assistants at Meghna and Sons form a lunch group and exchange book tips informally. The record places the group beside the formal structure and does not show it on the organization chart.

The lunch group is part of the informal organization. No manager created it. No chart records it. Yet it does useful work: when one sales assistant learns which new title a school teacher has asked about, the others hear it at lunch. The knowledge reaches the counter faster than any written notice would. This example shows an important point. The informal organization is neither good nor bad in itself. Its effect depends on what the group does and on how it relates to the goals of the shop.

Why do people join informal groups? The reasons are ordinary ones. Employees want company, because work fills much of the day and people dislike spending it alone. They want help, because a colleague can often explain a task faster than a manual. They want information, because the official channels are slow or incomplete. They want to feel that they belong and that their opinion counts. Each of these is a need that the formal structure rarely meets, and

The two patterns are therefore not rivals. They exist side by side inside one organisation, and each fills a gap in the other. The table below sets them side by side. Each row names a feature, and the two columns say how the feature appears in each kind. Read across a row to compare the two on one feature.

The two organizations are not rivals; they work together inside one organisation.

FORMAL AND INFORMAL ORGANIZATION COMPARED

FEATURE	FORMAL ORGANIZATION	INFORMAL ORGANIZATION
Origin	Set up by management	Grows among employees on its own
Record	Written: rules, job descriptions, chart	Unwritten: habits, friendships
Authority	Comes from the position held	Comes from the respect of the group
Main strength	Order and stability	Speed and goodwill
Main risk	Slowness and stiffness	Rumour and resistance

How can managers make an organisation effective, then? They begin with a sound formal structure: every task has an owner, every reporting line is clear, and the chart is kept up to date. They then treat the informal organization as a fact of the shop and not as an enemy. Managers share honest information, so that rumour has less room. They ask informal groups for ideas, because those groups often know where the work is going wrong. They also watch for harm, and they speak plainly when a group acts against the shop's goals.

Making an organization effective means designing the formal structure while working with the informal one.

Before moving on. The formal organization is the official structure that management sets up. The informal organization is the network of friendships that forms on its own. People join informal groups to meet needs the formal structure does not meet. Managers make an organisation effective by designing the formal structure well and by working with the informal groups.

The recap fixes the two kinds and the way they work together.

* * *

WORKED CASE: ORGANIZING THE DUTIES OF MEGHNA AND SONS

This section applies the process of section 12.2 and the distinction of section 12.3 to the shop's records. It is not a model answer. It shows one way of reasoning, so that you can follow each decision and then repeat it on a new case.

The worked case applies the four steps to the shop's ten duties.

Situation. The shop's record for 2026 lists ten duties: buying, receiving, shelving, selling, cashier, accounts, online orders, returns, school accounts and marketing. They are assigned to 7 people. The base record names the people: Rafiq Meghna, the store manager; Tariq Meghna, who handles accounts and online orders; 3 sales assistants; 1 cashier; and 1 stock clerk. A lunch group of 3 sales assistants exchanges book tips informally.

The situation gives the duties, the people and the lunch group.

Question. Group the ten duties into a small number of groups. Decide how many duties fall to each person on average. Then say

The question asks for a grouping, an owner for each

For Step three, the base record already gives one assignment, and the other assignments are not recorded.

The lunch group is informal, so it stays off the chart but not out of the manager's view.

The conclusion states the structure and the status of the lunch group in plain sentences.

The check tests the counting and guards against a hidden gap.

The summary restates the chapter in a few sentences.

Four situations test the chapter, from familiar to combined.

form a counter group of 2 duties. Accounts, online orders, school accounts and marketing all deal with records or with customers who are not at the counter, so they form a records-and-outside-customers group of 4 duties. The grouping is one reasonable choice, and the file does not say which choice the shop made.

For step three, the base record states that Tariq Meghna handles accounts and online orders, and both duties sit in the third group. That agrees with our grouping, which is a useful sign. The file does not record who holds the other duties, so we do not invent assignments. We can, however, work out how heavy the load is. Ten duties among seven people give an average of 10 divided by 7, which is about 1.4 duties per person. For step four, the store manager would take the reporting of the stock group and the counter group, and the owner would receive reports from the store manager and from Tariq Meghna.

The lunch group of 3 sales assistants is informal. It has no place on the organization chart, because no manager created it and it carries no authority. The store manager should still know of it, because the group already carries useful knowledge about which books customers ask for.

Conclusion. The shop's ten duties fall into three groups of 4, 2 and 4 duties. The load is about 1.4 duties per person on average, so some people hold more than one duty. The lunch group belongs to the informal organization, and managers should use it, not suppress it.

Check. We test the counting. The groups give 4 plus 2 plus 4, which equals 10, so no duty is lost or counted twice. The load check: 7 people times 1 duty each gives only 7 duties, so $10 - 7 = 3$ duties must be additional ones. This confirms that at least some people hold two duties. The shop should also confirm each person's duties in writing, because a task that is not assigned is the first step toward the puzzle that opened this chapter.

* * *

SUMMARY

Organizing is the function that arranges tasks, people and resources into a working structure. Its result is the organizational structure, the pattern of jobs, groups and reporting lines. The process has four steps: identify the tasks, group the tasks, assign the duties and set the relationships. The organization chart records the result. Beside the formal organization, which management sets up, an informal organization of friendships forms on its own. Effective managers design the formal structure well and work with the informal groups.

THINK IT THROUGH

Each question below changes the situation a little more than the one before. Answer them in order.

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12.T3 A small clinic has a doctor, two nurses and a receptionist. Three employees eat lunch together every day and share news about patients' appointments. Is this group part of the formal or the informal organization? Explain whether the manager should be pleased or concerned.

12.T4 A shop doubles its number of employees and keeps the same chart. Name two problems that may appear, using the ideas of structure and the formal and informal organization.

WHERE PEOPLE GO WRONG

Some students believe that the organization chart is the organisation. The chart is only a drawing of the formal structure at one moment. It leaves out the friendships and habits that also guide behaviour, and it may be out of date.

Some students believe that informal groups are always harmful. In the lunch group of the case file, the informal group carried useful knowledge. The correction has a limit: informal groups can also spread rumour or resist change, so a manager must judge each group by its effects.

Some students believe that organizing is only drawing boxes and lines. The drawing comes at the end. The real work lies in identifying tasks, grouping them, assigning duties with the means to perform them, and setting relationships.

Some students confuse a structure with its results. A structure that looks neat can still fail if people do not understand it or if it does not serve the goals of the organisation. A structure is judged by its results.

Mistake one: believing that the chart is the organisation.

Mistake two: believing that informal groups are always harmful.

Mistake three: believing that organizing is only drawing boxes.

Mistake four: confusing the structure with its effects.

IN HISTORY

In 1855 the New York and Erie Railroad was one of the longest railways in the United States. Its general superintendent, Daniel McCallum, faced the problem of the opening puzzle on a large scale: a company spread over many miles, in which messages travelled quickly by telegraph but responsibility for acting on them was unclear. McCallum worked with a civil engineer, George Holt Henshaw, to prepare a diagram of the company's plan of organization. It is dated by the returns of September 1855.

The diagram showed the division of duties and the number and class of employees in each department. It was drawn in the form of a tree, in which the highest authorities stood at the roots and the departments and staff spread out as branches. Divisional superintendents were responsible for the daily work of the lines under them, and regular reports travelled upward to the central office. Historians commonly regard the diagram as one of the first modern organization charts. McCallum also set down general principles of management, and he later worked on military railways during the American Civil War.

The case shows an idea from this chapter at work. A structure turns a large undertaking into duties with known owners and

Daniel McCallum drew an early organisation chart for the Erie Railroad in 1855.

The chart showed divisions, duties and the number of employees in each department.

The case shows that a drawn structure can make responsibility known across a

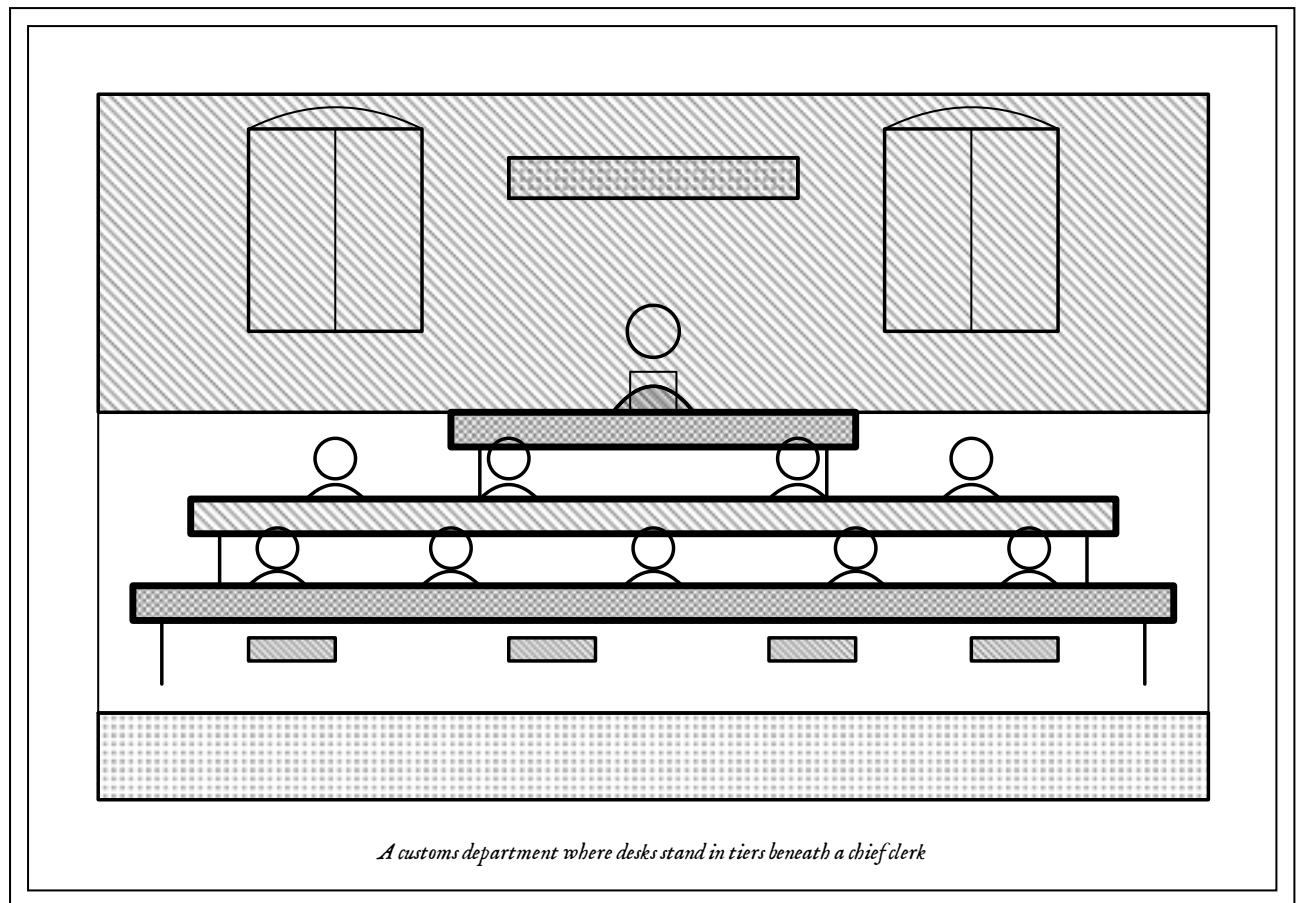
REVIEW QUESTIONS

12.1 Separate these into formal or informal: the job description of a cashier; a lunch group of friends; the reporting line from stock clerk to store manager; a habit of sharing book tips at noon.

12.2 State the four steps of the organizing process in order, and say which question each step answers.

12.3 In your own words, give the difference between the formal organization and the informal organization. Give one strength of each.

12.4 Explain why people join informal groups, and say how a manager can work with such groups.



CHAPTER XIII

Structure: Span, Departments and Forces

How many people can one manager lead, and how should the work be grouped?



Every structure answers two questions: how many people report to each manager, and how the jobs are gathered into groups. This chapter explains both choices, and then shows the forces that push managers toward one design or another.

Two ideas from Chapter 12 support this chapter.

A shopkeeper can watch three assistants, but no one can watch three thousand soldiers.

The puzzle has two parts: how many people to lead directly, and how to group the rest.

The chapter answers both parts and then names the forces that shape the answer.

A teacher with a class shows that the number of people led changes how well they are led.

Span of management. The number of employees who report directly to one manager.

The span of management is the number of employees reporting directly to one manager.

You will need two ideas from the previous chapter. The first is organizing, taught in Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is'.¹ The second is the formal structure, taught in Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.² Read those sections again if either idea feels uncertain.

Think of a shopkeeper who has three assistants. She sees every sale, hears every conversation and corrects a mistake within a minute. Now think of a general who has three thousand soldiers. He cannot see them all, and he cannot speak to each one in a single day. Somewhere between three and three thousand there lies a point where one person can no longer lead everyone directly. Where is that point, and what should the leader do when the point has been passed?

The puzzle has a second part. Suppose a Victorian customs department, with desks in tiers and a chief clerk above them, grows from twenty clerks to two hundred. The chief clerk must now decide how many officers will report to him, and how the clerks will be grouped beneath those officers. Some departments group clerks by the kind of task. Others group them by the ships they handle, or by the port where they work. Each choice changes how fast the work moves and how well the clerks understand their duties.

In this chapter we answer both parts of the puzzle. We begin with the number of people a manager leads directly. Then we study four ways of grouping jobs. Finally we examine the forces that push a manager toward one design or another, and we study the old idea that the work itself should be divided into small tasks. Throughout, we shall use the small Dhaka bookshop Meghna and Sons, whose records give us numbers to work with.

* * *

13.1 SPAN OF MANAGEMENT

Consider a teacher who gives private lessons to two students. She knows each student's weak points and can reply to every question at once. When the same teacher faces a class of sixty, she can no longer follow each student. Her skill has not changed. The number of people she leads has changed, and that number alters the quality of her leadership. Managers meet the same fact in every organisation.

Management gives this number a name. The **span of management** means the number of employees who report directly to one manager. The word *directly* matters. A manager with five employees and fifty more below them has a span of five, not fifty-five. A store manager who has three sales assistants, a cashier and a stock clerk reporting to him has a span of five.

¹ Recall: organizing, Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is'.

² Recall: formal organization, Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.

The shop has two different spans: 2 and 5. The owner has a small span, and the store manager has a larger one. Neither number is wrong. The question that matters is whether each manager can lead the people reporting to him well. To answer it, we need to know what makes a span suitable.

There is no single correct span for all managers. The number depends on factors, and a manager can reason about each one. First, the skill of the employees. Employees who are trained and experienced need little guidance, so one manager can lead many. Second, the similarity of the tasks. If all the employees do the same kind of work, one manager can supervise them together. Third, the clarity of plans and rules. When everyone knows the standards, fewer questions come to the manager.

Three more factors complete the list. Fourth, nearness: employees who work in one room are easier to lead than employees in different towns. Fifth, the manager's other duties. A manager who must also sell, buy and keep accounts has little time left for leading, so his span should be small. Sixth, the speed of change. Work that changes every week creates many questions, and a manager needs time to answer them. In general, the wider the span, the more the first three factors must be favourable.

These factors have exceptions. A manager with a very wide span may succeed if the employees are highly skilled and the rules are clear. A manager with a narrow span may still fail if he supervises too closely and gives employees no room to think. The factors are guides to judgement and not a formula.

PAUSE AND CHECK 13.P1 A store manager leads one cashier and one stock clerk, and both work in the same room on routine tasks. Name two factors from the list that suggest this manager could lead more people.

Spans in the whole organisation add together to give its shape. A **tall structure** means a structure with many levels and small spans of management. Imagine a company in which each manager has only two employees below. To cover a hundred employees, the company needs many levels of managers, one above another, and the structure grows tall.

The opposite design is the **flat structure**. A flat structure means a structure with few levels and wide spans of management. If each manager leads twelve employees, the same hundred employees need far fewer levels, and the structure is low and wide. The figure shows the idea with two small trees. To read it, begin at the top box, which holds the highest manager, and count the lines that leave each box. The number of lines leaving a box is its span, and the number of rows is the number of levels.

The two spans in the shop differ, and the difference has a reason.

Several factors decide how wide a span can be without harm.

More factors: distance, the manager's other duties, and how often things change.

An exception: a wide span can work when employees are skilled, and a narrow one can still fail.

Tall structure. A structure with many levels and small spans of management.

A tall structure has many levels and small spans.

Flat structure. A structure with few levels and wide spans of management.

A flat structure has few levels and wide spans.

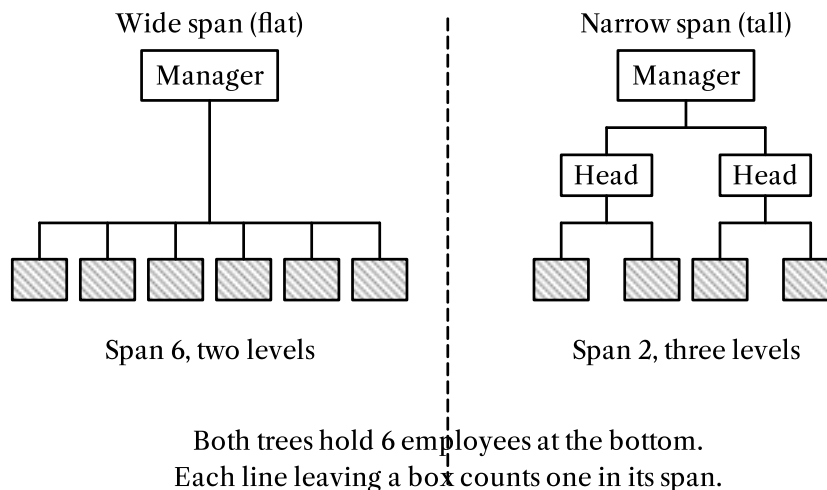


FIG. 13.1. *Wide and narrow spans shown as two trees*

Each design has benefits and costs, and neither is best in every case.

The recap fixes span, the factors, and tall and flat designs.

Each design has benefits and costs. A tall structure allows close supervision and gives employees many steps to climb, but messages must pass through many levels, so decisions are slow, and the many managers add to the cost of wages. A flat structure sends messages quickly and gives employees more freedom, but each manager is overloaded if the span is too wide. A manager who has too wide a span may miss mistakes. A manager who has too narrow a span may watch too closely and discourage initiative.

Before moving on. The span of management is the number of employees who report directly to one manager. The suitable span depends on factors such as skill, similarity of tasks, clarity of plans, nearness and the manager's other duties. A tall structure has many levels and small spans. A flat structure has few levels and wide spans.

* * *

13.2 DEPARTMENTALIZATION

Once a manager has many employees, the next question is how to gather them into groups.

Departmentalization.

Grouping jobs into departments according to a common basis, such as function or customer.

Departmentalization is the grouping of jobs into departments on a common basis.

A bookshop with three employees needs no groups. Everyone knows everyone, and the owner speaks to all of them daily. When the shop grows to seven employees, the owner must decide how to divide them. One idea is to put the people who buy and shelve books together, and the people who serve customers together. That is a division by the kind of work. Another idea is to place together everyone who deals with schools. That is a division by customer. The choice affects how the shop works every day.

Management has a name for this choice. **Departmentalization** means grouping jobs into departments according to a common basis, such as function or customer. A department is a group of jobs placed under one manager. The basis is the feature that the jobs in one group share. Four bases are common: function, product, customer and territory. We take each in turn.

By function. A function is a main kind of work that an organisation must perform, such as selling, stock handling or accounts. When jobs are grouped by function, all the people who perform the same kind of work belong to one department. The case file shows the shop's present design.

Case file, illustrative. The grouping used now at Meghna and Sons is by function: sales, stock and accounts.

Grouping by function has advantages. People who share a skill learn from one another, and the shop needs only one stock team and one accounts team. It also has a weakness. The functions can drift apart: the sales team may promise a delivery date that the stock team cannot meet, because neither group sees the whole customer. Function suits an organisation with a small range of products and customers.

By product. When jobs are grouped by product, each kind of good or service has its own department with all the skills it needs. A large bookshop might have one department for textbooks, one for storybooks and one for stationery. The head of each department can then be judged by the results of one product. The weakness is repetition: each department may need its own buyer and its own record keeper, which raises the cost.

By customer. When jobs are grouped by customer, each type of buyer has its own department. The shop's records show three types of customers, and the sales of 2025 can be set against them.

Case file, illustrative. The alternative grouping at Meghna and Sons is by customer type, using the 2025 sales: walk-in customers Tk 9,000,000; schools Tk 3,600,000; online Tk 1,800,000.

A customer grouping lets each department study the needs of its own buyers. A school department could learn when schools place their orders and what discounts they expect. Walk-in customers and online customers would also have their own teams. The cost is repetition, since each team may need its own stock handling and its own accounts, and a small shop may not have enough people to supply every team.

By territory. Territory means a geographical area, such as a district or a town. When jobs are grouped by territory, each area has its own department. A chain of bookshops with branches in several cities might give each city its own manager. Territory grouping suits an organisation that serves places far apart, because local managers can respond to local needs. The risk is that branches begin to follow different rules.

PAUSE AND CHECK 13.P2 A bookshop with two branches, one in Dhaka and one in Chattogram, gives each branch its own manager. Which basis of grouping is this?

The table below compares the four bases. Each row names a basis and says what it puts together. one strength and one weakness.

Department. Word in use. In daily life, a section of a large shop or a branch of government, such as a department of education. In management, any group of jobs placed under one manager, whatever name the organisation gives it.

Grouping by function puts together jobs that need the same skill.

The case file records the grouping now used at Meghna and Sons.

Grouping by function builds skill and avoids repeated effort, but can separate departments from one another.

Grouping by product puts together every job needed for one kind of good.

Grouping by customer puts together every job needed to serve one kind of buyer.

The case file records the alternative grouping by customer type, with the sales of 2025.

A customer grouping lets each department learn what its buyers need, but it repeats skills.

Grouping by territory puts together every job in one place.

The table compares the four bases by what they group and

FOUR BASES OF DEPARTMENTALIZATION

BASIS	PUTS TOGETHER	STRENGTH	WEAKNESS
Function	Jobs needing the same skill	Builds skill, saves effort	Departments may drift apart
Product	Jobs for one good	Focus on each product	Repeated skills, higher cost
Customer	Jobs for one kind of buyer	Knows the buyer well	Repeated skills, needs more people
Territory	Jobs in one place	Responds to local needs	Branches may differ in rules

Most organisations mix the bases, and the right mix depends on the size and the work.

The recap fixes departmentalization and the four bases.

Few organisations use only one basis. A shop may group by function at the top and by customer inside the sales department. A large company may group by territory, and inside each territory by product. The test is the same in every case: does the grouping help the organisation reach its goals, with a reasonable cost? A grouping that was right for a shop with five employees may fail for a shop with fifty.

Before moving on. Departmentalization is grouping jobs into departments on a common basis. The four common bases are function, product, customer and territory. Each has strengths and weaknesses, and most organisations mix them.

* * *

13.3 FORCES BEHIND STRUCTURE

Managers do not choose a structure freely; several forces push them toward one design.

The bureaucratic model asks for fixed rules, clear levels and impersonal treatment.

The behavioural model asks for a structure that fits the needs and habits of people.

A manager who has studied spans and departments may expect a single best design to exist. It does not. Two organisations of similar size can use different structures and both succeed. The reason is that managers work under forces, and the forces differ from place to place. We now name the main forces, beginning with two old ideas about what a structure should be.

The first idea is the bureaucratic model. Max Weber, a German sociologist who lived from 1864 to 1920, described an organisation built on written rules, fixed duties, a clear chain of command from top to bottom, and promotion by ability.³ Its aim was fairness and reliability: the same rule applies to everyone, and the work continues when a person leaves. Its weakness is stiffness. An organisation that follows rules without exception may respond slowly to new situations.

The second idea is the behavioural model. Writers of this school noticed that employees are not machines. People work better when they feel valued, when they have some say in decisions, and when the groups they belong to are respected. A structure built on this idea gives employees wider tasks, more freedom and more contact with managers. This model is linked to the informal organization that we studied in the previous chapter. Its weakness is that

³ Weber described this model in writings that were published after his death. The word *bureaucracy* has a neutral meaning in management, although in daily speech it is often used as a complaint about slow offices.

work and the people at hand. That question leads to the forces that decide the answer.

These forces are called situational factors, which means features of an organisation's own circumstances. Four are important. The first is size. A shop of seven employees can work with a flat structure and few rules, but a company of seven thousand needs levels and written procedures. The second is the kind of work. Routine work, which repeats every day, suits firm rules. Work that changes often, such as design, needs more freedom.

The third factor is the world outside the organisation. If customers, rivals and laws change slowly, a fixed structure can last for years. If they change quickly, a structure must be able to change with them. The fourth factor is the people. Employees who are skilled and keen to take responsibility can work with a wide span and few rules. Employees who are new or less trained need more guidance and closer supervision.

Behind every structure lies an older idea. **Division of labour** means dividing a large job into smaller tasks done by different people. The Scottish thinker Adam Smith described it in 1776, in his book *The Wealth of Nations*, by the example of a pin factory.⁴ He observed that when pin making is divided into separate operations, each done by a different worker, the workers together produce far more pins than the same people working alone. Division of labour is the reason structures exist at all: once a job is divided, someone must group the tasks and link the people.

Division of labour has benefits and limits. Each worker becomes skilled in a narrow task, time is saved because no one changes from one task to another, and training is short. The limits are real. Workers who repeat a single small task all day may become bored, and a chain of many small tasks needs someone to join the results. That joining is called coordination, and the next part of this book teaches it.⁵

The link between the two ideas is simple to state. The more finely a job is divided, the more coordination is needed to bring the pieces together. A bookshop in which one person does everything needs no coordination at all. A bookshop with ten duties spread over seven people needs a clear arrangement of who tells whom what. Structure is the means by which an organisation carries out both the division and the coordination.

Before moving on. The bureaucratic model values rules and clear levels. The behavioural model values the needs of people. Situational factors such as size, the kind of work, the world outside and the people decide which suits best. Division of labour means splitting a large job into smaller tasks, and it creates the need for coordination.

* * *

Situational factors such as size, the kind of work, and the world outside affect the best structure.

The other factors are the world outside and the people themselves.

Division of labour. Dividing a large job into smaller tasks done by different people.

Division of labour is the oldest force: splitting a large job into smaller tasks.

Division of labour raises output, but it creates the need for coordination and can bore workers.

The more finely the work is divided, the more coordination is needed.

The recap fixes the two models, the situational factors, and division of labour.

⁴ Adam Smith (1723 to 1790) was a Scottish thinker and an early writer on economics. The pin factory is the best-known illustration in his book.

⁵ Coordination is taught in Chapter 14, 'Authority, Delegation and Coordination'. Here it is needed only in its everyday meaning: making separate parts work together.

The situation gives the spans, the present grouping and the alternative.

The question asks whether the spans are reasonable and whether the shop should change its grouping.

Reasoning begins with the spans and the factors that affect them.

Next, the sales figures show how large each customer department would be.

The weakness of customer grouping is repeated skills, which a small shop can ill afford.

The conclusion states what the records support.

The check tests the arithmetic and counts the reports.

The summary restates the chapter in a few sentences.

Situation. In 2026 Nurul Meghna has 2 direct reports and Rafiq Meghna has 5. The shop groups its jobs by function: sales, stock and accounts. The alternative is to group by customer type, using the 2025 sales: walk-in Tk 9,000,000; schools Tk 3,600,000; online Tk 1,800,000.

Question. Are the two spans reasonable for a shop of this size? Would a grouping by customer type be a better choice than the present grouping by function?

Reasoning. First, the spans. Rafiq's span of 5 is the larger. The factors from section 13.1 help us judge it. His employees work in one shop, so nearness is favourable. Their tasks, which are selling, the counter and stock handling, differ somewhat but are routine. The record gives no sign of difficulty. Nurul's span of 2 is small, but the owner has other duties, such as dealing with the world outside the shop, so a small span is reasonable. Both spans seem suitable.

Second, the grouping. A customer grouping would create three departments. We first find how large each customer type is, by dividing each customer's sales by the total of Tk 14,400,000. Walk-in sales are Tk 9,000,000, which is 62.5% of the total. School sales are Tk 3,600,000, which is 25%. Online sales are Tk 1,800,000, which is 12.5%. A walk-in department would be the main part of the shop. An online department would serve only one-eighth of the sales, yet it would still need its own skills in stock handling and accounts.

With only 7 employees, three customer departments would leave very few people in each. Each would still need stock handling and accounts, so the work would be repeated, and the cost per sale would rise. The present grouping by function avoids repetition: one stock group and one accounts group serve all three kinds of customer. Its weakness, as section 13.2 explained, is that sales and stock may drift apart, and the shop's small size makes that risk low.

Conclusion. The two spans appear suitable, and the present grouping by function is reasonable for a shop of this size. A customer grouping becomes more attractive if school or online sales grow so that a customer department could keep several people busy.

Check. We test the numbers. The three sales figures add to Tk 9,000,000 + Tk 3,600,000 + Tk 1,800,000 = Tk 14,400,000, which equals the total sales of 2025. The three shares add to 62.5% + 25% + 12.5% = 100%. The spans add to 2 + 5 = 7, which equals the 7 employees on the staff list, so every employee reports to exactly one manager in our reading of the record.

* * *

SUMMARY

The span of management is the number of employees who report directly to one manager, and the suitable span depends on factors such as skill, similarity of tasks, clarity of plans, nearness and the

THINK IT THROUGH

Each question below changes the situation a little more than the one before. Answer them in order.

Four situations test the chapter, from familiar to combined.

13.T1 A bookshop owner has 12 employees who all serve customers at the counter. She asks them all to report to her. Give one reason this span may be too wide and one reason it may be acceptable.

13.T2 The shop places a new manager between the owner and the 12 employees, and each of two managers leads 6. Has the structure become taller or flatter? Name one benefit and one cost.

13.T3 A company runs three shops, one in each of three cities, and gives each city its own manager. Name the basis of grouping. Then say what would change if the company grouped by product instead.

13.T4 A small clinic has two doctors, three nurses and one receptionist. Suggest a span for the senior doctor and a basis of grouping, and give your reasons using at least two situational factors.

WHERE PEOPLE GO WRONG

Some students count every person below a manager as that manager's span. The span includes only those who report directly to the manager. A manager with five employees, each of whom leads others, has a span of five.

Mistake one: counting all the people below a manager as the span.

Some students believe that a flat structure is always better because it is faster. A flat structure can overload its managers. The correction depends on skill and on the work: skilled employees with clear rules can use a wide span, but untrained employees cannot.

Mistake two: believing that a flat structure is always better.

Some students believe that one basis of departmentalization is correct for all organisations. Function, product, customer and territory each suit different circumstances, and most organisations use a mixture.

Mistake three: believing that one basis of grouping is correct.

Some students praise division of labour without noticing its cost. Every division of a job creates a need to join the parts again, and a poor structure can lose the benefit of dividing the work.

Mistake four: forgetting that division of labour needs coordination.

IN HISTORY

General Motors, the American car maker, was formed in 1908 and grew by joining together several car companies. By 1920 it had many makes, but its structure was poorly organised, and some of its cars competed with one another. Ford's Model T was then the leading car in the United States. In 1923 Alfred P. Sloan became president of General Motors. He had joined the company as a director and vice-president in 1918.

Alfred Sloan reorganised General Motors into divisions in the 1920s.

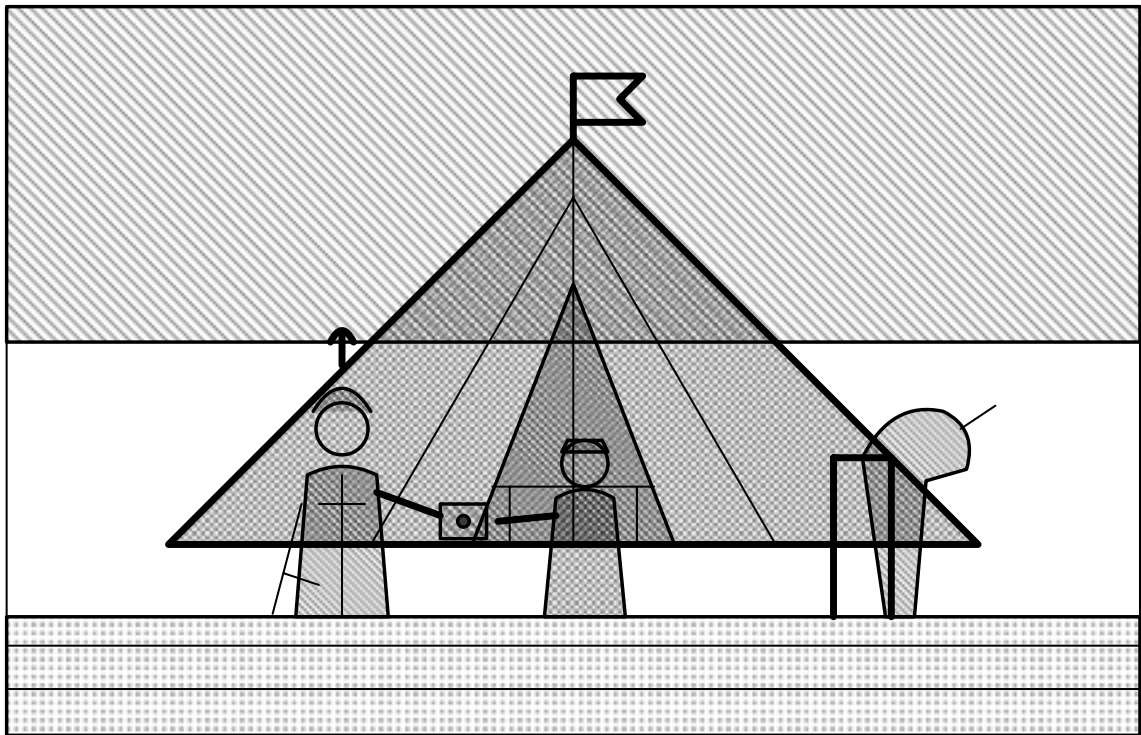
Sloan arranged the company in divisions, one for each car brand: Chevrolet, Pontiac, Oldsmobile, Buick and Cadillac. The brands were planned in a graded range of prices, from the lowest to the

Sloan grouped the company into divisions, each for a car brand, under central control.

The central office shows that grouping by product does not remove the need for coordination. Writers still debate how far central control should reach, and the debate has no final answer. The design became a pattern for many large companies in later years.

REVIEW QUESTIONS

- 13.1 A manager has 4 employees who report to him directly, and each of them leads 3 others. State his span and the total number of people below him.
- 13.2 Explain why a store manager who must also keep accounts should have a smaller span than one who has no other duties.
- 13.3 Name the four bases of departmentalization and match each to an example from a bookshop.
- 13.4 Explain, in your own words, the difference between a tall structure and a flat structure, and give one cost of each.



A general hands a sealed order to a messenger

CHAPTER XIV

Authority, Delegation and Coordination

Why does one person obey an order, and another only a request?



This chapter explains how the right to direct work moves through an organisation. It separates authority from power, shows the three kinds of authority, and teaches how a manager delegates a task, how decisions are spread or kept at the top, and how coordination holds separate work together. A small bookshop and a Brazilian factory serve as cases.

You will need: Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is'; Chapter 13, 'Structure: Span, Departments and Forces', section 13.1 'Span of Management'.

A sealed order raises the question of why anyone obeys it.

Two different reasons for obeying are easy to confuse.

The same puzzle appears in a bookshop on an ordinary morning.

The chapter has five parts and then applies them to one shop.

A general sits in his tent before dawn. He writes an order, seals it and hands it to a young messenger. The messenger bows and rides into the dark. Later that day, a thousand soldiers will move because of the paper in his pocket. None of them has met the general, and none of them has seen the paper. Why do they obey?

Perhaps they obey because they fear the general. Perhaps they obey because they respect him, or because the order comes from the office he holds. These are different reasons, and the difference matters. A soldier who obeys out of respect would stop obeying if the respect faded. A soldier who obeys the office would obey whoever held it.

The same puzzle appears in a bookshop. A store manager asks the cashier to count the till at six o'clock, and the cashier does it. A friendly colleague asks the cashier for a favour, and the cashier does that too. Both requests succeed, yet the reasons are not the same. One request rests on the manager's position. The other rests on liking and trust.

This chapter separates these reasons and gives each a name. It shows how the right to direct work is shared among managers, advisers and specialists. It explains how a manager hands a task to another person, how much decision-making stays at the top, and how an organisation keeps separate work moving toward one goal. We follow a Dhaka bookshop and, at the end, a Brazilian factory that went further than most.

* * *

14.1 AUTHORITY AND POWER

GO SLOWLY HERE.

Authority and power are easily confused, so we take them one at a time.

A traffic officer's raised hand stops cars because of the position, not the person.

Recall: Chapter 12, 'Organizing and Types of Organization', section 12.1 'What Organizing Is' — organizing.

Authority is a right that comes with a position and not with the person.

Authority. The right to give orders and to expect them to be obeyed, given by position.

Many students find these two words confusing, because in daily speech they mean nearly the same thing. In management they are different ideas, and a mix-up leads to wrong conclusions. We take authority first, then power, then compare them. Go slowly, and use the examples to test each idea.

Begin with a street in Dhaka. A traffic officer in uniform raises one hand, and the cars stop. The officer is not stronger than the drivers, and the drivers may not know the officer's name. The cars stop because of the officer's position. The uniform shows that the position exists, and the law supports it.

The traffic officer's right has a name. **Authority** means the right to give orders and to expect them to be obeyed, given by position. When a store manager tells a cashier to count the till, the manager uses authority. The shop owner gave the position its authority, and the cashier accepts that it belongs to the position. Chapter 12 showed that organizing creates positions. Authority is what makes a position able to direct other positions.

Authority has a size. A store manager has more authority than a sales assistant, and an owner has more than a store manager. It has a source, which is the organisation. It also has a limit: a store manager may direct the shop's work but may not order a customer to buy. Authority reaches only as far as the organisation has given it.

Here is a simple test. If the manager leaves and a new manager is appointed, who holds the authority? The new manager does. Authority stays with the post and passes to each new holder. This is why we say it is given by position.

Now consider the friendly colleague. The colleague holds no higher position than the cashier, yet the cashier does the favour. **Power** means the ability to influence other people's behaviour, whether or not one holds a position. The colleague has power because the cashier likes and trusts the colleague. If the colleague moved to another shop, the power would move too, since it belongs to the person.

Where does power come from? Writers on management name five common sources. The first is position, which is authority itself. The second is the ability to give rewards, such as a bonus or a better shift. The third is the ability to punish, such as a warning. The fourth is knowledge and skill: an employee who knows the stock system better than anyone else is listened to. The fifth is personal liking and respect. Chapter 17, 'Leadership', section 17.1 'Leader and Manager', studies the power of leaders in more detail.

The table below compares authority and power. Read each row from the left. The first cell gives a question, the second answers it for authority, and the third answers it for power.

AUTHORITY COMPARED WITH POWER

QUESTION	AUTHORITY	POWER
Where does it come from?	The position, given by the organisation	Knowledge, rewards, punishment, liking or position
Who holds it?	Whoever holds the post	A person, with or without a post
What happens when the holder leaves?	It stays with the post	It leaves with the person
Example	A manager orders a till count	A trusted colleague asks for a favour

A manager may hold both, and the strongest managers do. A manager may also hold one without the other. A newly appointed manager has full authority and may have little power, because the employees do not yet know or trust the new person. A senior employee may have much power and no authority. Employees often follow such a person, and a wise manager notices it.

Authority has a limit that is easy to miss. A right to give orders means little if the orders are not obeyed. In practice, authority

Authority can be small or large, but it always belongs to a post.

The test of authority is whether it would pass to a new holder of the post.

Power is a different idea: the ability to influence, with or without a position.

Power. The ability to influence other people's behaviour, whether or not one holds a position.

Power. Word in use. In daily life, physical strength, or electricity ("a power cut"). In management, the ability to influence what other people do.

Power has several sources that managers can name.

The two ideas are compared in a table with four questions.

A manager usually has both, and the two can work apart.

Authority works only when the people below accept it.

The recap of section 14.1.

A sentence to test the reader's grasp of the section.

A concrete picture of a shop shows why one kind of authority is not enough.

Line authority runs down the chain of command from the owner to each employee.

Line authority. The right of a manager to direct the work of employees in the chain of command.

Staff authority is the right to advise, not to command the line's employees.

Staff authority. The right to advise and support line managers, without commanding their employees.

The case file gives one example of staff authority in the shop.

Tariq's advice shows staff authority at work.

Functional authority is a narrow right to direct others in one specialised subject.

Before moving on. Authority is the right to give orders and expect them to be obeyed, and it belongs to the position. Power is the ability to influence others, and it belongs to the person. A manager may have both, or one without the other. Authority works best when the people below accept it.

You should now be able to say: authority comes from the position and stays when the holder leaves, and power comes from the person and leaves with the person.

* * *

14.2 LINE, STAFF AND FUNCTIONAL AUTHORITY

Picture a small shop with an owner, a store manager and a few employees. The owner cannot watch every employee, so the store manager does it. The owner also needs someone to advise on money, and that adviser does not run the shop floor. A second kind of authority is therefore needed. Management names three kinds, and we take them in order.

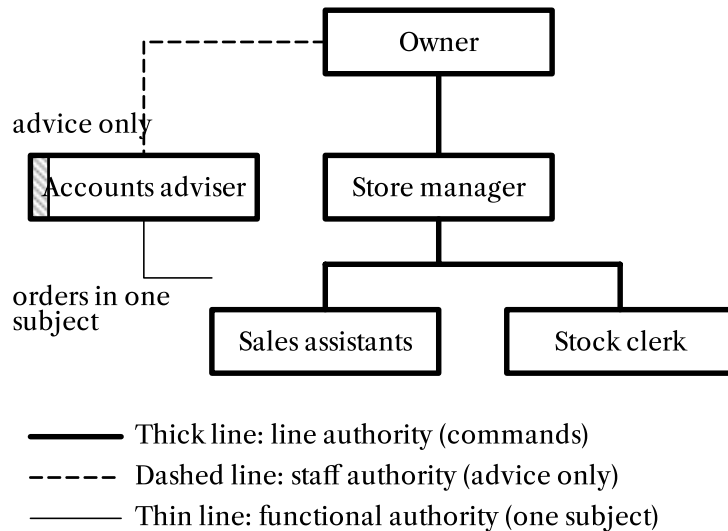
The first kind is **line authority**, which means the right of a manager to direct the work of employees in the chain of command. The chain of command is the line of positions from the top of the organisation to the bottom, along which orders travel. In a bookshop, the store manager directs the sales assistants, and an assistant answers to the store manager. Some books call these people subordinates, and this book calls them employees. Line authority is the backbone of every organisation.

The second kind is **staff authority**, which means the right to advise and support line managers, without commanding their employees. A staff person is a specialist who helps the line managers. Think of an adviser on money or law. The adviser studies a problem, gives an opinion and may be consulted before a decision. The line manager still decides, and the adviser's opinion carries weight only through its quality.

Case file, illustrative. Delegation from 1 March 2026, Meghna and Sons: Tariq holds staff authority. He advises on cash and credit.

The case file shows the idea. Tariq is responsible for accounts and online orders, and he advises on cash and credit. He may tell the store manager that a school's unpaid bill is large and that further credit is risky. He does not order the sales assistants to refuse the school. The line manager hears the advice and decides.

A third kind sits between the first two. We may call it functional authority. It means the right of a staff specialist to direct other people's work, but only in one specialised subject. For example, a specialist in safety may order every department to lock the store-room each night. The specialist commands in that one subject and in no other. Functional authority is used where a rule must be the same across the organisation.



The figure shows the three kinds of authority as three kinds of line.

FIG. 14.1. *Line, staff and functional authority*

Read the figure from the owner at the top downwards. The thick lines carry line authority down the chain of command. The dashed line carries staff authority, which is advice only, and it reaches the line manager but not the employees below. The thin line carries functional authority, which is a right to direct in one subject. The key at the bottom repeats these meanings in words, so the figure reads the same in grey.

The figure is read from the owner at the top, with the line types in a key below.

The three kinds can cause trouble. A staff person who begins to give orders to employees weakens the line manager, and the employee does not know whom to obey. An employee who receives two different orders is in a hard position. The remedy is to say plainly which kind of authority each person holds and in which matters. Where authority is stated in writing, such conflicts are fewer.

The three kinds cause trouble when they are not kept apart.

Before moving on. Line authority is the right to direct employees along the chain of command. Staff authority is the right to advise and support line managers without commanding their employees. Functional authority is a staff specialist's right to direct others in one subject only. In the case file, Tariq holds staff authority.

The recap of section 14.2.

You should now be able to say: line authority commands, staff authority advises, and functional authority commands in one subject only.

A sentence to test the reader's grasp of the section.

* * *

14.3 DELEGATION

Suppose Nurul, the owner of a bookshop, personally approved every return and every purchase. Each task would wait for Nurul to be free, and customers and suppliers would wait too. A manager has a limit on time and attention. Some tasks must therefore be handed to others, and the handing must be done with care.

GO SLOWLY HERE.

A manager cannot do all the work personally, so tasks must be handed on.

The word for this is **delegation**, which means giving an employee the right to make decisions and take action for a stated

Delegation hands a stated task to an employee together with the right to act.

The case file shows two delegated tasks, each with a money limit.

Two delegated tasks each carry a money limit.

Managers often fail to delegate, and the reasons are common.

Responsibility, authority and accountability must be kept in balance.

Accountability. The duty to explain and answer for results to the person who delegated the work.

A manager who delegates stays answerable to the person above.

Case file, illustrative. Delegation from 1 March 2026, Meghna and Sons: Rafiq may approve returns up to Tk 5,000, above that Nurul. Tariq may place purchase orders up to Tk 50,000.

The case file shows delegation in two forms. The task for Rafiq is approving returns, and the limit is Tk 5,000; above that, Nurul decides. The task for Tariq is placing purchase orders, and the limit is Tk 50,000. A limit in money keeps the manager's risk small, and it tells everyone where the right to decide ends.

Many managers find delegation hard. Some fear that the employee will make mistakes. Some believe they do the work better themselves. Some fear losing control or being needed less. Employees can also resist, because they fear blame or lack the skill. The cure is to match the task to the person, to state the limits plainly and to accept that early mistakes are part of learning.

Three ideas go together in delegation. Responsibility is the duty to do the task. Authority, as we saw, is the right to act so that the task can be done. **Accountability** means the duty to explain and answer for results to the person who delegated the work. Rafiq has the responsibility to handle returns, the authority to approve them up to Tk 5,000, and accountability to Nurul for how well he does it. If a person is given a duty without the matching authority, the task cannot be done.

One point surprises many students. A manager may hand over the task and the authority to do it. The manager still answers to the person above for the final result. Nurul, as owner, remains answerable for the shop as a whole. In this way delegation spreads the work but does not remove the answer owed to the owner or to the customer.

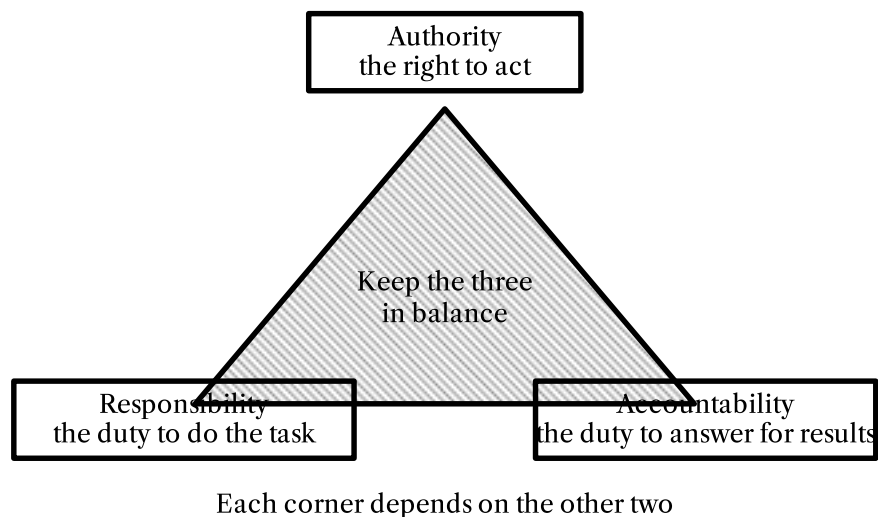


FIG. 14.2. *Authority, responsibility and accountability triangle*

The triangle is read corner by corner, with the centre giving the rule.

Read the triangle one corner at a time. Authority is the right to act, responsibility is the duty to do the task, and accountability is the duty to answer for the result. The sides join the corners because each depends on the others. The words in the centre give the rule: keep the three in balance.

PAUSE AND CHECK 14.P2 Rafiq is given the duty to approve returns

answer for results to the person who delegated the work. Responsibility, authority and accountability must be kept in balance. A manager who delegates still answers upward for the final result.

You should now be able to say: delegation hands over a stated task and the right to act, and accountability stays with the person who must answer for the result.

* * *

14.4 CENTRALIZATION AND DECENTRALIZATION

Delegation raises a larger question. Across the whole organisation, how many decisions should the top keep, and how many should it hand down? The answer is a matter of degree, and it ranges from almost all at the top to almost none.

Centralization means keeping most decision-making power at the top of the organisation. In a very centralized shop, the owner approves every return and every order. The opposite is decentralization, which means spreading decision-making power to lower levels. It is the sum of many acts of delegation. The two ideas mark the ends of a line, and real organisations sit between them.

Which should an organisation choose? Several things point the way. A large organisation cannot decide everything at the top and tends to spread decisions downward. Skilled and trusted employees make decentralization safer. Where speed matters, a decision made close to the customer is faster. Where a mistake is very costly, or where all parts must act in the same way, the top may wish to keep control. Tall structures, with many levels, slow decisions that must climb to the top.

Before moving on. Centralization keeps most decision-making power at the top, and decentralization spreads it downward. Neither is always best. Size, skill, speed and the cost of mistakes guide the choice.

* * *

14.5 COORDINATION

Division of work brings a problem. When a task is split among departments, each does its part, and the parts must still fit together. A shop whose buyer orders books the sales assistants cannot sell, or whose sales assistants promise books that the stock clerk has not received, is a shop whose parts do not fit.

The remedy is **coordination**, which means linking the work of different people and departments so that they pull toward the same goal. Coordination is not a separate task done once. It is a quality that good management gives to all the other tasks. The need grows with the number of people, since each added person adds more links to keep in order.

Two groups of methods help. The basic activities are shared

A sentence to test the reader's grasp of the section.

Every organisation must decide how many decisions to keep at the top.

Centralization keeps most decision-making power at the top.

Centralization. Keeping most decision-making power at the top of the organisation.

Which to choose depends on size, skill, speed and the cost of mistakes.

The recap of section 14.4.

When work is divided among many people, something must join it again.

Coordination links the work of separate people toward one goal.

Coordination. Linking the work of different people and departments so that they pull toward the same goal.

Basic activities and

minutes, or 26 hours. The case file does not list who attends. The cost is small, and the meeting lets the store manager, the accounts adviser and others share news before the week begins.

The recap of section 14.5.

Before moving on. Coordination links the work of different people and departments so that they pull toward the same goal. Basic activities include shared goals, communication and plans. Structural techniques include the chain of command, rules, liaison persons and meetings. The weekly meeting in the case file is one example.

* * *

WORKED CASE: WHO MAY DECIDE?

The worked case applies authority, delegation and accountability to the shop's limits.

We now apply the chapter to the decisions in the case file. The worked case has five parts: Situation, Question, Reasoning, Conclusion and Check. It applies the ideas to a situation. It is an application, not a model answer.

The situation lists the limits set from 1 March 2026.

Case file, illustrative. Delegation from 1 March 2026, Meghna and Sons: Rafiq may approve returns up to Tk 5,000, above that Nurul; Tariq may place purchase orders up to Tk 50,000; weekly meeting each Sunday at 9:00 for 30 minutes. Tariq holds staff authority and advises on cash and credit.

The question asks who may decide in four small situations.

Question. Suppose four things happen in one week: a return of Tk 4,800; a return of Tk 6,500; a purchase order of Tk 45,000; and a request for credit from a school. Who may decide each one, and what does each decision show about authority?

The first test compares each amount with the limit that was delegated.

Reasoning, part 1: returns. The limit for Rafiq is Tk 5,000. The first return is Tk 4,800, and 4,800 is less than 5,000, so Rafiq may approve it. The second return is Tk 6,500, and 6,500 is more than 5,000, so it goes to Nurul. The words "up to Tk 5,000" mean that a return of exactly Tk 5,000 is also within Rafiq's right.

The second test applies the same method to the purchase order.

Reasoning, part 2: the order. The limit for Tariq is Tk 50,000. The order is Tk 45,000, and 45,000 is less than 50,000. Tariq may place it. A further order of Tk 62,000 would exceed the limit by $62,000 - 50,000 = \text{Tk } 12,000$. The case file does not say who approves above the limit, so the shop should write this down. In an owner-managed shop, the owner is the natural choice, but the file does not state it.

The third test applies staff authority to the credit request.

Reasoning, part 3: the credit request. Tariq holds staff authority and advises on cash and credit. He may tell the store manager how large the school's unpaid bill is and whether credit is wise. He may not order the sales assistants to refuse. The line decision stays with the line manager, and Tariq's advice informs it.

The fourth test names who answers for each result.

Reasoning, part 4: accountability. Rafiq answers to Nurul for the returns he approves, because Nurul delegated that right. Tariq answers to Nurul for the orders he places. Nurul remains answerable to the shop's owners and customers for the whole. The Sunday meeting gives each person a place to explain results and to coordinate.

The conclusion and the check agree with the limits in the

Conclusion. Rafiq decides the return of Tk 4,800, Nurul decides the return of Tk 6,500, and Tariq may place the order of Tk

CHANGING THE SITUATION

The same ideas apply in new situations. Rung 1 is the case above. In rung 2, suppose a return is worth exactly Tk 5,000 and Rafiq is absent. Rafiq's right does not pass to anyone else, so the return waits for him, or goes to Nurul. This shows that authority belongs to the post and to its holder, and it does not pass without a decision.

In rung 3, a school head delegates the booking of class visits to a teacher, with a limit of 20,000 taka for each visit. The teacher has the task, the right to book up to the limit, and the duty to report to the head. The three parts match, and the head still answers to the school's governors.

In rung 4, an owner has 12 employees and delegates to three team leaders. Chapter 13 explained that the span of management is the number of employees a manager can direct well. With three leaders, each has a span of four employees, which is easy to manage. The owner holds a weekly meeting of the leaders to coordinate the three teams.

* * *

Rung 2 reshapes the case by changing the amount.

Rung 3 moves the idea to a new setting.

Rung 4 combines delegation with the ideas of the span of management and coordination.

SUMMARY

Authority is the right to give orders and to expect them to be obeyed, given by position. Power is the ability to influence others, with or without a position. Line authority directs work along the chain of command, and staff authority advises without commanding. Delegation gives an employee the right to decide and act for a stated task, and accountability is the duty to answer for results to the person who delegated. Centralization keeps decisions at the top, and coordination links separate work toward one goal.

A short summary of the chapter's eight defined terms.

THINK IT THROUGH

14.T1 Rafiq is away on a day when a customer returns books worth Tk 5,400. Explain who may approve the return under the case file, what Rafiq's authority does and does not cover, and one way Nurul could change the arrangement so that returns do not wait.

14.T2 Tariq has been advising Nurul to refuse credit to a school. A sales assistant says Tariq has told her to refuse the order herself. Which kind of authority has Tariq used, and is it within his right? Explain your reasoning.

WHERE PEOPLE GO WRONG

Mistake: "Authority and power are the same thing." Authority comes from a position and stays with the post. Power comes from the person and leaves with the person. A manager may have one without the other.

Mistake: "If I delegate, the result is no longer my concern." The employee is accountable to the manager who delegated. The man-

Mistake one: believing that authority and power are the same.

Mistake two: believing that delegating gives away

Mistake five: treating coordination as one meeting.

limits in the case file show. Control remains through limits, reports and meetings.

Mistake: "Coordination means holding a meeting." A meeting is one structural technique. Shared goals, communication, plans, rules and the chain of command also coordinate.

Ricardo Semler's changes at Semco show decentralization pushed far.

The case is famous, and it shows both the gain and the demands of the approach.

In the early 1980s, Ricardo Semler took over Semco, a Brazilian manufacturer of marine pumps and other industrial equipment, which had been founded by his father. Semco was in difficulty, and Semler changed how it was run. He gave workers wide rights to decide. Factory workers set their own production schedules and working hours, and elected committees took part in decisions. Some staff set their own pay, and the company shared its financial figures with employees.

Semler described these changes in a book, *Maverick*, published in 1993. The case became well known as an example of decentralization carried unusually far. It shows that spreading decisions can work when employees are trusted and well informed. It also shows how much the approach demands: information must be shared, limits must be understood, and managers must accept giving up control.

IN HISTORY

REVIEW QUESTIONS

14.1 Define authority and power in your own words, and state one difference between them.

14.2 Sort each example as line authority, staff authority or functional authority: (a) a store manager tells a sales assistant to restock a shelf; (b) an accounts adviser tells the store manager that a customer's credit is too high; (c) a safety specialist orders every department to lock its storeroom at night.

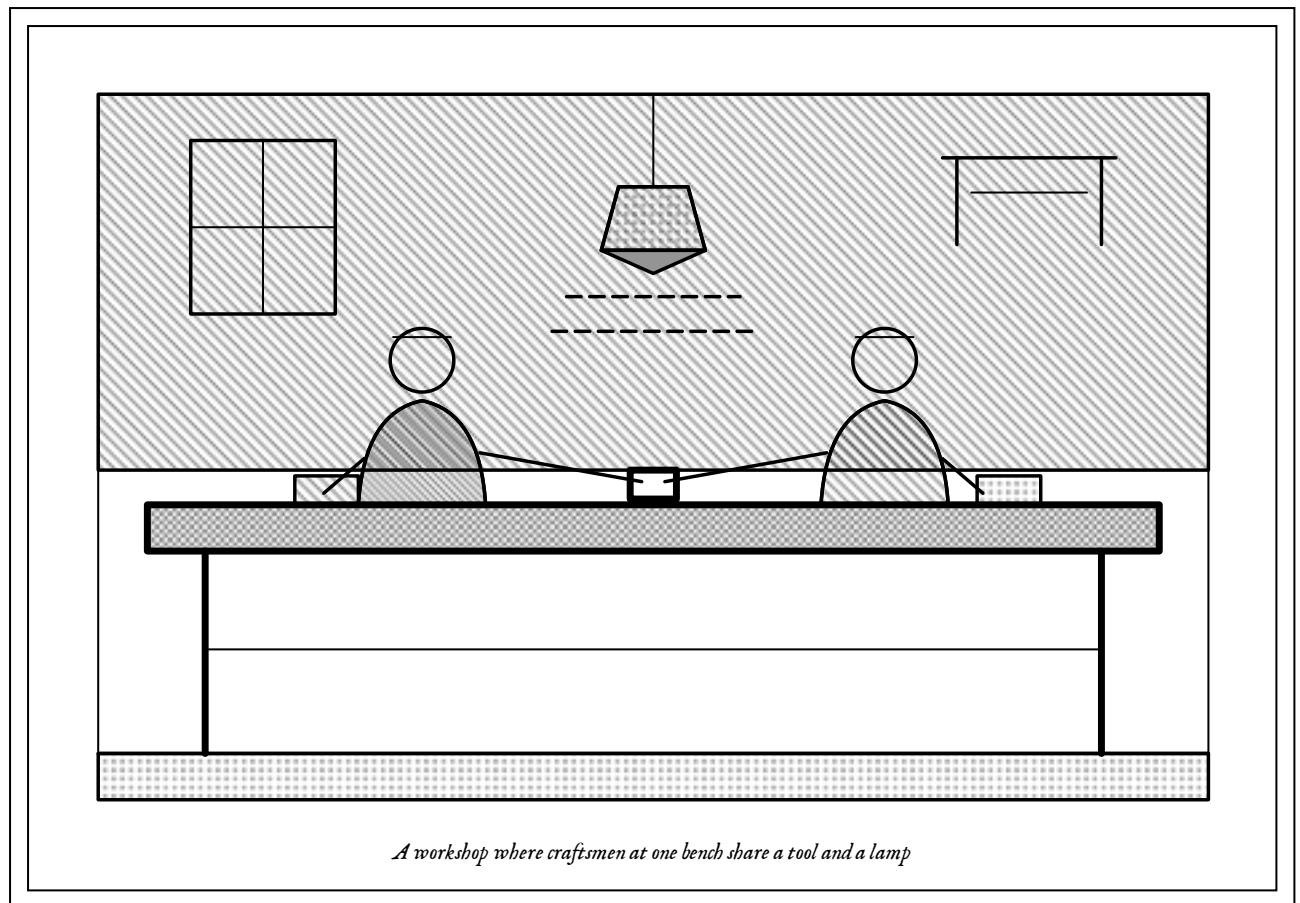
14.3 Define delegation and name the three ideas that must be kept in balance when a task is delegated.

14.4 Explain why accountability cannot be delegated away, using the shop's returns as an example.

14.5 Name two reasons why a manager may fail to delegate, and give one remedy for each.

14.6 Define centralization, and name two conditions that favour decentralization.

14.7 Define coordination. Name two basic activities and two structural techniques that support it.



CHAPTER XV

Human Factors, Groups and Creativity

If two people have the same tools, why does the work differ?



Machines behave the same way each day, but people do not. This chapter explains the human factors that shape work, shows how groups and teams form and set their own rules, and ends with the ways in which new ideas are born and put to use.

Two ideas from earlier chapters support this chapter.

A workshop shows two craftsmen with equal tools who still work in different ways.

The puzzle is that equal tools and equal pay do not give equal work.

The chapter studies the human side of work, from the single person to the group to the new idea.

A shop where two assistants react differently to the same instruction shows that people differ.

Human factors. The traits, feelings and relationships of people that affect how they work.

Human factors are the traits, feelings and relationships that affect how people work.

You will need two ideas from earlier chapters. The first is the informal organization, taught in Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.¹ The second is leading, one of the functions in the management process, taught in Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.² Read those sections again if either idea feels uncertain.

Picture a workshop of the nineteenth century. Two craftsmen stand at one long bench. They share a single lamp and pass one heavy tool between them. Both were trained by the same master, and both are paid the same wage. Yet by the end of each day, one has finished six pieces with great care, and the other has finished four, with more mistakes. The master, who watches them, does not blame the bench or the lamp. He knows that the difference lies in the two people.

This is the puzzle of the chapter. A manager can buy the same machine for every worker and arrange the same lamp for every bench, but the results will still differ, because people differ. A machine does not feel proud or tired. It does not ask what its neighbour is paid, and it does not form friendships. People do all of these things, and each of them changes the work. A manager who ignores this fact will find that careful plans fail without any visible cause.

In this chapter we study the human side of work in three steps. First we examine the traits and feelings that every person brings to a job. Then we study how people join into groups, and how those groups set rules that no one has written. Finally we look at the way new ideas arise and are put to use. We shall test each idea against the records of the small Dhaka bookshop Meghna and Sons.

* * *

15.1 HUMAN FACTORS IN MANAGEMENT

Suppose a store manager tells two sales assistants to place a new pile of books near the door. The first assistant finishes the task at once and asks whether anything else is needed. The second assistant does the task slowly and complains that the pile will block the customers. The instruction was the same, the books were the same, and the door was the same. The manager is left with the question that this section answers: what is it about people that makes the same instruction produce different results?

Management gives a name to the answer. **Human factors** means the traits, feelings and relationships of people that affect how they work. A trait is a lasting feature of a person, such as patience or shyness. A feeling is a state of mind at a given time, such as worry or pride. A relationship is a link between two people, such as friendship or rivalry. When the first assistant works quickly because

¹ Recall: informal organization, Chapter 12, 'Organizing and Types of Organization', section 12.3 'Formal and Informal Organization'.

² Recall: leading, Chapter 3, 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

belongs to the organisation, but it is felt first in the groups and then by each person.

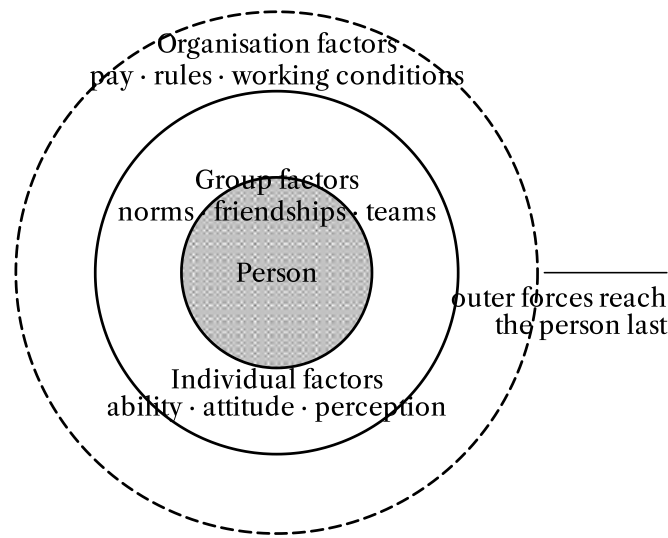


FIG. 15.1. *Individual, group and organisation factors as nested rings*

The innermost ring contains the individual factors, and these are the subject of the rest of this section. The group factors are studied in section 15.2. Organisation factors, such as pay, rules and working conditions, are treated in later chapters on motivation and leadership. We begin with the person because every group is made of persons, and a manager who understands one person at a time is better prepared to understand a group.

The first individual factor is the set of individual differences, which means the ways in which people vary from one another. People differ in ability: one can add a column of numbers quickly, and another can calm an angry customer. They differ in attitude, which is a person's settled way of judging something as good or bad. They differ in needs, in experience and in background. No two employees are alike, and a rule that suits one employee may confuse another.

Individual differences have a limit that managers must respect. They help to explain why people act differently, but they must not be used to label a person unfairly. A manager who decides, after one slow morning, that an employee is lazy has drawn a conclusion from too little evidence. The correction is to observe over time, to ask the employee, and to look for causes outside the person, such as unclear instructions or a broken tool.

The second individual factor is perception. Perception means the way a person selects, arranges and understands what he or she sees and hears. Two people can look at the same event and report different things, because each notices some parts and ignores others. The first assistant saw a pile of books as a simple task. The second assistant saw a pile that might block customers. Each assistant described what was real to him or her.

Perception depends on a person's experience, mood and

Individual factors come first, because every group is made of single persons.

Individual differences are the ways in which people vary in ability, attitude and background.

Individual differences have limits: they explain much, but they must not be used to label people.

Perception is the way a person selects and understands what he or she sees and hears.

Perception is shaped by experience, mood and

³ Many students find perception difficult to separate from opinion. A useful test is to ask: what did I observe, and what did I add from my own mind? Only the first is fact.

From four viewpoints, human factors matter to everyone in the shop.

PAUSE AND CHECK 15.P1 An employee arrives late twice in one week. The manager decides the employee does not care about the job. Name one piece of observed fact and one piece that the manager may have added from his own mind.

Human factors look different from each viewpoint. The owner sees them as a source of both cost and strength: good relationships keep employees, and poor ones send them away. The manager sees them as the main material of leading. The employee sees them as the feeling of being treated fairly and understood. The customer meets them in every conversation, because the mood of an employee reaches the buyer in a few seconds.

IN SHORT

Human factors are the traits, feelings and relationships that affect how people work. They act on three levels: the person, the group and the organisation. Individual differences and perception explain why one instruction can produce different results.

The recap fixes human factors, individual differences and perception.

Before moving on. Human factors are the traits, feelings and relationships of people that affect how they work. They act at the levels of the person, the group and the organisation. Individual differences show how people vary, and perception shows how each person understands the same event in a different way.

* * *

15.2 GROUPS AND TEAMS

People in a workshop quickly form small circles, and those circles shape the work.

Return to the workshop of the opening. After a few weeks, the two craftsmen at the bench find that they eat their midday meal together and compare their tools. A third craftsman joins them, then a fourth. Soon the four share advice, pass along rumours and help one another when the master is away. No one planned this circle. It formed because people who work near each other tend to join together. This is a group, and it is the second level of human factors.

A group is two or more people who interact, depend on each other and see themselves as members.

A group is two or more people who interact with each other, depend on each other in some way, and see themselves as belonging together. Passengers who wait at a bus stop are not a group, because they do not interact or depend on one another. The four craftsmen are a group. They talk, they rely on each other, and each of them knows that he belongs. The informal organization, studied in the previous chapter, is made of such groups.

People form groups for security, belonging, status, help with tasks and shared interests.

Why do people form groups? Several reasons are common. A group offers security, since a person in a group feels less alone when something goes wrong. It satisfies the wish to belong, which most people feel. It can give status, since membership of a respected group raises a person's standing. It gives help with the task, because members share knowledge and skill. Finally, people with a shared interest, such as a love of books, tend to join together. A manager who understands these reasons will not treat every group

Not every group is a team. A **team** means a small group of people with joined skills, working toward a common goal for which they are all accountable. To be accountable is to be answerable for the result. Three features separate a team from an ordinary group. The members have different skills that fit together. They share one goal. They are answerable together for whether the goal is reached. A stock clerk, a buyer and a cashier who work together to prepare a school order form a team. The four craftsmen eating lunch are a group but not a team.

The table below sets three kinds of gatherings side by side. Each row names a feature, and the columns show how the feature appears in a crowd, a group and a team. Read across a row to compare the three on one feature.

CROWD, GROUP AND TEAM COMPARED

FEATURE	CROWD	GROUP	TEAM
Interaction	None or little	Regular	Close and planned
Goal	None shared	Often personal	One common goal
Accountability	None	Each for self	All for the result
Example	Passengers at a bus stop	Colleagues at lunch	Staff preparing a school order

Every group, whether it is a team or not, develops rules of behaviour. The rules are rarely written, yet members know them well. A newcomer to the workshop learns, without being told, that one does not leave a mess on the common bench and that one does not criticise a colleague in front of the master. Such rules have a name in management.

A **group norm** means an unwritten rule that members of a group follow and expect others to follow. A norm may say how fast to work, how to speak to customers, or when to take a break. A norm is shown by what happens to someone who breaks it. The group may laugh, give a look, or leave the person out of conversation. Norms hold a group together, and they are an important part of the informal organization.

Norms can serve the organisation or oppose it. A norm that says *we help new employees learn the shelves* serves the shop. A norm that says *no one should work faster than the slowest member* works against it, because it limits what the group produces. Managers cannot abolish norms, but they can influence them. They can reward helpful norms, discuss harmful norms openly, and set an example in their own behaviour.

PAUSE AND CHECK 15.P2 The three sales assistants of a shop have an unwritten rule that no one sells a book to a customer who is already talking with another assistant. Is this a group norm? Say

Team. Word in use. In daily life, the players on one side in a sport. In management, a small group of people with joined skills who work toward a common goal.

Team. A small group of people with joined skills, working toward a common goal for which they are all accountable.

A team is a small group with joined skills, a common goal and shared accountability.

A table compares a crowd, a group and a team.

Groups develop rules that nobody writes, and these rules are called norms.

Norm. Word in use. In daily life, what is normal or average. In management, an unwritten rule of behaviour that a group expects its members to follow.

Group norm. An unwritten rule that members of a group follow and expect others to follow.

A group norm is an unwritten rule that members follow and expect others to follow.

Norms can help the organisation or work against it.

* * *

15.3 CREATIVITY AND INNOVATION

Most organisations are full of ideas, yet few ideas ever reach the customer.

Creativity. The ability to produce new and useful ideas.

Creativity is the ability to produce ideas that are both new and useful.

Creativity comes from knowledge, curiosity and freedom, and not only from special talent.

Innovation. Turning a new idea into a new product, service or method that is put to use.

Innovation is turning a new idea into something that is put to use.

Many creative ideas are not adopted, and a wise organisation chooses among them.

Encouraging ideas needs listening, time, safety to fail, and a way to turn ideas into action.

Some habits of managers stop ideas before they start.

Walk through any shop and ask the employees for ideas. Within an hour you will hear a dozen: a new way to arrange the shelves, a quicker way to check returns, a chair for readers near the window. Now ask a different question: how many of these ideas will ever be tried? In most shops the answer is very few. Between having an idea and using it there lies a gap, and the gap has a cost. This section explains both sides of it.

The first side is **creativity**. Creativity means the ability to produce new and useful ideas. Both words in the definition matter. An idea that is new but of no use, such as selling books by weight, is only a curiosity. An idea that is useful but not new is only a copy. A sales assistant who thinks of a quiet corner where customers can try a book before buying has produced an idea that is new to the shop and useful to its buyers.

Creativity is not a gift of artists alone. Every person can produce ideas, and the ideas grow from three sources. The first is knowledge: a person who knows the work well sees where it can be improved. The second is curiosity, the wish to ask *why is it done this way?* The third is freedom, which means room to try and to be wrong. A person who is afraid of being laughed at will keep ideas private. For this reason, creativity depends on the conditions that a manager creates.

The second side is **innovation**. Innovation means turning a new idea into a new product, service or method that is put to use. A method is a way of doing a task. Creativity is the seed, and innovation is the crop that grows from it. A suggestion on paper is creativity. A reading corner that stands in the shop and is used by customers is innovation. The step from one to the other needs money, time, a decision and the effort of several people.

Not every idea should be adopted. Some ideas cost more than they can earn. Some ideas fit poorly with the shop's goals. Some ideas need skills that the shop does not have. An organisation that adopts every suggestion would waste its money, and an organisation that adopts none would stop learning. The aim is to choose well, to test promising ideas on a small scale, and to explain the reasons for refusal to the people who offered them.

How can managers encourage ideas? Four practices help. First, listen: ask employees for suggestions and give each one a real hearing. Second, give time and freedom, because employees who are always hurried have no room to think. Third, make failure safe: if a tried idea does not work, treat it as a lesson and not as a fault. Fourth, provide a path from idea to action, such as a simple way to submit suggestions and a promise of an answer.

Some habits stop ideas. Quick ridicule teaches employees to keep silent. The remark *we tried that before* closes a discussion without examining whether conditions have changed. Silence in reply to a suggestion tells the employee that the effort was wasted.

The record shows the gap described above. Fourteen suggestions are creativity at work. Three adopted suggestions mark the step into innovation. The reading corner is an idea put into use, and the weekend sales figure records what followed. We now apply these figures in the worked case below.

Before moving on. Creativity is the ability to produce new and useful ideas. Innovation is turning a new idea into something that is put to use. Managers encourage ideas by listening, by giving time and freedom, by making failure safe, and by providing a path from idea to action.

* * *

WORKED CASE: SUGGESTIONS AND THE READING CORNER

This section applies the ideas of the chapter to the shop's record. It does not give a model answer. It shows one way of reasoning, step by step, so that you can follow each decision and then repeat it on another case.

Situation. In January to June 2026, Meghna and Sons recorded 14 suggestions from its staff, and 3 were adopted. A reading corner was built in April 2026 at a cost of Tk 45,000. Weekend sales rose from Tk 180,000 to Tk 207,000 a month.

Question. Which of the figures show creativity and which show innovation? What share of suggestions was adopted? What can be said about the reading corner and the rise in weekend sales?

Reasoning. First we separate the two ideas. The 14 suggestions measure creativity, because each is an idea offered by a person. The 3 adopted suggestions measure innovation, because they were put to use. The share adopted is 3 divided by 14, which is about 0.214, or 21%. So about one suggestion in five became part of the shop's work. This is not a poor result. It shows that the shop listens and also chooses.

Second, the reading corner. The record shows that it was built in April 2026 at Tk 45,000, so it is an idea put into use. The record does not say whether it came from one of the 3 adopted suggestions, and we shall not assume so. The weekend sales rose by Tk 207,000 – Tk 180,000 = Tk 27,000 a month. As a share of the earlier figure this is 27,000 divided by 180,000 = 0.15, or 15%, which agrees with the record. The corner may have helped, but weekend sales can rise for other reasons, such as a change in the weather or in the school year.

Third, a rough comparison. If the rise of Tk 27,000 a month continued, then in about 45,000 divided by 27,000 = 1.7 months the extra sales would equal the cost of the corner. This is only an illustration, because sales are not profit: the shop must still pay for the books it sells. The owner would need the profit on those extra sales to judge whether the corner paid for itself.

Conclusion. The record shows a shop that collects ideas and adopts one in about five. It shows one idea, the reading corner, put into use, followed by a rise of 15% in weekend sales. It does not

The record shows the gap between ideas and use, and one idea that crossed it.

The recap fixes creativity, innovation and the encouraging of ideas.

The worked case applies the chapter's ideas to the shop's record.

The situation lists the suggestions, the reading corner and the sales.

The question asks what the figures show about creativity and innovation, and what they cannot show.

Reasoning begins by separating ideas from ideas put to use.

The reading corner shows an idea put into use, but its link to the sales rise needs care.

A rough comparison of cost and extra sales helps the owner, but sales are not profit.

The conclusion states what the figures show and what they leave open.

* * *

SUMMARY

The summary restates the chapter in a few sentences.

Human factors are the traits, feelings and relationships of people that affect how they work, and they act at the levels of the person, the group and the organisation. Individual differences and perception explain why the same instruction can have different results. People form groups for security, belonging, status, help and shared interests. A team is a small group with joined skills, a common goal and shared accountability, and every group develops group norms. Creativity produces new and useful ideas; innovation puts them to use; and managers encourage both by listening and by making failure safe.

THINK IT THROUGH

Four situations test the chapter, from familiar to combined.

Each question below changes the situation a little more than the one before. Answer them in order.

15.T1 A new employee in a shop smiles at every customer. A second new employee rarely smiles but answers every question correctly. Explain, using individual differences, why a manager should not decide at once which is the better employee.

15.T2 The three sales assistants have an unwritten rule that they eat lunch at the same time every day. A manager asks them to eat at different times so that the counter is always covered. Name the norm, and explain one way the manager could win their agreement.

15.T3 Employees stop giving suggestions after a manager repeatedly answers *we tried that before*. Name the habit, explain its effect on creativity and innovation, and suggest one change.

15.T4 A clinic wants to find ways to reduce waiting time. It asks a nurse, a receptionist and a doctor to work together and report on the result. Is this group a team? Name the three features and say which of them the clinic must make clear for the group to be accountable.

WHERE PEOPLE GO WRONG

Mistake one: treating employees as if they were machines.

Some managers treat employees as if they were machines that respond in the same way to the same instruction. People differ in ability, attitude and background, and a method that works for one employee may fail for another.

Mistake two: believing that a group and a team are the same.

Some students believe that a group and a team are the same thing. A team is a small group with joined skills, a common goal and shared accountability. A group of friends at lunch may have none of these.

Mistake three: believing that norms are always harmful or always helpful

Some students believe that norms are always a problem for managers. A norm that helps new employees learn the work is useful.

IN HISTORY

The American company 3M, based in Minnesota, makes many products from adhesives to abrasives. In 1968, a scientist named Spencer Silver was working at 3M's headquarters in Maplewood, Minnesota, on a strong adhesive. He produced the opposite: a weak adhesive that held lightly and could be pulled off and stuck again. He believed that it was unusual and useful, but for several years he could not find a use for it.

Another 3M employee, Art Fry, sang in a church choir. His paper bookmarks kept falling out of his hymnbook. Fry remembered Silver's adhesive, and he saw that it could make a bookmark that stayed in place without damaging the page. He then saw that it would work better as a note. Sources date his idea to the early 1970s. Market tests in several cities gave mixed results, and 3M nearly stopped the product, but in 1980 Post-it notes were made available to customers in the United States.

The case shows both ideas of this chapter. Silver's adhesive was a creative result, new but without a known use. Fry's idea joined it to a need, and the work of testing, making and selling turned it into an innovation. The company allowed its scientists to share ideas with each other, and this freedom helped the idea to travel from one person to another.

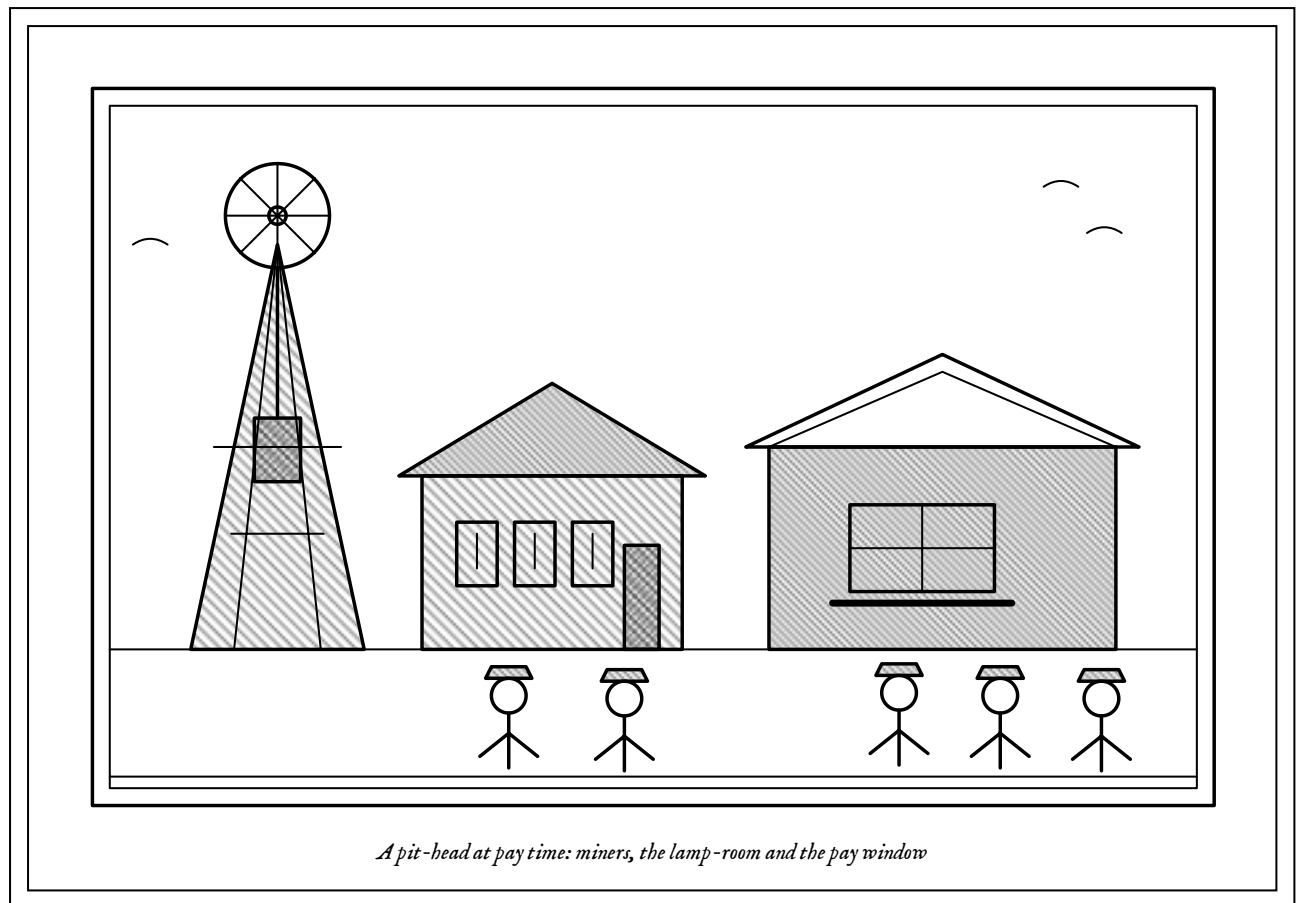
Post-it notes grew from a weak adhesive found at 3M in 1968 and were launched in 1980.

Art Fry linked the weak adhesive to a bookmark problem, and the company tested the product and sold it in 1980.

The case shows creativity first, then the long step to innovation.

REVIEW QUESTIONS

- 15.1 In your own words, state what is meant by human factors, and name the three levels at which they act.
- 15.2 Tell a group from a team. Give one example of a group that is not a team, and one example of a team.
- 15.3 Name two ways in which a manager can stop creative ideas, and two ways of encouraging them.
- 15.4 Explain the difference between creativity and innovation, using the reading corner at Meghna and Sons or another example.



CHAPTER XVI

Motivation

Why does one person work with care, and another only with the clock?



Two employees receive the same pay and the same tasks, yet one gives far more effort. This chapter explains the inner pushes behind work, compares the best-known need theories, and shows how a manager's assumptions shape the way people are led.

This chapter builds on two earlier sections.

Two employees with the same pay behave very differently.

Pay alone cannot explain the difference, so something else must be at work.

A person who works hard in one place and not in another has a reason that can be studied.

You will need: Chapter 15 'Human Factors, Groups and Creativity', section 15.1 'Human Factors in Management'; and Chapter 3 'The Functions of Management and How Management Is Judged', section 3.1 'The Management Process'.

Imagine a shop with two sales assistants. They were hired on the same day. They receive the same pay, work the same hours, and sell the same kinds of goods. Yet one of them arrives early, learns the names of regular customers, and tidies shelves without being asked. The other watches the clock, serves each customer correctly, and leaves at the minute the day ends. Neither person breaks a rule. Both are paid fairly. Why is the work so different?

The first answer that comes to mind is pay, but pay is equal here, so pay cannot be the whole explanation. A second answer is skill, yet both assistants are equally skilled. A third answer is character: one is hardworking, the other lazy. That answer is attractive, because it is short. But it explains nothing, and it gives the owner no action to take. Moreover, the "lazy" assistant may behave very differently at a cricket match or a family wedding, where the same person works hard for hours without pay.

This fact is the key. The same person can be eager in one setting and careless in another. The difference lies in what that setting offers: what the person wants, what the person hopes to gain, and what the person believes is possible. Managers cannot see these inner matters, but they can study them, and they can arrange work so that the right wants are met. Whole theories have been built on this idea, and this chapter introduces the best-known ones.

* * *

16.1 WHAT MOTIVATION IS

MOTIVATION

Start with an everyday case. A student has an examination in ten days. One student studies for two hours every evening. Another student, with the same books and the same teacher, leaves the books closed until the last night. The first student has a reason that is strong enough to start the work and to keep it going each evening. The second student has a weaker reason, or none. Both students are able to study. What differs is the push.

That push has a name. **Motivation** means the inner push that makes a person choose to act and keep acting toward a goal. The goal may be a pass in the examination, a higher income, or the respect of friends. The word *keep* matters. A short burst of effort that stops after one day is weak motivation. Strong motivation continues when work becomes tiring or dull.

Three points follow. First, motivation is inside the person. A manager cannot look at an employee and measure it. A manager can only observe what the employee does: the effort, the care, the time. Second, motivation is not the same as ability. A highly skilled employee may have little motivation, and a less skilled employee may have a great deal. Third, motivation is not force. A person who

GO SLOWLY HERE.

Before a word is defined, an everyday case shows the idea.

Motivation. The inner push that makes a person choose to act and keep acting toward a goal.

Motivation is the inner push that starts and sustains a person's action toward a goal.

Motivation lives inside a person, so a manager sees only its results.

only the behaviour it produces. It differs from ability, which is what a person can do, and from force, which is what a person is made to do.

NEED

Where does the push come from? Consider hunger. At midday, after a morning without food, you feel a discomfort, and you walk to the kitchen. The discomfort is a lack. Something is missing, and you want to fill it. The same pattern holds for other lacks, such as the lack of company when you are alone, or the lack of respect when others ignore your work.

This lack is called a **need**. A need is a lack that a person feels and wants to fill. A person who feels the need for income takes a job. A person who feels the need for friendship joins a club. Because needs differ from person to person, the same offer can move one employee and leave another untouched.

A need leads to a tension inside the person. Writers on motivation often call this tension a drive, and we use the word in that sense only. The tension is uncomfortable, so the person acts to remove it. The next section shows how need, drive, action and goal join in one loop. For now, remember only that a need comes first.

Before moving on. A need is a lack that a person feels and wants to fill. It creates a tension, called a drive, and the tension pushes the person to act. Needs differ between people, so one offer will not move everyone in the same way.

WHY MANAGERS MUST UNDERSTAND IT

Why should a manager study something that cannot be seen? The answer lies in the work of leading. In Chapter 3, leading was one part of the management process: it means guiding people so that they work toward the goals of the organisation. A manager who leads without understanding motivation is like a driver who does not know what the engine needs.

Motivation affects at least three results. It affects productivity, because motivated employees usually produce more from the same time. It affects effectiveness, because motivated employees are more likely to work on the right tasks, and not merely on any task. And it affects who stays. Employees whose needs are not met often look for work elsewhere, and replacing them costs time and money.

Case file, illustrative. In the 2025 staff record of Meghna and Sons, 2 of 7 employees left. Total wages in 2025 were Tk 2,940,000, an average of Tk 420,000 a year per employee. The record states the numbers and does not state the reasons.

Managers often respond to motivation with an incentive. An incentive means a reward offered to encourage a person to act in

A push begins with a lack, which we call a need.

Need. A lack that a person feels and wants to fill.

Need. Word in use. In daily life, to need something may mean that one must have it, as in "I need a pen". In this chapter, a need is the lack itself, felt by the person.

A need is a lack that a person feels and wants to fill.

Drive. Word in use. In daily life, to steer a vehicle, or a strong push toward a task. In the study of motivation, the inner tension that a need creates, which pushes a person to act.

A need creates a tension, called a drive, that pushes the person to act.

Before moving on: a need is a felt lack, and it creates the tension that starts action.

Managers get results through people, so people's motivation affects every result.

Recall: leading, Chapter 3, section 3.1 'The Management Process'.

Motivation affects productivity and effectiveness, and it affects who stays.

The Staff record shows that people left but not why.

Incentive. A reward offered to encourage a person to act

Before moving on: managers need motivation because it affects output, results and staff retention, and an incentive works only if it meets a need.

Every motivation theory starts from some assumptions about people.

The first two assumptions say that behaviour has causes and that people differ.

The last two assumptions say that managers shape conditions and that motives change.

Before moving on: the theories assume that behaviour has causes, that people differ, that managers shape conditions and that motives change.

Need theories explain motivation by asking which needs a person feels.

Recall: human factors, Chapter 15, section 15.1 'Human Factors in Management'.

If motivation grows from needs, a good theory should list the needs and show how they act.

GO SLOWLY HERE.

An everyday case begins the idea: a hungry person cares little about anything else.

Before moving on. Managers get results through people, so motivation affects productivity, effectiveness and the number of employees who stay. An incentive is a reward offered to encourage a person to act in a wanted way. It works only when it meets a need that the person feels.

ASSUMPTIONS ABOUT MOTIVATION

Every manager holds beliefs about what moves people, whether or not the manager has thought about them. These beliefs guide daily choices. The theories in this chapter make the beliefs explicit, so that we can examine them. Four assumptions run through most of the theories.

The first assumption is that behaviour has causes. People do not act for no reason, even when the reason is hidden. The second is that people differ. An owner may be moved by the wish to leave a legacy, a sales assistant by the wish to support a family, a customer by price. A manager must therefore look at each person, and not only at the group.

The third assumption is that managers cannot place motivation inside another person. They can only arrange the conditions in which a person's own motives are likely to work. The fourth is that motives change. An employee who first seeks a secure income may later seek responsibility. A manager who studied the person last year may need to study again.

Before moving on. Behaviour has causes. People differ. Managers shape conditions and do not insert motivation. Motives change over time. You should now be able to say: *motivation is an inner push from a felt need, and a manager can only arrange conditions in which that push is likely to appear.*

* * *

16.2 NEED THEORIES

In Chapter 15 you studied the human side of management. Now we ask a sharper question about that human side. If motivation grows from needs, which needs are there, and in what order do they matter? Three well-known answers follow. The first, and the most famous, is the work of Abraham Maslow. The second shows how any need leads to action. The third, by Frederick Herzberg, divides the features of a job into two groups.

MASLOW'S NEED HIERARCHY

Begin with a person who has eaten nothing for two days. Offer this person a place on a respected committee. Will the person care? Probably not. Hunger fills the mind, and the offer seems unimportant. Now offer a meal. The person eats, and the hunger fades. An hour later the same person starts to wonder about shelter for the night. Later again, about friends. The needs do not arrive all at once. They arrive in an order.

This idea is called the **need hierarchy**. The need hierarchy is Maslow's idea that human needs rank in five levels, and a higher need matters only after the lower ones are met. The word *hierarchy* means a ranking from lower to higher. Picture a stair: you stand on the first step before you climb to the second. The five levels are listed below, from the lowest to the highest.

To read the table, take one row at a time. The first cell gives the level, the second says what the person lacks at that level, and the third gives an example from the workplace.

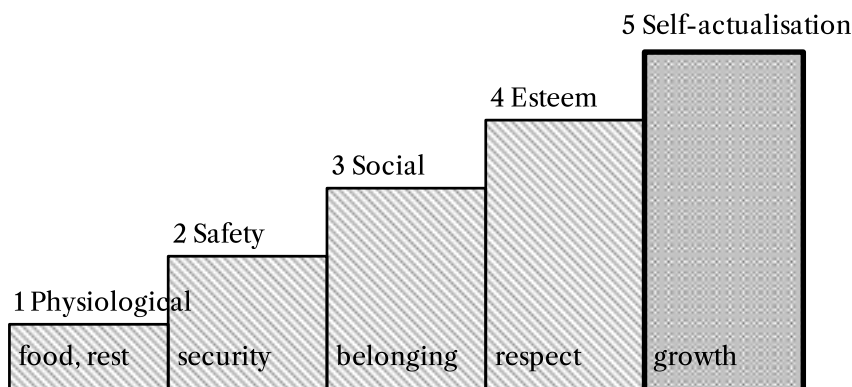
MASLOW'S FIVE LEVELS OF NEED

LEVEL	THE NEED	EXAMPLE AT WORK
1 Physiological	Food, water, shelter, rest	Wages large enough to buy food and pay rent
2 Safety	Security and freedom from danger	A steady job; a safe building; a fair notice period
3 Social	Belonging, friendship, acceptance	Being part of a friendly team
4 Esteem	Respect from others and self-respect	Praise, a title, trust with responsibility
5 Self-actualisation	Becoming all that one can be	Learning new skills; using one's abilities fully

The last level needs a plain explanation. *Self-actualisation* means becoming everything that one is able to become.¹ A musician who must make music, a teacher who wants every pupil to learn, and a trader who builds a business larger than anyone expected are examples. This level is never fully reached, because there is always more to learn. Maslow thought that it matters to a person only after the four lower levels are mostly met.

The figure shows the same levels as a stair. To read it, begin at the bottom step, which is the lowest need. Each step rises above the one before it, and the label on each step names the need. Moving upward means moving toward needs that matter more once the lower steps are secure.

Higher needs matter once lower needs are met
Read from the bottom step upward



Need hierarchy. Maslow's idea that human needs rank in five levels, and a higher need matters only after the lower ones are met.

The need hierarchy ranks five levels of need, from lowest to highest.

The table gives the five levels, the need at each level, and a shop example.

Self-actualisation means growing toward one's own full ability.

The figure shows the five levels as steps of a stair.

¹ Maslow wrote *self-actualization* with a z. This book uses British spelling.

Maslow's key claim is that a need that is met stops pushing, and a new need takes its place.

The cashier's training offer is a second kind of incentive.

The cashier's training touches esteem and growth, which sit high on the stair.

The hierarchy has limits, and Maslow himself did not claim a rigid order.

Before moving on: the need hierarchy ranks five levels of need and treats a met need as no longer pushing.

You should now be able to say what the five levels are and why an incentive may fail.

GO SLOWLY HERE.

A simple loop shows how a need becomes action and then returns.

Now the central claim, which many students misread. Maslow did not say that a person has only one need at a time. He said that an unmet lower need pushes most strongly, and that a need which is met stops pushing. When it stops pushing, the next need above it comes forward. The consequence for a manager is direct. An incentive works only when it touches the need that is pushing at that moment. A praise card means little to an employee who cannot pay the rent. A larger salary may mean less to an employee who already earns enough and wants respect.

Case file, illustrative. In the 2026 plan of Meghna and Sons, the cashier is offered training for a supervisor post.

Where does this offer sit on the stair? Training for a supervisor post offers a title and responsibility, which belong to esteem, and a chance to learn, which belongs to self-actualisation. Compare it with the bonus of one month's pay: that incentive speaks mainly to the lower levels, because income buys food, shelter and security. A shop owner who wants to keep a particular employee might offer either, but the theory tells the owner to ask which need is pushing that person now.

Now the exceptions. The order is not the same for everyone. Some people place respect above friendship, and some pursue a creative goal while poor. Needs also overlap, so a person may be partly secure and partly hungry for respect at the same time. Later researchers have found that the five levels do not always appear in the same order in every culture. Maslow himself said that the order is not fixed for every person. So the hierarchy is a useful guide to questions, not a rule that predicts one person's behaviour. **PAUSE AND CHECK 16.P2** A stock clerk has a steady job and a safe warehouse, and says the work feels pointless because nobody notices it. Which level of need is probably unmet?

IN SHORT

The need hierarchy ranks needs in five levels: physiological, safety, social, esteem and self-actualisation. An unmet lower need pushes hardest. When it is met, the next level comes forward.

Before moving on. The need hierarchy is Maslow's idea that human needs rank in five levels, and a higher need matters only after the lower ones are met. An incentive works when it touches the need now pushing. The order is a guide and not a law: it differs between people and cultures.

You should now be able to say: *needs rank in five levels, and an incentive works only when it meets the level that is pushing the person now.*

THE NEEDS-GOAL MODEL

Maslow told us which needs exist. A second idea, which we may call the needs-goal model, shows how any single need turns into action. Return to hunger. First comes the lack: the stomach is empty. Then comes tension, which is uncomfortable. The tension moves the person to act, so the person walks to the kitchen. The

To read it, follow the arrows from the need, around the loop, and back to the start. The words on each arrow tell you what moves the person from one part to the next.

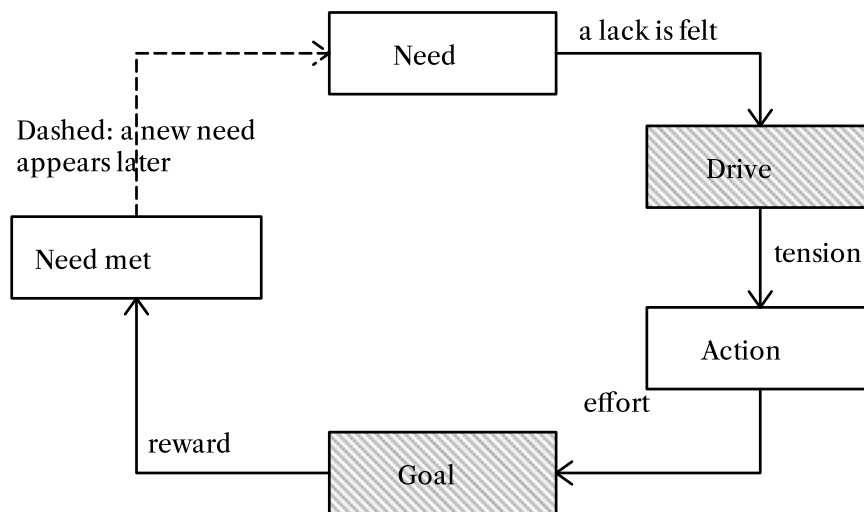


FIG. 16.2. *The needs-goal model as a loop*

Now place the loop in a workplace. An employee feels a need for more income. The tension leads to action: the employee works carefully to help the shop meet its sales target. The goal is the bonus of one month's pay. When the bonus is paid, the need is partly met, and a new need may come forward. If the goal is never reached, the loop breaks. The employee may try another action, lower the goal, or become frustrated and leave.

Before moving on. In the needs-goal model, a need creates a drive, the drive leads to action, and action leads to a goal. Reaching the goal meets the need and ends the tension. The loop breaks when the goal seems out of reach or the reward does not meet the need.

You should now be able to say: *a felt need creates tension, tension leads to action toward a goal, and reaching the goal ends the tension.*

At work, the same loop describes how an incentive leads to effort.

Before moving on: a need leads to a drive, to action and to a goal, and meeting the goal ends the tension.

You should now be able to describe how an unmet need becomes action.

THE TWO-FACTOR VIEW

Think about a classroom. Poor light and noise will make students unhappy. Good light and quiet, however, will not by themselves make them eager to learn. Removing the bad condition does not supply the push. Frederick Herzberg, an American psychologist, noticed a similar pattern at work. In a study with Bernard Mausner and Barbara Snyderman, published in 1959 as *The Motivation to Work*, he found that two different sets of job features were linked to feelings about work.

The first set deals with the conditions around the job: pay, working conditions, company rules, supervision and job security. Herzberg called these hygiene factors. When they are poor, people are unhappy. When they are good, people stop complaining, but they are not necessarily eager. The second set deals with the work itself: achievement, recognition, interesting tasks, responsibility and the chance to advance. He called these motivators. When they are

Herzberg found that some job features prevent complaints but do not create enthusiasm.

The first set, hygiene factors, prevents unhappiness; the second set, motivators, builds enthusiasm.

Critics say that the split is not sharp and that pay motivates some people.

Before moving on: Herzberg separated job features that prevent unhappiness from those that build effort.

Managers act on beliefs about people, and these beliefs have names.

Theory X. A set of assumptions that people dislike work and must be directed and watched.

Theory X assumes that people dislike work and must be directed and watched.

Theory X leads to close control, which can be right when tasks are dangerous or the workers are new.

Theory Y. A set of assumptions that people can enjoy work and seek responsibility.

Theory Y assumes that people can enjoy work and seek responsibility.

Both theories are assumptions held by managers, not facts.

upper levels, since recognition and growth serve esteem and self-actualisation. The match is rough, not exact. Both views agree on one lesson for managers: fixing the conditions around a job removes complaints, but it does not by itself create commitment.

The exceptions deserve attention. Later studies found that the split between the two sets is less sharp than Herzberg described, and that pay can motivate some people, especially when income is short. The sensible reading is to treat the two factors as a prompt for two questions. Are the conditions around the job fair? Is the work itself worth doing?

Before moving on. The two-factor view says that conditions around the job, such as pay and rules, prevent unhappiness, while the work itself, such as achievement and responsibility, builds effort. The split is rough, but it reminds managers to ask both questions.

* * *

16.3 THEORY X AND THEORY Y

THEORY X

Suppose two shop managers each find an employee arriving late. The first manager says that employees avoid work unless watched, and installs a clock-in system. The second manager says that the employee probably has a reason, and asks. Each manager acts on a belief about people. Douglas McGregor, an American management scholar, gave these beliefs names in his book *The Human Side of Enterprise* (1960).

Theory X is a set of assumptions that people dislike work and must be directed and watched. Under these assumptions, the average person avoids responsibility, wants security above all, and works hard only when pressed by rules, checks and the fear of punishment. A manager who holds these assumptions tends to give detailed orders, supervise closely, and use punishment and reward as the main tools.

The exception to note is that close control is sometimes the right choice. A new employee learning a dangerous task needs clear directions and checking. The weakness appears when a manager applies Theory X to everyone, always. Capable people then feel distrusted, and some live down to the low expectation.

THEORY Y

Theory Y is a set of assumptions that people can enjoy work and seek responsibility. Under these assumptions, effort at work is as natural as rest or play, and people direct themselves toward goals they accept. A manager who holds these assumptions shares goals, delegates tasks, and offers growth, as in the training offered to the cashier.

USING THE THEORIES

Use the two theories with care. They describe what a manager

Before moving on. Theory X assumes that people dislike work and must be directed and watched. Theory Y assumes that people can enjoy work and seek responsibility. Both are assumptions held by managers, and a good manager matches the approach to the person and the task.

Before moving on: Theory X and Theory Y are two sets of assumptions, and the best approach depends on the person and the task.

* * *

WORKED CASE: READING THE 2026 STAFF PLAN

Situation. Case file, illustrative. In 2025, 2 of 7 employees left Meghna and Sons. The 2026 plan has two measures: a bonus of one month's pay if the sales target is met, and training for the cashier for a supervisor post. *Question.* Which needs does each measure touch, which theory of management does each suggest, and what should the owner-manager do next?

The worked case applies the theories to the facts of Meghna and Sons.

Reasoning. The bonus is an incentive in money. Money buys food, shelter and security, so it touches the physiological and safety levels. In Herzberg's terms, pay is a hygiene factor, so it can reduce complaints but may not build lasting enthusiasm. The training offer gives responsibility and a chance to learn, so it touches esteem and self-actualisation. In Herzberg's terms, it is a motivator. As for assumptions, a reward tied to a target says little about whether the owner trusts people. The training offer assumes that the cashier can take on responsibility, which fits Theory Y.

The reasoning places each measure on the stair and then asks what is assumed.

Conclusion. The two measures reach different levels of need, and a plan that uses both is wider than a plan that uses only one. But the record does not say why two employees left, so the owner-manager cannot yet know which need was unmet. The next step is to ask the remaining employees what they want from their work, and not to assume. *Check.* Each statement uses only the case facts and the theories of this chapter. The needs-goal loop supports the conclusion: if the target seems impossible, the bonus will not start the loop.

The conclusion warns that the measures help only if they match the needs that employees really feel.

* * *

SUMMARY

Motivation is the inner push that makes a person choose to act and keep acting toward a goal. It begins with a need, a lack that a person feels and wants to fill. An incentive is a reward that works only when it meets a need. Maslow's need hierarchy ranks needs in five levels. The needs-goal model shows a loop from need to drive, action and goal. Herzberg separated job features that prevent unhappiness from those that build effort. Theory X and Theory Y are two sets of assumptions that managers hold about people.

The summary gathers the main ideas.

THINK IT THROUGH

16.T1 In the opening puzzle, two assistants have equal pay and skill, yet one works harder. Using the need hierarchy, suggest two dif-

16.T3 Meghna and Sons offered a bonus of one month's pay. Suppose instead that employees in a new setting, a hospital pharmacy, already earn comfortable salaries. Which kind of incentive might work better, and why?

A common error is to think that money always motivates.

Another error is to treat the five levels as a rigid ladder.

A third error is to treat Theory X as wrong and Theory Y as right in every case.

A last error is to confuse motivation with an incentive.

Maslow's paper of 1943 gave managers a way to ask what people want.

The idea spread from psychology into management, where it is still debated.

WHERE PEOPLE GO WRONG

Some students believe that more pay always raises motivation. Pay is powerful when income is the need that pushes. When it is not, extra pay may only remove a complaint.

Some students believe that a person must finish one level completely before the next appears. Needs overlap, and the order differs between people. The hierarchy is a guide to questions.

Some students believe that Theory X is always bad. It describes assumptions, and close direction is suitable where the task is dangerous or the person is new. The error lies in applying one set of assumptions to everyone.

Some students believe that a bonus and motivation are the same thing. The bonus is outside the person and is offered. Motivation is inside the person. The bonus works only if it meets a need.

IN HISTORY

In 1943 Abraham Maslow, then at Brooklyn College in New York, published *A Theory of Human Motivation* in the journal *Psychological Review*. The paper appeared in volume 50, issue 4, on pages 370 to 396. It arranged human needs in groups that rank one above another, and it said that when the strongest need is met, the next higher need comes forward. Maslow wrote in words, and the stair picture used in this chapter is a later way of showing his idea.

Managers later borrowed the idea, and Douglas McGregor used it in 1960 to explain his two sets of assumptions. Research has not confirmed the exact order of the five levels, and Maslow's paper itself says the order is not fixed for every person. Yet the central question, what does this person want now, remains one of the most useful questions in the book.

REVIEW QUESTIONS

16.1 Define motivation and need in your own words, and give one example of each from a shop.

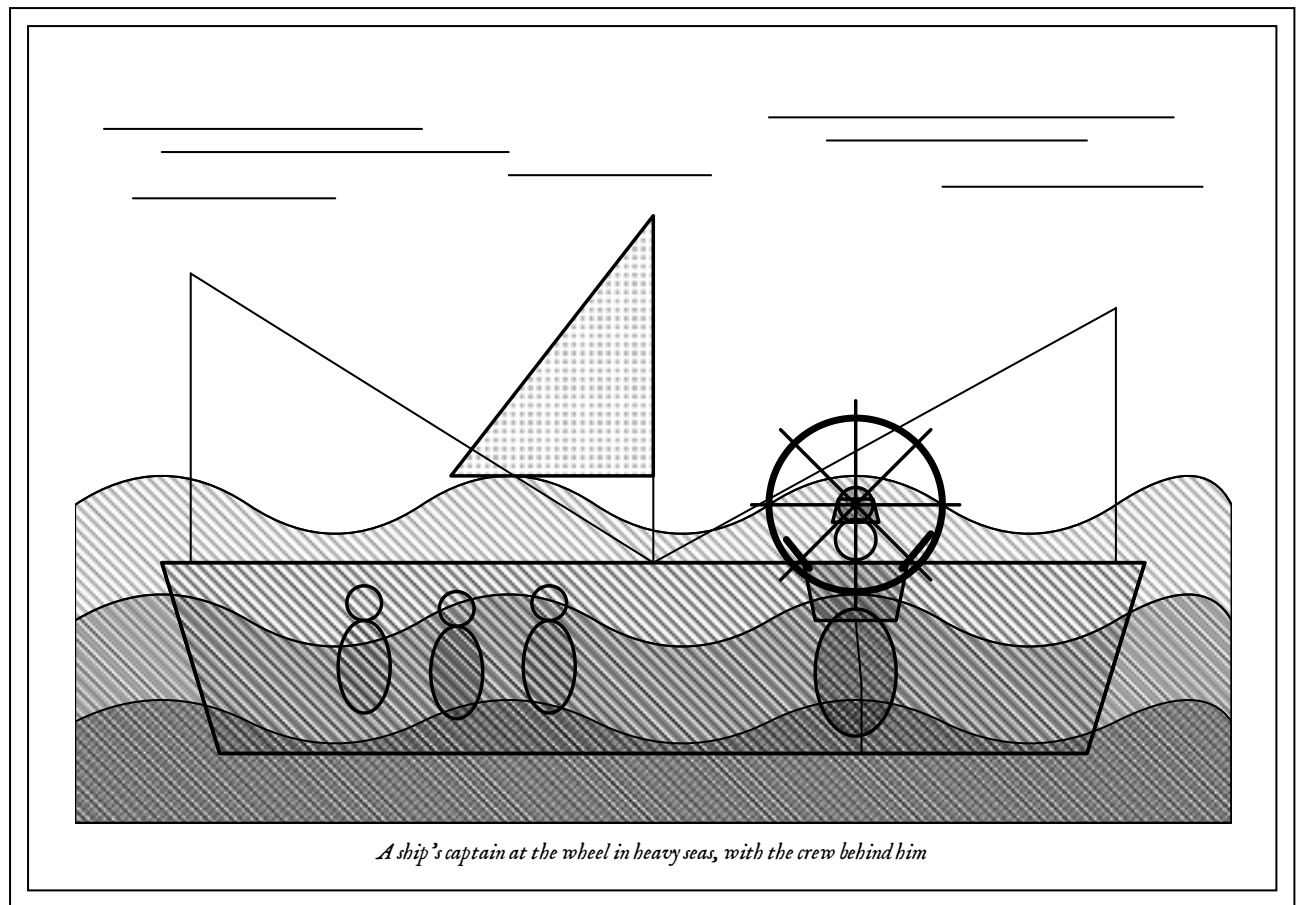
16.2 Explain the difference between motivation and an incentive. Why can an incentive fail?

16.3 List Maslow's five levels from lowest to highest, and say why the order matters to a manager.

16.4 Place each incentive on Maslow's levels: a monthly salary, safety shoes for the stock clerk, a team outing, the title of senior sales assistant, a chance to design the window display.

16.5 Describe the five parts of the needs-goal model, using an

-
- 16.7 State the assumptions of Theory X and of Theory Y. Describe one situation where Theory X suits better.
- 16.8 Meghna and Sons lost 2 of 7 employees in 2025. Why should the owner-manager ask employees before choosing an incentive?



CHAPTER XVII

Leadership

Why do some people attract followers, and others only obedience?



This chapter explains what leadership is, how a leader differs from a manager, and where a leader's power comes from. It compares the main approaches and styles of leadership and follows one bookshop through a change in its layout.

YOU WILL NEED

Chapter 16, 'Motivation', section 16.1 'What Motivation Is'; and Chapter 14, 'Authority, Delegation and Coordination', section 14.1 'Authority and Power'. Chapter 16 explains why people choose to act and keep acting. Chapter 14 explains authority, which comes with a position, and power, which is a wider idea. This chapter uses both ideas to explain why people follow one person and not another.

Two managers hold the same title and the same authority, yet their employees behave very differently.

Two tempting answers, talent and kindness, are each partly wrong.

A captain in a storm shows that authority puts a person in charge, while something more keeps the crew steady.

The chapter moves from the meaning of leadership, to the approaches, to the styles and modern ideas.

A group of friends planning a trip shows leadership appearing without any title.

Imagine two shops. Each has a store manager with the same title, the same pay and the same authority. In the first shop, employees do what the manager says, at the time he says it, and nothing more. When he leaves the room, the work slows. In the second shop, employees report problems before the manager asks, and they stay late before a busy week without being told. The title is the same, and the authority is the same. Something else differs. What is it?

One tempting answer is that the second manager was born with a gift. Another is that the second manager is a kinder person. Both answers contain a little truth, and both fail as a full explanation. People have studied this question for more than a hundred years, and their answers have changed several times. This chapter follows those changes, so that you can judge each answer for yourself.

Picture a ship in a storm. The captain at the wheel holds authority from the shipowner. In the storm, however, the crew do not obey because a contract says so. They obey because they trust him: he knows the sea, he stays calm, and he cares whether they live. His authority put him at the wheel. Something else holds the crew together. That something has a name, *leadership*, which section 17.1 defines.

Section 17.1 defines leadership, separates it from management, and shows where a leader's power comes from. Section 17.2 describes two approaches that explain why some people lead well, and a famous continuum of leader behaviour. Section 17.3 describes the styles of leadership and three modern ideas. Along the way, we follow a bookshop in Dhaka through a change that two of its managers handled in two different ways. Many students find this subject confusing, because the word *leader* is used loosely in daily talk. We shall use it with one meaning only.

* * *

17.1 LEADER AND MANAGER

Begin with a small, familiar case. A group of six friends plans a weekend trip. Nobody is the owner, nobody is the boss, and nobody has a title. Within an hour, one friend has proposed the dates, listened to objections, settled an argument and persuaded the group to share the costs. The others follow her plan, and they do so willingly. She holds no authority at all. What she holds is influence, and the group's decision to accept it.

This is the idea we call **leadership**. Leadership means the process of influencing people to work willingly toward group goals. Take the friend who planned the trip: she influenced the others, they worked willingly, and the goal belonged to the whole group. Each part of the definition matters. *Influencing* means that the leader changes what other people choose to do. *Willingly* means that people act from their own motivation, and not from fear alone. *Group goals* means that the aim is shared, and is not only the leader's private wish.

Leadership is a process, and a process has parts. The first part is the leader, the person who influences. The second is the followers, the people who are influenced; without followers there is no leadership, because influence needs someone to receive it. The third part is the situation: the task, the time available, and the dangers and chances around the group. The friend who plans a weekend trip faces a calm situation, so her gentle style works. The captain in a storm faces a dangerous one, and he needs a different way of influencing. Leadership changes when any of the three parts changes.

Two exceptions help to avoid mistakes. First, leadership does not depend on a title. An experienced sales assistant whom colleagues ask for advice may lead more than the store manager does. Second, the definition says nothing about whether the goals are good. A leader may influence people toward a harmful aim, and the process is still leadership. Leadership is therefore a description of what happens, and it is not a compliment. When we want to ask whether leadership is good, we look at its goals and its methods.

Now we separate two words that students often confuse. A manager is a person who holds a position in an organisation and carries the authority that comes with it. A leader is a person who influences others, whether or not that person holds a position. Some managers lead well. Some managers do not lead at all, and their employees obey the rules and no more. Some leaders are not managers: the friend on the trip, or the respected assistant at the counter. The best managers do both, because a manager who can also lead gets more from the same people. The table below sets the two ideas side by side. Read each row from left to right: the first column names a feature, and the other two columns say how it appears for a manager and for a leader.

A MANAGER AND A LEADER COMPARED

FEATURE	MANAGER	LEADER
Source of influence	Position and authority	Personal influence, with or without a position
Why people follow	The position requires it	They choose to follow
Main concern	Using resources well to	Moving people toward group

Recall: motivation, Chapter 16, 'Motivation', section 16.1 'What Motivation Is'.

Leadership is the process of influencing people to work willingly toward group goals.

Lead. Word in use. In daily life, "lead" can mean to be in front, as in a race, and "lead" is also a metal. In management, to lead is to influence people so that they follow willingly.

Leadership. The process of influencing people to work willingly toward group goals.

Leadership involves three things together: the leader, the followers and the situation.

Leadership has limits: it does not depend on a title, and it is not always used for good ends.

A manager holds a position, and a leader holds influence; one person can be both, or only one.

A leader's power can come from five sources, and one of them, liking and admiration, is called referent power.

Referent power rests on liking and admiration, and it often lasts longer than power from rewards or punishment.

Referent power. Power that comes from other people's liking and admiration for a leader.

People follow a leader who helps them meet their needs, which links leadership to motivation.

The owner, the manager, the employee and the customer each see leadership from a different position.

Before moving on: leadership is influence toward group goals, it differs from management, and it rests on several kinds of power.

People once believed that leaders are born with special

Why do people accept a leader's influence? The answer lies in power, the ability to influence the actions of others, which Chapter 14 separates from authority. In 1959 two American social psychologists, John French and Bertram Raven, listed five sources of power.¹ The first is *legitimate* power, which comes from a position. The second is reward power: the leader can give something people value, such as a bonus or a better shift. The third is coercive power: the leader can punish, for example by a warning. The fourth is expert power: people follow because the leader knows more. The fifth is referent power, which the next paragraph defines.

Referent power means power that comes from other people's liking and admiration for a leader. For example, a sales assistant whom colleagues respect and wish to resemble can influence them without any reward to offer. The first four sources can be taken away: a position can end, a bonus can be cancelled. Referent power belongs to the relationship, and it survives such changes. Even so, it has limits. A leader who relies on liking alone may avoid hard decisions to stay popular. Coercive power gets obedience, but it rarely gets commitment, because people who obey from fear do the minimum.

Leadership connects to motivation, which Chapter 16 explains. People follow willingly when they believe that the leader helps them to meet their needs. A leader who sees which need matters most to each person, whether pay, security, belonging or a chance to grow, can influence that person more easily. The need hierarchy is one way to think about this question. A leader who ignores people's needs may still hold authority, but the motivation of the group will be weak.

Each person sees leadership from a different position. The owner sees a way to turn a small number of managers into a strong team. The manager sees a skill that makes delegation safer, because people who follow willingly can be trusted with more work. The employee sees whether the person in charge understands and respects the work. The customer sees the result: a shop whose employees are led well shows it at the counter, in service that is steady and willing.

Before moving on. Leadership is the process of influencing people to work willingly toward group goals. It involves a leader, followers and a situation. A manager holds a position; a leader holds influence; one person may be both. Leader power has five sources: position, reward, punishment, expert knowledge and referent power, which is liking and admiration.

* * *

17.2 APPROACHES TO LEADERSHIP

In the early twentieth century, many writers held a simple belief: leaders are born, and one can recognise them by their qualities.

¹ French and Raven published their list in 1959. The names they gave to the five sources are still used in books on management.

alone. The trait approach still gives us something: it reminds us that a leader's character matters. Its weakness is that it ignores the situation, and a captain who commands well in a storm may command badly in a boardroom.

Researchers then asked a different question. Instead of asking "What qualities make a leader?", they asked "What should a leader do in this situation?" The situational approach holds that no single way of leading is best, and that the right way depends on the leader, the followers and the situation. A new employee who needs guidance, for example, may need a leader who gives clear directions. A skilled employee of many years may work best with a leader who gives freedom. The situational approach says that a good leader reads the situation and changes behaviour to fit it.

In 1958 two American writers, Robert Tannenbaum and Warren Schmidt, published an article in the *Harvard Business Review* called "How to Choose a Leadership Pattern". They described a continuum, which is a line along which a quantity changes in small steps. At one end, the manager uses a great deal of authority and the employees have little freedom. At the other end, the manager uses little authority and the employees have a great deal of freedom. Between the ends they described seven patterns. In the first, the manager decides and announces the decision. In the second, the manager decides and then tries to persuade the group. In the third, the manager presents a decision and invites questions. In the fourth, the manager presents a tentative decision that the group may change. In the fifth, the manager presents the problem, asks for suggestions and then decides. In the sixth, the manager sets limits and asks the group to decide. In the seventh, the group acts within limits set by the manager.

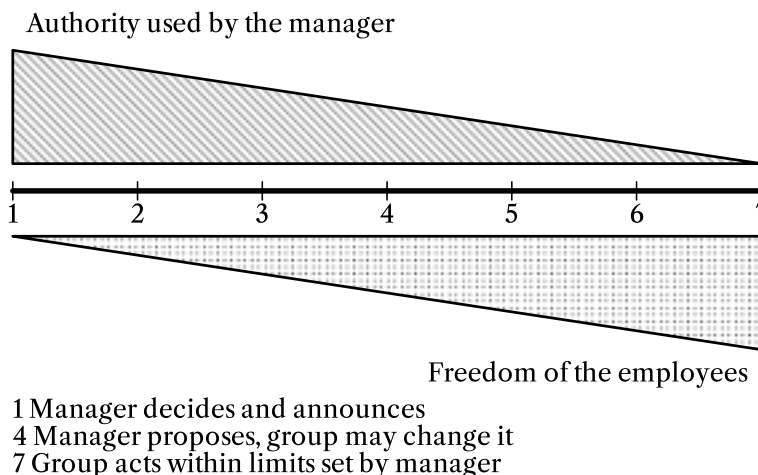
Figure 17.1 shows the continuum. To read it, begin at point 1 on the left and move along the line to point 7 on the right. The upper wedge shows the authority that the manager uses; it is widest at point 1 and narrows to the right. The lower wedge shows the freedom of the employees; it is narrowest at point 1 and widens to the right. A point on the line is one pattern of behaviour. Points 1, 4 and 7 carry short labels, and the paragraph above names the others.

The situational approach says that effective leadership depends on the leader, the followers and the situation together.

Recall: authority and power, Chapter 14, 'Authority, Delegation and Coordination', section 14.1 'Authority and Power'.

In 1958 Tannenbaum and Schmidt described seven patterns of leader behaviour along one line.

To read the figure, follow the line from left to right: the manager's authority falls and the employees' freedom rises.



The continuum offers no best point; the right point depends on forces in the manager, the employees and the situation.

Before moving on: the trait approach seeks the qualities of leaders, and the situational approach says the right way to lead depends on circumstances.

Two managers in one shop handled one change in two ways within a single week.

Why did one change bring two ways of deciding? The answer is that leaders have usual ways of behaving, called styles.

Recall: leadership, section 17.1 'Leader and Manager'.

A leadership style is a leader's usual way of making decisions and treating followers.

Style. Word in use. In daily life, a fashion or manner, such as a style of dress. In management, a leader's usual pattern of behaviour toward followers.

Leadership style. The usual way a leader makes decisions and treats followers.

Lewin and his colleagues

Tannenbaum and Schmidt did not say that one point is best. They said that the right pattern depends on three forces. The first is the forces in the manager: his values, his confidence in the employees and his comfort with uncertainty. The second is the forces in the employees: how much they wish for freedom, how well they know the work and how ready they are to take responsibility. The third is the forces in the situation: the type of organisation, the time available and the nature of the problem. When time is short, as in an emergency, a pattern near point 1 is often wise. When employees are skilled and time is available, a pattern nearer point 6 may bring better decisions and more commitment.

Before moving on. The trait approach searched for the personal qualities of leaders and found no list that works in every situation. The situational approach holds that effective leadership depends on the leader, the followers and the situation. Tannenbaum and Schmidt, in 1958, described a continuum of seven patterns, from the manager who decides and announces to the group that acts within limits. No point on the line is best in every case.

* * *

17.3 STYLES AND MODERN IDEAS

Case file, illustrative. Meghna and Sons, the family-run bookshop in Dhanmondi, Dhaka, changed the layout of its shop on 3 March 2026. Nurul Meghna, the owner-manager, decided on the change and told the staff. Rafiq Meghna, the store manager, asked the 3 sales assistants for their views first and then decided. Both ways of handling a decision were used within one week.

Why would two members of one family, in one shop, handle a similar change in two ways? One answer is that each person has a usual way of making decisions and treating followers. A leader's usual way has a name: the leadership style. The case file records two decisions and does not tell us why each manager chose as he did, so we cannot judge which choice was better. We can, however, describe what each choice was.

A **leadership style** means the usual way a leader makes decisions and treats followers. For example, a leader who nearly always decides alone and then informs the group has a different style from a leader who nearly always asks first. The word *usual* is important. A style is a habit and not a rule, and a skilled leader can step away from the habit when the situation calls for it.

In 1939 the German-born psychologist Kurt Lewin and two colleagues, Ronald Lippitt and Ralph White, reported their experiments with boys' clubs in 1939. The names of the three styles come from later textbooks, and they describe the results in simple form.

the participative style lies near the middle, and the free-rein style lies near point 7.

Each style has a time and a place. The autocratic style fits an emergency, a very new group, or a task where one wrong move is costly; its risk is that people stop offering ideas. The participative style fits a problem where employees know things the leader does not, and where commitment matters; its risk is that decisions take longer. The free-rein style fits a group of skilled and motivated people; its risk is that the group drifts, if people lack skill or goals are not clear. The table sets the three side by side. Read each row from left to right: the first column names the style, the second says who decides, and the third and fourth give a fitting case and the main risk.

THREE CLASSIC LEADERSHIP STYLES

STYLE	WHO DECIDES	FITS WHEN	MAIN RISK
Autocratic	The leader alone	Time is short or danger is high	Ideas stop coming
Participative	The leader with the group	Employees know the work well	Slow decisions
Free-rein	The group	People are skilled and motivated	Drift without direction

Look again at the layout change. Nurul decided and told the staff, which is the autocratic style, and the first point of the continuum. Rafiq asked the sales assistants first and then decided, which is the participative style and lies near the fifth point of the continuum. Both are patterns that a leader can choose. A good leader knows the range, and chooses the pattern that fits the people and the problem.

Emotional intelligence means the ability to notice, understand and manage one's own and other people's feelings. For example, a manager who notices that a tired employee is losing patience, and who responds calmly, shows emotional intelligence. Two psychologists, Peter Salovey and John Mayer, introduced the term in 1990, and the writer Daniel Goleman made it widely known in 1995. Its parts include knowing one's own feelings, controlling them under pressure, understanding other people, and handling relationships with care. The idea has limits. It does not replace knowledge of the work, and researchers still debate how well it can be measured.

Transformational leadership means leading by inspiring followers to look beyond self-interest and to change themselves and the organisation. The political scientist James MacGregor Burns drew the idea in 1978 from his study of political leaders. He contrasted it with a leader who trades, giving pay or praise in return for work done. The trading leader gets the agreed work. The transformational leader hopes for more: followers who accept a larger purpose and who grow in skill and confidence. For example, an owner who explains to the whole shop why good service matters to the neighbourhood, and who asks each person to help build that reputation, is leading in this way. The risk is that an inspiring leader

Each style fits some situations and fails in others, as the table shows.

The case file shows both ends of the range of behaviour in one shop: deciding and telling, and asking and then deciding.

Emotional intelligence is the ability to understand and manage feelings, one's own and other people's.

Emotional intelligence. The ability to notice, understand and manage one's own and other people's feelings.

Transformational leadership inspires followers to look beyond self-interest and to change themselves and the organisation.

Transformational leadership. Leading by inspiring followers to look beyond self-interest and to change themselves and the organisation.

Before moving on: a style is a leader's usual way; the best leaders can change style, and modern ideas add feelings, inspiration and development.

because people learn by doing real work. Coaching takes time, and it suits a person who wishes to learn. It fits poorly where the task is urgent and no time remains for teaching.

Before moving on. A leadership style is a leader's usual way of making decisions and treating followers. The three classic styles are autocratic, participative and free-rein, and each fits some situations. Emotional intelligence is the ability to notice, understand and manage feelings. Transformational leadership inspires followers to look beyond self-interest. Coaching leadership develops people through guidance and feedback.

* * *

WORKED CASE: MATCHING STYLE TO SITUATION

The worked case tests the choices in the case file against the three forces of the continuum.

The situation restates the facts of the case file and nothing more.

The question asks how to judge each choice with the three forces, using only what the file supports.

The reasoning applies the forces in the manager, the employees and the situation to each choice.

The conclusion and the check show what can and cannot be said from the file.

Case file, illustrative. We now apply section 17.2 and section 17.3 to the layout change. We take it in the order *Situation, Question, Reasoning, Conclusion, Check*, so that each step is announced and explained.

Situation. On 3 March 2026 the bookshop changed its layout. Nurul decided and told the staff. Rafiq asked the 3 sales assistants first and then decided. Both styles were used within one week. The file does not say what the change was, why each manager chose as he did, or what the results were.

Question. Using the three forces of Tannenbaum and Schmidt, under what conditions would each choice be wise?

Reasoning. First, name each choice. Nurul used point 1 of the continuum: decide and announce. Rafiq used point 5: ask for suggestions and then decide. Second, apply the three forces. For point 1, the situation should favour speed or a clear single view; the employees should accept direction; the manager should be confident in his own judgement. For point 5, the employees should know something the manager does not, such as how customers move through the shop; the time should allow a short discussion; and the manager should be willing to hear views he may not follow. Third, notice what the file cannot tell us. We do not know whether Nurul's change was urgent, or whether the assistants knew more about the shelves than Rafiq did. A judgement on either choice would therefore be a guess.

Conclusion. Both choices are acceptable patterns, and each is wise under some conditions. The file shows that the shop's leaders used two points of the continuum within one week. It does not show which was better. *Check.* A good test of the reasoning is that it names the missing facts: the purpose of the change and its results. A conclusion that picked a winner would claim more than the file supports. The next step for a manager is the one the continuum suggests: ask what the three forces say about this decision, and then choose the point.

* * *

the leader, the followers and the situation. Tannenbaum and Schmidt described seven patterns on a continuum. Autocratic, participative and free-rein are the classic styles. Emotional intelligence, transformational leadership and coaching leadership are modern ideas.

THINK IT THROUGH

A hospital ward has twelve nurses and one ward manager. The ward manager has authority from the hospital, but several nurses go to a senior nurse, who holds no position of authority, whenever they need advice or support. Use the ideas of this chapter to answer both questions.

17.T1 Is the senior nurse a leader, a manager, or both? Explain your answer using the definitions of leadership and management in section 17.1.

17.T2 A fire alarm sounds in the ward. Which point of the continuum should the ward manager use, and why? Then say which point suits a monthly meeting on improving the duty roster.

The problem asks you to apply the chapter to two situations.

WHERE PEOPLE GO WRONG

The first misunderstanding is that leaders are born and cannot be made. Correction: some qualities help, but studies found no list that works everywhere, and people can learn many leadership skills. The second is that a leader and a manager are the same. Correction: a manager holds a position; a leader holds influence; the two overlap but are not equal.

The third misunderstanding is that one style is always best. Correction: each style fits some situations and fails in others. The fourth is that the participative style is always better because it is kinder. Correction: in an emergency, or with a very new group, slow discussion may cause harm. The fifth is that power is only the ability to punish. Correction: power has five sources, and referent power often lasts longest. The sixth is that leadership is always good. Correction: the definition describes influence toward group goals and says nothing about whether the goals are good.

Six common misunderstandings about leadership are stated and corrected here.

Further misunderstandings concern the best style, the meaning of power, and the nature of a leader's goals.

IN HISTORY

Ernest Shackleton was a British explorer. On 8 August 1914, he sailed from Plymouth, in England, in the ship *Endurance*, with a crew of 28 men. His aim was to cross the continent of Antarctica. In January 1915 the ship became trapped in the ice of the Weddell Sea. The ice crushed her, and the men abandoned the ship on 27 October 1915. She sank in November 1915.

The men then lived for months on the ice, and in April 1916 they reached Elephant Island, an uninhabited island, in three small boats. On 24 April 1916, Shackleton and five others sailed one boat, the *James Caird*, about 800 miles to South Georgia to find help. He then crossed the mountains of that island on foot with two com-

Shackleton led the Endurance expedition from 1914 to 1916 and brought all 28 men home.

After the loss of the ship, Shackleton's choices kept his men together through months on the ice and in open boats.

rescue are well recorded, and they show a leader whose choices kept a group together under extreme danger.

REVIEW QUESTIONS

The review questions test each main idea of the chapter.

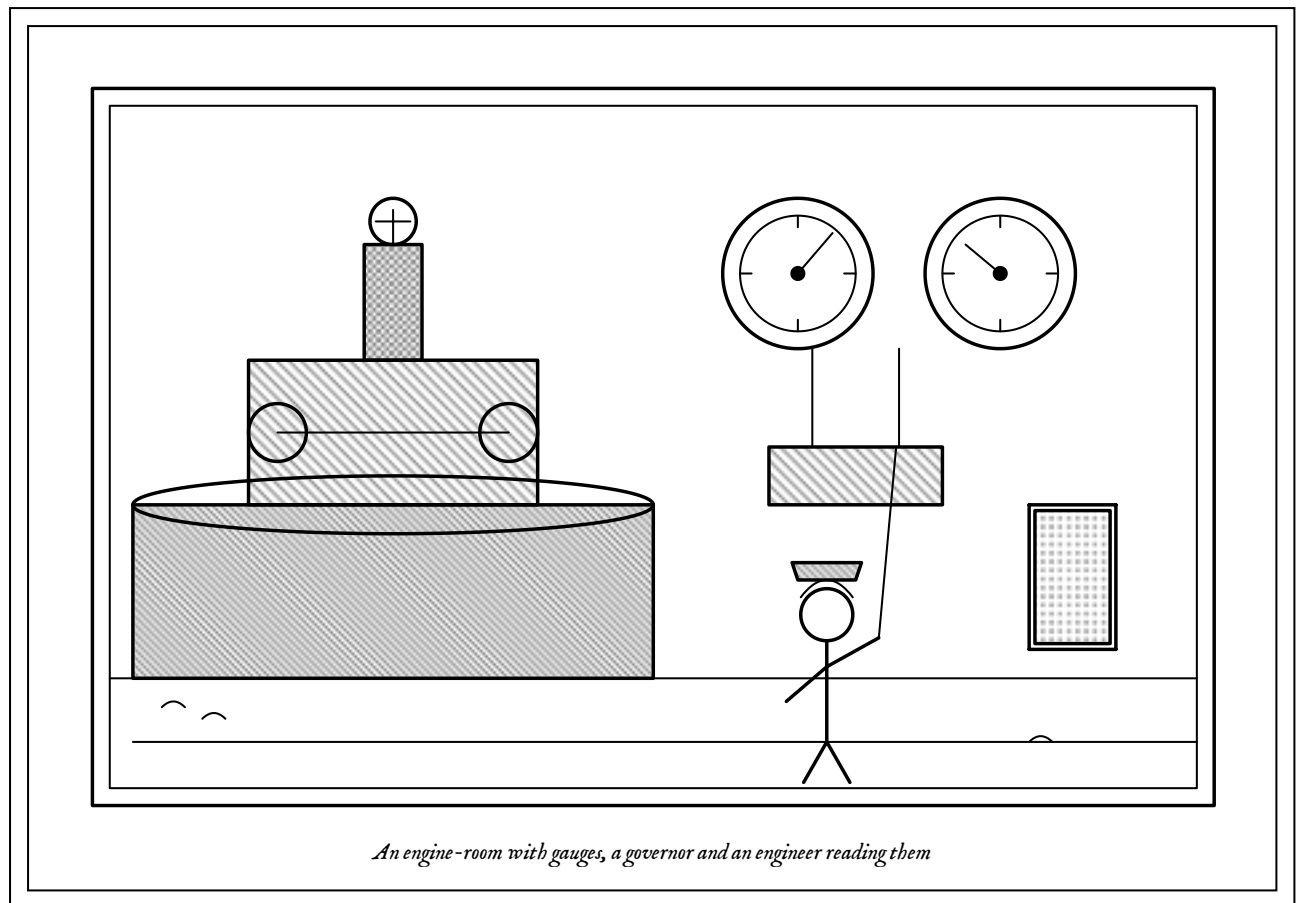
17.1 Define leadership in your own words, and name its three parts.

17.2 Give two differences between a manager and a leader, and explain how one person can be both.

17.3 Name the five sources of a leader's power, and explain referent power with an example of your own.

17.4 Match each situation to the style that fits best, and give one reason: (a) a flood warning reaches a warehouse; (b) a team of skilled designers needs to choose a new logo; (c) a group of experienced accountants works on a long project with clear goals.

17.5 Explain the difference between the trait approach and the situational approach, and say which one the Tannenbaum and Schmidt continuum supports.



CHAPTER XVIII

Controlling

How does a manager know that the plan is working?



A plan is only a promise until its results are checked. This chapter explains how managers set standards, measure results, compare them with the plan and correct the difference, and it shows the three moments at which control can act.

This chapter builds on two earlier sections about planning and objectives.

A plan was made, a year passed, and the result was a surprise.

The question has an answer: she could have known early, if she had checked.

Engineers solved this problem for machines long before managers solved it for people.

A business needs the same pattern, and this chapter explains it.

Plans stay useful only when results are compared with them.

Controlling. Checking results against plans and taking action to correct any difference.

Control. Word in use. In daily life, to control a person or a thing often means to have power over it. In management, control means checking results and

You will need: Chapter 7 'Planning: Meaning, Nature and Steps', section 7.1 'What Planning Is'; and Chapter 9 'Objectives and Management by Objectives', section 9.1 'Objectives'.

Imagine a shopkeeper who plans carefully in January. She decides how much stock to buy, how much to sell, and how much profit to earn. Twelve months later she counts the stock, and she finds that goods worth far more than expected have disappeared. The plan was sensible. The shopkeeper worked hard. Yet nothing in the year warned her, and now the loss cannot be recovered. She asks herself a painful question: when could I have known?

The answer is: much earlier. If she had counted a part of the stock each week, the first small shortage would have appeared in the first month. She could have found the cause, such as a door left open or a mistake in recording sales, and she could have acted while the loss was still small. The plan did not fail. The checking failed, because there was none.

Engineers met the same problem with steam engines. In a Victorian engine-room, a steam engine that runs too fast may damage itself, and one that runs too slowly cannot do its work. So engineers fitted a device called a governor. The governor senses the speed and adjusts the flow of steam, so that the engine returns to its right speed without anyone touching it. A gauge shows the engineer how the engine is doing. The pattern is: a level to aim for, a reading, a comparison, and an adjustment.

A business needs the same pattern. This chapter shows how managers apply it to stock, money, time and people. It also shows that the pattern has three moments: before work, during work and after work. Last, it explains why even a good system fails when people resist it or when no one truly checks. This is the final function of management, and it closes the cycle that began with planning.

* * *

18.1 WHAT CONTROLLING IS

CONTROLLING

Think about driving a car. You choose a destination (the plan). During the drive you watch the road, the speed and the fuel. If the car drifts toward the edge, you turn the wheel. You do not wait until you arrive to find out that you left the road. This everyday habit of watching and adjusting is exactly what managers do with an organisation.

This function is called **controlling**. Controlling means checking results against plans and taking action to correct any difference. A shop manager who compares weekly sales with the sales target, and who changes the display when sales are low, is controlling. Notice that controlling has two halves: the check, and the action. A check without action is only an observation, and action without a check is guessing.

Viewpoints differ. The owner sees control as protection of money and name. The manager sees it as a way to learn early where help is needed. The employee may see it as distrust, unless the purpose is explained and the checks are fair. The customer meets control as reliable quality, such as the right book on the shelf at the right price. A good manager keeps all four views in mind when designing a check.

Different people see control from different sides.

Before moving on. Controlling means checking results against plans and taking action to correct any difference. It has two halves, the check and the action. It is the last function of management, and its findings feed the next plan.

Before moving on: controlling is checking results against plans and correcting differences.

NATURE AND PRINCIPLES

Five features describe the nature of control. It is *continuous*, because results change every week, and one check at year end is not enough. It is *forward-looking*, because its purpose is to improve future results, not to blame the past. It is *universal*, because every manager at every level controls something, from the owner who watches profit to the cashier who counts the till. It is *tied to plans*, because without a plan there is nothing to check against. And it is *connected to people*, because people carry out the checks and people are checked.

Control has certain features that hold in every organisation.

Five principles guide the work. First, control should follow the plan, so that it checks what matters to the plan. Second, it should watch key points. A manager cannot watch everything, so the manager watches the key result areas, the few areas where results decide success. Third, it should act quickly, because a late correction is costly. Fourth, it should be economical: its cost must stay below the loss it prevents. Fifth, it should be flexible, so that it still works when the plan changes.

Good control follows a few simple principles.

The economical principle deserves an example. Suppose a stall loses about Tk 500 a year to small errors. A system that costs Tk 2,000 a year to prevent this is poor value, since it spends four times what it saves. Controlling every item with equal care is therefore wasteful. Wise managers spend most effort where the possible loss is largest.

A check costs money, so its cost must be compared with the loss it prevents.

Before moving on. Control is continuous, forward-looking, universal, tied to plans, and connected to people. It should follow the plan, watch key points, act quickly, cost less than the loss it prevents, and remain flexible.

Before moving on: control is continuous, forward-looking and universal, and it follows a few clear principles.

THE LINK WITH PLANNING

Planning and controlling are like two sides of a coin. Planning sets the objectives and chooses the course of action. Controlling checks whether the objectives are being reached. Without plans, control has nothing to measure against, so it cannot begin. Without control, planning is only hope, because no one learns whether the plan worked. Control also supplies facts for the next plan: a premise that proved wrong in this year can be corrected in the next.

Planning and controlling depend on each other.

Recall: planning, Chapter 7, section 7.1 'What Planning Is'; objectives, Chapter 9, section 9.1 'Objectives'.

Plans give control something to measure, and control tells planners if the plan works.

Before moving on. Plans give controlling its measures, and controlling shows whether the plan works. The findings return to

Before moving on: planning and control are a pair, and each makes the other useful.

18.2 THE CONTROL PROCESS

The control process has four steps, and the first is to decide what counts as good.

Standard. A level of performance set in advance as the measure of what is acceptable.

Standard. Word in use. In daily life, standard may mean ordinary or average, as in a standard model. In control, a standard is a level that has been chosen in advance as the measure.

A standard is a level of performance chosen in advance.

A standard must be measurable and fair, or it cannot guide anyone.

The case file's standard is a small percentage of the stock value.

The second step is to find out what actually happened.

The case file measures loss over the year and also counts a sample every week.

A weekly sample gives an early signal at low cost.

The third step compares the actual result with the standard.

A worked example with small numbers makes the method easy to follow.

SETTING STANDARDS

How does a teacher know whether a student passed? The teacher first decides the pass level, and then compares the work with it. Control follows the same order. Before work begins, the manager decides what level of result is acceptable. Only then can results be judged.

Such a level is called a **standard**. A standard is a level of performance set in advance as the measure of what is acceptable. A stall owner may set a standard that no more than 2 in every 100 loaves are wasted. Standards come from objectives and plans, and they are often stated in four ways: in quantity (units sold), in quality (errors allowed), in time (hours to finish), and in cost (taka spent).

A good standard is measurable, so that everyone can tell whether it was met. It is also fair, so that employees believe it can be reached. A standard that nobody can meet teaches people to ignore it. The key result areas tell the manager where standards matter most.

Case file, illustrative. The control record of Meghna and Sons sets a stock loss standard of 1% of stock value. Stock at cost is Tk 3,500,000, so the standard allows a loss of Tk 35,000.

MEASURING PERFORMANCE

A standard is useful only if real results are measured. A manager must decide what to measure, how often, and by whom. Results may be counted, weighed, timed or reported. Good measurement is accurate and timely: a figure that arrives six months late can no longer change the result.

Case file, illustrative. In 2025 the actual stock loss of Meghna and Sons was Tk 56,000. The record also shows a weekly count of 200 titles.

The weekly count of 200 titles is a sample: a small part of the stock, counted often. It would be costly to count every title every week, but a sample gives an early signal. This is the economical principle at work.

COMPARING

Comparing means placing the actual result beside the standard and finding the difference. This difference is called the *deviation*. A deviation is the amount by which the actual result differs from the standard. We calculate it in one way, so that signs never confuse us: actual minus standard.

Given. A stall holds stock worth Tk 100,000. The standard loss is 2% of stock value. The actual loss is Tk 3,000. *Find.* The deviation. *Formula.* In words: deviation equals actual result minus standard.

percentage points. *Answer.* The loss is Tk 21,000 above the standard, which is 1.6% against 1%. *Check.* $0.6\% \text{ of } 3,500,000 = 0.006 \times 3,500,000 = \text{Tk } 21,000$, which agrees.

Three slips are common. The first is to subtract the wrong way and so reverse the meaning of the sign. The second is to confuse a percentage with a percentage point: from 1% to 1.6% is a rise of 0.6 percentage points, but it is also a rise of 60 per cent of the standard rate. The third is to compare figures for different periods, such as a week's loss with a year's standard.

Three slips are common when comparing.

CORRECTING

A deviation calls for a question: why? The manager looks for the cause before choosing the action. Possible actions are to improve performance (stop the cause), to change the plan, or to revise the standard if it was unrealistic. If the deviation is small and temporary, the manager may do nothing, and this too is a decision.

The fourth step is to act on the difference.

Case file, illustrative. As the correction, Meghna and Sons fitted security tags at Tk 30,000 in 2026.

The tags are the correction chosen for the stock loss.

A correction also has a cost, so it must be weighed against the benefit. Suppose, as an illustration, that the tags remove the whole excess loss of Tk 21,000 a year. Then $\text{Tk } 30,000 \div \text{Tk } 21,000$ is about 1.4, so the tags repay their cost in a little under a year and a half. If they remove only half the excess, the repayment takes about three years. The cost and the benefit must both be judged.

A correction has a cost, so it must be weighed against its benefit.

The figure shows the process as a loop. To read it, start at the box *Set standard* and follow the arrows: measure, compare, correct. The dashed arrow shows that what is learned returns to the first box and to planning.

The figure shows the four steps as a loop that returns to planning.

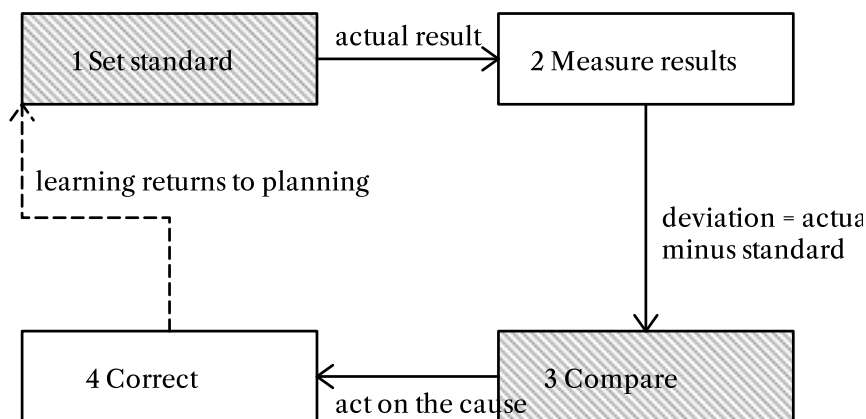


FIG. 18.1. *The control process as a loop*

PAUSE AND CHECK 18.P1 A cafe sets a standard of 5 spoiled cups in a week. In week 6 it has 9. What is the deviation, and what should the manager do first?

IN SHORT

The four steps are: set a standard. measure performance. compare

Before moving on: the control process sets a standard, measures, compares and corrects, and the findings return to planning.

Before moving on. The control process has four steps: set a standard, measure performance, compare it with the standard to find the deviation, and correct. The deviation is actual minus standard. A correction must be weighed against its cost.

* * *

18.3 TYPES OF CONTROL

THREE MOMENTS OF CONTROL

Control can act before, during or after the work.

Picture a cook. Before cooking, the cook checks that the rice is clean and the oil is fresh. While cooking, the cook tastes the food and lowers the flame if it burns. After cooking, the cook judges the finished dish and remembers to use less salt next time. The cook has controlled at three different moments. Managers do the same, and each moment has a name.

Feedforward control.
Control that acts before work begins, to prevent problems.

Feedforward control acts before the work begins, to prevent problems.

The first is **feedforward control**. Feedforward control acts before work begins, to prevent problems. It checks the inputs: people, materials, money and plans. Training new employees before they serve customers is an example, and so is checking that an order is within budget before it is placed. Its strength is that a problem prevented costs nothing to repair. Its weakness is that it needs good forecasts, and a forecast can be wrong.

Concurrent control.
Control that acts while the work is being done.

Concurrent control acts while the work is going on.

The second is **concurrent control**. Concurrent control acts while the work is being done. It checks the process as it happens, and it corrects at once. A supervisor who watches a cashier in the first week and gives help on the spot is using it. A security tag that sounds an alarm as a book leaves the shop is another example, since it acts during the event. Its strength is speed. Its weakness is cost, because someone or something must watch all the time.

Feedback control. Control that acts after the work is done, using the results to correct later work.

Feedback control acts after the work is done and uses the results to improve later work.

The third is **feedback control**. Feedback control acts after the work is done, using the results to correct later work. The year-end stock count of the case file is of this kind, and so is a review of last month's sales. It is the easiest control to apply, because results are plain to see. But it has a clear limit: it cannot undo what has already happened. A loss found at year end stays lost, and only future losses can be reduced.

The table compares the three types on the moment, the target and the limit.

The table below compares the three types. Read each row across: the first cell names the type, and the other cells say when it acts, what it checks and what limits it.

THREE TYPES OF CONTROL

TYPE	WHEN IT ACTS	WHAT IT CHECKS	MAIN LIMIT
Feedforward	Before work	Inputs and plans	Depends on forecasts
Concurrent	During work	The process	Costs constant attention
Feedback	After work	Results	Too late to undo the past

The choice among the types

How should a manager choose? When a mistake is costly or cannot

mixture: the year-end count is feedback, the tags act during the event, and the weekly count of a sample gives feedback often enough to behave almost like a live check.

PAUSE AND CHECK 18.P2 A manager reads last month's complaints and changes the rota. Is this feedforward, concurrent or feedback control? Give one reason.

MONITORING

Monitoring means watching progress regularly through reports, visits and counts, so that deviations appear early. It is not a fourth type of control but a habit that supports all three. A weekly report on sales, or a short daily meeting at the shop door, are forms of monitoring.

Before moving on. Feedforward control acts before work begins, to prevent problems. Concurrent control acts while the work is being done. Feedback control acts after the work is done, using the results to correct later work. Organisations use all three, with monitoring to keep watch.

Monitoring is regular watching of progress, and it supports all three types.

Before moving on: control acts before work (feedforward), during work (concurrent) or after work (feedback), and monitoring supports all three.

* * *

18.4 MAKING CONTROL EFFECTIVE

REQUIREMENTS OF EFFECTIVE CONTROL

For control to help, it must be timely, so that the information arrives while action is still useful. It must be accurate, because wrong figures lead to wrong actions. It must be understood by the people who are checked, and it must be economical, costing less than it saves. It should also focus on key points, and it should suit the plan and the people it serves.

Effective control meets a short list of requirements.

BARRIERS TO CONTROL

Several barriers stand in the way. Employees may resist, because they fear blame or feel distrusted. Some results are hard to measure, such as the quality of service or the spirit of a team. Measuring everything is costly and can slow the work. And too much control can harm: when every step is checked, employees may stop thinking for themselves and wait for orders.

People and costs can stop control from working.

POWER AND CONTROL

A person who checks the work of others holds power: the power to praise, to blame and to correct. Power is a separate term, studied elsewhere in this book. The danger is that one person controls too much alone. If the same person does the work and also checks it, no one else can see a mistake or a deceit. For this reason good organisations separate the doing from the checking.

Control gives power to those who check, and that power needs its own limits.

Timing deserves a closer look. A shop that learns of a shortage in the first week can search the till records, ask the staff and check the door while memories are fresh. A shop that learns in the twelfth month can do none of these. The same figure has very different value on different days, and so the speed of reporting is part of the

Timing matters because information loses value as it ages.

WORKED CASE: CLASSIFYING THE CONTROLS AT MEGHNA AND SONS

The worked case applies the three types to the control record.

Each action is placed by asking when it acts.

Rates give a fair comparison between the standard and the result.

The conclusion shows a balanced system with one gap.

Situation. Case file, illustrative. The control record of Meghna and Sons shows: a stock loss standard of 1% (Tk 35,000 of Tk 3,500,000); an actual 2025 loss of Tk 56,000, which is 1.6%; security tags at Tk 30,000 fitted in 2026; and a weekly count of 200 titles. **Question.** Place each action as feedforward, concurrent or feedback control, and say what is still missing.

Reasoning. The standard of 1%, set in advance, guides decisions before the year begins, and in that sense it belongs with feedforward control. The year-end figure of Tk 56,000 is a result found after the work, so it is feedback. The tags act at the moment an item leaves the shop, so they are concurrent. The weekly count of 200 titles comes after each week's trading, so it is feedback, but it is frequent and so gives early signals.

Notice that the record states both taka amounts and rates. The rate (1.6% against 1%) compares loss with the value of stock, so the result can be judged fairly in a year when stock is larger or smaller. The taka amount (Tk 21,000 above the standard) shows how much money is at stake.

Conclusion. The record has all three moments, but its feedforward element is thin: it has a standard and no preventive step, such as training staff in stock handling before they work. A shop that adds one would stop some losses before they begin. **Check.** The deviation was Tk 56,000 – Tk 35,000 = Tk 21,000, and each classification follows from when the action acts.

* * *

SUMMARY

The summary gathers the main ideas.

Controlling is checking results against plans and taking action to correct any difference. The control process has four steps: set a standard, measure performance, compare it with the standard, and correct. A standard is a level of performance set in advance as the measure of what is acceptable. Feedforward control acts before work, concurrent control acts during work, and feedback control acts after work. Effective control is timely, accurate, understood and economical, and it needs checks on those who check.

THINK IT THROUGH

18.T1 Meghna and Sons lost Tk 56,000 against a standard of Tk 35,000. Find the deviation. If the tags remove Tk 15,000 of the yearly loss, how many years do they take to repay their Tk 30,000 cost?

18.T2 A shop owner asks every employee to write down each action in a daily log, and checks the log every evening. Describe how this control could go too far, and one change that keeps the benefit.

Some students believe that any difference from the standard must be corrected at once. A small, temporary deviation may need only watching. A manager who reacts to every small change wastes effort and may unsettle the employees.

Some students believe that measuring is the whole of control. Without action to correct a deviation, a check changes nothing.

Some students believe that feedback control is enough because its figures are exact. It arrives after the loss, and it cannot undo it. Feedforward and concurrent control act earlier.

Some students believe that checking everything is wise. Control costs money and can discourage employees. The aim is control that costs less than the loss it prevents.

Another error is to think that every deviation needs a strong response.

Another error is to think that a check alone is enough.

A third error is to think that feedback control is the best because it is accurate.

A last error is to think that more control is always better.

IN HISTORY

Barings was Britain's oldest merchant bank, founded in 1762. In 1992 it sent a young trader, Nick Leeson, to run its futures trading in Singapore. He also oversaw the records of his own trades, so the person who traded was also the person who reported the trades. When a discrepancy in the accounts was raised in January 1995, it was Leeson who gave the explanation, and for a time it was accepted.

By February 1995 the hidden losses amounted to about £827 million, more than the bank's capital. Barings went into administration on 26 February 1995, and the Dutch group ING bought it on 6 March for one pound, taking on its debts. Leeson was sentenced in Singapore in December 1995 to six and a half years in prison. A Bank of England inquiry reported in July 1995 and found failings by management and weak control. The lesson for this chapter is plain: where doing and checking sit in one pair of hands, no standard, report or comparison can protect the organisation.

Barings Bank, in 1995, shows how weak control can destroy a strong organisation.

The losses grew in secret until the bank collapsed on 26 February 1995.

REVIEW QUESTIONS

- 18.1 Define controlling and explain its two halves.
- 18.2 Name the four steps of the control process in order, and say what each does.
- 18.3 Define standard, and give a standard in quantity, in time and in cost for a bookshop.
- 18.4 A stall has stock worth Tk 200,000 and a standard loss of 1.5%. The actual loss is Tk 4,200. Find the standard in taka and the deviation.
- 18.5 Place each action as feedforward, concurrent or feedback control: training a new cashier before the first day; watching a new cashier at the till; reviewing last month's sales figures; an alarm on a door.
- 18.6 Give one strength and one limit of each of the three types of control.
- 18.7 State three requirements of effective control and two barriers to control.

GLOSSARY

Accountability. The duty to explain and answer for results to the person who delegated the work. It stays with the person who must answer, even when the task has been handed on. For example, a store manager reports returns to the owner. (*Chapter XIV*)

Alternative. One of several possible courses of action from which a choice is made. A manager lists alternatives and weighs each one before choosing. Example: order all the books at once, order half now and half later, or order only what schools have promised. (*Chapter VII*)

Authority. The right to give orders and to expect them to be obeyed, given by position. It belongs to the post and passes to each new holder. For example, a store manager may order a till count. (*Chapter XIV*)

Budget. A plan that states expected income and spending in money terms for a period. It turns plans into figures. Example: a book fair budget with planned spending of Tk 240,000 and expected sales of Tk 900,000. (*Chapter VIII*)

Bureaucracy. A system of management built on written rules, clear levels of authority and appointment by skill. Max Weber described it as a design for fairness. Example: a shop with a written rule list and clear roles for each employee. (*Chapter IV*)

Business ethics. The moral rules that decide what is right and wrong in business conduct. A moral rule is a belief about how people ought to treat one another, such as "do not deceive". Ethics asks for more than the law requires. (*Chapter VI*)

Centralization. Keeping most decision-making power at the top of the organisation. Its opposite is decentralization. For example, an owner who approves every return and every order runs a centralized shop. (*Chapter XIV*)

Certainty. A condition in which the result of each alternative is known in advance. A manager can compare the known results and choose the best. For example, a shop that signs a lease knows its rent of Tk 38,000 a month. (*Chapter X*)

Conceptual skill. The ability to see the organisation as a whole and to understand how its parts connect. It is about working with ideas and with the links among them. A manager who sees that late books delay school orders and harm the shop's name shows conceptual skill. (*Chapter II*)

Concurrent control. Control that acts while the work is being done. It checks the process as it happens and corrects at once, but it costs constant attention. For example, a supervisor helping a new cashier at the till, or an alarm that sounds as an item leaves the shop. (*Chapter XVIII*)

Contingency approach. The view that the best way to manage depends on the situation. It grew from the 1960s. Example: a fixed method suits unloading cartons, while a flexible style may suit a team inventing a new service. (*Chapter IV*)

and meetings. For example, a weekly Sunday meeting lets a shop share news before the week begins. (*Chapter XIV*)

Corporate social responsibility. An organisation's duty to act for the good of society as well as for profit. It applies to every organisation, large or small. A bookshop that pays its employees fairly and recycles its waste paper is acting on this duty. (*Chapter VI*)

Creativity. The ability to produce new and useful ideas. Both words matter: an idea that is new but useless is a curiosity, and an idea that is useful but not new is a copy. A quiet corner where customers try a book before buying is an example. (*Chapter XV*)

Decision criterion. A standard used to judge which alternative is best. A criterion is a test that every option must pass or be scored against. For example, a student sets two criteria for a journey: arrive by 8:55 and spend no more than Tk 50. (*Chapter X*)

Decision making. Choosing one course of action from several, in order to solve a problem or use an opportunity. It begins when a manager sees a gap to close or a chance to use. For example, a shop owner picks one site for a second shop from two possible sites. (*Chapter X*)

Decision support system. A computer-based tool that helps managers study data and compare alternatives. It stores data, analyses it and answers what-if questions. It supports the manager and does not decide. For example, sales software that lists the top 20 titles each week helps a shop plan its stock. (*Chapter XI*)

Decision tree. A diagram of choices and chance events, drawn from left to right as branches. Squares mark choices, circles mark chance events, and the branch ends show results in taka. We calculate from right to left, finding the expected value at each circle, and then choose the highest value at the square. (*Chapter XI*)

Delegation. Giving an employee the right to make decisions and take action for a stated task. The task has limits, and the manager still checks the result. For example, an owner lets a store manager approve returns up to Tk 5,000. (*Chapter XIV*)

Departmentalization. The grouping of jobs into departments according to a common basis, such as function, product, customer or territory. A department is a group of jobs under one manager. A shop may group by function into sales, stock and accounts. (*Chapter XIII*)

Division of labour. The dividing of a large job into smaller tasks done by different people. Each worker becomes skilled in a narrow task, but the divided tasks must be joined again, which creates the need for coordination. (*Chapter XIII*)

Effectiveness. How far an organisation reaches its goals, whatever the cost. It asks whether the goal was met and ignores the input used. Example: a cook who serves all 200 guests is effective, even if food was wasted. (*Chapter III*)

Efficiency. How little input an organisation uses to reach a given result. For the same result, less input means higher efficiency. Example: relabelled shelves cut the daily search for stock from 3 hours to 1 hour. (*Chapter III*)

Environmental scanning. The regular collection and study of information about outside changes that could affect an organisation. A shop owner who reads rivals' online prices every week is scanning. Its value lies in meeting change early, when many responses remain open. (*Chapter V*)

Ethical dilemma. A choice in which every option has a cost to some person's rights or interests. For example, a shop owner may have to dismiss one of two employees or cut the pay of both. It differs from a temptation, where one option is plainly wrong. (*Chapter VI*)

Expected value. The average result of a choice, found by multiplying each possible result by its probability and adding the products. Gains count as positive and losses as negative. It need not equal any single result. For example, a game that pays Tk 100 or costs Tk 20 with equal chances has an expected value of Tk 40. (*Chapter XI*)

External environment. All the forces outside an organisation that affect it, such as customers, rivals, laws and technology. Managers cannot command these forces, but they can study them, prepare for them and sometimes influence them. (*Chapter V*)

Feedback control. Control that acts after the work is done, using the results to correct later work. It is easy to apply because results are known, but it cannot undo what has happened. For example, a year-end stock count. (*Chapter XVIII*)

Feedforward control. Control that acts before work begins, to prevent problems. It checks inputs such as people, materials, money and plans. Its strength is prevention, and its weakness is reliance on forecasts. For example, training a new cashier before the first day. (*Chapter XVIII*)

First-line manager. A manager who directly supervises employees doing the day-to-day work. The people supervised are not themselves managers, so this level is closest to the daily work and to the customer. The supervisor of the cashiers at a bank counter is a first-line manager. (*Chapter II*)

Flat structure. A structure with few levels and wide spans of management. Messages travel quickly and employees have more freedom, but each manager may be overloaded when the span is too wide. (*Chapter XIII*)

Forecast. A reasoned estimate of what will happen in the future, used as a base for a plan. It rests on evidence such as past records. Example: estimating next season's sales from the sales of the last three seasons. (*Chapter VII*)

Formal organization. The official structure of jobs, rules and lines of authority that management sets up on purpose. It is usually written down in job descriptions, rules and the organization chart, and it gives order and stability. (*Chapter XII*)

Function of management. One of the four main kinds of work managers do: planning, organising, leading or controlling. The word means a main kind of work. Example: arranging who does which job is the function of organising. (*Chapter III*)

member who breaks it may be laughed at or left out. Norms can help or harm the organisation. (*Chapter XV*)

Human factors. The traits, feelings and relationships of people that affect how they work. A trait is a lasting feature, such as patience. A feeling is a state of mind, such as worry. A relationship is a link between people, such as friendship. They act on the person, the group and the organisation. (*Chapter XV*)

Human relations school. A school of thought holding that workers' feelings and social bonds strongly affect output. It grew from the Hawthorne studies. Example: an employee who feels respected by the group may work harder, even when pay is the same. (*Chapter IV*)

Human skill. The ability to work with, understand and motivate other people. It includes listening and explaining. A manager needs it at every level, because results come through people. A supervisor who notices a tired cashier and explains a new rule well shows human skill. (*Chapter II*)

Incentive. A reward offered to encourage a person to act in a wanted way. It comes from outside the person and works only when it meets a need the person feels. For example, a bonus of one month's pay for meeting a sales target. (*Chapter XVI*)

Informal organization. The network of friendships and social groups that forms among employees on its own. No manager designs it and no document records it. A lunch group that shares book tips is an example. It can help or harm the organisation. (*Chapter XII*)

Innovation. Turning a new idea into a new product, service or method that is put to use. Creativity is the seed and innovation is the crop. A suggestion on paper is creativity; a reading corner built and used by customers is innovation. (*Chapter XV*)

Internal environment. The conditions inside an organisation that its managers can change directly, such as its people, its structure and its culture. When a shop hires a new employee or changes who works the counter, it is changing its internal environment. (*Chapter V*)

Key result area. A part of the organisation's work where results are vital to success. A bookshop might name sales, stock, customer service, returns and profit as its key result areas, and set objectives in each. (*Chapter IX*)

Leadership. The process of influencing people to work willingly toward group goals. It involves a leader, followers and a situation, and it does not depend on a title. Example: a friend who persuades five others to share the costs of a trip. (*Chapter XVII*)

Leadership style. The usual way a leader makes decisions and treats followers. It is a habit and not a rule. Example: a leader who nearly always decides alone and then informs the group has a different style from one who nearly always asks first. (*Chapter XVII*)

Line authority. The right of a manager to direct the work of employees in the chain of command. Orders travel down the line of positions from the top. For example, a store manager directs the

Management by objectives. A system in which managers and employees agree objectives together, then judge performance against them. Its cycle runs from agreeing objectives, to acting with self-control, to review and appraisal, and back to agreement. Peter Drucker described it in 1954. (*Chapter IX*)

Management process. The continuing cycle of planning, organising, leading and controlling that managers repeat. When one round ends, the information it produces feeds the next round. Example: a shop owner sets a target, arranges jobs, guides employees, checks results and then plans again. (*Chapter III*)

Management science approach. The use of mathematical models and data to solve management problems. It grew after the Second World War. Example: using past sales data to decide how many copies of a title to order. (*Chapter IV*)

Manager. A person who is responsible for the work of other people and for the results of their work. A manager does not only do tasks personally, but sees that others do them well. The person who directs the sales assistants and answers for the shop's sales is a manager. (*Chapter I*)

Middle manager. A manager between top and first-line managers, who turns the direction into plans for departments. The middle manager receives a direction and returns a plan. The manager of a bank's loans department, who decides how many loan officers are needed, is a middle manager. (*Chapter II*)

Motivation. The inner push that makes a person choose to act and keep acting toward a goal. It is inside the person, so managers see only the behaviour. For example, an assistant who tidies shelves each day to earn trust shows strong motivation. (*Chapter XVI*)

Need. A lack that a person feels and wants to fill. A need creates tension, called a drive, that pushes the person to act. Needs differ between people. For example, one employee lacks income and another lacks respect. (*Chapter XVI*)

Need hierarchy. Maslow's idea that human needs rank in five levels: physiological, safety, social, esteem and self-actualisation. An unmet lower need pushes most strongly, and a higher need matters only after the lower ones are met. The order is a guide, not a fixed law. (*Chapter XVI*)

Objective. A specific result that an organisation or person intends to reach within a stated time. It names a result, a measure, a number and a date. Example: selling 500 copies of one textbook before 30 June. (*Chapter IX*)

Organisation. A group of people working together in a planned way to reach goals they could not reach alone. The group has a purpose, shared tasks and some rules. A bookshop with its owner and staff is an organisation, and so are a school and a hospital. This book uses one word for all such groups. (*Chapter I*)

Organization chart. A diagram that shows the jobs in an organisation and who reports to whom. Boxes stand for jobs and lines stand for reporting lines. It records the formal structure at one moment and can become out of date. (*Chapter VII*)

accounts is organizing. It follows planning in the management process. (*Chapter XII*)

Performance appraisal. A review of how well an employee has done against agreed standards. In management by objectives, the standard is the objective agreed at the start. Example: an employee who agreed 14 training hours and received 10 is told the gap. (*Chapter IX*)

Plan. A written or spoken statement of what will be done, by whom and by when, to reach an objective. A plan names the action, the person and the time. Example: "Place the book order by 12 December; the manager will do it." (*Chapter VII*)

Planning premise. An assumption about future conditions on which a plan is built. The manager writes it down so that everyone shares it. Example: a shop assumes that publishers raise prices by 5%, and builds its order budget on that assumption. (*Chapter VII*)

Policy. A general guide for decisions, setting limits within which choices are made. It guides but leaves room for judgement. Example: "We exchange books within 7 days", where the employee decides about a damaged book on the last day. (*Chapter VIII*)

Power. The ability to influence other people's behaviour, whether or not one holds a position. It comes from the person, through knowledge, rewards, punishment or liking, and leaves with the person. For example, a trusted colleague persuades others to follow a plan. (*Chapter XIV*)

Principle of management. A broad guide to action drawn from experience, to be adapted and not applied like a law. It may suit one organisation and fit another badly. Example: dividing work among specialists suits a factory more than a shop of seven people. (*Chapter III*)

Procedure. A fixed series of steps to be followed for a repeated task. It says how to do the work, in a fixed order. Example: ask for the receipt, check the date, check the book, give the new book, record the exchange. (*Chapter VIII*)

Productivity. The amount of output an organisation obtains for each unit of input it uses. It equals output divided by input. Example: 41,000 copies sold by 7 employees is 5,857 copies per employee. (*Chapter III*)

Profession. An occupation that requires long training, a body of knowledge and a code of conduct enforced by an association. Example: medicine, where doctors normally register with a medical council. Managing a small shop does not require a licence. (*Chapter III*)

Referent power. Power that comes from other people's liking and admiration for a leader. It belongs to the relationship, not the position. Example: a sales assistant whom colleagues respect and wish to resemble can influence them without any reward. (*Chapter XVII*)

Resource. Anything an organisation uses to reach its goals, such as people, money, materials and information. A bookshop's resources include its books, cash, shelves, staff and list of school customers.

Risk. A condition in which the result is not known but its chances can be estimated, usually from past records. The result may be good or bad. For example, a stall owner whose records show rain on 4 of 10 days estimates the chance of rain as 0.4. (*Chapter X*)

Scientific management. The study of work by measurement, aimed at finding the one best method and training workers in it. Frederick Taylor began it in the 1900s. Example: timing several ways of unloading a carton and choosing the fastest. (*Chapter IV*)

Single-use plan. A plan made for one occasion or project and not used again. It has a start and an end. Example: the plan for a 28-day book fair stall, which will not be repeated in the same form. (*Chapter VIII*)

Span of management. The number of employees who report directly to one manager. Only direct reports count. A store manager with three sales assistants, a cashier and a stock clerk has a span of five. The best span depends on skill, nearness and the kind of work. (*Chapter XIII*)

Staff authority. The right to advise and support line managers, without commanding their employees. The line manager still decides. For example, an accounts adviser tells a store manager that a school's credit is too high. (*Chapter XIV*)

Standard. A level of performance set in advance as the measure of what is acceptable. It can be stated in quantity, quality, time or cost. A good standard is measurable and fair. For example, a stock loss of no more than 1% of stock value. (*Chapter XVIII*)

Standing plan. A plan made once and used again and again for situations that repeat. It saves thought and gives fair, equal treatment. Example: a shop's returns policy, applied to every customer who asks to exchange a book. (*Chapter VIII*)

Sustainability. Running an organisation so that it meets today's needs without harming later generations' ability to meet theirs. It covers the natural environment, people and the organisation's own money. A farmer who plants seedlings after picking fruit follows the same rule. (*Chapter VI*)

Tall structure. A structure with many levels and small spans of management. Close supervision is possible, but messages pass through many levels and the many managers raise the cost of wages. (*Chapter XIII*)

Task environment. The outside groups that deal directly with an organisation, such as its customers, suppliers and rivals. For a bookshop, the schools that order books and the publishers that supply them belong to its task environment. (*Chapter V*)

Team. A small group of people with joined skills, working toward a common goal for which all are accountable. Accountable means answerable for the result. Staff who prepare a school order together form a team; colleagues who only eat lunch together are a group. (*Chapter XV*)

Technical skill. The ability to use the tools, methods and knowledge of a specific job. It is mainly about working with things and

and punishment. For example, a clock-in system for every employee. (*Chapter XVI*)

Theory Y. A set of assumptions, named by Douglas McGregor in 1960, that people can enjoy work and seek responsibility. A manager holding it tends to share goals and delegate. For example, offering training for a supervisor post. (*Chapter XVI*)

Time and motion study. Timing each movement of a task to remove wasted effort and set a fair standard. The tool of scientific management. Example: timing a carton at 14 minutes, changing the method, and timing it again at 9 minutes. (*Chapter IV*)

Top manager. A manager at the highest level, who sets the direction of the whole organisation. The top manager decides what the organisation will do and what it will not do. In a bank this is the managing director, and in a small family shop it is the owner-manager. (*Chapter II*)

Transformational leadership. Leading by inspiring followers to look beyond self-interest and to change themselves and the organisation. It differs from a leader who trades pay or praise for work done. Example: an owner who explains why good service matters to the neighbourhood. (*Chapter XVII*)

Triple bottom line. A way of judging an organisation by its profit, its effect on people and its effect on the natural environment. John Elkington proposed it in 1994. It adds two more "last lines" to the single last line of the accounts, which shows profit. (*Chapter VI*)

Uncertainty. A condition in which neither the results nor their chances can be estimated, often because no records or comparisons exist. A manager must rely on judgement. For example, a stall owner opening in a town with no records cannot say what sales to expect. (*Chapter X*)

ANSWERS TO QUESTIONS

CHAPTER I · WHAT MANAGEMENT IS

1.p1 Planning: deciding in advance what to cook and buy. Organising: sharing the tasks among family members. Leading: guiding and encouraging the members. Controlling: checking at the month's end whether the money lasted, and correcting any difference.

1.p2 Yes. The owner employs two drivers, tells each one what to do and answers for the results of their work, which matches the definition of a manager.

1.t1 Nurul Meghna, the owner-manager of a shop with seven staff. The resources are stock Tk 3,500,000, fittings Tk 600,000 and the seven staff.

1.t2 Rafiq answers for the assistants' work that week, because a manager answers for the work of the people he directs. Nurul remains the owner.

1.t3 The stall still has resources and goals. Its owner must decide what to buy, arrange the stall and check the takings, which is management on a very small scale. Only the leading of other people is missing.

1.t4 The organisation is the club. Two resources are the members and the fund (or the ground). The manager is the person who is responsible for the members' work and its results, such as the person chosen to direct the club.

1.1 Doctors and nurses (people), money from fees or public funds (money), medicines and equipment (materials), patient records (information).

1.2 Responsibility is a duty to perform a task and to answer for how well it is done. Authority is the right to give orders and make decisions.

1.3 Resources are limited, people differ, conditions change and tasks depend on one another, so someone must direct resources toward the organisation's goals.

CHAPTER II · MANAGERS AND THEIR SKILLS

2.p1 Nurul works at the top level. Rafiq's weekly stock plan is middle-level work, and his direct supervision of five employees is first-line work.

2.p2 Human skill. The manager is working with and understanding a person, and giving the employee a reason to speak and to keep learning.

2.t1 Nurul works at the top level. Rafiq stands at the middle level (weekly stock plan, supervising 5 staff) and at the first-line level (directly supervising 3 sales assistants, 1 cashier and 1 stock clerk).

2.t2 The new manager stands at the middle level, below Nurul, and is a general manager because the manager answers for a whole shop. If the manager directly supervises employees, the work also stands at the first-line level.

parts connect becomes the main ability. A first-line manager sees the work directly.

2.3 People (the sales assistants), money (the day's takings), materials (stock and shelves), information (sales records or the list of school customers).

2.4 A functional manager is in charge of one kind of work, such as the accounts department. A general manager is in charge of several kinds of work together, such as the head of a whole branch.

CHAPTER III · THE FUNCTIONS OF MANAGEMENT AND HOW MANAGEMENT IS JUDGED

3.p1 Week 1: $600 \div 3 = 200$ books per employee. Week 2: $800 \div 5 = 160$. Productivity fell, although the total rose.

3.p2 It is effective, because every planned parcel was delivered. It is not efficient, because it used twice the budget.

3.t1 The firm reached its goal of 300 parcels, so it was effective. The vans used twice the distance needed, so input was wasted and it was not efficient. Better planning of routes would help most.

3.t2 $45,100 \div 8 = 5,637.5$, so about 5,638 copies per employee. This is lower than the 5,857 of 2025, because input grew while output stayed at the forecast level.

3.1 Planning, organising, leading, controlling. Examples: set the sales target; assign jobs to the 7 employees; encourage the sales assistants; compare actual sales with the target.

3.2 Before: $1,800 \div 4 = 450$ loaves per baker. After: $2,100 \div 4 = 525$. Change: $75 \div 450 = 0.1667$, a rise of about 16.7%.

3.3 Productivity is a ratio. If input grows faster than output, the ratio falls. Example: 600 books with 3 employees is 200 each; 800 books with 5 employees is 160 each.

3.4 Effectiveness is how far goals are reached, whatever the cost. Efficiency is how little input a result needs. A firm that does a wrong job very cheaply is efficient but not effective.

3.5 Long training, a body of knowledge, and a code of conduct enforced by an association. For a small shop, no licence or governing body is required, so the enforced code and association are missing.

3.6 A law of physics holds every time. A principle is drawn from experience and must be adapted, because it may suit one organisation and fit another badly.

CHAPTER IV · THE EVOLUTION OF MANAGEMENT THOUGHT

4.p1 Unity of command is broken, because the employee has two managers giving different orders. The shop should give each employee one manager, and let the two managers agree on the work before they speak to the employee.

4.t1 The first principle: replace rule-of-thumb methods with a science. The fixed shelf-position list replaced habit with one tested method, and the timing showed that it saved 5 minutes a carton.

4.t2 The saving per carton would be $14 - 11 = 3$ minutes. For 10 car-

study for repeated tasks and the human relations school for team spirit in the rush.

4.t4 Timing every task can feel like constant watching, which weakens team spirit. A manager can explain the reason for the timing, share the saving with employees, and ask them to suggest better methods.

4.1 Taylor: one best method found by timing. Fayol: fourteen principles for the whole organisation. Weber: written rules and levels of authority. Mayo: group bonds affect output.

4.2 Replace rule of thumb with a science for each job; select, train and develop each employee; cooperate with employees so work follows the science; divide work and responsibility between managers and employees.

4.3 The contingency approach is the view that the best way to manage depends on the situation. Example: a fixed method suits unloading cartons, while a flexible style suits a team inventing a new service.

4.4 Structure and command (for example, unity of command); fairness and people (for example, equity); order and purpose (for example, discipline).

4.5 Strength: it solves problems that can be measured, such as how many copies to order. Limit: it cannot measure feelings or judgement.

CHAPTER V · THE ENVIRONMENT OF ORGANISATIONS

5.p1 External. The owner of Meghna and Sons cannot direct a rival, so the change lies outside the boundary.

5.t1 External, because the shop cannot direct the school. It fits the task environment, since the school is a customer, and it also reaches the shop through the political and legal group if an education authority required the change.

5.t2 The stock clerk is an employee of the shop, so a proposal from the clerk is an internal matter that managers can accept or refuse directly. The school's later choice would still be external. The boundary follows control.

5.t3 Political and legal group, and outside the boundary because the shop cannot command a government. Adapting: change the price list or the range of books kept. Influencing: join a trade association and present the shop's case to the public body, honestly and lawfully.

5.t4 The departure of 2 of 5 nurses is internal. The new clinic is external, in the task environment as a rival. The new rule is external, in the political and legal group. Watching them is scanning, which supports planning, the first function of the management process, because plans must fit the conditions in which they will be used.

5.1 A rival's price cut: external. A new store manager: internal. A change in school book rules: external. The shop's own opening hours: internal.

5.4 Scanning is regularly collecting and studying information about outside change. Adapting is changing the organisation to fit an outside force. Influencing is acting on the outside force itself. The price-match list is adapting, because the shop changed its own prices.

CHAPTER VI · SOCIAL RESPONSIBILITY, ETHICS AND SUSTAINABILITY

6.p1 (a) is an ethical dilemma, because both options cost someone: dismissing one employee harms that person, and cutting both pays harms both. (b) is only a temptation, because the right answer is known and only the will is tested.

6.t1 The people affected are the employees, the owner, the customer who ordered, and the families of the employees. Step 2 says power creates a duty of equal size, and the owner has power over the employees' safety. Step 4 says the cost of danger must be counted, not left with the employees. The owner should refuse the unsafe night shift and ask the customer for more time.

6.t2 Profit: Tk 750,000 in 2025. People: 7 employees, which is a first fact only. Natural environment: 380 kg of unsold paper sent to recycling in 2025. Two missing facts are, for example, measures of pay and of safety for the employees, and a measure of energy or waste use.

6.1 Corporate social responsibility is an organisation's duty to act for the good of society as well as for profit. Example: a bookshop refuses to sell stolen books and sends waste paper to recycling.

6.2 For: society gives organisations permission to operate, so harm to society returns as lost trust. Against: the money belongs to the owners, who did not choose to spend it on society. Owners would more likely make the argument against; a manager or an employee would more likely make the argument for.

6.3 The steps are: 1 social power; 2 responsibility matches power; 3 open dealing with society; 4 social costs counted and priced; 5 help with social problems. The iron law warns that those who do not use power in a way society considers responsible will tend to lose it.

6.4 In a dilemma every option has a cost to someone, such as dismissing one of two employees or cutting both pays. In a temptation one option is plainly wrong and only the will is tested, such as selling stolen books for profit.

6.5 (a) Both: the law requires it and it is right. (b) Neither: it is forbidden and wrong. (c) Ethical but not required by law: it is a kind act that no law demands.

6.6 The rival puts employees and customers at risk, and society bears the cost if harm occurs, so Davis step 4 is ignored. The rival also risks its good name and a stricter law. A manager at our shop should keep our safety standard, not copy the rival, and be ready to explain why to customers.

CHAPTER VII · PLANNING: MEANING, NATURE AND STEPS

7.t2 The order costs Tk 216,000 more (5% of Tk 4,320,000). The manager should have forecast the price change, stated it as a premise, and weighed alternatives such as ordering earlier or ordering part of the stock.

7.t3 Awareness: what chance or problem do we see? Objectives: what exact result, by what date? Premises: what do we assume about customers and costs? Alternatives: what other ways could we reach the aim? Choosing and carrying out: who does what, with what money and by when?

7.t4 Two alternatives: hold the sports day indoors, or fix a spare dry date. The indoor option removes the weather risk but limits the activities; the spare date keeps the activities but needs a second booking and a second notice to families.

7.1 Awareness of opportunity (notice a chance or problem); setting objectives (state the exact result and date); planning premises (write down what is assumed); listing and weighing alternatives (compare the possible ways); choosing and carrying out (select one and put it into action).

7.2 A plan is a statement of what will be done, by whom and by when, to reach an objective; example: "Rafiq places the book order by 12 December". A planning premise is an assumption about future conditions on which a plan is built; example: publishers raise prices by 5%.

7.3 A forecast is a reasoned estimate of what will happen, such as next season's sales from past records. A planning premise is the assumption written into the plan; a forecast is a good source for a premise.

7.4 The supplier needs 21 days to deliver, so books ordered after the deadline arrive after the season starts. Waiting for empty shelves means customers go elsewhere.

7.5 Wrong premises: write them down and check them. Vague or unshared objectives: state the result exactly and tell every employee. Lack of commitment: name a person for each task and review progress. Other reasons are cost, rigidity and an unknown future.

CHAPTER VIII · KINDS OF PLANS AND MAKING PLANNING WORK

8.p1 A one-day sale to clear old stock is a single-use plan, because it is made for one occasion. If the shop held the same sale every year with the same steps, it would become a standing plan.

8.p2 Communication is weak, because the employees who carry out the plan do not know it. The owner should explain the plan to everyone affected, in writing, and answer their questions.

8.t1 The limit "within 7 days" decides the case, because day 8 is outside it. That part is a policy, so the manager has room for judgement, for example to help if the book was damaged. The manager should record the exception.

8.t2 The expected sales would exceed the spending by 600,000 – 240,000 = Tk 360,000. The event still seems worth planning, but the margin is smaller, and the cost of the books must be counted

8.1 Standing: returns policy and yearly budget (renewed in form each year). Single-use: book fair stall and the opening of a new branch.

8.2 A policy is a general guide that sets limits for choices (exchange within a set number of days). A procedure is a fixed series of steps for a repeated task (check receipt, check book, record exchange). A budget states expected income and spending in money for a period (the book fair budget of Tk 240,000).

8.3 A long-range plan looks several years ahead and a short-range plan a year or less. A strategic plan sets the main direction of the organisation; an operational plan states how one part does its daily work.

8.4 If senior managers ignore a plan, employees ignore it too. Support gives the plan money, time and example, so people follow it.

8.5 Review plans on a fixed date and check the premises; plan in proportion, with more detail where an error costs much; keep a margin of money or time.

CHAPTER IX · OBJECTIVES AND MANAGEMENT BY OBJECTIVES

9.p1 For example, sales and customer service. In sales, a shop that does not sell cannot survive; in customer service, poor service loses customers.

9.p2 A cycle suits MBO because results from one period teach what objectives to agree for the next. A straight line would end after one year and waste what was learned.

9.t1 Sample: (1) key result area: results; measure: share of pupils passing the year-end tests; target 85%; period: the next school year. (2) key result area: staff development; measure: hours of training per teacher; target 14 hours; period: the next school year.

9.t2 Two ways: objectives imposed by the head teacher, so teachers feel no ownership (fix: agree each objective in a meeting); results judged that a teacher cannot control (fix: agree objectives only for results the teacher can influence, and review them during the year).

9.1 An objective is a specific result that an organisation or person intends to reach within a stated time. An activity is something done, such as holding a sale; an objective is what is reached, such as selling 2,000 copies.

9.2 Sample: "Sales to schools of Tk 4,000,000 in 2026"; "Stock turnover of at least 3.2 times in 2026"; "12 hours of customer-service training per employee in 2026". Each names a measure, a target and a period.

9.3 Average stock = cost of sales ÷ turnover = 9,800,000 ÷ 3.2 = Tk 3,062,500. Check: 3,062,500 × 3.2 = 9,800,000.

9.4 (1) Organisation states objectives; (2) manager and employee agree individual objectives; (3) employee acts with self-control; (4) progress is reviewed; (5) performance appraisal; (6) results guide the next period. Performance appraisal is stage five.

10.p2 The price is certain, because the list is fixed for a year and the result is known. The number of customers is uncertainty, because nobody can say the result or estimate its chance. If past records of similar books existed, it would be risk.

10.t1 Supplier A's price is certainty. Supplier B's delivery times are uncertainty, since no records exist to estimate a chance. Step 1 and step 2 come first, but the manager learns what to find out at step 4, evaluating, where the missing delivery facts stop the test.

10.t2 The criteria stay the same, but the figures in them change: the expected sales fall, so criterion A (sales per taka spent) and criterion B (sales less rent) fall for Mirpur. Rent and fitting costs do not change. The owner should re-evaluate the options with the corrected figure and compare again.

10.1 Decision making is choosing one course of action from several, in order to solve a problem or use an opportunity. Example: a shop owner chooses one of two sites for a second shop.

10.2 A problem is a gap between what is and what should be, and an opportunity is a chance to do better than before. An opportunity is easier to miss, because nobody complains about a chance that was lost.

10.3 The steps are: name the problem or opportunity; set the criteria; develop the alternatives; evaluate them; choose; implement and follow up. Criteria come first so that they cannot be bent to favour the option already liked.

10.4 A decision criterion is a standard used to judge which alternative is best. Two examples are the price per copy and the days a supplier takes to deliver.

10.5 (a) Certainty: the lease fixes the amount. (b) Risk: past records allow the chance to be estimated at 4 in 10. (c) Uncertainty: no records exist, so neither the results nor their chances can be estimated.

10.6 It is risk, because the chance can be estimated from records. The estimated chance of a late order is 3 out of 12, which is $1/4$.

10.7 She must go back to step 2 and rank the criteria, because no alternative now passes both tests. She must decide whether cost or arrival time counts more.

CHAPTER XI · DECISION TOOLS AND GROUP DECISIONS

11.p1 $0.8 \times 500 = 400$ and $0.2 \times (-1,000) = -200$, so the expected value is $400 - 200 = \text{Tk } 200$.

11.t1 Worth it: an important, unclear decision that affects many people, or one that needs the support of employees to be carried out. Not worth it: an urgent or simple choice, or one where a single person holds both the knowledge and the authority.

11.t2 $0.4 \times 900,000 = 360,000$ and $0.6 \times (-300,000) = -180,000$, so the expected value is Tk 180,000. This is still above Tk 0, so opening still beats waiting on expected value.

11.t3 $900,000p - 300,000(1 - p) = 0$ gives $1,200,000p = 300,000$, so $p = 0.25$. Check: $0.25 \times 900,000 = 225,000$ and $0.75 \times 300,000 =$

11.3 $0.2 \times 100,000 = 20,000$; $0.5 \times 40,000 = 20,000$; $0.3 \times (-60,000) = -18,000$. The sum is Tk 22,000, and the probabilities add up to 1.

11.4 X: $0.6 \times 200,000 = 120,000$ and $0.4 \times (-100,000) = -40,000$, so Tk 80,000. Y is Tk 70,000. X is higher by Tk 10,000, but it can lose Tk 100,000, so a manager would also ask whether the firm can bear that loss.

11.5 A square shows a decision point and a circle shows a chance event. We draw the tree from left to right and calculate from right to left, which is called folding back.

11.6 It stores and analyses data quickly, and it answers what-if questions. One limit: wrong data give wrong answers, and the system supports the manager but does not decide.

11.7 Advantages: more information and ideas, and stronger support from those affected. Disadvantages: it is slower and uses the time of several employees; it can also let one member dominate or blur responsibility.

11.8 Any three of: time, cost, importance, the manager's values and attitude to risk, and authority.

11.9 A tree works when the possible results and their probabilities are known, which describes risk. Under certainty each choice has one known result, so the manager compares the results directly and needs no probabilities.

CHAPTER XII · ORGANIZING AND TYPES OF ORGANIZATION

12.p1 Step four, set the relationships, because it decides who answers to whom.

12.p2 Formal. The job description and the reporting line were set up by management and are written down.

12.t1 One grouping: stock tasks (ordering books) and counter tasks (serving customers, counting cash). The reason is that the counter tasks happen at the same place and use the same dealing with customers and money.

12.t2 It could join the group with ordering books if the delivery depends on stock, or it could join a group of tasks for customers away from the counter. Either answer is reasonable, which shows that grouping needs judgement guided by the purpose of organizing.

12.t3 Informal, because no manager set it up. The manager should be pleased that staff share useful information, and concerned only if private patient details are passed on carelessly or if rumour spreads.

12.t4 The chart may be out of date, with tasks that nobody owns and reporting lines that are too long. The informal organization may also grow in new groups that spread rumour, because the formal structure no longer links the larger staff.

12.1 Formal: the cashier's job description and the reporting line from stock clerk to store manager. Informal: the lunch group and the habit of sharing book tips at noon.

12.2 Identify the tasks (what work must be done?); group the tasks

12.4 People join to meet needs such as company, help, information and belonging. A manager can learn which groups exist, listen to them, share honest information and use their strengths, while watching for harm.

CHAPTER XIII · STRUCTURE: SPAN, DEPARTMENTS AND FORCES

13.p1 The tasks are similar (both routine, in one room) and the employees are near the manager. Either of these, or the small number of other duties, suggests a wider span.

13.p2 Territory, because each branch is a place with its own manager.

13.t1 Too wide: one owner cannot follow 12 people and answer all their questions. Acceptable: all 12 do the same counter work in one place, so the tasks are similar and near, and the rules may be clear.

13.t2 Taller, because a new level has been added. The owner's span falls from 12 to 2 and each manager has 6. A benefit is closer supervision; a cost is the added wages and slower messages through the extra level.

13.t3 Territory. Grouping by product would give each product its own department, and each department would work in all three cities, so the managers would be responsible for a product everywhere and not for a place.

13.t4 One reasonable answer: a span of about 3 to 5 for the senior doctor, since the clinic is small and all work in one place (size and nearness), and a grouping by function (doctors, nurses, reception) because the clinic is small and its work is routine. Other reasoned answers are acceptable.

13.1 His span is 4, because only the 4 report to him directly. The people below him number $4 + (4 \times 3) = 16$.

13.2 The accounts work takes his time, so he has less time to lead. A smaller span lets him give each employee enough attention.

13.3 Function (sales, stock, accounts); product (textbooks, story-books, stationery); customer (walk-in, schools, online); territory (a branch in each city).

13.4 A tall structure has many levels and small spans; a flat structure has few levels and wide spans. A cost of tall: slow messages and high cost of many managers. A cost of flat: managers may be overloaded.

CHAPTER XIV · AUTHORITY, DELEGATION AND COORDINATION

14.p1 The new manager holds the authority, because it belongs to the post. The ten-year employee holds power, because the other employees follow that person's influence. Power can outweigh authority when employees do not yet accept the new manager.

14.p2 Authority is missing. Rafiq has the duty but not the right to act, so every return waits for Nurul and the task cannot be done as delegated. The three ideas must be in balance.

14.t1 Tk 5,400 is above Rafiq's limit of Tk 5,000, so Nurul must approve it. Rafiq's authority covers returns up to Tk 5,000 only.

14.1 Authority is the right to give orders and expect them to be obeyed, given by position. Power is the ability to influence others, whether or not one holds a position. Authority stays with the post, while power leaves with the person.

14.2 (a) Line authority: it is a manager directing an employee in the chain of command. (b) Staff authority: it is advice and does not command. (c) Functional authority: a specialist commands in one subject only.

14.3 Delegation is giving an employee the right to make decisions and take action for a stated task. Responsibility, authority and accountability must be kept in balance.

14.4 Rafiq answers to Nurul for returns he approves, but Nurul still answers to the owners and customers for the shop as a whole. Handing over the task does not remove the answer owed upward.

14.5 Two reasons are fear of mistakes and fear of losing control. Remedies: match the task to the person and state limits plainly; accept early mistakes and check progress through reports and meetings.

14.6 Centralization is keeping most decision-making power at the top of the organisation. Conditions favouring decentralization include a large organisation and skilled, trusted employees, or a need for speed near the customer.

14.7 Coordination is linking the work of different people and departments so that they pull toward the same goal. Basic activities: shared goals, regular communication. Structural techniques: the chain of command, rules, liaison persons or meetings.

CHAPTER XV · HUMAN FACTORS, GROUPS AND CREATIVITY

15.p1 Observed fact: the employee arrived late twice. Added by the manager: the belief that the employee does not care about the job.

15.p2 Yes, it is an unwritten rule that members follow and expect others to follow. It probably helps, because it avoids quarrels over customers, though it could harm the shop if a waiting customer is left unserved.

15.t1 People differ in traits, abilities and attitudes. A smile shows one trait, and correct answers show another, so one observation is too little. The manager should observe over time and compare impressions with records of work.

15.t2 The norm is that all three assistants eat lunch together at the same time. The manager could explain the reason, ask the group to propose a rotation that keeps the counter covered, and respect the wish to eat together on some days.

15.t3 The habit is a quick answer that closes discussion. It teaches employees that suggestions are wasted, so creativity falls and innovation stops. One change: ask what has changed since the last attempt and give a reason for any refusal.

15.t4 Yes, if the three have different skills that fit together, share one goal (less waiting) and are answerable together for the result. The clinic must make the shared accountability clear, so that all

group example: colleagues at lunch. A team example: staff preparing a school order.

15.3 Stopping ideas: ridicule, saying we tried that before, or silence in reply. Encouraging ideas: listening, giving time and freedom, making failure safe, or providing a path from idea to action.

15.4 Creativity is producing a new and useful idea; innovation is putting the idea into use. The 14 suggestions show creativity, and the reading corner, built in April 2026 at Tk 45,000, shows an idea put into use.

CHAPTER XVI · MOTIVATION

16.p1 Answers vary. A good answer names a goal, such as a skill, a result or the respect of others, that kept the person working without orders.

16.p2 Esteem. The job is steady and safe, so levels 1 and 2 are met, but the clerk feels unnoticed and wants respect for the work.

16.t1 Possible answers: the first assistant may feel a need for esteem and belonging (known by regular customers), while the second may feel mainly a need for income and safety, which the pay already meets. The same pay meets one need and leaves the other untouched.

16.t2 Theory X: it assumes employees must be watched. A result could be distrust, or careless work that matches the low expectation. A Theory Y change: share the till targets with staff and give them responsibility for checking their own sales.

16.t3 Their lower needs are already met, so extra pay may remove no complaint and add little push. Incentives that touch esteem and growth, such as training, a senior title or responsibility for a project, may work better.

16.1 Motivation is the inner push that makes a person choose to act and keep acting toward a goal, such as an assistant who tidies shelves each day to be trusted. A need is a lack a person feels and wants to fill, such as the wish for more income.

16.2 Motivation is inside the person; an incentive is a reward offered from outside. An incentive fails when it does not meet a need the person feels, or when the goal seems out of reach.

16.3 Physiological, safety, social, esteem, self-actualisation. The order matters because the need pushing now is the one an incentive must meet.

16.4 Salary: physiological (and safety). Safety shoes: safety. Team outing: social. Title of senior sales assistant: esteem. Designing the window display: self-actualisation (and esteem).

16.5 Need (wish for income), drive (tension), action (working carefully toward the target), goal (the bonus), need met and tension falls. The loop may restart with a new need.

16.6 Hygiene factors are the conditions around the job (pay, rules, supervision, security); they prevent unhappiness. Motivators are in the work itself (achievement, recognition, responsibility, advancement); they build effort.

CHAPTER XVII · LEADERSHIP

17.p1 Point 5: the manager presents the problem, asks the employees for suggestions, and then makes the decision. The manager keeps the final decision, but only after hearing the group.

17.t1 Both parts are possible. She is a leader, because colleagues seek her advice and follow it willingly, which is influence without a position. She is not a manager unless she also holds a position of authority.

17.t2 In the emergency, a point near 1: time is short and danger is high, so the manager decides and announces. For a monthly meeting on the duty roster, a point near 5 or 6, because nurses know their own needs and time allows discussion.

17.1 Leadership is the process of influencing people to work willingly toward group goals. Its three parts are the leader, the followers and the situation.

17.2 Any two of: a manager holds a position, a leader holds influence; followers must obey a manager but choose to follow a leader; a leader needs no title. One person can be both when a manager also influences people to follow willingly.

17.3 Legitimate (position), reward, coercive (punishment), expert and referent power. Referent power comes from liking and admiration: an assistant whom colleagues respect and copy can influence them with no reward to offer.

17.4 (a) Autocratic: danger and short time. (b) Participative or free-rein: skilled people with knowledge the leader lacks. (c) Free-rein: experienced people with clear goals need little direction.

17.5 The trait approach looks for personal qualities of leaders; the situational approach says the right way to lead depends on leader, followers and situation. The continuum supports the situational approach, because no point on it is best everywhere.

CHAPTER XVIII · CONTROLLING

18.p1 Deviation = actual – standard = 9 – 5 = 4 spoiled cups above the standard. The manager should first find the cause (for example a machine fault or a new employee) before choosing the correction.

18.p2 Feedback control. It uses last month's results, after the work is done, to correct later work.

18.t1 Deviation = 56,000 – 35,000 = Tk 21,000 above the standard. Repayment = 30,000 ÷ 15,000 = 2 years.

18.t2 It could go too far by taking much of the employees' time and showing distrust, so they may write the log to please the owner and stop using their own judgement. A change that keeps the benefit: log only the key items, such as cash and stock, and check by sample.

18.t3 The same person does and checks the work, so an error or a theft can pass unseen. The risk falls if another person counts the cash, or if the owner checks the count.

18.3 A standard is a level of performance set in advance as the measure of what is acceptable. Examples: sell at least 120 books a day (quantity); serve each customer within 5 minutes (time); keep the stock loss to Tk 35,000 a year (cost).

18.4 Standard = $0.015 \times 200,000 = \text{Tk } 3,000$. Deviation = $4,200 - 3,000 = \text{Tk } 1,200$ above the standard.

18.5 Training before the first day: feedforward. Watching at the till: concurrent. Reviewing last month's sales: feedback. An alarm on a door: concurrent.

18.6 Feedforward: prevents problems, but depends on forecasts. Concurrent: corrects at once, but costs constant attention. Feedback: easy to apply since results are known, but too late to undo the past.

18.7 Any three of: timely, accurate, understood, economical, focused on key points. Barriers, any two of: resistance, results hard to measure, cost, too much control.

18.8 Planning needs controlling to learn whether the plan works. Controlling needs objectives to supply the standards against which results are compared.

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What does a manager actually do, and how is it done well? This treatise answers from the beginning. It sets out the nature and functions of management, then follows the manager's work through planning, organising, staffing, directing and controlling. Each chapter opens with a puzzle, closes with review questions, and needs no earlier knowledge.